Caring for the Landscapes of Aotearoa New Zealand

Synthesis Report

Raewyn Peart, Cordelia Woodhouse, Shay Schlaepfer,
Deidre Koolen-Bourke and Lara Taylor

EDS
ENVIRONMENTAL DEFENCE SOCIETY

Caring for the Landscapes of Aotearoa New Zealand

Synthesis Report

Raewyn Peart, Cordelia Woodhouse, Shay Schlaepfer, Deidre Koolen-Bourke and Lara Taylor

February 2021

First published February 2021

Published by:

Environmental Defence Society Incorporated
P O Box 91736 Victoria St West
Auckland 1142

Phone (09) 302 2972

manager@eds.org.nz

www.eds.org.nz

www.environmentguide.org.nz

ISBN 978-0-9951186-3-8

© Environmental Defence Society Incorporated 2021

Photographic images: Raewyn Peart unless otherwise indicated

Cover image: Kenepuru Sound

Copies can be downloaded from www.eds.org.nz

Design: Neale Wills, Wilsy Design & Production Ltd

Printed by: Milne Print, Wellington

This document is printed on an environmentally responsible paper, produced using Elemental Chlorine Free (ECF), FSC® certified, Mixed Source pulp from Responsible Sources, and manufactured under the strict ISO14001 Environmental Management System.

TABLE OF CONTENTS

1 Introduction 1

2 What is landscape in Aotearoa New Zealand? 5
- 2.1 Te ao Māori view of landscape 6
- 2.2 Early New Zealand European view of landscape 8
- 2.3 Reflecting two world views 8

3 Threats to Aotearoa New Zealand's landscapes 11

4 Landscape protection under the RMA 15
- 4.1 Natural landscapes 17
- 4.2 Cultural landscapes 24
- 4.3 Heritage landscapes 29
- 4.4 Amenity landscapes 32
- 4.5 Coastal landscapes 33

5 Assessing landscape values 45
- 5.1 Early development of landscape assessment 46
- 5.2 Impact of the RMA on landscape assessment 47
- 5.3 Incorporation into planning documents 50
- 5.4 Refining assessment approaches 50

6 Current landscape protection tools 57
- 6.1 RMA regulatory tools 58
- 6.2 Other regulatory tools 64
- 6.3 Treaty-based legal tools 69
- 6.4 Non-regulatory approaches 74

7 Summary of case studies 77
- 7.1 Te Manahuna Mackenzie Basin 78
- 7.2 Te Pātaka o Rākaihautū Banks Peninsula 84
- 7.3 Hauraki Gulf islands 93
- 7.4 Waitākere Ranges 103
- 7.5 Tourism and landscape protection 108

8 International protected landscape models 113
- 8.1 Bicultural and co-governance models 114
- 8.2 Biosphere reserves 120
- 8.3 IUCN Category V protected areas 122
- 8.4 Urban areas and landscape protection 124
- 8.5 Bespoke landscape protection models 126
- 8.6 Lessons learned from international models 131

9 A new landscape protection model 137
- 9.1 Pulling the threads together 138
- 9.2 Is the currently regulatory framework adequate? 141
- 9.3 Key elements of a new landscape protection model 142
- 9.4 Prospective RMA reforms 143
- 9.5 Heritage Landscape Order 150

10 Conclusions and recommendations 153
- 10.1 Short-medium term recommendations 154
- 10.2 Longer-term recommendation 155

References 156

TABLE OF FIGURES

2.1	Different conceptual relationships with whenua	9
4.1	Relationship between landscape and natural character	33
6.1	The influence of iwi planning documents under the RMA	63
6.2	Main types of land covenants	68
8.1	Summary of governance arrangements for bicultural and co-governance models	115
8.2	Structure of a model biosphere reserve	120
8.3	Stockholm's Green Wedge compared with London's Green Belt	125
8.4	Summary of key features of bespoke landscape protection models	127

ACRONYMS

AONB	Areas of Outstanding Natural Beauty
BPCT	Banks Peninsula Conservation Trust
DOC	Department of Conservation
EDS	Environmental Defence Society
HGMPA	Hauraki Gulf Marine Park Act 2000
King Salmon	Environmental Defence Society v New Zealand King Salmon Limited
LINZ	Land Information New Zealand
NBEA	Natural and Built Environments Act
NESPF	National Environmental Standard for Plantation Forestry
NPS	National policy statement
NZCPS	New Zealand Coastal Policy Statement 2010
NZILA	Tuia Pito Ora New Zealand Institute of Landscape Architects
ONF	Outstanding natural feature
ONL	Outstanding natural landscape
PC13	Plan Change 13 of the Mackenzie District Plan
RMR Panel	Resource Management Review Panel
QEII Trust	Queen Elizabeth the Second National Trust
RMA	Resource Management Act 1991
RPS	Regional policy statement
UNESCO	United Nations Educational, Scientific and Cultural Organisation
WESI	Wakatipu Environmental Society Inc v Queenstown Lakes District Council
WRHAA	Waitakere Ranges Heritage Area Act 2008

GLOSSARY OF KUPU MĀORI – MĀORI WORDS AND TERMS

Aotearoa	New Zealand
Awa	River
Hapū	Subtribe
Kaupapa	Topic, issue
Iwi	Tribe
Kaitiakitanga	Guardianship, stewardship
Karakia	Prayer
Korowai	Traditional Māori cloak
Kōrero	Speech, narrative, story, news, account, discussion or conversation
(Ngā) kōrero a kui mā a koroua mā	Speech, stories from our elders
Mahi toi	Art, craft
Mātauranga Māori	Māori knowledge – the body of knowledge originating from Māori ancestors, including the Māori world view and perspectives, Māori creativity and cultural practices
Mana	Prestige, authority, control, power, influence, status, spiritual power – a supernatural force in a person, place or object
Manaakitanga	Hospitality, kindness, generosity and support; the process of showing respect, generosity and care for others
Mana whenua	Territorial rights, power from the land, authority over land or territory, jurisdiction over land or territory
Marae	Courtyard – the open area in front of the **wharenui**, where formal greetings and discussions take place. Often also used to include the complex of buildings around the **marae**
Mātaitai	Customary seafood gathering site, shellfish bed, concept used for protection of customary rights
Mauri	Life principle, life force, vital essence – the essential quality and vitality of a being or entity. Also used for a physical object, individual, ecosystem or social group in which this essence is located
Maunga	Mountain
Moana	Ocean, sea
Ngākau	Seat of affections, heart, mind, soul
Pākehā	English, foreign, European, exotic. New Zealander of European descent
Pepeha	Tribal saying, personal introduction including tribal affiliations
Pou	Post, cultural marker
Rāhui	To put in place a temporary ritual prohibition, closed season, ban, reserve – traditionally a **rāhui** was placed on an area, resource or stretch of water as a conservation measure or as a means of social and political control
Rangahau, kairangahau	Research, Māori researcher
Rangatiratanga	Chieftainship, right to exercise authority, chiefly autonomy, chiefly authority, ownership, leadership of a social group, domain of the rangatira, noble birth, attributes of a chief
Rohe	Tribal boundary, district, region, territory, area or border (of land)
Rūnanga	Council, tribal council, assembly, board, boardroom or iwi authority
Takutai moana	Coast, foreshore and seabed

Taiāpure	A stretch of coast, reef or fishing ground set aside as a reserve for inland kinship groups to gather shellfish or to fish
Taonga	Treasures
Taonga tūturu	Protected treasures under the Crown's Protected Objects Act
Tauparapara	Incantation
Te ao Māori	The Māori world
Te Mana o te Wai	An overarching principle and concept adopted by the Crown for freshwater management interpreted as the integrated and holistic wellbeing of the water
Te Tiriti o Waitangi	The Treaty of Waitangi – constitutional, founding agreement between tribal leaders and the British Crown, which underpins the bicultural partnership between Māori and Pākehā in Aotearoa New Zealand
Teina	Younger siblings
Tikanga	Correct procedure, custom, habit, lore, method, manner, rule, way, code, meaning, plan, practice, convention or protocol – the customary system of values and practices that have developed over time and are deeply embedded in the social context
Tūpuna	Ancestors
Tūpuna maunga	Ancestral mountains
Tūrangawaewae	Place where one has the right to stand – place where one has rights of residence and belonging through kinship and whakapapa
Wāhi tapu	Sacred place, sacred site – a place subject to long-term ritual restrictions on access or use, for example a burial ground, a battle site or a place where tapu objects were placed
Wai, ngā wai	Water, water bodies
Waiata	Song, sing
Waipuna	Freshwater spring
Whaikōrero	Oratory, oration, formal speech-making, address
Whakapapa	Māori genealogical connections, lineage, descent
Whanau	family
Whanaungatanga	Relationships, kinships, sense of family connection
Whenua	Land

ACKNOWLEDGEMENTS

We would like to acknowledge the support of the Department of Conservation and Land Information New Zealand for this project including the preparation of this Synthesis Report and the Te Manahuna Mackenzie Basin case study. We would also like to acknowledge the additional financial supporters of the case studies which have been published separately who are: Environment Canterbury (Te Pātaka o Rākaihautū Banks Peninsula case study); Auckland Council (Hauraki Gulf Islands case study); Waitākere Ranges Protection Society and Foundation North (Waitākere Ranges case study) and the Ministry of Business, Innovation and Employment and the Tourism Industry New Zealand Trust (Tourism and Landscape Protection case study). In addition, we would like to thank the many people who have participated in the project as interviewees and peer reviewers.

With respect to the rangahau Māori aspects, though research and writing was led by Lara Taylor, she discussed and worked up the ideas, direction, and technical expertise related to Māori cultural landscapes with a small rōpū of experienced and talented Māori planning and policy practitioners. We would like to acknowledge Nathan Kennedy, Edward Ashby, Siani Walker and Jamie Forsman for your time, whakaaro and insights to this mahi, and your commitment to the overall kaupapa of landscape protection and management. He mihi miharo ki a koutou mo ō koutou mahi kia tiaki i ēnei taonga, ngā maunga, ngā whenua, ngā awa, me ngā moana katoa.

Plimmerton, Wellington

Introduction

Ōhau, Mackenzie Basin

Aotearoa New Zealand's distinctive natural landscapes are an integral component of our individual and national wellbeing. For Māori, they provide a deep cultural connection with the past and future through whakapapa. Landscapes provide physical and spiritual respite, historical links and havens for biodiversity. They are a source of creative endeavour, a key component of the Aotearoa New Zealand brand, and a major drawcard for tourists.

Despite the importance of landscapes to the country, and this being recognised in law for close to 30 years, we are still seeing poor landscape outcomes. This indicates that there is institutional and regulatory failure in protecting these high value places in the public interest.

In order to help address the loss of landscape values, the Environmental Defence Society (EDS) has initiated a project to investigate how existing legislative and policy tools could more effectively be deployed to protect important natural landscapes, as well as how a new 'protected landscapes' model could be adopted to achieve better landscape protection throughout the country.

The project has involved case studies of landscape protection in Te Manahuna Mackenzie Basin, Te Pātaka o Rākaihautū Banks Peninsula, the Waitākere Ranges and the Hauraki Gulf Islands. A fifth case study investigated potential linkages between tourism and landscape protection. The case studies were deliberately chosen to provide a wide range of different contexts, histories and pressures. All the case studies involved reviews of literature, policies and planning documents as well as in-depth interviews with people directly engaged with each landscape area. The studies have been published in full and are available on the EDS website (www.eds.org.nz). They are summarised in Chapter 7 of this report.

Due to capability, budget and time constraints, an in-depth investigation of Māori interests, values and aspirations associated with each case study landscape was not undertaken for this project. However, we have endeavored to acknowledge respective mana whenua (tribes and subtribes with authority in a particular area) and their broad interests through drawing on secondary resources. In addition, we commissioned work specifically documenting Māori perspectives on landscape and landscape protection for this Synthesis Report. Due to the importance of natural landscapes to tangata whenua (people of the land), relevant aspects and insights have been integrated throughout the report rather than being limited to a single section.

Two further dedicated pieces of research were undertaken for the project, the first being an analysis of the Court's interpretation of landscape and landscape assessment methodologies (included in Chapters 4 and 5 respectively); and the second a review of international landscape protection models (summarised in Chapter 8). A full report on international landscape protection models is also available on the EDS website.

Landscape is a complex and multi-dimensional concept.[1] The term 'landscape' has variously been described as important, slippery, ambiguous or just plain difficult.[2] It comes as no surprise, then, that there is no easily distilled description that can encapsulate all of its nuances and meanings. To some, landscape is the culturally modified environment, that which specifically reflects the patterns and structures of its human creators.[3] To others, landscape is an idea, a way in which people make sense of their world.[4] It can be seen as the symbolic environment created by a human act of conferring meaning on nature and the environment.[5]

'Landscape' is, indeed, an all-encompassing term. The Tuia Pito Ora New Zealand Institute of Landscape Architects (NZILA) Charter describes landscape as being "the cumulative expression of natural and cultural elements, pattern and processes in a geographical area."[6] Moreover, the Charter's Preamble offers the following, slightly more fulsome, description of landscapes:

> Landscapes are the result of unique combinations of biophysical, cultural and social processes, evolving over time and interwoven with memory, perception and tradition. They include land, water systems and marine areas, and play a vital role in human nurture, fulfilment and in shaping individual and collective identity. Landscapes range from the outstanding and the memorable, to the familiar and commonplace...

So as to not artificially constrain our investigation, in this project we have adopted a broad concept of landscape, which focuses on human interactions with the natural environment in Aotearoa New Zealand at multiple scales.

Ultimately, whatever meaning is ascribed to landscape, it is clear that natural landscapes are very important to tangata whenua and all New Zealanders. The landscape is who we are. As a young nation, it is the flora and fauna, the mountains, lakes, wetlands and coastlines that are our tūpuna, our ancestors, and it is these natural forces that form our view of 'home'.[7] The landscape, in all its dimensions, is alive. It moves with us as we ebb and flow through life, forming the interface between humans and place.

Aotearoa New Zealand's modern landscape is an open book of our history. Its pages tell a story of the inherent tension between, on the one hand, the need to preserve and protect nature and our natural capital for our identity and wellbeing and, on the other hand, exploiting that same resource for our economic prosperity. The ability to 'get outdoors', catch a breath of fresh air and connect with nature is a key pillar of our spiritual and mental wellbeing. Yet our sprawling settlements, infrastructure and primary industries pose a serious threat to that, encroaching as they have on our coasts, forests, wetlands and tussock grasslands.

We need our natural landscapes, whether that be as a connection to our past, as a place to stand – our tūrangawaewae and our 'home', a source of spiritual, cultural and metaphysical relativity, a source of recreation, a source of artistic inspiration, a grounding for our wellbeing or as an economic resource – a place to farm or a place to call '100% Pure'.[8] The management of our landscapes is crucial to balancing and satisfying these requirements. Yet the evidence reveals an unhappy story with significant loss of native ecosystems, destruction of heritage and degradation of mauri and mana. This report explores that history through localised case studies and spotlights on specific case law, policy and plans. It reflects on changes necessary to rebalance our relationship with our landscapes.

In Chapter 2 we explore what 'landscape' means in Aotearoa New Zealand, reflecting on the two main paradigms held within the country – te ao Māori (indigenous Māori worldview) and a Pākehā (English or European) view of landscape derived from European heritage. We then explore threats to the country's landscapes in Chapter 3. Chapter 4 explores how various types of landscapes have been defined under the Resource Management Act 1991 (RMA) and interpreted by the Courts. In Chapter 5 we chart the development of landscape assessment approaches by landscape architects which have influenced how the regulatory framework under the RMA has been implemented. The various landscape protection tools that are currently applied, including regulatory and non-regulatory approaches, are described and assessed in Chapter 6. In Chapter 7 we provide a summary of the findings of the five case studies undertaken for the project. Chapter 8 summarises our review of international models for landscape protection and distils relevant lessons from overseas experience. In Chapter 9 we bring together the key threads from the previous chapters to develop proposals for a new landscape protection model for Aotearoa New Zealand. In Chapter 10 we provide our conclusions and key recommendations for future policy change.

Akaroa Harbour

ENDNOTES

1 Read, 2005, 19
2 Swaffield, 1993, 58-64
3 Read, 2005, 19
4 Read, 2005, 19
5 Grieder and Garkovich, 1994
6 New Zealand Institute of Landscape Architects, 2010a
7 Brown, 2019
8 https://www.newzealand.com/nz/

What is landscape in Aotearoa New Zealand?

Taieri Plains

As described in Chapter 1, 'landscape' is an interpretative concept that has different meanings according to culture and context.[1] It has been suggested that the 'language of landscape is our native language'.[2] This is because our interpretation of, and connection with, landscape reflects who we are. This section seeks to describe the concept of 'landscape' in Aotearoa New Zealand from both a te ao Māori and early Pākehā context. It then discusses differences between the two world views that have converged within Aotearoa New Zealand and explores how they might relate to each other.

2.1 Te ao Māori view of landscape

There is no single, clear translation into English of a te ao Māori perspective of 'landscape'. A general interpretation, such as the one offered in this section, is just that – a generalised view or interpretation, not a definitive Māori view. It is the prerogative of each iwi, hapū and whānau to define what 'landscape' means to, and for, them. Hence a critical component of any landscape management initiative is involving tangata whenua from the outset, so they are empowered and enabled to articulate their cultural landscapes and associated issues, interests, apsirations and values for management purposes.

For Māori, there is no separation between the land (whenua) and themselves (tangata whenua). The term 'tangata whenua' indicates a metaphysical relationship in which the land and its people are intertwined. It denotes an inseparable connection that conveys particular rights and obligations.[3] Māori and the natural environment function together as a single ecosystem, comprising visible and invisible, and tangible and intangible dimensions.[4] In a broad sense, then, the rohe (tribal region) of each iwi (tribe) or hapū (sub-tribe) serves as their cultural landscape. Māori connect to a specific landscape through the whakapapa (genealogical) connections of iwi, hapū and whānau to specific features of the landscape within their rohe – whenua, maunga (mountain), awa (river), moana (sea) and other pou (markers within the landscape).

Indigenous scholar Hirini Moko Mead described ancestral cultural landscapes as follows:

> A mountain is part of the landscape, it is a reference point, a known landmark to which is attached some cultural meaning… [mountains] have special significance to members of tribes for whom these names are immediately recognisable as symbols of their people. Together with other named features of the land – rivers, lakes, blocks of land, promontories, holes in the ground, fishing grounds, trees, burial places, the islands – they form a cultural grid over the land which provides meaning, order, and stability to human existence. Without the fixed grid of named features we would be strangers on the land – lost souls with nowhere to attach ourselves.[5]

Mead's definition reflects the importance and relevance of landscape to Māori culture, context and identity. The ngākau (heart, mind, soul) of the Māori world view is whanaungatanga (relationships, kinships, sense of family connection),[6] a term that encapsulates the whakapapa (genealogical) connections between tangata whenua and their wider ecosystem, including the spiritual world. The mauri (life force) of tangata whenua is interconnected with the mauri of the natural environment through ancestral connections.

Hence, tangata whenua refer to mountains, and other natural features, in the same manner that they refer to humans, and elders feel comfortable speaking directly to those features.[7] Humans are teina (younger siblings) to the mountains, rivers, seas and lands to which they whakapapa. In Tāmaki Makaurau (Auckland), for example, the maunga (mountains) are referred to as the Tūpuna (ancestral) Maunga. This is also the reason why, in Māori culture, a person always begins their pepeha (personal introduction) to others by referring first to their whakapapa connections to the wider landscape, and then to their name.[8] A pepeha is not a simple connection to *where* they live, but describes *who* they are.[9]

'Landscape' is now a common term used within Aotearoa New Zealand and internationally, and it has planning and policy implications that affect tangata whenua. Therefore, in more recent times, iwi/Māori have engaged with the concept. Both 'cultural landscapes' and 'ancestral landscapes' are terms used by Māori to describe their *particular* landscapes as opposed to natural landscapes in general.

The World Heritage Convention (which Aotearoa New Zealand signed) was the first legal instrument to recognise and protect cultural landscapes. The United Nations Educational, Scientific and Cultural Organisation (UNESCO) World Heritage Committee promotes the recognition and protection of 'cultural landscapes of outstanding universal value'. UNESCO defines cultural landscapes as falling into three main categories:

- Landscapes designed and created intentionally by humans
- Organically evolved landscapes – often associated with and in response to the natural environment
- Associative cultural landscapes – with powerful religious, artistic or cultural associations with the natural elements

The 'associative cultural landscapes' category emerged largely as the result of input by indigenous peoples, including Māori.[10] Tongariro National Park, in the rohe (tribal region) of Ngāti Tūwharetoa in the central North Island, was scheduled

in 1993 as the first United Nations World Heritage Cultural Landscape. This was on the basis that:

> The cultural qualities of Tongariro are intimately linked with its natural qualities, which were recognized by its inscription on the World Heritage List in 1990 as a natural property ... In the case of Tongariro the natural landscape plays a fundamental role through oral tradition in defining and confirming the cultural identity of the Māori people: the two are indissolubly linked. A basic sense of continuity through tūpuna (ancestors) is manifested in the form of profound reverence for the peaks. The natural beauty of Tongariro is the spiritual and historical centre of Māori culture.

Tongariro National Park

> ... the mountains ... are potent symbols of the fundamental spiritual connections between this human community and its natural environment.[11]

The scheduling of Tongariro National Park under 'cultural criteria' was no simple feat. According to the World Heritage Committee Report on the application, there was detailed discussion and careful consideration about the matter before the Committee decided that it fitted the stipulation in the Operational Guidelines that the cultural criterion be only applied "in exceptional circumstances or in conjunction with other criteria".[12] To date, only 114 properties have been scheduled on the World Heritage List as cultural landscapes. Tongariro remains the only site within Aotearoa New Zealand. However, the Tāmaki Makaurau Tūpuna Maunga Authority for the volcanic cones of Auckland has put in a bid to schedule both the ancestral maunga and associated cultural landscape.

2.2 Early New Zealand European view of landscape

The genesis of the early New Zealand European concept of 'landscape' in Aotearoa New Zealand traces back to the early days of European settlement.[13] Settlers brought with them an understanding of landscape centred around the concept of 'beauty'. This was drawn from a rich cultural heritage where 19th century poets, writers and social commentators, such as William Wordsworth and John Ruskin, wrote about the beauty of the landscape in England and the need to protect it from the ravages of industrialisation.[14] So it is no surprise that colonial accounts of landscape in Aotearoa New Zealand are dominated by scenic narratives, drawing on prevailing discourses in Britain of the sublime, the beautiful and the picturesque.[15]

These notions were employed to great effect when marketing Aotearoa New Zealand land for sale to the international market. The New Zealand Company, established in 1839, led this endeavour with glowing accounts of the country's landscapes as superiorly aesthetically pleasing compared to their European counterparts.[16] Aotearoa New Zealand, seen as the very incarnation of the picturesque and another Arcadia, was often captured in early writings and paintings where Aotearoa New Zealand scenes were considered attractive, especially if they emulated the English countryside in some way.[17]

The reality for European colonists was something entirely different. Early settlers spoke of the rugged, barren, desolate environment – untamed and lonely.[18] The landscape underwent dramatic modification during this time. With the rise of pastoralism came extensive bush clearance through logging and burn-offs. 'Landscape' was used during this time to describe the 'improved land'.[19]

Towards the end of the 19th century, and continuing into the first half of the 20th century, new uses of the term landscape emerged. These were based largely around concepts of unity; both in the sense of unity in appearance (in terms of ecological and biophysical features) and emotional and spiritual unity.[20] European notions of organisation and unity in landscapes pervaded early Aotearoa New Zealand ecological and geomorphological assessments.

Although the term 'landscape' gained traction in post-colonial Aotearoa New Zealand, it was by no means widely used.[21] By the late 19th century an increasing concern for the wholesale loss of native plants and animals, and the impacts of introduced pest species, spurred a campaign for 'scenic preservation'.[22] The resulting Scenery Preservation Act 1903 was Aotearoa New Zealand's first law specifically designed to protect sites of scenic and historical interest. In a Parliament more accustomed to promoting legislation to clear areas for farming, the Act represented a new way of looking at the land, one in which aesthetics were important.[23] However, notwithstanding the close association of landscape and scenery, no references to 'landscape' were found in the parliamentary and public debates surrounding the introduction of the Act.[24]

2.3 Reflecting two world views

The Environment Court has observed that one must step deeply inside Māori thinking to be able to understand Māori views of the landscape. A person has to see the world through Māori eyes, and assess Māori values within a Māori world view,[25] to be able to truly understand it. Therefore a culture or a 'cultural landscape' cannot be fully understood in terms of the world view of another.[26] This means that it can be difficult to breach the divide between the Māori and Pākehā conception of landscape.

The first settlers who arrived on the shores of Aotearoa around 800 years ago, brought with them a rich cultural heritage from eastern Polynesia. A distinct Māori culture progressively evolved as the people lived in and interacted with the new land. This means that Māori culture developed within and in relation to, the landscapes of Aotearoa. As a result, it reflects a depth of association with those landscapes that is unmatched by the European landscape tradition which evolved in the context of a very different place. An early article on Aotearoa New Zealand's landscapes reflected this difference when it stated:

> To match in any way the power and riches of Māori tradition, Pākehā New Zealanders have to dig deep into their own cultural store. To match the richness of the landscape itself they are wise to draw also on the cultural store of the tangata whenua, the people of the land themselves.[27]

In her research on the human dimension to landscape protection in international law, Amy Strecker posits that the European concept of 'landscape' emerged in response to the separation of humans from nature. This occured due to many 'developments' including the Enclosure movement, end of communal land tenure, dispossession of the peasant classes, urban migration and the Industrial Revolution. Strecker argues that, in some ways, the visual and scenic landscape subsituted for the loss of connection with the land. Where there was once a 'use and dependency' value, this became an 'aesthetic' value.[28]

Moewaka Barnes and McCreanor similarly argue in their research on colonisation, hauora (health), and whenua (land) in Aotearoa, that the disconnection between land and people emerged due to the British enshrinement of land as property in Aotearoa New Zealand by the English Laws Act.[29] As illustrated in Figure 2.1, this view of land as property contrasts with the fundamental Māori worldview that whenua and people are symbiotic. Figure 2.1 serves to highlight the different conceptual relationships with land according to cultural worldview.[30]

Whenua as the determinant of wealth Whenua as property and for resource extraction	**Whenua as the determinant of health** Whenua as a 'person' with rights and mana
Property How much of the land, foreshore, river and/or lake can I own and control?	**Mauri** How can we protect the whenua and retain its integrity?
Extraction How much can I get out of it as cost effectively as possible?	**Kaitiakitanga** How can we work with and for the whenua for healing and sustenance?
Profit How much capital gain and profit can I make?	**Manaakitanga** How can we uphold and practice respectful relationships with the whenua?

Figure 2.1: *Different conceptual relationships with whenua*[31]

In contrast to the dominant view of land as the determinant of wealth, and a 'resource' to own and to extract from, which emerged through colonisation,[32] the reciprocal relationship of people with the land underpins the Māori worldview.

This dictates that people must recognise, respect and protect the land's own mana, in order to safeguard the mauri of the land and the people connected to it. Mauri is described as the "life force or essence that derives from a common centre, binding us together".[33]

As with the evolution of Māori culture, the European tradition of landscape has changed within the Aotearoa New Zealand context, as the European settlers engaged with new landscapes. Chapter 4 chronicles how the concept of landscape has more recently broadened and evolved through case law. However, navigating the two world views and devising effective protection mechanisms that encompass, empower and enable both te ao Māori and Pākehā approaches, is still some distance away. It has been the European tradition that has largely dominated institutional and legal settings. This means that the generic term 'landscape', as it has been applied in the Aotearoa New Zealand context, does not comfortably fit with te ao Māori and necessitates the use of the terms 'cultural' or 'ancestral' when referring to landscapes in relation to Māori.

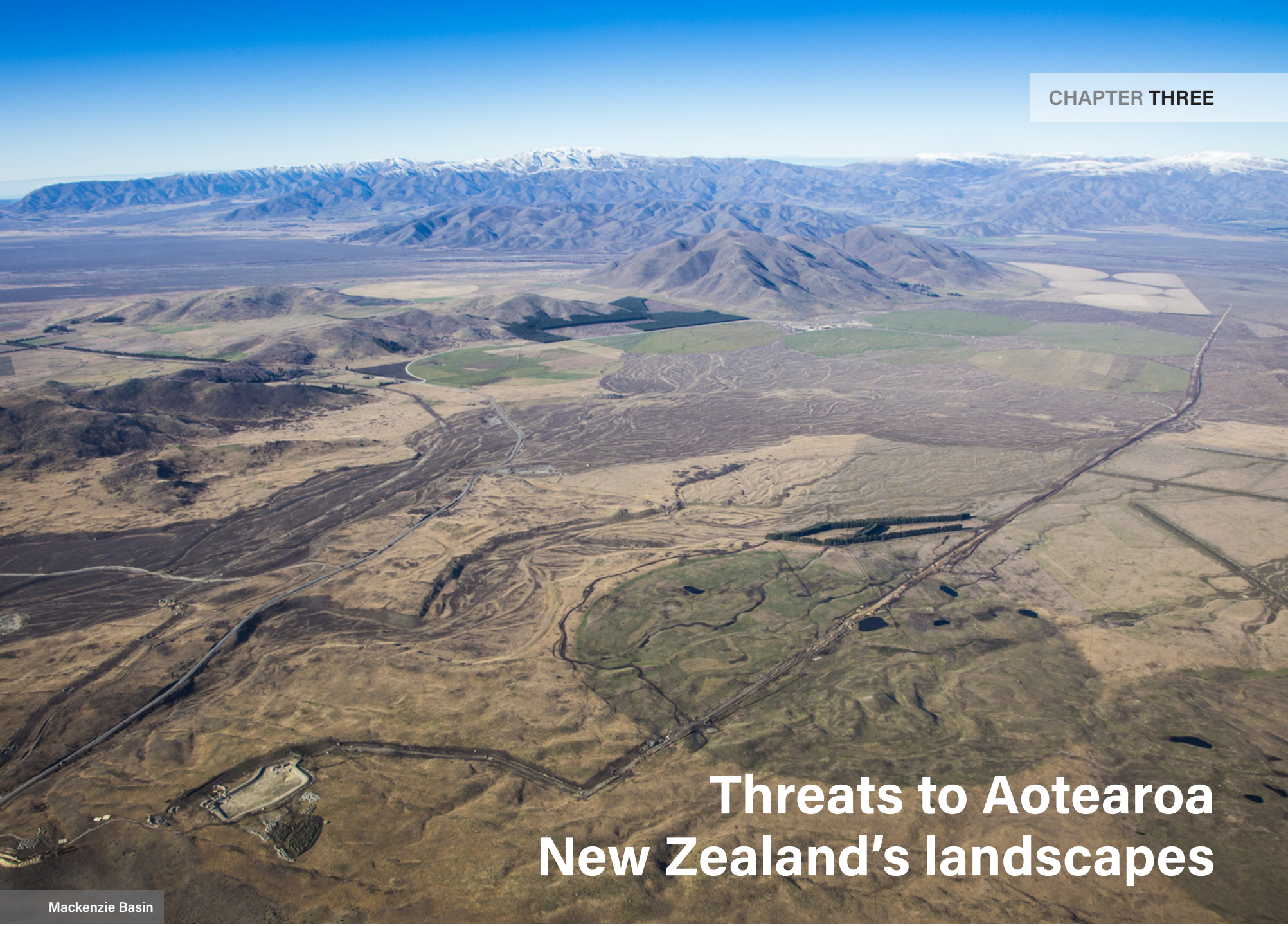
NZILA has been developing new Landscape Asssessment Guidelines that seek to develop a biculturally inclusive landscape concept and approach to landscape assessment which could serve to usefully inform future policy development in this area.

KEY MESSAGES

- **The te ao Māori view of 'landscape' has evolved over centuries within Aotearoa and is relational, with land and people being intertwined over time. It is also specific to each iwi, hapū and whanau.**
- **The early Pākehā view of landscape, based on the European tradition, emphasised the concepts of beauty and the picturesque.**
- **It is the European tradition which has largely dominated legal and institutional settings for landscape protection in Aotearoa New Zealand, although this is now changing.**
- **The country's landscape protection system needs to be broad enough to include both world views and value systems. It should build on their respective strengths and protect their respective views, interests and values.**

ENDNOTES

1 Menzies, 2017
2 Spirn 1998, 15 in Menzies 2017
3 Marsden, 2003
4 Moewaka Barnes and MCreanor, 2019
5 Mead, 1984, 20
6 Independent Māori Statutory Board, 2016
7 Independent Māori Statutory Board, 2016
8 Pere and Barnes, 2009
9 Hoskins et al, 2008; Mead, 1984, 20
10 Murdoch, 2015, 4
11 United Nations Educational, Scientific and Cultural Organisation, 1993
12 United Nations Educational, Scientific and Cultural Organisation, 1993
13 Swaffield, 1999a, 3
14 Reynolds, 2016
15 Fairclough, Sarlöv Herlin and Swanwick, 2018, 189
16 Falconer, 2015
17 Phillips, 2007
18 Phillips, 2007
19 Swaffield, 1993, 58
20 Swaffield and O'Connor, 1986, 13-17
21 Swaffield, 1993, 59
22 Nightingale and Dingwall, 2003, 16
23 Nightingale and Dingwall, 2003, 5-6
24 Swaffield, 1993, 58
25 *Ngāti Hokopu ki Hokowhitu v Whakatāne District Council* (2002) 9 ELRNZ 111 (NZEnvC)
26 Joseph, 2018
27 Swaffield and O'Connor, 1986, 16
28 Strecker, 2012, 333
29 Moewaka Barnes and McCreanor, 2019
30 Moewaka Barnes, 2019
31 Adapted from Moewaka Barnes, 2019
32 Reid, Taylor-Moore and Varona, 2014; Moewaka Barnes and McCreanor, 2019
33 Moewaka Barnes and McCreanor, 2019, 27



Threats to Aotearoa New Zealand's landscapes

Mackenzie Basin

When EDS undertook its first investigation of landscape protection in Aotearoa New Zealand, in 2004, the country was in the grip of rampant subdivision of rural farms and along previously undeveloped coastlines and lakesides. The report described the impact of this development as follows:

> Lifestyle development usually involves the construction of large houses and associated infrastructure such as driveways and garages. In important landscapes, this type of development can result in the intrusion of large man-made structures into predominantly natural areas. This can be in the form of large houses dotted across the valley floor and hillsides, as has occurred in the Wakatipu Basin; development extending along the coastline, as is happening in the Coromandel Peninsula and in the Whangarei District; or the transformation of small bach settlements into intensively developed suburbs, as is happening at Piha Beach in the Waitākere Ranges.[1]

In the wake of the report, and the New Zealand Landscape Conference that was convened by EDS in 2003, there was heightened awareness of the importance of landscape and the need to protect landscape values from urban development. Rules were toughened in many district plans. The revised New Zealand Coastal Policy Statement (NZCPS), which came into effect in 2010, enshrined in law a much more protective approach for coastal landscapes. This was strengthened by the 2014 Supreme Court decision in *Environmental Defence Society v New Zealand King Salmon Limited (King Salmon)* which confirmed that the NZCPS sets a firm environmental bottom line for coastal landscapes which needs to be given effect to.[2] In the wake of the 2007-08 Global Financial Crisis, pressure for lifestyle and holiday home development has eased somewhat. But other landscape pressures have been growing since that time.

In 2004, Aotearoa New Zealand had a population of just over 4 million and there were around 5 million dairy cows. During the relatively short time since the first

Waiheke Island

EDS report was written, the country's population has increased by a million people and the number of cows by more than 1.5 million.[3] Arguably, a tipping point has been reached in terms of the ability of the landscape to accommodate these increases, particularly in the context of a continuation of historic approaches to managing urban development and rural land use change such as dairy intensification.

The *2019 Environment Aotearoa* state of environment report chronicled landscapes in decline. Thousands of indigenous species are threatened with extinction and there is continuing loss of native vegetation and wetlands, pollution of waterways and growing carbon emissions. The report stated that:

> Our unique native biodiversity is under significant pressure from introduced species, pollution, physical changes to our landscapes and coast, harvesting of wild species, and other factors. Almost 4,000 of our native species are currently threatened with or at risk of extinction...
>
> Logging native forests, draining wetlands, and clearing land have degraded a range of benefits provided by native vegetation, accelerated our naturally high rates of soil loss, and affected our waterways...
>
> Waterways in farming areas are polluted by excess nutrients, pathogens, and sediment. This threatens our freshwater ecosystems and cultural values, and may make our water unsafe for drinking and recreation...
>
> Our per-person rate of greenhouse gas emissions is one of the highest for an industrialised country. Most of our emissions in 2016 came from livestock and road transport.[4]

The pressures on the landscape are now much more multi-faceted than in the past and they play out in different ways in different places. In our cases studies we found that some of the biggest threats to landscape values were:

- Pastoral intensification and irrigation (Mackenzie Basin and Kaitorete Spit)
- Urban development (Waitākere Ranges and Waiheke Island)
- Tourism (Waiheke Island and Akaroa)
- Exotic plantation forestry (Mackenzie Basin and Banks Peninsula)
- Wilding pines (Mackenzie Basin)
- Plant pathogens (Waitākere Ranges)
- Weeds (Waitākere Ranges)
- Predators (most places but particularly Aotea Great Barrier Island and Banks Peninsula)

The common denominator behind these pressures is people. This means that, to be effective, any landscape management system needs to address a wide range of pressures and most importantly human attitudes, values and behaviours. A future system also needs to be flexible and adaptive so it can be tailored for particular localities and contexts.

KEY MESSAGES

- **Pressures on Aotearoa New Zealand's landscapes have increased markedly over the past couple of decades with a fast-growing population and intensification of agriculture.**
- **Our landscapes are still in decline with thousands of indigenous species threatened with extinction, continuing loss of native vegetation and wetlands, pollution of waterways and growing carbon emissions.**
- **Human pressures on landscapes are multi-faceted and they play out in different ways in different places.**
- **Any landscape management system needs to address a wide range of pressures and be able to be tailored for specific localities and contexts.**

Port Underwood

ENDNOTES

1 Peart, 2004, 6

2 *Environmental Defence Society v New Zealand King Salmon Company Ltd* [2014] NZSC 38

3 Statistics New Zealand

4 Ministry for the Environment and Statistics New Zealand, 2019

CHAPTER **FOUR**

Landscape protection under the RMA

Jacks Point, Queenstown

The RMA is currently the main piece of legislation in Aotearoa New Zealand that sets out how the environment should be managed. In this chapter we do a deep dive into how the legislation has been interpreted and applied to landscape management matters by the courts. This material will be of particular interest to practitioners and regulators.

The RMA is underpinned by the principle of sustainable management and seeks to "promote the sustainable management of natural and physical resources".[1] Under Part 2 of the RMA, 'sustainable management' is broadly defined as follows:

> (2) In this Act, sustainable management means managing the use, development, and protection of natural and physical resources in a way, or at a rate, which enables people and communities to provide for their social, economic, and cultural wellbeing and for their health and safety while—
>
> (a) sustaining the potential of natural and physical resources (excluding minerals) to meet the reasonably foreseeable needs of future generations; and
>
> (b) safeguarding the life-supporting capacity of air, water, soil, and ecosystems; and
>
> (c) avoiding, remedying, or mitigating any adverse effects of activities on the environment.

Previously an 'overall broad judgment' approach was adopted in interpreting the sustainable management purpose of the RMA. It was considered that the word 'while', in the chapeau, acted to separate the environmental effects contained in subparagraphs (a)-(c) from the economic considerations in the chapeau; and that these effects had to be balanced against one another. In 2014, the Supreme Court in *King Salmon* rejected this approach. It determined that the definition of sustainable management must be read as an integrated whole and the environmental effects described in subparagraphs (a)-(c) must be observed in the course of the management referred to in the opening part of the definition.[2]

The Supreme Court also determined that, although section 5 does not create primacy for environmental protection (through the setting of environmental bottom lines), in certain circumstances sustainable management may require that particular environments be protected from the adverse effects of inappropriate activities. These circumstances will be defined by other planning documents, which themselves may set strict environmental bottom lines. In this case, the NZCPS set strict bottom lines by requiring the avoidance of adverse effects *on* outstanding natural landscapes (ONLs) in the coastal environment (including from effects from activities outside an ONL).

King Salmon represented a sea change in resource management law. It held that, where provisions in higher order policy documents were sufficiently directive and included strict bottom lines, those bottom lines must be complied with. Prior to this, these policy documents had been important, but not necessarily determinative. It was common for decision-makers to resort back to Part 2 of the RMA in making their decisions, which involved a balancing of environmental bottom lines against other considerations.

The Supreme Court decision has been subsequently applied in a number of cases. Of particular importance is the case of *RJ Davidson Family Trust v Marlborough District Council* which concerned an application for resource consent. In that case, the Court of Appeal held that the Supreme Court's reasoning in *King Salmon* did not apply to the resource consent process and that the statutory language contained in section 104 of the RMA clearly contemplates consideration of Part 2 matters, when appropriate. However, the extent to which reference to Part 2 is available depends on the applicable planning instruments. If the policies in the relevant planning instruments are coherent and clear-cut (eg require avoidance of effects) then resort to Part 2 would not be justified. However, if the planning documents do not clearly indicate whether a resource consent should be granted or refused, then regard could be had to Part 2. This highlights the importance of clear and directive provisions in planning documents if landscape values are to be protected.

Section 5 contains the overall purpose of the RMA but is not intended to be "the primary operative decision-making principle".[3] Sections 6, 7 and 8 supplement the purpose and identify matters that are of special significance for resource management. These sections set out obligations that those administering the RMA must adhere to.

Matters of national importance are listed in section 6 and there is a direction for decision-makers to 'recognise and provide' for them. In contrast, decision-makers must only 'have particular regard to' the matters in section 7, and only 'take into account' the matters in section 8 (the principles of Te Tiri o Waitangi). Out of the three sets of principles, section 6 provides the strongest direction – decision-makers must make actual provision for the matters listed. However, this remains subject to the overall purpose of the Act.

ONLs are identified under section 6(b) and their protection is a matter of national importance. Other matters of national importance include the protection of significant indigenous vegetation (section 6(c)), the relationship of Māori and their culture with taonga (section 6(e)) and the protection of historic heritage from inappropriate subdivision, use and development (section 6(f)). Sections 7(c) and 7(f) apply to landscapes that contribute to amenity values or the quality of

the environment and create an obligation on decisions makers to have particular regard to the maintenance and enhancement of them.

Below we describe how the courts have dealt with different categories of landscapes under the RMA: natural, cultural, heritage, amenity and coastal. These are not discrete categories and there is typically a large degree of overlap between them. For example, a single landscape will often have significant natural, cultural, heritage and amenity values. While a 'cultural' category is specified, which pertains to Māori in particular, Māori have interests, views and values across all of these different types of landscapes.

As will be illustrated in the following sections, the Courts have played a seminal role in interpreting what 'landscape' is, defining different categories of landscape, and determining how they should be assessed and what degree of protection they are afforded. The Courts have intervened in a manner that is quite unusual in the Aotearoa New Zealand legal context. For example, the Environment Court's intervention in the Mackenzie District Council's Plan Change 13 (PC13) resulted in the plan, in effect, being written by the Court. As such, it is important to assess the case law in some detail.

4.1 Natural landscapes

Interpretative issues

Section 6(b) of the RMA provides for "the protection of outstanding natural features and landscapes from inappropriate subdivision, use, and development" as a matter of national importance, which must be recognised and provided for when making decisions under the Act.

The meaning of various terms within section 6(b) have sparked rigorous debate since the RMA's inception. With expert assistance from the landscape profession, the Court has spent considerable time and effort determining what a 'landscape' and 'feature' are, what makes them 'outstanding', and how 'natural' fits into these concepts. Discourse has also focused on what is 'inappropriate' within ONLs and outstanding natural features (ONFs) and how they should be protected.

Section 6(b) concepts remain keenly contested, as evidenced by the number of High Court, Court of Appeal and Supreme Court cases addressing landscape-related matters over the last five years. The NZILA Best Practice Note defines

Mimiwhangata

certain landscape terms, and while this has been useful for Court's grappling with these issues, their preference is that the text of section s6(b) should be considered in terms of principles rather than rules or definitions.[4]

Definition of landscape

How we address landscape issues depends on how the landscape is defined.[5] It is commonly asserted that landscape used to be regarded in visual or visibility terms only,[6] that it was something to look at and admire, something over 'there' separate from the viewer. Jurisprudence in the early 1990s certainly referred regularly to visual impacts or visual effects on the landscape, or discussed visual and landscape values as one. However, while there may have been a preference for understanding landscapes in this manner, the idea that landscape is more than just its visual attributes is a long-held view by at least some divisions of the Court. From as far back as 1997 the Environment Court was accepting that landscape, when considered under the RMA and NZCPS, was not restricted to visual aspects and could be interpreted to include both the physical and perceptual.[7]

It is now roundly accepted by the Courts that expressions of landscape that focus primarily on visual aspects, and assessments that only address effects on visual perception, are outdated in terms of Aotearoa New Zealand law and practice.[8] In particular, it is too narrow an approach for the identification of ONLs.[9]

So, if landscapes are more than just aesthetic areas of land to be observed and admired, what other elements are important for their identification? In 2000, the Environment Court released a decision called *Wakatipu Environmental Society Inc v Queenstown Lakes District Council*[10] (*WESI*) which became pivotal to landscape jurisprudence over the following decade. It created the following list of landscape components, termed the 'amended Pigeon Bay criteria', that provided a framework for assessing and defining landscapes:[11]

- Natural science factors – the geological, topographical, ecological and dynamic components of the landscape
- Aesthetic values including memorability and naturalness
- Expressiveness (legibility) – how obviously the landscape demonstrates the formative processes leading to it
- Transient values – occasional presence of wildlife or its values at certain times of the day or year
- Whether the values are shared and recognised
- Value to tangata whenua
- Historical associations

The list is widely framed to reflect the Court's understanding that 'landscape' is a large subset of the 'environment'.[12] It involves both natural and physical resources and also various factors relating to viewers and their perception of the landscape(s).[13]

At the time, the Court emphasised that its list of criteria was not to be regarded as frozen, and that it may be improved with further use and understanding.[14] Subsequent decisions have consequently attempted to refine the amended Pigeon Bay criteria.[15] The assessment framework has been further summarised by the Court as having three sets of components:[16]

- the biogeographical elements, patterns and processes;
- the associative or relationship contributions; and
- the perceptual aspects.

The Court made it clear that the matters discussed above cannot be divided into technical (being biogeographical or perceptual aspects) and non-technical (being associative or relationship aspects) groups.[17] Such distinctions may import different standards of expert evidence which is not warranted or appropriate.[18] Further, it is inconsistent with case law which states that any of the range of factors may be so significant in relation to a particular landscape or feature as to justify a finding that it is outstanding, without the need for expert analysis.[19]

While the amended Pigeon Bay criteria are still mentioned by the Court when grappling with landscape issues, they have become less prominent. The language of landscape architects and the Courts has developed into discussion about what is now the commonly accepted attributes of a landscape – its biophysical, sensory/perceptual and associative values.[20] These groupings have recently been described as the 'dimensions' of the assessment of landscapes and features, all of which are essential to a full understanding of landscape.[21]

The most appropriate view for assessing a landscape or feature has also been considered. The Courts have stated that this involves a viewpoint not only in space, but also in time, given the four-dimensional existence of a landscape:

> Just as a viewer can see a landscape from close up, or in the fore- or middle ground or from a long distance, so the time dimension may be fleeting, or last for few years, or the life of the relevant plan, or for a generation, or over a much longer term: the process elements of a landscape or feature may be appropriately considered over geological epochs.[22]

The Courts have held that it would seem unlikely that there will ever be a single viewpoint or viewing time. Conceptually you must stand back and bring together the full range of views. together with the relevant processes and associations that inform an understanding of those views.

With assistance from the landscape profession, the Court has laid the foundations for undertaking this conceptual challenge which evokes the sense of standing in an area and being bathed in the multifaceted layers of its past, present and future. Although Courts now readily accept that landscape is more than just its biophysical elements, this is still seen as the starting point for any assessment.[23] While an evaluative opinion of a landscape expert should be informed by the best available factual/scientific or other foundations for an evaluative opinion, contextual assessment should follow to elicit how people would perceive its relative naturalness, given the associations they may have with a landscape or feature.[24]

Tekapō, Mackenzie Basin

A spotlight on public input to landscape assessment

The natural and physical attributes of a landscape can be both objectively and subjectively analysed. Likewise, change to the natural environment can be described and assessed. In undertaking these assessments, it is important to take into account people's values.[25] Landscapes are often described as a cultural construct, and as discussed in Chapter 2, different cultures hold different views about landscape and values may change over time and according to context. A landscape can convey different meanings to the same or different people.[26] In this way we can understand the landscape through the values and perceptions held by people, be they expert landscape architects, people who have an attachment to a place, or those who have knowledge and experience of a region, area or site and its natural and physical resources.[27]

The Court has referred, with apparent approval, to public perception studies which uncover two main paradigms that help explain most New Zealander's responses to landscape and their assignment of values to different landscapes:[28]

- "The 'wild nature' paradigm, repeatedly identified in the research, is strongly correlated with the native endemic character of landscape scenes and the predominance of natural elements and patterns within them.
- The 'cultured nature' paradigm is more accepting of exotic vegetation and productive rural uses, but again shows a strong aversion to obvious signs of development and buildings in the landscape."

The Environment Court expects that ascertaining public perception of a proposed landscape change necessarily involves public input. Further, not just any public input is acceptable. To gain a true representation of the public's perception of a proposal on landscape values, one must ask the local community. The people surveyed on their perception of landscape values must be either sufficiently informed of localised landscape effects or qualified to assess them.[29]

Features v landscapes

Features and landscapes are not the same thing. Generally speaking, a feature can be described as a single element of natural and physical resources while a landscape is usually a collection of such elements.[30]

Historically, the Environment Court held that a feature is a distinctive or characteristic part of a landscape[31] and therefore an ONF is a distinctive part of a larger landscape which is an ONL.[32] This relationship is no longer accepted as being so fixed. The scale of elements is necessarily relative and a feature may be so large, as in the case of a mountain, that it can encompass one or more landscapes while retaining its integrity as a feature.[33] Conversely, a feature may be relatively small, at a scale not ordinarily characterised as an area of landscape.[34] In some cases, an ONF can occur in isolation to an ONL, while in others it may be outstanding because of its relationship to other features or landscapes, be they outstanding or not.[35] Further, there can be landscapes within landscapes, such that the classification of a large coastal bay does not preclude a finding that a smaller area within it is also an ONL rather than a feature.[36]

Te Mata Peak

Scale

The appropriate geographical scale for evaluation is one that treats a landscape or feature as a whole.[37] This can be a small area or a large one, but the ONL and ONF boundaries should be legible and coherent to the community.[38] Given the splendour of Aotearoa New Zealand's landscapes, large tracks of land in certain areas has been classified as ONL or ONF, for example 97 per cent of the Queenstown Lakes District rural area.[39] As discussed further below, the presence of human occupation and land management within these areas are not necessarily a reason to exclude areas from protection.

Landscape assessments that divide landscapes into too many small units can result in surface features (such as human occupation and uses) and vegetation patterns (exotic or native, grass or bush etc) taking on greater significance in distinguishing the character of one landscape unit from another.[40] Further, underlying geomorphological processes cannot be discussed logically in terms of individual units as these processes span and inter-connect many units.[41] Conversely, if the spatial area is too broad, this can result in an inappropriate discounting of localised landscape values and dilution of associated effects on those values.[42] The Court has rejected landscape assessments that use the wrong scale of assessment as these may materially overstate (or understate) the true impacts of a proposal.[43]

Spatial boundaries

Although the spatial boundaries of natural landscapes are often readily mapped in plans, fixing a line boundary in an area of transition from one landscape to another must inevitably be somewhat arbitrary.[44] Environments (or landscapes) often cannot be readily identified by lines on a map and there will frequently be grey areas and blurred edges.[45] There is even more complexity with cultural landscapes. However, the need to identify specific landscapes is somewhat a necessity as it is the only way the Court can make a finding of 'fact' sufficient to identify the resource management issues.[46] When the Court is tasked with making a delineation, it will most often identify a 'practical boundary' of 'best fit'.[47]

A spotlight on landscape delineation in the Queenstown Lakes District

The Queenstown Lakes District Plan is an interesting case study in dividing and categorising rural landscapes. As a result of the structure of the plan, areas of landscape have been categorised as visual amenity landscapes simply because there is no other category to put them into. The is best described in the excerpt below:

> [53] Case law has developed the application of the landscape categorisations in Chapter 4 of the district plan. Importantly the High Court stated in *Queenstown Lakes District Council v Trident International Ltd* that there is no fourth category. Fogarty J stated: "The analysis of whether the site falls within outstanding natural, visual amenity or other rural landscape … is exhaustive". *Trident* was about an application to subdivide Rural-General zoned land immediately adjacent to a hillside suburb in Queenstown. The Environment Court found that the site was in a townscape (as a fact) but then held that the district plan "… does not require that urban landscapes be assigned to a class of rural landscape rather than be recognised for what they are". That is the point which the High Court held was wrong. In other words even if heavily influenced by an adjacent urban zone (or area) any rural area must be categorised into one of the three categories identified in Chapter 4.2 of the district plan. While crude, at least that approach does have the advantage of concentrating minds on clean urban edges as required by a policy I will come to.
>
> [54] Before and after *Trident*, a number of attempts were made to define small areas of the Rural-General Zone with reduced naturalness as "other rural landscapes" rather than as VALs [visual amenity landscapes]. In a sequence of cases starting with *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* the Environment Court has reiterated that landscapes are not "landscape units", they must be approached as a whole: see *Parkins Bay Preserve Inc v Queenstown Lakes District Council.* In other words, a small area of a VAL may have very low visual amenity values indeed (because of its proximity to existing development and because it possesses low natural values) but it still needs to be categorised as part of a VAL because there is no other pigeonhole to put it into as a landscape. This is, in effect, a "wash over" principle where a small area of low landscape value looked at by itself may be in a higher category landscape simply because most of the surrounding land is of higher landscape value. The quality of a landscape varies within a continuum.[48]

What is 'natural'?

The adjectives 'outstanding' and 'natural' apply to both landscapes and features such that, to qualify as an ONL or an ONF, a landscape or feature must be both outstanding and natural.[49] Taking heed of the Court's caution against searching for 'outstanding natural landscapes' when the RMA requires recognition of natural landscapes which are outstanding,[50] the first question to ask is whether the landscape is a 'natural landscape' or the feature a 'natural feature'?

'Natural' can be defined as existing in or caused by nature or not artificial (natural landscape) or uncultivated and wild (existing in its natural state).[51] However, this definition has been dismissed as being too simplistic[52] with the Courts noting that a purely 'natural landscape' is likely an oxymoron under the RMA.[53]

'Natural' does not equate to endemic or pristine, and can include such things as pasture, exotic trees and wildlife (as opposed to human-made structures, roads and machinery). The fact that humans have intervened does not make a landscape unnatural, and both working farms and plantation forestry have been included within ONLs.[54] Rather, it is the nature of that intervention, and most notably the presence and use of buildings and infrastructure, which tends to make a landscape look unnatural.[55]

The *WESI* decision developed a set of criteria for naturalness as follows:[56]

- Physical landform and relief
- The landscape being uncluttered by structures and/or 'obvious' human influence
- Presence of water (lakes, rivers, sea)
- Vegetation (especially native vegetation) and other ecological patterns

The absence or presence of one or more of these criteria does not mean that the landscape or feature is non-natural, just that it is less natural.[57] There is a spectrum of naturalness from pristine, to natural but modified, to highly modified.

As is the case for the criteria for 'landscape', the *WESI* list of 'natural' has also been re-developed by the Courts. The Environment Court has rephrased the first point above to state "relatively unmodified and legible physical landform and relief" and introduced the possibility of a 'cultured landscape' being part of an ONL.[58] As noted above, the Court has considered that it is the extent of human or cultured modification – on a continuum – that determines whether a landscape is natural or not.

Exactly where on the spectrum of 'naturalness' a landscape or feature must sit in order to meet the threshold for ONL or ONF also depends on context. An early

case gave approval for somewhere between the moderate-high range.[59] More commonly, however, the Courts accept that there is no threshold of moderate-high to be met in order for an area of land to qualify as part of an ONL or ONF.[60] To do so would artificially construct a threshold for what is inherently a judgment call guided by contextual evaluation.[61]

Very recently, the Environment Court has accepted that 'moderate' natural character can be included within an ONL. This approach requires that the landscape be read as a whole. Land that has comparatively less naturalness through human modification, but which reads as part of an ONL or ONF, is still capable of being treated as part of it. The overriding consideration must be to ensure that the overall legibility of the landscape is maintained. Ultimately the question that must be asked is whether the area of land in issue is too modified or inappropriately developed such that including it in the ONF or ONL would detract from, or undermine, the values of the ONL or ONF when considered as a whole.[62]

What is 'outstanding'?

Once a landscape or feature is confirmed as 'natural' for the purpose of section (6)(b) of the RMA, the next question is whether it is 'outstanding'? The RMA contains no threshold for 'outstandingness'. Thus, whether a landscape or feature is outstanding is a vexed issue subject to heated debate. This is particularly so, as Aotearoa New Zealand is full of beautiful or picturesque landscapes which are not necessarily ONLs.[63]

It is often quoted that the meaning of 'outstanding' means "conspicuous, eminent, especially because of excellence; remarkable";[64] it is a strong adjective that imports some 'special quality'.[65] But in what context? Does the landscape or feature have to exhibit these attributes and 'stand out'[66] among other landscapes or features within the district, region or across all of Aotearoa New Zealand?

The comparative scale for assessment of 'outstanding' has recently been assessed by the Courts. Restricting what is outstanding to a district-wide assessment has been rejected, with the Courts noting that while a local authority may be confined to its knowledge of its district, the Environment Court need not be as it has built up its knowledge of Aotearoa New Zealand's landscapes.[67] The contention that a comparison with other landscapes that may be described as outstanding on a national basis has also been rejected.[68] Overall, the scale of assessment for ONLS is generally accepted to be those that are outstanding in terms of the region's natural environment.[69]

In accordance with its accumulated knowledge the Environment Court has, on a number of occasions, determined that a landscape or feature is a ONL or ONF notwithstanding a lack of recognition in relevant planning maps[70] or in contradiction of expert evidence. As *WESI* noted, ascertaining an area of ONL should not normally require experts.[71] Usually, it should be so obvious in general terms, that there is no need for expert analysis. The converse can also apply, where the Environment Court determines that an area is not part of an ONL even though it is classified as such in the district plan.[72]

Who's paying for all this splendour?

In her opening address to the Resource Management Law Association Conference in 2012, Judge Borthwick stated that:

> [I]t is a poverty of imagination (and understanding) to value an outstanding natural landscape solely because it is aesthetically pleasing. When landscapes are discussed in this way the public discourse becomes polarised and descends into an enquiry as to 'who is to pay for all this splendour'?

It is interesting to consider the final question posed by Her Honour in the context of ONLs in particular. Much of the opposition to identification of ONLs within regional and district planning instruments originates from landowners (predominately farmers)[73] who feel aggrieved that an ONL classification unfairly restricts their ability to use and develop their land. They feel they are paying for a public benefit.

The first point to note is that the identification of an ONL is not informed by or dependent upon, the protection afforded to that landscape, whether that be under the RMA or the relevant planning instrument. The issue of whether land has attributes sufficient to make it an outstanding landscape within the ambit of s 6(b) of the Act is a factual assessment based upon the inherent quality of the landscape itself. The question of what restrictions apply to the land is a secondary consideration to be determined once the ONL has been identified.[74] Thus, if a landscape is an ONL it should be classified as such, irrespective of whether the relevant plan imposes onerous restrictions on the affected land.[75] This is clearly the direction afforded by section 6(b) of the RMA and it would be illogical and contrary to conclude that the outstanding area should only be so classified if it were not suitable for a range of other activities.[76]

The second point to note is that an ONL classification is not akin to a blanket restriction on land use. Indeed section 6(b) does not provide for absolute protection of ONLs and ONFs, but rather protection against 'inappropriate' subdivision, use and development. New farming activities in a rural environment which is subject to an ONL may, in some instances, give rise to adverse effects and be properly described as inappropriate, but this is to be assessed on a case by case basis.[77] As the Supreme Court in *King Salmon* stated, much turns on "what it is that is sought to be protected".[78]

A natural landscape's 'outstanding' status does not preclude development. Rather what is to be protected are those values which qualified the landscape as outstanding.[79] Mapping only assists in identifying the geographic extent of what is sought to be protected. Listing the values that inform why a feature or landscape is an ONL or ONF is an important further element of setting this out.[80]

In that regard, the Courts support clear planning frameworks that assist plan users in assessing whether and how any proposal might affect an ONL or ONF (commonly referred to as scheduling ONL and ONF values). Such frameworks require more than just descriptive text.[81] Focus should instead be on what makes the landscape or feature outstanding, what in it needs to be protected, or what needs to be controlled to protect it from inappropriate subdivision, use and development.[82] As an example, this could be done by analysis of the way the landscape is likely to respond to change, including (taking concepts from the NZILA Best Practice Guide) its resilience, capacity, sensitivity and vulnerability. Plans could then identify the opportunities, risks, threats, costs and benefits arising from potential change, and their relative magnitude and importance.[83] This analytical process could then inform the judgement to be made about what activity is inappropriate.[84]

Of course, there are issues associated with scheduling ONL and ONF values. Such undertakings can be 'herculean' tasks,[85] well outside the budget of many local authorities. Further, doing so risks being overly prescriptive or missing things.[86] In the absence of a particular value being scheduled, submitters can be left with the burden of calling evidence on further values (the applicant not having the incentive and the Council choosing not to do so).[87]

Finally, on the issue of who is paying for all this splendour, there is nothing in the RMA that gives farmers a privileged place in the scheme of things.[88] Just as there can be no blanket approach requiring that all use and development be prohibited, there can be no doctrinaire approach that says where a farm's future financial viability is affected, there must be no protection of the ONL.[89] Local authorities are best advised to avoid that consequence occurring, but it may not be possible in some situations. If landowners are not able to comply with plan rules, they will have to make a decision to seek a resource consent, to change the land use practice, or to move somewhere else.[90]

Queen Charlotte Sound

A spotlight on landscape protection in the Mackenzie Basin

The issue of landowners' (primarily farmers') rights and landscape protection came to the fore in 2017 in a lengthy decision of the Environment Court on PC13 of the Mackenzie District Plan. As described in Chapter 7, the purpose of PC13 is to "provide greater protection of the landscape values of the Mackenzie Basin from inappropriate subdivision, development and use". The Court was tasked with deciding whether regulation (PC13) was an efficient and effective means of achieving the desired social outcomes,[91] balancing competing interests of private landowners and the public good.

Although neither of the two main economic experts in the case attempted to quantify the cost of the potential loss of the inherent (ie not tourism related) value of the landscape of the Mackenzie Basin (if pastoral intensification was not regulated) the Court held an attempt would have been useful.[92] The decision sets out the following quote from text by D Moran on The Economic Valuation of Landscapes:[93]

> ... the production of landscape falls under the rubric of market failure. In essence the public cannot easily transact to satisfy a demand for landscape as a good. In the absence of a demand backed by a willingness to pay, land owners, predominantly but not exclusively farmers, may not be motivated to provide the features that might match demand. This is because landscape is a public good and they cannot capture benefits from all forms of users. Accordingly, and provided landscape is valuable to the public, there is a rationale for government intervention to stimulate the supply of features that are deemed to be in the public interest.

The Court goes on to accept economic evidence which acknowledges that land which has a significant public benefit (for example, land that is an ONL) will incur private costs associated with maintaining or enhancing that value. Although regulation may highlight that tension, it does not have to solve it.[94]

On balance, the Court held that PC 13 was a more efficient use of the land and landscape of the Mackenzie Basin than the status quo.[95] Ultimately, in the absence of analysis quantifying the externalities, which are the adverse effects of pastoral intensification and agricultural conversion on the scenic qualities of the ONL, the Court was satisfied that regulation would better manage these externalities than no regulation.

4.2 Cultural landscapes

Although the RMA does not refer to 'cultural landscapes', it has been common parlance in case law for decades. Court references to cultural landscapes use the phrase as a description of a landscape that is no longer natural but is one that has been 'cultured' in some form by human endeavours. Other references are to a collection of historic sites or places (of both Māori and European importance) that comprise a cultural landscape.

However, indigenous peoples including Māori traditionally passed down landscape associations through oral and artistic traditions, including speeches, story-telling, songs and carving:

> As Māori we have a unique sense of our 'landscape'. It includes past, present, and future. It includes both physical and spiritual dimensions. It is how we express ourselves in our environment. It connects whānau, flora and fauna, through whakapapa. It does not disconnect urban from rural. It transcends the boundaries of 'land'scape into other 'scapes'; rivers, lakes, ocean and sky. It is enshrined in our whakapapa, pepeha, tauparapara, whaikorero, karakia, waiata, tikanga, ngā korero a kui mā a koroua mā, and our mahi toi. It is not just where we live – it is who we are![96]

This strong oral and creative tradition means that very little of that information is often recorded in books, scripts or reports.[97] The lack of documented information is not a measure or reflection of the amount or strength of cultural associations that Māori have with their landscapes. However, the paucity of easily accessible information in a written form has become an issue in a modern planning and policy context, where decision-making about proposed land use activities and development is based on the information available to those decision-makers.[98]

Therefore, in the context of planning, policy-making or any other process that affects landscape matters, engagement of mana whenua and development of partnerships between councils and other responsible agencies and organisations are critical. This is so those agency partners can develop an understanding through korero (talking) about mana whenua issues, rights and interests in relation to landscapes and incorporate them into planning documents.

Many iwi and hapū are beginning to address the 'knowledge-gap' in written documents by developing their own planning documents (often referred to as iwi management plans) and adopting the concept of landscape within those. Planning mechanisms such as these are now strategically employed by some iwi and hapū to help identify, document and/or direct protection and management of their cultural landscapes. Through their own terminology

and expression, mana whenua can articulate their values and relational connections with their ancestral taonga (mountains, rivers, sea, wāhi tapu) and wider landscapes.

A spotlight on the protection of cultural landscapes on Banks Peninsula

Through a Ngāi Tahu submission on the proposed Christchurch District Plan, a cultural landscape approach has been incorporated into the Plan. As explained in the relevant iwi management plan:

> A cultural landscape approach is used by Papatipu Rūnanga to identify and protect tangata whenua values and interests from the effects of subdivision, land use change and development. While many specific sites (e.g. pā sites) are protected as recognised historic heritage, the wider contexts, settings or landscapes in which they occur are not. A cultural landscape approach enables a holistic identification and assessment of sites of significance, and other values of importance such as waterways, wetlands and waipuna.[99]

A range of cultural layers have now been incorporated into the District Plan. At the broader level, they include Ngā Tūranga Tūpuna – a cultural landscape layer which identifies specific landscapes of cultural importance. The Plan provides a set of policies that apply to the area but there are no rules attached to the layers. However, if a resource consent is required by other rules in the Plan, then this needs to address cultural values which in turn triggers consultation with rūnanga. The rūnanga can then provide practical advice on how their values can be protected.

Below these broader cultural landscape layers more discrete culturally-important sites are identified in the Plan as Wāhi Tapu/Wāhi Taonga or silent files. Rather than identifying the precise location of a site, a silent file gives a general indication of its location and triggers the need for engagement with mana whenua over a development proposal that might affect that locality.

The Plan also includes Nga Wai which cover estuaries, rivers and the coastal margin around the Peninsula. These reflect the statutory acknowledgement areas under the Ngāi Tahu Claims Settlement Act 1998. "Nga wai are water bodies, including wetlands, tupuna (springs) and coastal waters which are significant areas of mahinga kai or other customary use for Ngāi Tahu."[100]

The role of a landscape architect in articulating cultural landscapes is debated. Some Environment Court divisions have criticised landscape experts for overreaching into this domain, while other divisions have preferred landscape evidence that incorporated mana whenua values over those that did not. One landscape architect posits that landscape architect contributions might be in "linking with planning policy, mapping, making connections and describing how such intangible aspects might be expressed, protected and enhanced, as well as including in assessment".[101] This could occur by listening, locating, explaining and incorporating, with landscape architects serving to include those stories in planning records and documents.

While the concept of a cultural landscape has been accepted for a long time as being relevant to consideration of Part 2 provisions relating to Māori (ie sections 6(e), 7(a) and 8),[102] the same cannot be said with respect to cultural landscapes and section 6(b) of the RMA. Historically, Courts appear to have dabbled with, but not entirely embraced, the idea that a landscape may be worthy of protection under section 6(b) primarily due to its Māori cultural connections. Hesitation appears to have arisen out of desire to avoid the 'double counting' of Part 2 issues. As the Court has repeatedly stated, Māori issues are specifically provided for in sections 6(e), 7(a) and 8 of the Act.[103]

One issue with identifying and recognising cultural landscapes under section 6(b) arises with the word 'natural' in ONLs and ONFs. This issue also arises in the context of heritage landscapes. Not all cultural (or heritage) elements of a landscape are natural – some represent built form, which can detract from the naturalness of an environment.

Ōnuku Marae

A spotlight on cultural landscape protection on Matakana Island

Western Bay of Plenty District Council v Bay of Plenty Regional Council is a 'game changer' for cultural landscapes and their relevance to identification of ONLs and ONFs. It brought Māori cultural landscape values to the fore when considering whether to extend the outstanding natural features and landscapes category on Matakana Island to include parts of the island which comprise production forestry on a sand barrier.

In that case, the Court accepted that while it needs to examine details which may be wholly within the scope of only one of the matters of national importance, it must still "keep the wood in our mind's eye as we examine particular trees".[104] This is especially so where the matter of landscape axiomatically involves associative matters such as cultural heritage.[105] The Court dismissed arguments of 'double counting' factors that may be common to two or more matters of national importance, noting that the risk was not so great as to inhibit the general (ie integrated) approach to Part 2.[106]

The hapū evidence in *Western Bay of Plenty* demonstrated very longstanding connections to the sand barrier which made the shared historical associations of tangata whenua a significant component of the way in which the sand barrier was perceived.[107] These associational elements of the landscape were considered to be very high[108] and were an important, arguably the most important, factor in the Court's overall assessment under section 6(b) to include it in the 'outstanding natural features and landscapes' category. Although the decision does not explicitly state that the forested sand barrier is an ONL or ONF by virtue *only* of its cultural landscape, it is very clear that the cultural landscape 'tipped' the landscape into a section 6(b) matter.

In the coastal environment, Policy 15(c)(viii) of the NZCPS states that cultural landscapes are a relevant consideration when identifying natural features and landscapes. In a recent case, Te Akitai Waiohua successfully proved the significance of their cultural landscape and importance of its protection over and above development. The Environment Court accepted the cultural landscape as a reason not to provide for urban development in an ONF in the coastal environment (see spotlight).[109] The planning context, the fact that this was the last chance for Te Akitai Waiohua to have its relationship recognised and provided for, and the directives of section 6(e) and 8 of the RMA all supported the Court's decision.

For cultural landscape values to be protected, it is important that they are identified and mapped in statutory plans. Plans also need to contain supporting objectives, policies and assessment criteria.

A spotlight on the protection of a Māori cultural landscape within the Puhinui Precinct

An approach that has shown promise in recognising and protecting Māori cultural landscapes is the use of precinct plans. Precincts are a planning mechanism that allow local differences to be recognised through detailed place-based provisions. Provisions in the Auckland Unitary Plan were co-developed to specifically identify and map Te Akitai Waiohua's cultural landscapes within the Puhinui Precinct.

The Puhinui area, of approximately 1,100ha, is located next to the Manukau Harbour and Auckland International Airport. It has significant Māori cultural heritage and landscape values including the Pūkaki Crater and Crater Hill (identified as ONFs in the Auckland Unitary Plan) and the Puhinui Stream, identified as a Significant Ecological Area in recognition of its high habitat values. The Pūkaki Crater and lagoon (Te Pūkakitapu o Poutukeka) is ancestral Māori land of particular spiritual value to tangata whenua, and ownership is held by the Pūkaki Māori Marae Committee.

The precinct approach was taken at Puhinui to resolve competing land use issues and priorities between mana whenua, transport links – the significance of Puhinui Road (State Highway 20B) as a direct transport link between the Auckland International Airport and the South-Western Motorway – and private landowner interests, which were represented by the Southern Gateway Consortium.

In 2010, four months before the Auckland Council amalgamation, the Consortium lodged an application under the Auckland Council District Plan – Operative Manukau Section 2002 that sought to rezone 150ha of rural land outside the Metropolitan Urban Limits to a business zoning. The council sought further information on significant outstanding matters relating to mana whenua engagement on historic heritage, cultural values, transport and stormwater management matters.

In 2013, Council accepted the private plan change for public notification. However, it deferred its consideration of a change to the Metropolitan Urban Limits until a comprehensive planning process covering the whole Puhinui area had been completed. This process evolved into the Puhinui Structure Plan, the preparation of which was approved by the Council in February 2014.

Te Ākitai Waiohua opposed the Consortium application and, as a result, the Council worked in partnership with the hapū to address the concerns raised in their submission. This included commissioning a Māori cultural heritage assessment and identifying the cultural landscape values of importance to them. The Council then co-developed provisions with Te Ākitai Waiohua for the Puhinui area that included objectives, policies and assessment criteria to manage the effects of future development on the cultural landscape values of the hapū. This was to help ensure that an integrated approach was taken to managing potential adverse effects on cultural values.[110] The Puhinui precinct provisions evolved from the structure planning that the Council undertook for the Puhinui area. The provisions were included in the Auckland Unitary Plan which became operative (in part) in 2016. They state:

> The primary purpose of the Puhinui Precinct is to enable a transition from rural to urban development, while recognising the cultural, spiritual and historical values and relationships that Te Ākitai Waiohua have with the land and sea in Puhinui as part of the Māori cultural landscape. The precinct also recognises the relationship which exists between Māori cultural landscape values and the management of natural and physical resources.[111]

The cultural landscape applies to the entire precinct. Four maps included in the Plan identify respective values for protection, with Precinct Plan 1 being the most comprehensive. It spatially identifies the following elements: Important site or place; Puhinui heritage gateway; Mana whenua management area; Acknowledgement areas; Views and spiritual connections; Pā/settlement; View/linkage; Portage route; Coastal edge; Ephemeral streams; Intermittent streams; Permanent streams; Creek; and Wetland. The other plans provide more information about specific features (such as streams and viewshafts). This planning approach reflects an understanding of the broad but integrated nature of Te Ākitai Waiohua's interests, which encompass the geological, coastal, archaeological, and ecological features within the precinct.

The maps can be used to gauge whether a potential project or application could have potential effects on the identified values. The consent process then provides a direct opportunity for the hapū to engage and prepare a cultural impact assessment if Council concludes that a proposal may cause adverse effects on Māori cultural landscape values. This enables mana whenua to be actively engaged in the consideration of proposed land use activities and potential developments, and provides the opportunity for Council, mana whenua and applicants to work together to avoid, remedy, or mitigate potential adverse effects. It could even provide opportunity for those involved to consider how they might enhance or elevate Māori cultural landscape values for any particular project.

The clear articulation of mana whenua values associated with a specific, identifiable landscape played an important part in an Environment Court decision[112] to deny a request by a private landowner collective to rezone their land so that development rules were more permissive.

Ōtuataua Stonefields, Auckland

The recent report from the Resource Management Review Panel (RMR Panel) proposes that "protection and restoration of the relationship of iwi, hapū and whanau and their tīkanga and traditions with their ancestral lands, *cultural landscapes*, water and sites" be provided for by decision makers. The report states:

> We propose cultural landscape to be "a defined area or place with strong significance for mana whenua arising from cultural or historical associations and includes connected natural, physical or metaphysical markers or features". This is an important change. Recognition of interconnections and that a cultural landscape can be 'more than the sum of its parts' will enable the multi-faceted relationships that mana whenua have with land and water to be adequately protected and restored.[113]

A spotlight on landscape provisions in the Thames-Coromandel Proposed District Plan

To date, no strategic (district or region-wide) landscape study has comprehensively addressed associative values (although this has been undertaken on a site or project basis). The Thames-Coromandel Proposed District Plan attempted to do this but, to date, has not been successful.

The Proposed District Plan was notified in December 2013. Its landscape section was informed by a 2011 landscape study undertaken by one of Aotearoa New Zealand's most experienced landscape architects. The study methodology employed the amended Pigeon Bay factors, in conjunction with public perception research from Lincoln University, to identify section 6(a) and (b) landscapes. It included a detailed inventory of the characteristics and values of the district's ONLs and ONFs and areas of outstanding and high natural character.

The landscape inventory was subsequently translated into a proposed appendix to the Proposed District Plan. The appendix is lengthy (some 108 pages long) and addresses each of the amended Pigeon Bay factors, with the noticeable exception of associative values, which is left blank. Identifying and understanding the associative characteristics of a landscape requires input from tangata whenua. Accessing such information can be challenging and, although attempts were made to obtain it for the Thames-Coromandel district landscape assessment, these were not successful.

In June 2016, the Ngāti Maru Runanga Trust filed an appeal against the Thames-Coromandel District Council's decision on the Proposed District Plan.[114] The appeal stated that the areas of ONL and ONF and areas of outstanding and high natural character did not include mana whenua world views and would 'sterilize' Māori land and Treaty settlement land. It explained, this was because much Treaty settlement land was undeveloped and of high value and, in accordance with a Eurocentric world view, worthy of protection. It also stated that the outcome collided with the Māori world view that the land was productive and should be available for tangata whenua use. It also collided with Waitangi Tribunal decisions that consistently return unproductive Crown land to iwi – land that is often within coastal and hill country margins that are captured by the ONLs identified across Aotearoa New Zealand.

In February 2020, a consent order was issued for the landscape topic of the Proposed District Plan. To address part of Ngāti Maru's appeal, the Order provided for more permissive rules relating to buildings, dwellings and subdivision on Māori Land or Treaty settlement land within ONL and ONF areas. Relief sought by the Ngāti Maru appeal, to amend the proposed appendix of values and characteristics of ONLs and ONFs, remains outstanding. This relief seeks to fill in the 'blanks' of the proposed landscape appendix with respect to associative values.

In December 2017, the Thames-Coromandel District Council passed a resolution to initiate a variation to the Proposed District Plan which would complete the associative values section of the landscape appendix for all 88 sites listed.[115] To date, seven years after the Plan was first notified, and three years after the Council resolved to initiate a variation, it is still not equipped with the information necessary to complete the landscape appendix.

Effective engagement with tangata whenua is critical to incorporating associative values of a landscape into regional and district plans. Failure to successfully achieve that risks undermining professional landscape assessments, and the plan provisions that implement them, particularly as:

- Associative or relationship contributions are an accepted 'dimension' of landscapes;[116]
- Cultural and spiritual values for tangata whenua must be had regard to when identifying and assessing natural landscapes and features in the coastal environment;[117] and
- Recourse back to Part 2 remains appropriate for resource consent applications in certain situations.[118]

4.3 Heritage landscapes

The heritage landscape construct rose to prominence in Aotearoa New Zealand largely as a result of a 2004 study led by the Department of Conservation (DOC) called the Bannockburn Heritage Landscape Study.[119] The study was initiated due to concerns that heritage management agencies tended to focus on individual heritage sites and features. It sought to trial a newly developed methodology for investigating heritage at a landscape scale.[120] The authors of the study entered into a complex and detailed interdisciplinary methodology of spatial analysis, using the connection between superimposed layers of history.[121]

Heritage Landscapes as defined in the Bannockburn Study

A heritage landscape is a landscape, or network of sites, which has heritage significance to communities, tangata whenua, and/or the nation. The landscape methodology uses the concept of layered webs to analyse and highlight key relationships between physical remains, key stories and contemporary associations. As development and subdivision make their own marks on the landscape, the older continuities become fainter, and their cohesion as a physical aspect of the past become more difficult to establish. A landscape approach, recognising the interconnectedness of physical remains and stories associated with the land, can help to bring together understanding about the different traces of the past on the landscape, and how and why it is valued by people today.

Two main questions arise with the concept of heritage landscapes under the RMA.[122] First, can a landscape can be classified as an ONL or ONF by virtue of its heritage elements? Second, can heritage landscapes can be protected under section 6(f), which provides for the protection of historic heritage from inappropriate subdivision, use, and development?

Underpinning both issues are concerns as to double-counting on the basis that heritage landscapes are already protected by other sections of the RMA. For example, Māori values are recognised and protected under sections 6(e), 7(a) and 8 of the RMA. Heritage values also inevitably overlap with landscape and often heritage values will contribute to the assessment that a landscape is outstanding in terms of section 6(b) of the RMA. While the need for caution is justified, it is artificial to attempt to segregate or 'quarantine' matters such as landscape and heritage values, consideration of which must inevitably overlap.[123]

For example, historical features can give particular character to a landscape such that together they make the wider landscape outstanding as a natural landscape.[124] If a natural landscape is outstanding for its heritage, the extent of heritage items and associations must be such as to give the landscape its particular character.[125]

An impediment to widespread acceptance of heritage landscapes within section 6(b) is that often items cited as contributing to the historical ambience of an area are of human construction and do not contribute to the 'natural' landscape to qualify for consideration under section 6(b).[126] As a result, individual buildings and sites are often left marooned while the historic heritage landscape of which they are part is devastated.[127]

In that regard, commentators consider the institutional and philosophical divide in New Zealand between those who care for nature and those who care for heritage to be dysfunctional.[128] Falling between these two ideals are heritage landscapes and as a result they are being destroyed and degraded. High country landscapes are an example of this, where DOC is purchasing areas of outstanding 'natural' value, while land which does not meet this description is being freeholded and sometimes subdivided, breaking up historic stations.[129]

If a landscape does not qualify under section 6(b) as being natural, whether it can qualify under section 6(f) is also relevant.[130] Section 6(f) provides for the "protection of historic heritage from inappropriate subdivision, use, and development". The definition of historic heritage in section 2 includes "historic sites, structures, places, and areas" and "sites of significance to Māori, including wāhi tapu", but does not explicitly refer to landscapes.

Akaroa Wharf

The Environment Court has since developed three lines of thinking on the concept of whether section 6(f) encompasses heritage landscapes. This has ranged from acceptance to full rejection of the concept. [131]

Taking a purposive interpretation of section 6(f), the Courts have accepted that a collection of historic 'sites', 'places' or 'areas' can be described as a heritage landscape.[132] While not accepting *per se* a heritage landscape, the High Court has also accepted that the protection afforded by section 6(f) may extend to "the land on which a heritage building is sited, its precincts and the relationship of the heritage item with its built context and other surroundings".[133]

The challenge with this line of thinking is that section 6(f) contains no threshold qualifier such as 'outstanding' or 'significant'. Therefore what must be protected is to be drawn from the introductory words of section 6; that it is a matter of national importance.[134] In this regard, the protection afforded by s 6(f) should only be extended to the surrounding land of the historic sites or places (eg the wider heritage landscape) if there is sufficient intensity of heritage fabric woven into the landscape to warrant the application of section 6 of the RMA.[135] This necessitates probative evidence to that effect, which in a world of ever changing heritage, is a difficult exercise

While in some cases the Courts have accepted the concept of a heritage landscape, there are few cases which have found one to actually exist. A lack of probative evidence is one reason for this. The Environment Court appears to have set the bar for 'probative evidence' at the level of the Bannockburn Study (or similar);[136] putting the identification of heritage landscapes beyond the expertise of landscape architects alone and beyond the financial capabilities of most councils and applicants.

Ihumātoa Peninsula

Underlying the Court's need for this depth of study and analysis is an apparent concern that extending heritage planning provisions to landscapes would 'lock up' large areas of private land.[137] That may be the case for provisions which seek to preserve site specific areas. However, neither section 6(f) nor the Bannockburn Report require preservation of heritage landscapes. Instead, section 6(f) requires protection from 'inappropriate' subdivision, use and development.

Providing for some areas of conservation and sympathetic development can, therefore, go hand in hand with a heritage landscape. They need not be mutually exclusive. There are many creative ways in which heritage landscapes in private ownership might be protected.[138] The question at the end of the day is whether the proposal would overwhelm the heritage landscape.

In places where there is uncertainty or a lack of information regarding mana whenua values associated with a heritage landscape, courts are reluctant to provide explicit protection.[139] But, perhaps most peculiarly, there has also been resistance to accepting the concept, *in spite of* strong evidence demonstrating the presence of a historical landscape.

A spotlight on the Ihumātoa Peninsula

The case of *Gavin v Auckland Council*, contrary to the Council's decision, extended the metropolitan urban limit to include the now infamous Ihumātao Peninsula. In that case, both landscape architects (who were very experienced and highly regarded) acting for the Council and the landowners, considered that the Peninsula was a heritage landscape.[140] The Council's landscape architect was also supported by a heritage expert and an archaeologist.

The Court, after accepting that it was open to it to find on sufficiently probative evidence, that a landscape is a heritage landscape under section 6(f), found difficulty endorsing the concept because:[141]

> [a] Heritage landscape is not a concept referred to in the Act;
>
> [b] Outstanding landscapes and features are protected from inappropriate subdivision use and development by section 6(b) of the Act;
>
> [c] Māori values are recognised and protected by sections 6(e), 7(a) and 8 of the Act;
>
> [d] Historic heritage is protected from inappropriate subdivision use and development by section 6(f) of the Act; and
>
> [e] There are also other important matters provided for in the Act that would apply, such as matters relating to amenity, indigenous vegetation, natural character and coastal environment, that may at times be relevant to a given situation.

The Court held that to introduce a new concept not recognised explicitly by the RMA would add to the already complex web of the Act and make matters more confusing.[142] However, while not explicitly recognised by the RMA, the Act already provides for the concept of heritage landscapes, both in section 6(b) and 6(f). As now retired Environment Court Judge Kenderdine said in her address to the NZILA conference in 2005:

> Nevertheless, when considered alongside the broad definition of 'historic heritage', the 2003 amendment will require greater weight to be given to heritage sites and areas. The provision of s2(a)(iii) 'cultural', (iv) 'historic' and s2(b)(i) 'historic sites' and 'areas' together with (b)(iii) 'sites of significance to Māori' and 'surroundings' in s2(b)(iv) as well as the inclusive nature of the provision overall, provides considerable scope for asserting an area, place, feature, as a 'landscape' contributes to historic heritage in planning and resource consent processes.
>
> In fact, taken in conjunction with the existing s6(e) relating to the relationship of Māori and their culture and traditions with their ancestral lands, water, sites, wahi tapu, and other taonga, the presenters of the Law Society Seminar on the 2003 Amendment to the RMA consider the possible ramifications of s6(f) may be widespread in cultural terms.[143] I agree. As a result I do not see why historic heritage landscapes cannot be a consideration in regional and district plans. There is sufficient leeway in the definition of 'Historic Heritage'. And its elevation to a matter of national importance gives that added distinction.[144]

Further, the majority of the land on the Ihumātao Peninsula in *Gavin* is within the coastal environment. Policy 15(c)(viii) of the NZCPS expressly includes cultural landscapes as a matter that must be considered when identifying and assessing the natural landscape.

Aotearoa New Zealand's provisions for the protection of heritage landscapes are weak compared with many other countries[145] and one cannot help but wonder if *Gavin* was the country's chance to significantly recognise and start providing for its heritage landscapes. The jurisprudence was there to support the concept, as was the evidence. Sadly, however, the Court did not take the reins on that occasion.

After the Court's decision in *Gavin*, the newly zoned Future Development land at Ihumātoa Peninsula was sold to Fletcher Residential and designated as a Special Housing Area. The "tragedy of Ihumātoa"[146] was extensively canvassed by the media as protestors, mobilised by the Save Our Unique Landscape campaign led by mana whenua members whose families have resided in Ihumātao for many generations, gathered *en masse* to oppose the move. Prime Minister Jacinda Ardern entered the fray vowing that no building would take place until the parties brokered a solution.[147] Mana whenua have recently reached a consensus over what they want to do with the land – they want it back.[148] At the time of writing, a deal had been reached on the future of Ihumātao – with the government purchasing the disputed land from Fletchers for almost $30 million.[149]

It is notable that mana whenua chose to strategically and politically use the concept 'cultural landscape', when asserting their rights and interests in their ancestral whenua, with Save Our Unique Landscape stating on its website "The entire Ihumātao area is a cultural landscape, embedded with identity, meaning, and significance."[150]

4.4 Amenity landscapes

Section 6 is not the end of the story for landscape consideration under the RMA. While section 6(b) provides protection for ONLs, all landscapes need to be considered under sections 5(2) and 7(b), (c), (d), (f) and (g).[151]

Sections 7(c) and 7(f) apply to landscapes that contribute to amenity values or the quality of the environment and create an obligation on decision-makers to have particular regard to the maintenance and enhancement of them. 'Amenity values' are defined by the RMA as "those natural or physical qualities and characteristics of an area that contribute to people's appreciation of its pleasantness, aesthetic coherence, and cultural and recreational attributes".[152]

There is a significant degree of overlap between 'landscape' and 'amenity'.[153] Like landscape, amenity values are multi-dimensional. Subject to any directions and priorities as specified in planning instruments, they encompass more than just visual effects, and extend to sensory or other ways in which people enjoy their living environments.[154] Like landscape values they involve human perception of the physical environment, the attachment of values to different locations and places, and the shaping of both values and identity by cultural mores and associations.[155]

Yet amenity has remained the 'little brother'[156] of landscapes. This is reflected in cases which relegate section 7 landscapes to a 'second tier' of landscapes, denoted on planning maps or described in plans as, variously, 'amenity landscapes', 'rural amenity landscapes', 'visual amenity landscapes' or 'significant amenity landscapes'.

So, what is the difference? The *WESI* decision described the ONLs of the Queenstown Lakes District as the romantic landscapes – the mountains and lakes, and the visual amenity landscapes as those that 'wear a cloak of human activity'.[157] They are pastoral or Arcadian landscapes with more houses and trees and greener (introduced) grasses.[158]

In essence, amenity landscapes are significant landscapes, but are not quite natural enough to qualify under section 6(b). Often amenity landscapes form a buffer between ONLs and landscapes with no particular values meriting management or urban areas.

When considering amenity values of a landscape, the Courts typically start with the residents' views.[159] However, variability of perception commonly arises in cases[160] and the topic of amenity can be emotionally charged.[161] People tend to feel very strongly about the amenity they perceive they enjoy[162] and often resist

Muriwai *(Craig Potton)*

change simply because it is different to what they know.[163] As such, individual perceptions of the effects of a proposal on their future amenities will not usually be a sufficient guide to reasonableness of the effects.[164]

The Courts then look to the district plan for context,[165] and having ascertained the amenity values, seek to have them objectively tested by landscape experts who will advise whether the basis for those values are reasonably held and then assess whether the proposal gives rise to landscape and visual effects.[166] Finally, the Court assesses the landscape and effects on amenity values in light of the outcomes for the relevant resources and values specified under the district plans.[167]

Although sections 7(c) and (f) require that decision-makers have particular regard to the maintenance and enhancement of amenity values and the quality of the environment (respectively), the RMA does not necessarily protect the status quo of section 7 landscapes; change is provided for. Section 7 imparts no preference, for example, for grass over pine or farms over lifestyle blocks (which could include restoration of indigenous bush, grasses or wetlands).

Care must be taken with this line of thinking. While it is true that some development proposals add to the naturalness of an area through extensive planting, the Court has expressed scepticism about whether screening is an effective method to avoid, remedy or mitigate adverse amenity effects. That is particularly so within the coastal environment where one of the prime motivations for people to buy sections and later residences will be the presence of views, which screening vegetation can block.[168]

4.5 Coastal landscapes

*Land*scape protection and management in Aotearoa New Zealand implicitly includes *sea*scapes too, which indicates that the term 'landscape' may not be fully fit for purpose. For Māori in particular, coastal and freshwater features of significance are usually included in cultural landscape descriptions. The term 'coastal environment' encompasses both land and sea and their coastal interface.

The preservation of the 'natural character' (as opposed to 'landscapes') of the coastal environment has been a matter of national importance in environmental legislation since 1977.[169] Drafted in response to rampant coastal subdivision and development, legislative recognition of the natural character of the coastal environment was considered to be a more workable option for strengthening the management of the coast than an alternative proposal to place a national moratorium on coastal development.[170]

The RMA brought forward the earlier legislative recognition, expanding it slightly so that section 6(a) of the RMA now provides for the "preservation of the natural character of the coastal environment (including the coastal marine area), wetlands, and lakes and rivers and their margins and the protection of it from inappropriate subdivision, use and development". At the same time, as already mentioned, a new section 6(b) was introduced into the RMA which referenced natural landscapes. There is not a tidy distinction between the two concepts and they often overlap with, for example, ONLs often include areas with high natural character and vice versa.

Both landscape and natural character involve biophysical as well as experiential aspects (including sounds, smells and movement as well as scenic considerations). However, they are not the same thing.[171] The associative values (how societies have associated with a place, such as heritage aspects), is only addressed in terms of natural features and landscapes. They are not part of natural character.[172]

Boffa Miskell has produced a diagram illustrating the relationship between how its practice interprets the natural character/landscape interface (see Figure 4.1). This is in direct response to Policies 13 of the NZCPS which states that "natural character is not the same as natural features and landscapes or amenity values". In accordance with its approach, natural character sits on the natural science end of landscape and includes a small part of the human sensory aspects (ie experiential). Historic and heritage values are not included in the assessment, as they are not listed under the matters to be included within Policy 13.

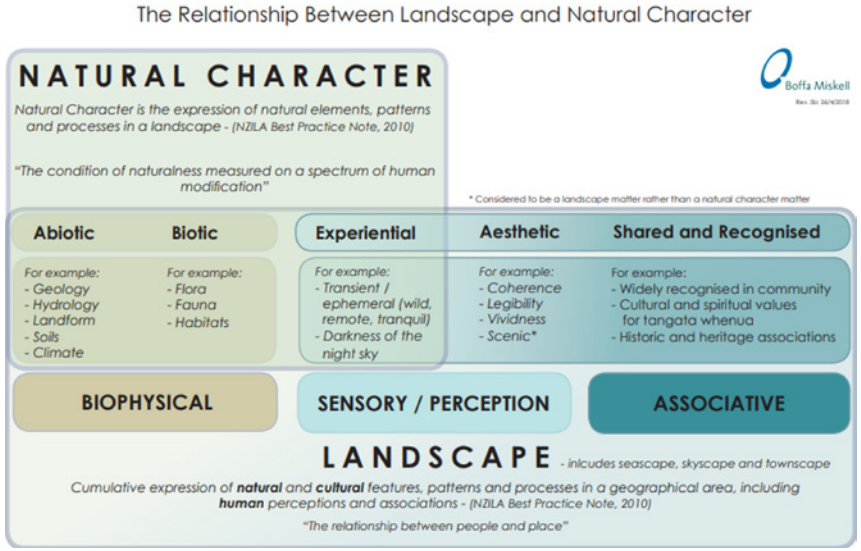


Figure 4.1: *Relationship between landscape and natural character (Source: Boffa Miskell)*

Whilst this diagram has been helpful to many, there is some disagreement among practitioners about the approach, with some experts interviewed considering that natural character also includes associative values with, for example, Māori cultural values being an aspect of natural character.

The identification of landscapes or features, and natural character in the coastal environment is guided by the NZCPS, which is discussed below and further in Chapter 6.

Identifying coastal landscapes

For any landscape or feature in the coastal environment, Policy 15(c) of the NZCPS displaces the list for landscape assessment set out in case law (as described above).[173] While the assessment must be against the policy itself, case law can be useful in elucidating the text and purpose of the policy, and tracing its evolution.[174]

NZCPS, Policy 15 Natural features and natural landscapes

(1) To protect the natural features and natural landscapes (including seascapes) of the coastal environment from inappropriate subdivision, use, and development:

a. avoid adverse effects of activities on outstanding natural features and outstanding natural landscapes in the coastal environment; and

b. avoid significant adverse effects and avoid, remedy, or mitigate other adverse effects of activities on other natural features and natural landscapes in the coastal environment; including by:

c. identifying and assessing the natural features and natural landscapes of the coastal environment of the region or district, at minimum by land typing, soil characterisation and landscape characterisation and having regard to:

i. natural science factors, including geological, topographical, ecological and dynamic components;

ii. the presence of water including in seas, lakes, rivers and streams;

iii. legibility or expressiveness – how obviously the feature or landscape demonstrates its formative processes;

iv. aesthetic values including memorability and naturalness;

v. vegetation (native and exotic);

vi. transient values, including presence of wildlife or other values at certain times of the day or year;

vii. whether the values are shared and recognised;

viii. cultural and spiritual values for tangata whenua, identified by working, as far as practicable, in accordance with tikanga Māori; including their expression as cultural landscapes and features;

ix. historical and heritage associations; and

x. wild or scenic values;

d. ensuring that regional policy statements, and plans, map or otherwise identify areas where the protection of natural features and natural landscapes requires objectives, policies and rules; and

e. including the objectives, policies and rules required by (d) in plans.

The matters to have regard to largely mirror the amended Pigeon Bay criteria, with some changes. For example, there are additional references in Policy 15(c) to the presence of water and vegetation (including exotic vegetation) which appear to have their source in the Court's criteria of naturalness; wild and scenic values; cultural values being identified in accordance with tikanga Māori; and the expression of cultural and spiritual values as cultural and landscape features.

Purerua Peninsula

Identifying natural character

The NZCPS also attempted to distill the approaches adopted by the profession and the Courts, to codify the case law on what natural character is, in Policy 13(2). The Environment Court has stated that 'natural character' in section 6(a) of the RMA needs to be read in light of the particularisation in Policy 13(2).[175] A common broad brush description is that natural character is the level of actual (abiotic and biotic) and perceived (perceptual and experiential) naturalness of an area.[176] The natural character of the coastal environment is to be assessed holistically, without artificial distinctions between terrestrial and water components.[177]

NZCPS, Policy 13 Preservation of natural character

(1) To preserve the natural character of the coastal environment and to protect it from inappropriate subdivision, use, and development:

(a) avoid adverse effects of activities on natural character in areas of the coastal environment with outstanding natural character; and

(b) avoid significant adverse effects and avoid, remedy or mitigate other adverse effects of activities on natural character in all other areas of the coastal environment;

including by:

(c) assessing the natural character of the coastal environment of the region or district, by mapping or otherwise identifying at least areas of high natural character; and

(d) ensuring that regional policy statements, and plans, identify areas where preserving natural character requires objectives, policies and rules, and include those provisions.

(2) Recognise that natural character is not the same as natural features and landscapes or amenity values and may include matters such as:

(a) natural elements, processes and patterns;

(b) biophysical, ecological, geological and geomorphological aspects;

(c) natural landforms such as headlands, peninsulas, cliffs, dunes, wetlands, reefs, freshwater springs and surf breaks;

(d) the natural movement of water and sediment;

(e) the natural darkness of the night sky;

(f) places or areas that are wild or scenic;

(g) a range of natural character from pristine to modified; and

(h) experiential attributes, including the sounds and smell of the sea; and their context or setting.

Natural character cannot be determined via scientific analysis uncontaminated by matters such as public opinion.[178] Rather, determining the natural character values of an area involves a high degree of evaluative judgment.[179] Although a natural character assessment properly commences with consideration of a factual and science-focused enquiry of the biophysical status of an area, this is only the first step in understanding the degree of naturalness of an area. As character is a perceived value, the second step is to evaluate how people would sense and experience the naturalness of that environment.

In response to interpretation and implementation issues, DOC undertook workshops on the NZCPS during 2010 and 2011. Participants agreed that the Policy 13(2) list above provided direction on the essential elements of natural character based on case law, but that it was not definitive:[180] The following definition was agreed upon by most of the workshop participants:

> Natural character is the term used to describe the natural elements of all coastal environments. The degree or level of natural character within an environment depends on:
>
> 1. The extent to which the natural elements, patterns and processes occur
> 2. The nature and extent of modification to the ecosystems and landscape/seascape. The degree of natural character is highest where there is least modification.
>
> The effect of different types of modification upon natural character varies with context and may be perceived differently by different parts of the community.[181]

This definition brings together quantitative biophysical parts of natural character and perceptual elements – an approach now supported by the Courts[182] and most practitioners. Consensus is more precarious when discussing the weighting that should be applied between 'science' factors and 'perceptual' factors of natural character.

Natural character thresholds

Once natural character is identified, it must then be assessed against a scale of naturalness. Natural character is commonly rated on a seven-point scale from very low (ie highly modified), low, moderate-low, moderate, moderate-high, high and very high (ie virtually unmodified).

Section 6(a) of the RMA and Policy 13(1) seek to preserve all the natural character of the coastal environment and protect it from inappropriate subdivision, use and development. However, somewhat confusingly, the policy then requires councils to identify areas of 'high' natural character but requires that effects only be avoided in areas of 'outstanding' natural character. In other areas not classified as outstanding, the direction is to avoid 'significant adverse effects' and avoid, remedy or mitigate other adverse effects.

The confusing nature of the Policy may be due to changes made to its wording after the Board of Inquiry considering the proposed NZCPS had reported. The Board of Inquiry had worded Policy 13(1)(a) as avoiding "adverse effects of activities on areas of the coastal environment with *high* natural character".[183] This qualifier was changed to 'outstanding' by the Minister for the Environment before the NZCPS was adopted but without a similar change being made to the wording of Policy 13(1)(c).

It is recognised that 'outstanding' is not equivalent to 'high' natural character but rather a sub-set of it, being at the very top end of 'high'.[184] A combination of attributes that are individually just 'high' can push natural character into the 'outstanding' category.[185]

Whitireia Park, Wellington

A spotlight on windfarms

Situated within the 'roaring forties' and remote from large land masses, Aotearoa New Zealand is a windy country,[186] perfectly positioned for harnessing wind energy. The New Zealand Wind Energy Association website states that Aotearoa New Zealand has 20 windfarms that are either operating or under construction,[187] with a further 10 that have been consented but not yet developed.[188] The accompanying interactive map on the website shows a scattering of windfarms across the country, with congregations around Palmerston North, Wellington and Southland.

Windfarms pose a unique challenge to landscapes and amenity values. Their need for high wind means that they are ideally located along ridgelines and in the coastal environment, where most of the suitable wind resource is found.[189] More often than not, windfarm locations are either within ONLs or ONFs or in close enough proximity to them to result in adverse effects. Windfarms do not have to located within an identified ONL or ONF to compromise it.[190] Nor do turbines have to be located within the coastal environment to be seen as part of it.[191] This is important as often individual turbines within a windfarm proposal are located outside of, but adjacent to or nearby, identified section 6(a) and (b) areas.

For efficiency, windfarms are also located in proximity to necessary infrastructure, which brings them in conflict with coastal and (most often) rural communities. Some people have a positive predisposition to windfarms and others do not.[192] An individual's philosophical opinion about power generation is one factor in determining preference.[193] However, the single most important factor in determining whether an individual accepts a windfarm or not is proximity. The closer the view is to the turbines, the greater the perception of visual dominance or effect on amenity.[194]

Wind turbines are tall structures, with some reaching above 150 metres into the sky.[195] Elevation above potential viewers enhances their visual prominence,[196] as does the juxtaposition of an unnatural vertical form in motion and a relatively contiguous area of low-lying vegetation. Some windfarms emulate a 'picket line', where turbines march across the landscape in sequence, while others are more diverse, with turbines located at varying heights across different landforms.

It is generally desirable to choose locations where complaints about adverse effects are minimised. But care must be taken. An 'out of sight out of mind' attitude may lead to a race to the mediocre, where outstanding resources are constantly having large bites taken out of them.[197]

In 2005, the Environment Court heard its first wind farm case.[198] Since then, a multitude of proposals have been assessed by the Court, with the majority of those being granted. The primary issue in almost every case was the balancing of adverse effects on landscape and/or amenity values against consideration of climate change and renewable energy. Windfarms are particularly confronting in that regard; their inherent nature brings section 6 and 7 matters into direct competition.

Important as the issues of climate change and the use of renewable sources of energy unquestionably are, under the RMA they do not dominate all other values.[199] The RMA does not currently attach any greater emphasis to climate change and renewable energy matters than it does, for example, to the maintenance and enhancement of amenity values and the quality of the environment.[200]

Nor does the National Policy Statement (NPS) for Renewable Electricity Generation 2011 necessitate that greater weight be given to sections 7(i) (the effects of climate change) and 7(j) (the benefits to be derived from the use and development of renewable energy) of the RMA. The NPS gives effect to these provisions through its policies which, *inter alia*, direct decision-makers to "recognise and provide for" the benefits of renewable electricity generation activities. It does not follow from this that the benefits are necessarily given greater weight than other matters addressed in section 7.[201] Rather, it is the statutory planning documents that will indicate the weight to be given to these matters.[202]

The Environment Court is yet to assess a windfarm proposal against operative plan provisions which give effect to both the NPS for Renewable Electricity Generation and the NZCPS. Such a case is likely to present obvious challenges as decision-makers grapple with having regard to objectives, policies and rules which both provide for renewable energy sources and seek to avoid adverse effects on ONLs and ONFs.

By their very nature, wind farm turbines are likely to be highly visible and prominent in any landscape. That is unavoidable. The issue is the significance of that visibility in any particular case.[203] The Court has refused windfarm proposals ranging from a single turbine[204] to over 170 turbines on the basis of adverse effects on landscape and amenity values.[205] Cumulative adverse landscape effects, and the relationship of Māori to the landscape, are other reasons for refusal.[206]

Notwithstanding the above, in the majority of windfarm cases, the positive effects of wind farms have outweighed adverse effects on landscape and natural character values. The country's first windfarm proposal to be heard by the Court was the Awhitu windfarm in Franklin. Consent was granted even though the Court concluded that, although not within an ONL, the wind farm would "dominate the surrounding area and undermine the visual integrity of the natural character and landscape of the coastal environment", and that would be a "significant adverse effect."[207]

The Makara windfarm (Project West Wind) in Wellington was granted consent on the basis that "the natural character of the coastal environment and the preservation of the outstanding landscape are not to be preserved at the cost of what is an appropriate development in this location."[208] In that case, the proposed windfarm generated adverse effects on natural character of the coastal environment,[209] which the Court considered to be an ONL.[210]

Similarly, when assessing the neighbouring Mill Creek windfarm, the Court held that although parts of the coastal environment adjacent to the windfarm had high natural character,[211] and the turbines would have significant adverse effects on that,[212] the proposal should be granted consent. Decided before *King Salmon*, the Court held that section 6(a) or the NZCPS do not "trump" other Part 2 matters.[213]

In *Unison*, the Court held that there would be a significant change in the landscape for the life of the two windfarms under consideration, but they would not result in any irreversible effect on the features contributing to the significance of the ONL.[214] The Court accepted that the effects on landscape and visual amenity were more than minor,[215] but the issue is how those effects can, and should, be balanced against the other Part 2 factors.[216]

It does not matter that, on a world scale, the contribution of one windfarm to a global reduction in greenhouse emissions is *de minimis*.[217] The supply of electricity through the use of wind energy is a renewable energy supply. As the Court stated in *Unison*, the generation of a substantial output of electricity from a perpetually renewable source which emits no pollution, particularly greenhouse gases, is of such national importance and benefit that it clearly outweighs site-specific adverse effects.[218]

The ultimate question for the Courts was whether each proposal met the sustainable management purpose of the RMA. Determination required an overall assessment as to whether the benefits of the proposal, when seen in the national context, outweighed the site-specific effects and effects on the local surrounding area.

In most cases the decision was stark; to maximise access to wind, turbines need to be on or near ridgelines and will often be on skylines. This means there is no real prospect of remedying or mitigating their adverse landscape and visual effects. Either the activity proceeds, or the effects are avoided by refusing consent.[219]

King Salmon was a sea change for resource management in Aotearoa New Zealand. The Environment Court has not yet tested a windfarm proposal against this 'new law'. Its applicability is not only to windfarms within ONL and ONFs in the coastal environment, but also to proposals that impact on those areas, and on natural character more generally. In future, the Environment Court will not be able to so readily adopt an overall judgment approach, with recourse back to section 5 when assessing a windfarm proposal that triggers the NZCPS. As the Court of Appeal in *RJ Davidson* set out:[220]

> Where the NZCPS is engaged, any resource consent application will necessarily be assessed having regard to its provisions. This follows from s 104(1)(b)(iv). In such cases there will also be consideration under the relevant regional coastal plan. We think it inevitable that *King Salmon* would be applied in such cases. The way in which that would occur would vary. Suppose there were a proposal to carry out an activity which was demonstrably in breach of one of the policies in the NZCPS, the consent authority could justifiably take the view that the NZCPS had been confirmed as complying with the Act's requirements by the Supreme Court. Separate recourse to pt 2 would not be required, because it is already reflected in the NZCPS, and (notionally) by the provisions of the regional coastal plan giving effect to the NZCPS. Putting that another way, even if the consent authority considered pt 2, it would be unlikely to get any guidance for its decision not already provided by the NZCPS. But more than that, resort to pt 2 for the purpose of subverting a clearly relevant restriction in the NZCPS adverse to the applicant would be contrary to *King Salmon* and expose the consent authority to being overturned on appeal.
>
> On the other hand, if a proposal were affected by different policies so that it was unclear from the NZCPS itself as to whether consent should be

granted or refused, the consent authority would be in the position where it had to exercise a judgment. It would need to have regard to the regional coastal plan, but in these circumstances, we do not see any reason why the consent authority should not consider pt 2 for such assistance as it might provide. As we see it, *King Salmon* would not prevent that because first, in this example, there is notionally no clear breach of a prescriptive policy in the NZCPS, and second the application under consideration is for a resource consent, not a plan change.

Although depressed electricity prices and the announced closure of the Tiwai Aluminium Smelter[221] have created economic challenges for windfarms, the New Zealand Wind Energy Association still lists six projects under investigation and Meridian has recently announced its intention to proceed with stage two of its Turitea wind farm in the Manawatū.[222] As the coastal environment is a high wind area, an opportunity to test a windfarm proposal against 'bottom line' policies (13 and 15) of the NZCPS looks inevitable, and is likely to be highly scrutinised.

West Wind Farm, Wellington

KEY MESSAGES

Landscapes in general

- Jurisprudence has played an integral role in developing landscape assessments and methodologies.
- The implications of recent of case law means that planning documents need to have clear and directive provisions if important landscape values are to be protected.
- The Courts have long recognised that the concept of landscape is broader than just visual matters.
- More recent case law has confirmed that landscape is comprised of biophysical, sensory/perceptual and associative values.
- Features and landscapes are not the same thing. Generally speaking, a feature can be described as a single element of natural and physical resources and a landscape is usually a collection of such elements.
- The appropriate geographical scale for evaluation varies and is the one that treats a landscape or feature as a whole in a given context.

Natural landscapes

- A 'natural' landscape or feature does not need to be pristine (and can include such things as pasture, exotic trees and introduced wildlife) but should be uncluttered by structures or 'obvious' human influence.
- An ONL should have some 'special quality' and its 'outstanding' nature should usually be so obvious that there is no need for expert analysis.
- Identification as an ONL does not preclude development, particularly if it does not impact on what is sought to be protected.

Cultural landscapes

- The RMA does not refer to 'cultural landscapes', but they are explicitly referred to in the NZCPS, and in case law.
- Due to the strong Māori oral tradition, and obligation for meaningful engagement of tangata whenua in natural resource matters, obtaining information on cultural landscape values requires direct engagement with mana whenua.
- Iwi planning documents developed by iwi and hapū can help articulate their values and relational connections with ancestral taonga and wider landscapes. They can provide useful insight and background information for statutory agencies but are not a substitute for direct engagement with mana whenua.
- Plan provisions co-developed between councils and mana whenua can help protect important cultural landscape values. Several examples now exist which can inform and assist councils and mana whenua with this task.

Heritage landscapes

- Historic heritage is broadly defined to mean natural and physical resources that contribute to an understanding and appreciation of Aotearoa New Zealand's history and cultures and include sites of significance to Māori.
- The Courts have been reluctant to protect heritage landscapes and the protection of these landscapes is weak compared to many other countries.

Amenity landscapes

- Amenity landscapes are significant landscapes that are not quite natural enough to qualify as ONLs. They often provide a buffer between ONLs and heavily developed areas.

Coastal landscapes

- The existence of the NZPCS means that different policies apply to landscapes within the coastal environment than those located elsewhere and these provide a higher level of protection.

Aotea Great Barrier Island *(Craig Potton)*

ENDNOTES

1 Resource Management Act 1991, s5(1)
2 *Environmental Defence Society v New Zealand King Salmon Company Ltd* [2014] NZSC 38 at [24]
3 *Environmental Defence Society v New Zealand King Salmon Company Ltd* [2014] NZSC 38 at [130]
4 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [114]
5 *Bayswater Marina Holdings Ltd v North Shore City Council* Decision A 018/2009 at [121]
6 *Bayswater Marina Holdings Ltd v North Shore City Council* Decision A 018/2009 at [121]
7 *New Zealand Marine Hatcheries (Marlborough) Ltd v Marlborough District Council* W129/97at 18
8 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [216]
9 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [216]
10 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59
11 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [80]
12 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [77]
13 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [77]
14 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [80]
15 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [160]
16 *Upper Clutha Tracks Trust & ors v Queenstown Lakes District Council* [2010] NZEnvC 432 at [51]-[62]
17 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [135]
18 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [135]
19 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [136]
20 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [215]
21 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [114]
22 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [137]
23 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [80(c)]
24 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [80(c)]
25 *Mainpower NZ Ltd v Hurunui District Council* [2011] NZEnvC 384 at [298]-[299]
26 *Bayswater Marina Holdings Ltd v North Shore City Council* A 018/2009 at [122]
27 *Mainpower NZ Ltd v Hurunui District Council* [2011] NZEnvC 384 at [292]
28 *Long Bay-Okura Great Park Society v North Shore City Council* A078/2008 at [124]-[140]
29 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [231]
30 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [82]
31 *Wakatipu Environmental Society v Queenstown-Lakes District Council* Decision No. C 129/2001 at [33]
32 *Queenstown Bungy Ltd v Queenstown-Lakes District Council* Decision No. C 35/2002 at [18]
33 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [82]
34 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [82]
35 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [82]
36 *Friends of Nelson Haven and Tasman Bay Inc v Tasman District Council* [2018] NZEnvC 46 at [91]
37 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [80]
38 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [80(b)]
39 *Upper Clutha Environmental Society Inc v Queenstown Lakes District Council* [2019] NZEnvC 205 at [5]
40 *Lake McKay Station Ltd v Queenstown Lakes District Council* [2019] NZEnvC 206 at [256]
41 *Lake McKay Station Ltd v Queenstown Lakes District Council* [2019] NZEnvC 206 at [255]
42 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [156]
43 *Ohau Protection Society Inc v Waitaki District Council* [2018] NZEnvC 243 at [101] and [110]
44 *H.I.L. Ltd v Queenstown Lakes District Council* [2014] NZEnvC 177 at [26]
45 *Kaupokonui Beach Society Inc v South Taranaki District Council* W030/2008 at [46]
46 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [94]
47 *H.I.L. Ltd v Queenstown Lakes District Council* [2014] NZEnvC 177 at [36]
48 *Staufenberg Family Trust No. 2 v Queentown Lakes District Council* [2013] NZEnvC 100 at [53]-[54]
49 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [81]; *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [84]
50 *Upper Clutha Tracks Trust v Queenstown-Lakes District Council* [2010] NZEnvC 432 at [65]
51 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [87]-[89]
52 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [87]
53 *Upper Clutha Tracks Trust v Queenstown-Lakes District Council* [2010] NZEnvC 432 at [60]
54 *Upper Clutha Environmental Society Inc v Queenstown Lakes District Council* [2019] NZEnvC 205 at [411]; *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147
55 *Upper Clutha Tracks Trust v Queenstown Lakes District Council* [2010] NZEnvC 432 at [61]
56 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [89]
57 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [89]
58 *Long Bay-Okura Great Park Society v North Shore City Council* A078/2008 at [135]
59 *High Country Rosehip Orchards Ltd v MacKenzie District Council* [2011] NZEnvC 387 at [93]-[94]
60 *Hawthenden Ltd v Queenstown Lakes District Council [*2019] NZEnvC 160 at [61]; *Upper Clutha Tracks Trust v Queenstown Lakes District Council* [2010] NZEnvC 432 at [62]
61 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [61]
62 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [63]
63 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [82]
64 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [82]
65 *Man O'War v Auckland Council* 19 ELRNZ 662 at [86]
66 *Waiareka Valley Preservation Soc Inc v Waitaki District Council* C 058/2009 at [135]
67 *Maniototo Environmental Society Inc v Central Otago District Council* C103/2009 at [323]
68 *Man O'War v Auckland Council* 19 ELRNZ 662 at [87]
69 *Man O'War v Auckland Council* 19 ELRNZ 662 at [93]
70 *R J Davidson Family Trust v Marlborough District Council* [2018] NZCA 316 at [82] as authority that the Court can have recourse to Part 2 of the RMA in the case of resource consent applications and *Unison Networks Ltd v Hastings District Council* [2011] NZRMA 394 at [81] as authority that the Court can properly make its own judgment on Part 2 matters to conclude that an area is an ONL even though it is not identified as such in the plan
71 *Wakatipu Environmental Society Inc v Queenstown-Lakes District Council* [2000] NZRMA 59 at [99]
72 *Whangaroa Maritime Recreational Park Steering Group v Northland Regional Council* [2014] NZEnvC 92 [78]
73 A case in point being the Far North District Plan, Hurunui District Plan and Banks Peninsula District Plan which are notable for the level of outrage generated and the intensive public campaigns carried out by Federated Farmers against councils' perceived taking of their property rights
74 *Man O'War v Auckland Council* 19 ELRNZ 662 at [59]-[69]
75 *Man O'War v Auckland Council* 19 ELRNZ 662 at [63]
76 *Man O'War v Auckland Council* 19 ELRNZ 662 at [62]
77 *Man O'War v Auckland Council* [2017] NZHC 3217 at [93] and [97]
78 *Environmental Defence Society v New Zealand King Salmon Company Ltd* [2014] NZSC 38 at [101]
79 *Granger v Dunedin City Council* [2018] NZEnvC 250 at [151]
80 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [30]

81 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2019] NZEnvC 110 at [35]

82 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2019] NZEnvC 110 at [35]

83 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2019] NZEnvC 110 at [36]

84 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2019] NZEnvC 110 at [38]

85 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [56]

86 *Hawthenden Ltd v Queenstown Lakes District Council* [2019] NZEnvC 160 at [49]

87 *Long Bay-Okura Great Park Society v North Shore City Council* A078/08

88 *Day v Manawatu-Wanganui Regional Council* [2012) NZEnvC 182

89 *Federated Farmers of New Zealand v MacKenzie District Council* [2017] NZEnvC 53 at [521]

90 *Day v Manawatu-Wanganui Regional Council* [2012) NZEnvC 182

91 *Federated Farmers of New Zealand v MacKenzie District Council* [2017] NZEnvC 53 at [475]

92 *Federated Farmers of New Zealand v MacKenzie District Council* [2017] NZEnvC 53 at [471]

93 *Federated Farmers of New Zealand v MacKenzie District Council* [2017] NZEnvC 53 at [468]

94 *Federated Farmers of New Zealand v MacKenzie District Council* [2017] NZEnvC 53 at [476] – [477], quoting Dr Fairgray's economic evidence

95 *Federated Farmers of New Zealand v MacKenzie District Council* [2017] NZEnvC 53 at [477]

96 Hoskins et al, 2008

97 Renata et al, 2015

98 Renata et al, 2015

99 Ngāi Tūāhuriri Rūnanga et al, 2013, 105-106

100 http://www.chchplan.ihp.govt.nz/wp-content/uploads/2015/12/3722-Ngāi-Tahu-S.32AA-Report-corrected-tracked.pdf

101 Menzies, 2017

102 For example, *The Sterling Trust v Whakatane District Council* W028/08

103 *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211 at [185] and *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120 at [64]

104 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [131]

105 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [131]

106 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [132]

107 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [149]

108 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [158]

109 *Self Family Trust v Auckland Council* [2018] NZEnC 49

110 Lau and Wong, 2016

111 Auckland Unitary Plan (Operative in Part 2016)

112 *Self-Family Trust v Auckland Council Decision No. [2018] NZEnvC 49*

113 Resource Management Review Panel, 2020, 99

114 ENV-2016-AKL-000120, Appeal 38

115 Ordinary meeting, Thames-Coromandel District Council, Public Excluded Minutes, 12 December 2017, section 5.5. Ngati Maru Appeal to Proposed District Plan.

116 *Upper Clutha Tracks Trust v Queenstown Lakes District Council* [2010] NZEnvC 432 at [51]-[62]

117 New Zealand Coastal Policy Statement, Policy 15(c)(viii)

118 *R J Davidson Family Trust v Marlborough District Council* [2018] NZCA 316

119 Stephenson, Bauchop and Petchey, 2004

120 Stephenson, Bauchop and Petchey, 2004, 10

121 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [251]

122 This section does not address heritage landscapes in the context of other heritage legislation

123 *T W Reed Estate v Far North District Council* [2014] NZHC 3328 at [96] and [100]

124 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [175]-[177]

125 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [177]

126 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [170]

127 Salmond, 2005, 5

128 Salmond, 2005, 4-5

129 Salmond, 2005, 6

130 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [227]

131 See *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [230]-[231]; *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211 at [185]; *T W Reed Estate v Far North District Council* [2014] NZHC 3328 at [65]; *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120

132 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [230]-[231]; *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211 at [185]

133 *New Zealand Transport Agency v Architectural Centre Inc* 19 ELRNZ 163 at [381]-382]

134 *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009 at [232]

135 *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211 at [193]

136 *Harrison v Whangarei District Council* W 034/2005 at [32]; *Waiareka Valley Preservation Society Inc v Waitaki District Council* C058/2009; *Clevedon Cares Inc v Manukau City Council* [2010] NZEnvC 211

137 *Harrison v Whangarei District Council* W 034/2005 at [33]

138 *MacPherson v Otorohanga District Council* W25/2007 at [230]

139 Menzies and Ruru, 2011

140 *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120 at [55]-[59]

141 *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120 at [66]

142 *Gavin H Wallace Ltd v Auckland Council* [2012] NZEnvC 120 at [67]

143 Nolan and Somerville, 2003

144 Kenderdine, 2005, 39

145 Salmond, 2005, 18

146 Wilson, 2019

147 https://www.rnz.co.nz/news/national/395318/ihumatao-protests-no-building-while-a-solution-is-sought-pm

148 McLachlan, 2019

149 https://www.rnz.co.nz/national/programmes/middayreport/audio/2018777610/agreement-reached-for-crown-to-purchase-ihumatao-land

150 https://www.protectihumatao.com/ihumatao.html

151 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [90]

152 Resource Management Act 1991, s2

153 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [77]

154 *Beadle v Queenstown Lakes District Council* [2019] NZEnvC 92 at [127]

155 Brown, undated

156 Brown, undated

157 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [93]

158 *Wakatipu Environmental Society Inc v Queenstown Lakes District Council* [2000] NZRMA 59 at [93]

159 *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [158]

160 *SKP Incorporated v Auckland Council* [2018] NZEnvC 081 at [205]

161 *Schofield v Auckland Council* [2012] NZEnvC 68 at [51]

162 *Schofield v Auckland Council* (2012] NZEnvC 68 at [51]

163 *Port Gore Marine Farms v Marlborough District Council* [2012] NZEnvC 72 at [213]

164 *Port Gore Marine Farms v Marlborough District Council* [2012] NZEnvC 72 at [213]

165 *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [158]

166 *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [158]

167 *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [158]

168 *Long Bay-Okura Great Park Society v North Shore City Council* A078/2008 at [485]-[486]

169 See the Town and Country Planning Act 1977

170 Peart, 2009, 204-205

171 New Zealand Coastal Policy Statement, Policy 13(2)

172 Department of Conservation, undated(b), 12

173 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [123]

174 *Western Bay of Plenty District Council v Bay of Plenty Regional Council* [2017] NZEnvC 147 at [123]

175 *KPF Investments Ltd v Marlborough District Council* [2014] NZEnvC 152 at [92]

176 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [125]

177 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [192]

178 *Briggs v Christchurch City Council* C045/08 at [143]-[145]

179 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [154]. This decision was appealed but the Environment Court's approach to assessment of natural character was not challenged. See *Clearwater Mussels Ltd v Marlborough District Council* 21 ELRNZ 67 at [114]

180 Department of Conservation, 2011b, 10

181 Department of Conservation, 2011b, 16

182 *Clearwater Mussels Limited v Marlborough District Council* [2018] NZEnvC 88 at [125]

183 Board of Inquiry into the proposed New Zealand Coastal Policy Statement, 2009

184 Department of Conservation, 2011b, 18

185 Department of Conservation, 2011b, 16

186 *Maniototo Environmental Society Incorporated v Central Otago District Council* C 103/2009 at [26]

187 http://www.windenergy.org.nz/operating-&-under-construction

188 http://www.windenergy.org.nz/wind-energy/nz-windfarms

189 *Genesis Power Ltd v Franklin District Council* (2005) 12 ELRNZ 71 at [84]

190 *Unison Networks Ltd v Hastings District Council* W058/06 at [42]

191 *Meridian Energy Ltd v Wellington City Council* [2011] NZEnvC 232 at [144]

192 *Unison Networks Ltd v Hastings District Council* W058/06 at [61]

193 *Unison Networks Ltd v Hastings District Council* W058/06 at [61]

194 *Upland Landscape Protection Society Inc v Clutha District Council* C085/08 at [102] and [109]

195 *Maniototo Environmental Society Incorporated v Central Otago District Council* C 103/2009 at [10]

196 *Motorimu Wind Farm Limited v Palmerston North City Council* W 067/2008 at [203]

197 *Maniototo Environmental Society Incorporated v Central Otago District Council* C 103/2009 at [748]

198 Prior to 2005, wind farms had been developed in New Zealand (see, for example, Hau Nui in the Wairarapa and Tararua in the Manawatu), but the 2005 Awhitu wind farm was the first to be examined by the Environment Court

199 *Outstanding Landscape Protection Society Inc v Hastings District Council* [2008] NZRMA 8 at [116]

200 *Motorimu Wind Farm Limited v Palmerston North City Council* W 067/2008 at [361]

201 *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [323]

202 *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [323]

203 *Motorimu Wind Farm Limited v Palmerston North City Council* W 067/2008 at [151]

204 The most recent wind application (a single 110m turbine at Blueskin Bay, Dunedin) to come before the Court was declined on the basis that the turbine would have significant adverse landscape and visual effects and significant adverse effects on amenity values. See *Blueskin Energy Ltd v Dunedin City Council* [2017] NZEnvC 150 at [351]

205 Adverse effects on the ONL of the Lammermoor in Central Otago weighed heavily in the Court's decision to refuse consent to the Project Hayes windfarm proposal, which was later abandoned by Meridian after planning and legal processes spanning five years. See *Maniototo Environmental Society Incorporated v Central Otago District Council* C 103/2009

206 *Outstanding Landscape Protection Society Inc v Hastings District Council* [2008] NZRMA 8 at [115]-[116]

207 *Genesis Power Ltd v Franklin District Council* (2005) 12 ELRNZ 71 at [108]

208 *Meridian Energy Ltd v Wellington City Council* W031/07 at [452]

209 *Meridian Energy Ltd v Wellington City Council* W031/07 at [283]

210 *Meridian Energy Ltd v Wellington City Council* W031/07 at [412]

211 *Meridian Energy Ltd v Wellington City Council* [2011] NZEnvC 232 at [170]

212 *Meridian Energy Ltd v Wellington City Council* [2011] NZEnvC 232 at [172]

213 *Meridian Energy Ltd v Wellington City Council* [2011] NZEnvC 232 at [174]

214 *Unison Networks Ltd v Hastings District Council* W058/06 at [44]

215 *Unison Networks Ltd v Hastings District Council* W058/06 at [46]

216 *Unison Networks Ltd v Hastings District Council* W058/06 at [46]

217 *Genesis Power Ltd v Franklin District Council* [2005] NZRMA 541 at [222]-226]; *Unison Networks Ltd v Hastings District Council* W058/06 at [74(j)]

218 *Unison Networks Ltd v Hastings District Council* W058/06 at [79]

219 *Unison Networks Ltd v Hastings District Council* W058/06 at [68]

220 *R J Davidson Family Trust v Marlborough District Council* 20 ELRNZ 367 at [71]-[72]

221 Which uses approximately 15 per cent of New Zealand's electricity

222 Rankin, 2019

Assessing landscape values

Te Kouma Harbour

The application of the provisions of the RMA to protect landscape values in Aotearoa New Zealand has been strongly influenced by the landscape architect profession and the way that landscape assessment has been approached. In this chapter we describe the evolution of landscape assessment approaches within Aotearoa New Zealand.

NZILA is currently developing new Landscape Assessment Guidelines which, if adopted, will provide the most up-to-date articulation from the profession on how landscape assessment should be undertaken within the current context. The Guidelines have been developed with input from Te Tau-a-Nuku, the collective of Māori landscape architects under the korowai of Ngā Aho the mandated collective body of Māori design professionals. As already indicated, they seek to promote assessment methodologies which recognise the country's unique bicultural context and histories. The Guidelines, when finalised and assuming they are adopted by the Institute's membership, should make a significant contribution to landscape management and protection in Aotearoa New Zealand into the future. They are seen as a living document which will continue to evolve in relation to changing context and practice.

5.1 Early development of landscape assessment

Landscape architecture is the design of outdoor space. Its origins in Aotearoa New Zealand stem from the many plant nurseries, in the early 1900s, offering garden design and landscaping services in addition to plant and seed supply.[1] American practitioners led the way in landscape architects moving beyond the garden and into broader public space in the late 19th century. New ideas about landscape architecture began to filter through to Aotearoa New Zealand during the 20th century.[2]

The main impetus for the formal development of the profession of landscape architecture in Aotearoa New Zealand came from the movement for scenic beauty preservation during the 1960s.[3] During this time, concern arose among conservationists over the environmental impacts of proposed public works. The Government's proposal to raise the water level in Lake Manapouri, as part of a hydroelectric generation scheme, accelerated the movement. The 'Save Manapouri' campaign was the first to attain national reach and precipitated the dawning of the 'environmental era' in Aotearoa New Zealand that was to follow in the 1970s.[4]

J.T. Salmon's influential book 'Heritage Destroyed: The Crisis in Scenery Preservation in New Zealand', published in 1960, highlighted the environmental impacts of state development projects[5] and gave significant stimulus to the development of visual landscape assessment and design in rural areas.[6]

Landscape architects were called upon to help 'fit' public works into the environment.[7] A major focus was the expert evaluation of landscape as scenery, using principles derived from fine art.[8]

By the late 1960s, Aotearoa New Zealand's first body of landscape practitioners was established as the New Zealand Association of Landscape Designers.[9] The group comprised mainly unqualified enthusiasts who were occupationally interested in landscape matters, including town planners.[10] The first two-year landscape architecture course subsequently commenced at Lincoln College in 1969.[11] Graduates were accredited by the Australian Institute of Landscape Architects. However, neither the Australian Institute nor the New Zealand Association were adequate to meet the needs of Aotearoa New Zealand's newly trained landscape architects.[12] Something more specialised was required for the profession.

In 1972, NZILA was born. In stating the case for its formation, it was argued that "the expression 'landscape architecture' is now recognised throughout the English-speaking world and accordingly the public in New Zealand are entitled to expect the words to be used as indicative of a recognised professional group with established standards".[13] The Institute held its first general meeting in November 1973 and declared as its primary objective the promotion of landscape architecture as "a fundamental design, planning and conservation profession, and a conscious departure from its former largely horticultural and remedial role".[14]

The 1970s was a busy decade for the environment in Aotearoa New Zealand. A number of conferences were held and various committees, commissions and portfolios were established to give due consideration to the environment, at least at face value.[15] In 1972, the Commission for the Environment was established, which prompted the formalisation of environmental assessment and enhancement procedures for major projects. This led to the implementation of a systematic framework for impact assessment.[16]

During the 1970s and early 1980s, most of the landscape architecture profession worked in the public sector, undertaking large landscape assessments under the patronage of central government agencies such as the Ministry of Works and the Department of Lands and Survey.[17] Two strands of landscape assessment evolved during this time. The first was a parametric approach, in which different biophysical and visual phenomena were separately mapped and valued before being synthesised. The second was a landscape approach, which emphasised delineation and comparative evaluation of homogenous areas of landscape character.[18] There was very little review of associative landscape attributes – the values people ascribed to the landscape and, when sensory attributes were considered, it was from a Eurocentric perspective.

The landscape profession was highly regarded during this period and a number of large-scale public works projects did not proceed on the advice of landscape architects. For example, a proposed motorway going north from Hamilton was scrapped after landscape advice concluded the effects would be too great. These decisions were significant in terms of the future environment; if the Hamilton motorway had proceeded, it is likely that the urban sprawl between Auckland and Hamilton would now be continuous.

The transformation of government ideology and policy, during the 1980s, saw the devolution of responsibility and disestablishment of key central agencies which had sponsored many of the landscape studies of the early 1980s.[19] The focus of professional landscape architecture thus shifted to the private sector.[20]

5.2 Impact of the RMA on landscape assessment

The introduction of the RMA in 1991 was a shot of adrenaline to the landscape profession. The Act elevated the protection of ONLs and ONFs and referenced amenity values and natural character as described in Chapter 4. Local authorities were now tasked with drafting policies and plans to recognise and provide for these landscape concepts. Landscape architects were called upon to undertake district and region wide studies to inform them.

At the same time, requirements for assessing the overall effects of development at a project level were strengthened. Landscape and visual effects were specifically listed as matters that must be addressed by assessments of

Lake Benmore hydro scheme

environmental effects (a progression of the earlier environmental assessment and enhancement procedures) which led to an increase in the number of site-specific landscape assessments.[21]

In 1993, the first regional landscape assessment under the RMA was undertaken, commissioned by the Canterbury Regional Council.[22] This assessed the characteristics and values of identified areas from six perspectives: natural science, legibility, transient, aesthetic, shared and recognised, and tangata whenua.[23] Unbeknown to the authors at the time, these six matters would subsequently shape the development of landscape assessment in Aotearoa New Zealand for decades to come. With minor amendment, they became known as the 'amended Pigeon Bay criteria'.

Throughout the 90s, a variety of landscape assessment methodologies evolved, but two were prominent. Practitioners, primarily in the South Island and fuelled by the Court's acceptance of the Canterbury Regional Landscape Study methodology, tended towards assessing landscapes according to the Pigeon Bay criteria. Notable to that assessment method was a lack of scoring or ranking of features and landscapes. Instead, attempts were made to build up patterns of value by analysing overlaid material.[24] Further, the method was not dominated by visual aspects and attempted to reflect community values, including those of Māori.

Running in parallel were a number of landscape assessments undertaken primarily in the North Island. These adopted a methodology largely based on the 1984 Auckland Regional Landscape Assessment. That study drew on the psychology of landscape perception and sought to gauge the general community's responses to the full range of landscapes found throughout the region. It adopted a method for determining landscape quality as follows:[25]

> The assessment of landscape quality throughout the Auckland Region was based around 633 landscape units. All of these lie outside the Region's metropolitan and other urban areas, and capture areas which display a consistency of landscape character. These units provided a platform for the identification of 85 'landscape types'. Large scale photos of each landscape unit were then used in a public preference study carried out over the summer of 1983-4. Some 1091 respondents from throughout the Region were polled in the study, with each being asked to categorise different landscape types on the basis of the visual quality which they felt each landscape conveyed (through the photos). The categorisation process resulted in seven classifications or groupings of visual quality – from low to high. Each landscape type was then allocated to one of these quality groupings. These findings were then extrapolated to all 633 units based on the visual similarities

Lyttelton Harbour

between each landscape type and the individual landscape units found throughout the Region.

ONLs and ONFs were then determined by an overall 'sensitivity' classification that was based on landscape values *and* landscape vulnerability.[26] This meant that the assessment factors used were primarily concerned with the perception of visual landscape values and the predicated visual effects of assumed types of landscape change on landscape quality.[27]

Subsequent landscape assessments[28] following this approach similarly divided an area into landscape units which were grouped together based on their landscape character. The units were then assessed for their landscape values, vulnerabilities, elements and patterns and assumed viewing audiences. A score or rating was attributed to each of these factors. These were then aggregated before final evaluation and calculation of a sensitivity score (the accumulation of ratings). Generally speaking, lower scores equated to lower sensitivities and were an indication of no impact of change, whereas higher scores indicated vulnerability to change. The focus on visual aspects of the landscape meant that spiritual or traditional associations were not considered. Further, no community consultation was undertaken as part of these assessments.

During 2002 and 2003, the Auckland Regional Landscape Assessment was updated. The study adopted the Q-Sort methodology but was criticised for not having regard to the 'amended Pigeon Bay' factors. As a result, the authors of the study undertook a second round of analysis which added layers based on the 'amended Pigeon Bay' factors, with the exception of Māori cultural considerations which were considered too difficult to incorporate at the time.

An emphasis on quantification can oversimplify landscape evaluation. It can mean that scoring becomes a focus of the landscape assessment, rather than the landscape characteristics and management issues that the assessor is attempting to 'rate'.[29] Practitioners who adopted this methodology speak of getting caught up in a numbers game, which did not fit comfortably with the concept of landscape, that crosses the divide between science and art. The Environment Court has subsequently cautioned against reducing landscape to a mathematical formula.[30]

Nowadays, landscape attributes are more commonly assessed and weighted using descriptions, not numbers. While more meaningful, the process raises its own set of issues. Since the Supreme Court's decision in *King Salmon* that 'inappropriate' subdivision, use and development depends on 'what is sought to be protected', there has been a conscious effort to 'beef up' plans by explicitly articulating landscape characteristics and values. This has added significant length to some plans and can be a costly exercise beyond the reach of small local authorities.

A spotlight on public preference surveys

Public preference surveys, like those used in the 1984 Auckland Regional Landscape Assessment, were popular in the late 90s and early 2000s with studies being undertaken by academics at Kaikoura (1998),[31] the Coromandel (1999),[32] Rotorua (2000),[33] Westland (2001)[34] and Auckland (2004).[35] The methodology of these 'Q-Sort' assessments, where the public was asked to view and sort photos, originated from America and aimed to decipher people's perceptions of landscape.

The analysis revealed a remarkable degree of consistency in the appreciation of, and attachment of values to, Aotearoa New Zealand's landscapes. It led to the identification of key traits that consequently relate to high preference including:[36]

- Naturalness – correlated with apparent levels of development or lack of development
- Endemic values or 'NZness' – related to sense of place
- Strong landscape structure – related to landform and the interaction of land with freshwater and the sea
- Strong landscape patterns – typically related to vegetation and land uses
- Visual drama or memorability
- Visual diversity

While helpful in understanding where 'natural' sits for New Zealanders on the nature-culture continuum, the studies relied significantly on appearance as a basis for evaluation. As a result, the research had a bias towards landscape as a visual experience, with other aspects being less important.[37] In effect, the Q-sort method of assessment reinforced a Eurocentric approach to landscape; with it largely being perceived as a visual construct, something to be admired 'over there'. This was an approach which the authors of the Canterbury Regional Landscape Study hoped to move away from as understanding of the Māori approach to landscape evolved.[38]

5.3 Incorporation into planning documents

Excitement over the RMA was short lived as tension arose over the interpretation of key landscape provisions, including how landscape protections afforded by the RMA should be translated into policy, and the consequential regulation of activities.[39] Farmers attempting to adapt to failing global commodity prices within a deregulated economy, strongly resented what they perceived to be attempts by urban-based landscape consultants to constrain their freedom of action under the guise of policies to protect landscape quality.[40]

As a result, a number of first generation plans that imposed landscape protections generated significant outrage resulting in co-ordinated opposition campaigns by the likes of Federated Farmers.[41] Despite plan provisions being based on landscape assessments undertaken by prominent landscape architects, the rural sector was largely successful. Many of the plans, like the Hurunui District Plan and the Banks Peninsula District Plan, were significantly modified to lessen landscape protections. Although landscape architect professionals were only infrequently involved in translating the results of their work into policy,[42] the lack of a clear and consistent approach to landscape assessment put the profession under intense scrutiny and criticism.

By the late 1999s, after almost a decade of plan preparation under the RMA, landscape assessment was routinely blamed (arguably unfairly) for hindering policy formation and plan preparation.[43] At the time, a pre-eminent academic posed the question that maybe it was the formal delineation of ONLs, and the associated rules, that created the opposition rather than a fundamental difference in perception of the quality of the landscape in question. Whatever the case, as the landscape profession entered the new millennium, there was undoubtedly an urgent need for it to carefully review its landscape assessment procedures and practice and to adopt improved guidelines if it was to regain credibility and policy relevance.[44]

5.4 Refining assessment approaches

In 1999, at the height of the profession's identity crisis, NZILA held an annual conference entitled 'Landscape Assessment: Means and Ends'. Research undertaken in association with the conference demonstrated that the predominant approach adopted by NZILA consultants in their landscape assessments was either visual only, or a combined visual/biophysical assessment.[45] Very few reported using experiential or solely biophysical approaches.[46]

The dichotomy between these two approaches, visual versus a more comprehensive assessment, originated largely from a difference in opinion as to the scope of a professional landscape architect's expertise. A feature of early landscape architecture, was an attempt to define landscape broadly and to claim professional expertise over the whole landscape, in order to establish a significant planning role for the new profession. However, claiming wider expertise by virtue of their professional affiliation left some practitioners exposed in the Environment Court setting. As practitioners at the time made clear, it was important to distinguish between the expertise needed to interpret the significance of biophysical and ecological patterns and processes for landscape character and quality, and the scientific expertise needed to analyse biophysical and ecological phenomena in themselves.[47]

Three distinct landscape assessment methods were identified in the 1999 research: assessment that followed the environmental assessment and enhancement procedures; assessment that focused on design outcomes; and a discursive method of assessment focussing on issues.[48] According to experts at that time, valuation of landscape was dominated by a qualitative objective approach, with very little community input being considered.[49]

As a result of this analysis, it was concluded that the profession needed to address the 'community' dimension more in landscape assessments, and focus more on change management as opposed to static protection (as the RMA only protects against 'inappropriate' subdivision, use and development).[50] There was also an ongoing call for consistent methodology.

Significant differences of opinion between experienced practitioners contributed to the continuing diversity of approach to landscape assessment practice.[51] For example, one practice evaluated landscape quality with a significant emphasis on underlying ecological patterns and processes. Extensive landscape settings were identified as being outstanding and there was an emphasis on design and management guidelines as the policy approach. In contrast, another practice limited 'outstanding' to tightly defined landscape features, with a strong cultural emphasis, but then favoured more prescriptive policy.[52]

A series of follow up workshops were undertaken in 2008 to discuss the problem of inconsistent landscape assessment methodology. Out of those came NZILA's 2010 'Best Practice Note for Landscape Assessment and Sustainable Management'.[53] The Practice Note provided a guide for how landscape assessments should be undertaken and set out best practice for the assessment stage of landscapes studies.[54]

The process adopted in the Best Practice Note was dependent on the issue under consideration, with the greatest distinction being drawn between landscape assessments undertaken as part of policy and plan making, and assessments associated with specific projects.[55] However, there were common fundamental

processes that could be adopted when undertaking landscape assessments. The Best Practice Note recommended that assessments included:

(a) A systematic account of landscape attributes in the assessment area such as biophysical elements, patterns and processes, sensory qualities and spiritual, cultural and social associations[56]

(b) Expert interpretation of landscape character based on classification of different types of landscape through identification of patterns of natural and cultural features, processes and influences and analysis of their characteristics and spatial location and the extent to which they are distinctive, representative or typical at the different scales[57]

(c) An evaluation of the landscape values of the existing landscape, including within the statutory framework[58]

(d) An analysis of the way the landscapes are likely to respond to change, including landscape resilience and capacity; sensitivity and vulnerability; and opportunities, risks and threats[59]

(e) Identification of ways and opportunities to ensure and enable sustainable landscape management, including considerations of alternatives; ways to enhance or create values; and actions to avoid, remedy or mitigate adverse landscape effects.[60]

Perhaps the most helpful part of the Best Practice Note was its list of defined terms that sought to bring some degree of consistency to the profession. Terms such as 'landscape', 'landscape value', 'natural character' and ONL were defined. Notions of landscape resilience, capacity, sensitivity and vulnerability were also included which the Environment Court has found useful when describing a landscape.[61]

Although highly anticipated and providing some structure to the profession via its defined terms, the Best Practice Note did not go far enough in ensuring consistency across experts.[62] It resulted in virtually no unified landscape assessment methodology being adopted.

During 2012, Marion Read as part of her Master's thesis, interviewed a number of expert landscape architects to uncover their interpretation of landscape. She found that the most common approach to defining landscape was to separate landscape into biophysical and perceptual aspects.[63] While most considered both aspects important, there was a spectrum of opinion on their weighting, with some placing greater weight on biophysical aspects and others on perceptual aspects.[64]

Interestingly, when comparing expert landscape assessment with people's perceptions, studies have demonstrated that emotion, everyday experience and intimate knowledge of places are more important to locals.[65] The visual criteria used by experts – and also proposed by the Environment Court – were of less importance to residents, and more akin to that of a tourist passing through the landscape.[66] Indeed, this highlights concerns about focusing on landscapes that are regarded as the 'best of the best', often at the expense of those that are important (at least in part) because of familiarity, local knowledge and even the perception of shared ownership.

Notwithstanding a general shift in thinking away from assessing landscapes on a purely visual basis, the significant divergence in approach to landscape assessment continued to trouble practitioners and others attempting to interpret landscape studies. Inconsistent methodologies continue to pose a serious risk to the profession. Practitioners themselves acknowledge that the use of different definitions and criteria for evaluation, by different landscape witnesses, results in confusion and ultimately a loss of credibility.[67] As a now retired Judge commented in the late 1990s, this may have the effect of cancelling out landscape evidence in Court or undermining the credibility of the profession.[68]

Judges have, on a number of occasions, expressed concern about the lack of consistency in the assessment methodology employed by different landscape architects.[69] In 2011, only one year after the promulgation of the profession's Best Practice Note, the Environment Court seemingly criticised the profession for its divergent approach to interpreting landscape factors:[70]

> Recent divisions of the court have encouraged landscape architects to move beyond description when giving evidence in relation to the modified Pigeon Bay factors. The difficulty is that no robust methodology has been developed for their application. Mere repetition of these factors without further methodological development is a barrier, to better understanding the complex construct that is landscape.

The Court went on to state that "[d]evelopment of methodology for analysis to address the three groups of aspects we outline below may produce more useful outcomes for decision-making."[71] The three groups referred to were the biogeographical elements, patterns and processes; the associative or relationship contributions; and the perceptual aspects as described in Chapter 4.

In the absence of a consistent methodology from the profession, Marion Read concluded in her 2012 thesis that the Environment Court has had a significant influence on the practice of landscape assessment in Aotearoa New Zealand.[72] She states that the profession has retained some control over the concept of landscape, but that control relies on the more 'slippery' nature of the concept than on a clear definition to which all ascribe.[73]

Compared to other countries,[74] Aotearoa New Zealand's landscape assessment methodologies have developed in an unorthodox manner. The uncomfortable tussle between the profession and the Environment Court has, at times, made it difficult to determine who is leading the vanguard in landscape assessment methodology. This freedom of expression, encouraged by the RMA, is not necessarily a bad thing. It has, as predicted back in 1986, enabled the concept of landscape to develop its richness of meaning bringing in associative attributes that might not have otherwise evolved in a more rigid system.

However, within that context, more rigor over the process of landscape assessment has occurred. Refreshingly, there is increasing agreement over approach and, to some extent, method. Most practitioners (especially those that present evidence in the Environment Court) agree that the factors relevant to assessing landscapes are the three groups of landscape attributes that have been accepted by the Court.

Experts also generally agree on a seven-point scale for rating those factors as very high, high, moderately high, moderate, moderately low, low and very low. Likewise, with respect to naturalness, the construct of 'moderate to high' as a threshold for qualification under section 6(b) is reasonably common among landscape architects. Once the attributes of a landscape have been described and rated, an overall judgement is then made of the significance of the landscape or

Marlborough

feature. Differences among experts predominantly arise in respect to the scale of the landscape to which each assessment should apply and the weighting to be afforded to the different factors.

The classification of the environmental effect of a proposal on landscape values is a secondary area of disagreement between landscape architects. It is common practice to rank an effect against the values of a landscape or area of natural character, the categories of which are relatively settled (the effects rating scale); negligible, very low, low, moderate, high, very high. Difficulties arise with the translation of that level of effect into the RMA language of 'minor'. Some professionals contend that a 'moderate' effect equates to 'minor' under the RMA,[75] although the Court has warned against conflating these two words.[76] Others contend that, to be minor, an effect on the landscape value must be low.

In 2017, NZILA reported that the Environment Court had taken a significant interest in seeking consistent methodologies among witnesses involved in landscape assessment.[77] It noted that a number of decisions had been critical of landscape experts and the 'unwillingness' of the profession to adopt a unified approach to landscape values assessment.[78]

As already noted, the Institute is now embarking on a process to provide significantly more detailed guidance.[79] The guidelines are intended to be a significant improvement on the 2010 version and are expected to clarify the seven-point scale for landscape assessment, provide terminology about what those ratings mean and provide guidance on how those ratings relate to effects. Importantly, however, the guidance is just that, guidance and practitioners will still have the freedom to adopt different approaches to assessment.

If a unified approach to landscape assessment cannot be achieved via guidance developed by the landscape profession, there is scope for a NPS to address landscape evaluation which, as noted by Environment Court Judge Borthwick, might provide a more "sturdier platform on which to mediate properly informed but competing interests".[80]

A preeminent legal practitioner has described landscape methodology as an area "literally demanding" national direction. In his opinion, such direction could set out what attributes must be considered, how each attribute is rated, the weighting to be applied between attributes and how the final evaluation should be made as to whether a landscape meets the outstanding threshold.[81] He considers that it could also cover process issues such as to what extent a multi-disciplinary exercise is required for the task and requisite degree of community and stakeholder engagement. To that end, he states that a clear and concise record of landscape characteristics and qualities that led to its rating would aid in resolving the management of landscape change.

Tapeka Point, Russell

A spotlight on landscape assessment of Banks Peninsula

Differences in methodology can result in very different outcomes for landscape protection. A prime example is the identification of ONLs on the Banks Peninsula, a highly summarised history of which follows.[82]

In the 1993 Canterbury Regional Landscape Study, the entire Banks Peninsula was identified as an ONL. The 1997 Banks Peninsula Proposed Plan subsequently identified around half (some 50,000ha of approximately 96,000ha) of the rural land on Banks Peninsula as either landscape protection areas or coastal protection areas. There was extensive public reaction to the extent of these classifications and, in response, the Council chose not to proceed to decisions on submissions to these provisions. Instead, in 2002 it notified Variation 2, which defined landscape and coastal protection areas of about 30,000ha.

As part of the resolution of appeals on Variation 2, a landscape study was undertaken by Boffa Miskell which the Court described as the "most comprehensive approach to a district landscape undertaken in New Zealand to date".[83] The study included three phases: character descriptions, landscape values and management mechanisms. It sought to maximise the objective information which can be provided into the layers before reaching a value judgment and integrating those layers into the various landscape protection categories.[84]

The protected landscape and coastal areas of the Boffa Miskell study encompassed an area of some 23,000ha (7,000ha less than Variation 2) limited mostly to prominent ridgelines and the coastal edge.[85] The central issue for the Court's determination was where the landscape and coastal protection lines on the maps should go.

There were seven landscape witnesses in the case, with "as many opinions as to what represents outstanding natural landscape and coastal natural character landscapes as there are experts".[86] The Court accepted that the Banks Peninsula landscape is an ONL at a regional scale given the geomorphology of the area, but considered the question it had to address was whether it constituted an ONL at a district level.[87] In the end, the Court accepted the delineation of ONL areas undertaken by Boffa Miskell. Important to that finding (as opposed to accepting a more comprehensive coverage of ONL and coastal protection areas) was the presence of human modification on Banks Peninsula, specifically prominent houses and forestry. As the Court stated:[88]

> One of the unfortunate consequences of being a member of the Court is that repeated analysis of landscapes leads to a critical evaluation of landscapes that many take for granted. For the general population there appears to be a conceit of the mind which occurs when it views pleasant landscapes. This appears to subtract from the view the incongruent elements such as large square forestry blocks up to and including ridges, prominent buildings situated on high points, roads, telephone poles and trees and other clear indications of extensive modification with the result that the individual reads the environment as natural.

In this manner, the scale at which the Court 'read' the landscape was important to its (and Boffa Miskell's) conclusions. Other landscape experts 'reading' the landscape within a wider (regional) context did not ascribe the same significance to the cultural modifications of the land (in terms of their detraction from its naturalness). The differences in approach dictated where the landscape and coastal protection lines on the maps went and, ultimately, what landscapes were afforded protection under section 6(b). Landscapes that are afforded protection can evolve in a significantly different manner to those that are not.

As an aside, the Banks Peninsula case study is interesting when compared to the Queenstown Lakes District. Both areas are cultured landscapes in that they include large areas of pastoral farming and signs of human occupation. On the Banks Peninsula, landscape protection has been limited to prominent ridgelines and the coastal edge. However, in the Queenstown Lakes District, approximately 97 per cent of the rural environment is classified as ONL. Indeed, rather than simply highlighting different assessment methodologies, this casts a spotlight on variability within the Environment Court and its divisions in relation to the implementation of strategic landscape assessments. The Court's view on assessment is frequently more important – or at least has more legal weight – than the assessments undertaken by specialists in the field.

KEY MESSAGES

- NZILA is currently developing new Landscape Assessment Guidelines which will provide the most up-to-date articulation from the profession on how landscape assessment should be undertaken within the current context.
- The methodology used for the 1993 Canterbury Regional Landscape Assessment shaped landscape assessment in Aotearoa New Zealand for decades and was the basis for the 'amended Pigeon Bay criteria' that were adopted by the Courts. The Assessment included natural science, legibility, transient, aesthetic, shared and recognised and tangata whenua perspectives.
- A variety of landscape assessment methodologies evolved during the 1990s and coalesced into two main approaches: the first being based on the amended Pigeon Bay criteria and the second focusing more strongly on the perception of visual landscape values.
- When councils incorporated the findings of landscape assessments into first generation RMA plans they often encountered concerted push back from rural landowners and watered down protective provisions resulted.
- Inconsistent approaches taken by members of the landscape profession to landscape assessment has contributed to the Environment Court having significant influence on landscape concepts and categorisation.
- There has been a more recent convergence in approach, with the differences amongst landscape experts now primarily focused on the scale of the landscape to which the assessment should apply and the weighting to be afforded different factors.
- Compared to other countries, Aotearoa New Zealand's landscape assessment methodologies have developed in an unorthodox manner. However, this has enabled the concept of landscape to develop a richness of meaning, including bringing in associative values which may have been excluded in a more rigid system.
- A NPS on Landscape could usefully provide national guidance on a consistent methodology to be adopted for landscape assessment.

Ocean Beach, Hastings

ENDNOTES

1 Pollock, 2014, 'Landscape architecture', Te Ara - the Encyclopedia of New Zealand, at http://www.TeAra.govt.nz/mi/landscape-architecture/print
2 Pollock, 2014
3 Swaffield, 1993, 59
4 Knight, 2018, 11
5 Knight, 2018, 33
6 Swaffield, 1999a, 3
7 Swaffield, 1993, 59
8 Swaffield, 1999a, 4
9 https://nzila.co.nz/about/history
10 https://nzila.co.nz/about/history
11 https://nzila.co.nz/about/history
12 https://nzila.co.nz/about/history
13 https://nzila.co.nz/about/history
14 https://nzila.co.nz/about/history
15 Knight, 2018, chapter 2
16 Swaffield, 1999a, 4
17 Swaffield, 1999a, 4
18 Swaffield, 1999a, 4
19 Swaffield, 1999a, 4
20 Swaffield, 1999a, 4
21 Swaffield, 1999a, 5
22 Boffa Miskell and Lucas Associates, 1993
23 Boffa Miskell and Lucas Associates, 1993, 28
24 Boffa Miskell and Lucas Associates, 1993, 35
25 Auckland Council Regional Policy Statement – Appendix F: Auckland regional landscape assessment
26 Auckland Council Regional Policy Statement – Appendix F: Auckland regional landscape assessment
27 Boffa Miskell, 2010, 11-12
28 For example, three assessments undertaken by LA4 Landscape Architects – Whangarei District landscape assessment report 1995, Far North District Council landscape assessment 1995 and Kaipara District landscape assessment 1999
29 Boffa Miskell, 2010, 24
30 *Unison Networks v Hastings District Council* Decision C11/2009 at [96]
31 Fairweather, Swaffield and Simmons, 1998
32 Fairweather and Swaffield, 1999
33 Fairweather, Swaffield and Simmons, 2000
34 Fairweather et al, 2001
35 Fairweather and Swaffield, 2004
36 Brown, 2011, 4
37 Read, 2012, 14
38 Boffa Miskell and Lucas Associates, 1993 at 35
39 Swaffield, 1999a, 5
40 Swaffield, 1999b, 1-2
41 Read, 2005, 7
42 Swaffield, 1999a, 6
43 Swaffield, 1999a, 7
44 Swaffield, 1999a, 6
45 Swaffield, 1999a, 6
46 Swaffield, 1999a, 6
47 Swaffield, 1999a, 9
48 Swaffield, 1999a, 6
49 Swaffield, 1999a, 6
50 Swaffield, 1999b, 1-2
51 Swaffield, 1999a, 7
52 Swaffield, 1999a, 8
53 https://www.nzaia.org.nz/marion-read.html
54 New Zealand Institute of Landscape Architects, 2010b
55 New Zealand Institute of Landscape Architects, 2010b, 3
56 New Zealand Institute of Landscape Architects, 2010b, 3
57 New Zealand Institute of Landscape Architects, 2010b, 3
58 New Zealand Institute of Landscape Architects, 2010b, 3
59 New Zealand Institute of Landscape Architects, 2010b, 4
60 New Zealand Institute of Landscape Architects, 2010b, 4
61 *Friends of Nelson Haven and Tasman Bay Inc v Tasman District Council* [2018] NZEnvC 46 at [99]
62 Borthwick, 2016, 15
63 Read, 2012, 34
64 Read, 2012, 34
65 Read, 2012, 16
66 Read, 2012, 16
67 Swaffield, 1999a, 10
68 Swaffield, 1999c, 45
69 Borthwick, 2016, 15; Swaffield, 1999c, 45
70 *Mainpower NZ Ltd v Hurunui District Council* [2011] NZEnvC 384 at [294]
71 *Mainpower NZ Ltd v Hurunui District Council* [2011] NZEnvC 384 at [294]
72 Read, 2012, 80
73 Read, 2012, 80
74 The United Kingdom, in particular, has a very structured approach to landscape assessment
75 *Li v Auckland Council* [2018] NZEnvC 87 at [557]
76 *Li v Auckland Council* [2018] NZEnvC 87 at [557]
77 https://nzila.co.nz/news/2017/09/project-landscape-assessment-methodology
78 https://nzila.co.nz/news/2017/09/project-landscape-assessment-methodology
79 Borthwick, 2016, 16
80 Borthwick, 2016, 16
81 Williams, undated
82 See *Briggs v Christchurch City Council* C045/08 at [9]-[12] for a history
83 *Briggs v Christchurch City Council* C045/08 at [122]
84 *Briggs v Christchurch City Council* C045/08 at [130]
85 *Briggs v Christchurch City Council* C045/08 at [22] and Annexure 1:E
86 *Briggs v Christchurch City Council* C045/08 at [123]
87 *Briggs v Christchurch City Council* C045/08 at [125]
88 *Briggs v Christchurch City Council* C045/08 at [84]

Current landscape protection tools

Mackenzie Basin

There is a wide range of regulatory and non-regulatory tools which can be deployed to protect landscape values in Aotearoa New Zealand. We review in this chapter national direction under the RMA, regional policy statements (RPSs), regional and district plans, iwi recognised planning documents, Crown pastoral lease discretionary consents, concessions under the Conservation Act 1987, Treaty-based legal tools, protective covenants, community trusts and collaborative initiatives.

6.1 RMA regulatory tools

RMA national direction

National direction under the RMA can establish a strong policy framework for decision-making under the Act. As demonstrated by the *King Salmon* case, it can even go so far as to establish 'environmental bottom lines' for landscape values that need to be protected. However, despite the RMA being in force for nearly 30 years, there is still no NPS specifically on landscape management. This has meant that the 78 councils around the country have been largely left to work out for themselves how to apply the landscape provisions of the RMA. As discussed in Chapters 4 and 5, it has also meant that the Environment Court has been in the driving seat in terms of how the provisions in Part 2 of the RMA are applied to various localities and activities, but only if those matters reach the Court.

Despite there being no national direction specifically on landscape protection, there is a raft of national direction on other matters that impact landscape outcomes, most particularly the NZCPS which applies to landscapes within the coastal environment.

New Zealand Coastal Policy Statement

The RMA has a mandatory requirement for a NZCPS which is to set objectives and policies to achieve the purpose of the RMA in relation to the coastal environment.[1] The existence of the NZCPS means that different policies apply to landscapes within the coastal environment than to those located elsewhere.

The first NZCPS, adopted in 1994, contained high level policy and suffered from poor implementation.[2] Its 2010 successor is more directive and has more clout (particularly since the Supreme Court's decision in *King Salmon*). It seeks to preserve and restore natural character (Policies 13 and 14) and protect natural features and natural landscapes (Policy 15) in the coastal environment, amongst other things.

The NZCPS is an instrument at the top of the country's planning hierarchy. It gives substance to Part 2 of the RMA in the coastal environment.[3] Initially, the RMA required that regional and district plans not be inconsistent with the NZCPS. However, this has since been strengthened, and now RPSs and regional and district plans must give effect to it.[4] 'Give effect to' is a strong directive creating a firm obligation to take action.[5]

As noted in Chapter 4, Policies 13 and 15 of the NZCPS direct that adverse effects on outstanding natural character and ONLs and ONFs in the coastal environment be avoided. 'Avoid' means 'not allow' or 'prevent the occurrence of'.[6] In that context, and as enunciated in *King Salmon*, Policies 13 and 15 (the 'avoidance policies') provide something of an environmental bottom line.[7] As the policies must be given effect to, areas of outstanding natural character, ONLs and ONFs in the coastal environment are arguably better protected than analogous areas elsewhere where each local authority is left to its own devices in determining how protection is to be implemented.

A similar issue arises with respect to wetlands, and lakes and rivers and their margins. Section 6(a) provides for the protection of them from inappropriate subdivision, use and development. However, the 2010 NZCPS only applies to those areas if they fall within the coastal environment. In effect, that means that only coastal wetlands, lakes and rivers are afforded the protection of the 'avoidance policies'. Outside of the coastal environment, their fate is similarly left to local authorities.

The implications of the avoidance policies on land use necessitates transparent assessments underpinned by robust methodologies. However, where an area does justify identification as an ONL, ONF or area having outstanding natural character, the strong direction of the NZCPS policies should not be used to adopt an even higher threshold in the coastal environment than would normally apply, such as a need to be 'unique'.[8] Doing so risks squeezing areas of outstanding natural character to isolated sections of the coast and retreating ONLs in the coastal environment to ridgelines.

In 2017, DOC produced its review of the 2010 NZCPS. It recorded a key issue relating to the effectiveness of Polices 13 and 15 as being the lack of consistent methodology.[9] It noted that, while there had been significant effort at the regional level to address this, the issue was still problematic for NZCPS implementation, particularly in relation to identification of 'outstanding' natural character, natural landscapes and natural features.[10] The review stated that:

> Prioritising work on developing consistent assessment methodologies would be valuable, particularly methods for identifying outstanding natural character, natural landscapes and natural features. Consistent methodologies would also greatly assist with the mapping and identification of the coastal environment...[11]

It is understood that an updated NZILA Guidelines will cover natural character issues.

Other national direction

The revised National Policy Statement on Freshwater Management, which came into force in September 2020, provides a strong framework for freshwater management within landscapes. In particular, freshwater managers must give effect to 'Te Mana o te Wai' which prioritises the health and wellbeing of water bodies above uses of them. Every regional council is directed to engage with communities and tangata whenua to determine how Te Mana o te Wai applies to water bodies and freshwater ecosystems within their region.[12] This approach should support collaborative and contextualized management approaches being applied at the catchment and landscape scales. Overall, the national direction should help improve water quality and protect wetlands and other water bodies within our broader landscapes, whilst simultaneously restoring and protecting mana and mauri.

Other national direction is not so supportive. For example, the National Environmental Standard for Plantation Forestry (NESPF) severely restricts the ability of councils to regulate and manage plantation forestry. Coupled with financial incentives for the planting of exotic species like *Pinus radiata* through the emissions trading scheme and government's 'One Billion Trees' programme, the NESPF seems likely to result in an expansion of plantation forest in sensitive landscapes such as the Mackenzie Basin and Banks Peninsula.

Plantation forestry surrounding Lyttelton Harbour

A spotlight on the National Environmental Standard for Plantation Forestry

The NESPF came into effect in May 2018 and applies to any commercial forest greater than 1ha in size. It seeks to control environmental outcomes associated with plantation forestry activities nationally, while also increasing certainty and efficiency for foresters.[13] The NESPF prevails over the rules in a district or regional plan except where it specifically allows more stringent plan rules. The NESPF does not allow for such plans to be more lenient than its standards.[14]

The NESPF sets a permissive baseline for afforestation and replanting. This means that, subject to compliance with the listed conditions, planting plantation forestry trees on land where there is no existing plantation forestry does not require resource consent. If the conditions cannot be complied with, afforestation and replanting are restricted discretionary activities.

To provide for the protection of ONLs, an exception to the permitted activity status is carved out. The NESPF draws a clear distinction between the establishment of plantation forestry (afforestation) and undertaking other activities that are part of the process of a plantation forest.[15] Afforestation within an ONL must not occur as a permitted activity.

In order to fall within this exception, the landscape must be identified in a RPS, regional plan or district plan as outstanding, and that planning document must include the identification of its location by way of mapping, a schedule or description. This requirement for ONLs to be identified in planning documents, and the variable quality of that identification process (when it comes to determining borders), may result in the lack of protection for some ONLs that have not been fully identified.

For example, the extent of the Mackenzie Basin ONL as identified in the Canterbury RPS, is not fully identified and mapped in the Waitaki District Plan. In addition, the Christchurch District Plan has only identified small areas of ONL on the Banks Peninsula despite the Canterbury RPS identifying the whole area as outstanding. The effect of the NESPF is that the councils now have virtually no control over the establishment of plantation forestry in these important landscapes.

While there are stricter regulations for afforestation within ONLs, other activities associated with plantation forestry do not have the same controls. For example, in the majority of the Mackenzie Basin, harvesting and replanting can occur as a permitted activity.

The NESPF also offers quite weak controls in respect of wilding pine invasion, which is a major threat to ONLs including those in the Mackenzie Basin. The Wilding Tree Risk Calculator must be used by foresters when planting new forests or replanting forests with a new conifer that has not previously been planted. Calculations are based on five indicators: species growth, species palatability (to stock), siting of the new planting, downwind land use and downwind vegetation cover. A requirement to obtain resource consent is only triggered when the calculator gives a rating of 12 ('high risk') or higher.

Despite there being risks of wilding spread for 'relatively high risk' plantations, these are deemed a permitted activity. Replanting of the same species does not require a resource consent, regardless of what species was initially planted.[16] This is in contrast to the district plan rules applying in the Mackenzie Basin that had previously prohibited the planting of some pine species to prevent further wilding pine spread.

Overall, the NESPF has increased the likelihood that important landscape values will be degraded through exotic plantation forestry activities.

Regional policy statements

RPSs are the 'heart of resource management in each region.'[17] They provide an overview of resource management issues specific to the region and include policies and methods (but not rules) to achieve the integrated management of those resources. An RPS must give effect to national direction and Part 2 of the RMA.

An RPS could be a strategic region-wide document which identifies landscapes and landscape values of importance to the region and sets out how they should be managed and protected. Unfortunately this opportunity is not always utilised. Two of our case studies (Mackenzie Basin and Banks Peninsula) are within the Canterbury region. Although Environment Canterbury had been proactive in commissioning a regional landscape study as early as 1993, this has not translated into clear policy in the Canterbury RPS. In particular, the ONLs have not been mapped (although they are described in the second generation RPS which became operative in 2013) and it is unclear how they are to be protected. This has led to virtually no protection being provided for the ONLs on a region-wide basis, with Environment Canterbury having no regional plan that addresses landscape issues. The matter has effectively been delegated to territorial

authorities which have provided only partial protection in their district plans (with smaller areas being identified as 'outstanding'). In particular, on Banks Peninsula less than a quarter of the rural land which is identified as an ONL in the Canterbury RPS is identified as an ONL in the Christchurch District Plan (see spotlight above).

Our other two case studies (Waitākere Ranges and Hauraki Gulf Islands) are within the Auckland region where there is a Unitary Plan. The ONLs across the region are clearly mapped. This has enabled the provisions of the Auckland RPS to directly flow down into the policies and rules of the regional and district plan component of the Unitary Plan in an integrated manner. This avoids the disjunct that can occur when district and city councils are left to implement vague regional policies with little guidance.

In our Mackenzie Basin case study, we recommended that a new sub-regional chapter be developed as part of the Canterbury RPS, which would focus on the Mackenzie Basin. Such location-focused chapters:

> ...can address more detailed objectives and policies within the framework of region-wide provisions ... they need not cover the entire region but could focus on spatial areas where there are significant regional issues to resolve. Such an approach enables the RPS to be 'grounded' in the different physical characteristics of sub-regional areas.[18]

Aotea Great Barrier Island *(Craig Potton)*

In respect of the Mackenzie Basin our case study report stated:

> We think there is a strong case for developing a new sub-regional chapter of the Canterbury RPS for the Mackenzie Basin. This would enable the development of a robust integrative framework for the management of the area, including integration between landscape, ecology and water issues which are particularly closely interlinked across the Basin.
>
> The incorporation of sub-regional chapters has been recognised as a way in which integration across separate domain-based chapters of a RPS (such as fresh water, ecosystems and biodiversity, landscape and historic heritage) can be effectively achieved. Mapping is an important tool to incorporate into such a sub-regional chapter in order to enable the strategic direction of the RPS to be clearly articulated. Maps provide a greater degree of certainty, particularly for key issues such as the location of outstanding natural landscapes and significant natural areas and where key objectives are sought to be achieved...
>
> A sub-regional chapter for the Mackenzie Basin should set out clearly what is sought to be achieved where and how it will be achieved, with measurable outcomes identified which can be monitored and reported on. Maps should be included to identify areas where values will be strongly protected and other areas where greater development might be permissible.[19]

The development of such subchapters for important landscape areas, such as the Mackenzie Basin and Banks Peninsula, would enable more nuanced policies to be developed in the RPS that are tailored to the context of each area and provide clearer direction for the development of regional and district plan policies and rules.

Regional and district plans

Regional councils and territorial authorities develop plans under the RMA which have as their purpose to assist the councils to carry out their functions "in order to achieve the purpose of the Act."[20] Policies and rules in these plans are the prime regulatory mechanism to control land and natural resource use that has the potential to impact landscape values. The functions of regional councils and territorial authorities are set out in sections 30 and 31 of the Act. Landscape is not mentioned in either of these sections. In terms of territorial authorities, landscape management is implied to be within their general function which is:

> ...the establishment, implementation, and review of objectives, policies, and methods to achieve integrated management of the effects of the use, development, or protection of land and associated natural and physical resources of the district.[21]

Because territorial authorities manage land use – which includes many activities which can degrade natural landscape values such as subdivision (or more accurately its consequences), vegetation clearance, earthworks, the building of houses and other structures, and plantation forestry – district plans are typically where rules that protect landscape values are found. And without strong regional policy, landscapes which are managed by more than one territorial authority can be subject to inconsistent management regimes across similar landscapes. This is the situation in the Mackenzie Basin where there is now strong protection in place for landscape values in the Mackenzie District, but lax rules applying to the Basin floor in the Waitaki District. There can also be inconsistency between resource consenting at the regional and district levels, and a lack of bundling of multiple consents applying to the same proposal, such as with the Simons Pass Station irrigated dairy development.

Territorial authorities managing parts of Aotearoa New Zealand with high natural landscape values are often poorly resourced and therefore not well placed to undertake this task on behalf of the nation. For example, the Mackenzie District Council, which manages most of the remaining intact natural landscape in the Mackenzie Basin has a budget of less than $9 million and fewer than 40 employees.

In the absence of any clarity within the RMA as to the respective roles of regional councils, territorial authorities and tribal authorities in protecting landscapes, a NPS on Landscape would be helpful. This could set out at what scale landscape should be assessed, the assessment methodology to be used (as discussed in Chapter 5) and roles of the respective councils and their planning documents in providing protection. A Landscape NPS could also establish bottom lines for the management of landscapes outside the coastal environment.

Iwi planning documents

Relevant planning documents, recognised by an iwi authority and lodged with councils, must be taken into account when RPSs and regional and district plans are being developed or updated.[22] They can provide councils with an indication of tribal issues, values and interests related to cultural landscapes as well as broader environmental aspects. Figure 6.1 indicates how such planning documents fit within the RMA system.

Iwi planning documents are useful, but they are not a substitute for council engagement with mana whenua.[23] The prime purpose of the documents, in relation to councils, is to assist with the recognition of iwi rights and interests within their applicable rohe, and to provide councils with some background and understanding of their issues, values and interests prior to proper engagement.

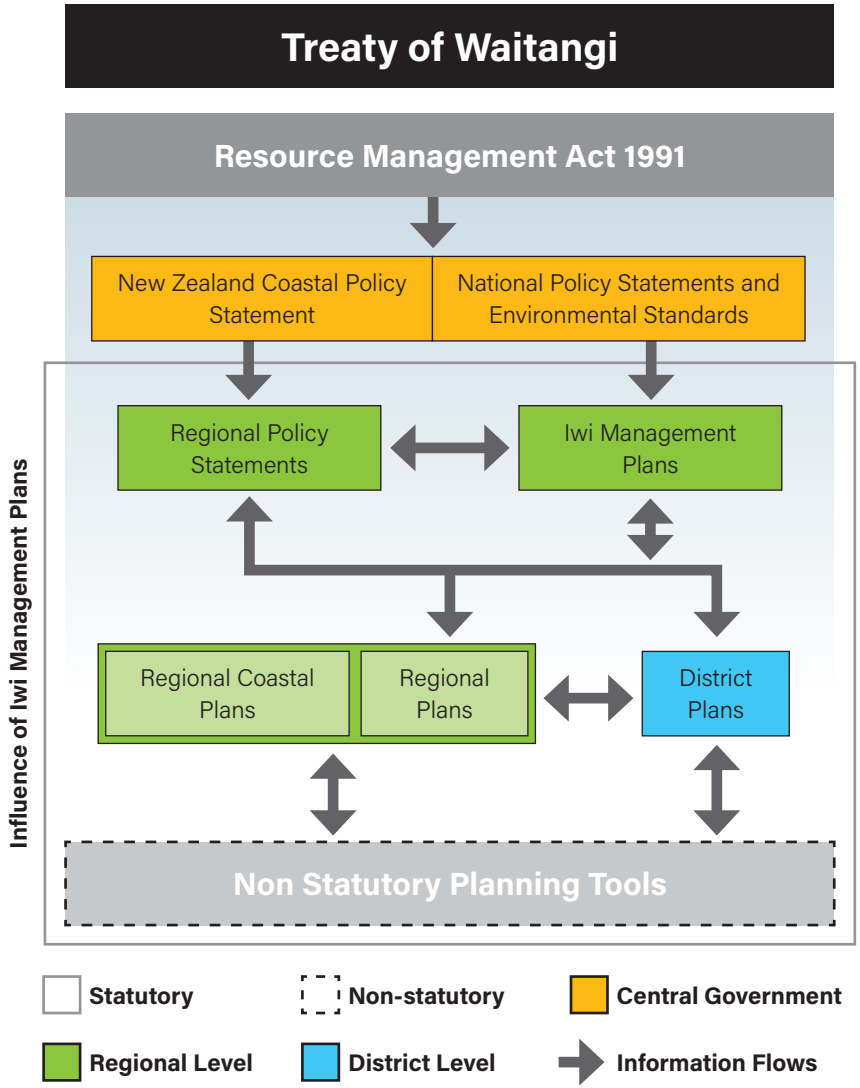


Figure 6.1: *The influence of iwi planning documents under the RMA*[24]

There are many examples of iwi planning documents including (most commonly) broad-scoping iwi management plans, more environmentally-focused management plans, and joint management plans between tribal entities and government agencies. There is no standard approach for such planning documents and not all include cultural landscapes.

A spotlight on the Te Arawa Lakes Trust environmental plan

The Te Arawa Lakes Trust recently developed He Mahere Taiao mō ngā wai o Te Arawa (Te Arawa Lakes Environmental Plan) which gives effect to Te Tuapapa o ngā Wai o Te Arawa (Te Arawa Cultural Values Framework) 2019–2030. This plan has been recognised through several awards (including the 2020 Nancy Northcroft Planning Practice Award and Best Practice-Non-statutory Planning Award) for its effective integration of cultural values, mātauranga Māori and contemporary policy development.

The Plan provides clear advice, guidance and support to councils, Crown entities and community partners, in a language that resonates with those organisations. It was intentionally written as an accessible resource for agencies to consult before engaging with the Trust.

Included in the Plan is a policy framework which has a concise articulation of issues, objectives and policies for resource management and describes iwi relationships with their cultural landscapes. The implementation of Te Tūāpapa has already resulted in increased involvement of Te Arawa whānau, hapū and iwi in restoration projects for lakes, wetlands and streams in the catchments of the Te Arawa Rotorua Lakes, and cultural mapping of significant sites around the Te Arawa Rotorua Lakes.

Section 7 of the Plan specifically addresses 'Wai Ata' – Te Arawa's special places, the theme of which is "matters which affect or enhance our special places and our connection to those places". The following excerpts illustrate how the Trust articulates cultural landscapes and the interconnections not only between people, place and wellbeing, but also between culture, context and identity.

7. Wai Ata | Waiata: Our special places

Within Te Tūāpapa, Wai Ata | Waiata relates to the "rhythm of wai as embodied and transmitted through waiata"...

Every Te Arawa hapū and iwi have associations with specific waterbodies that are reflected in their whakapapa (ancestral lineage), waiata (song), and whaikorero tuku iho (stories of the past) ... We use waiata to transmit and share our stories about our special places and ancestors. They form part of our identity and are of absolute relevance to contemporary resource and environmental management...

7.1 Cultural heritage and identity

The lakes are part of us. Every nook and cranny we have a name for, they are named after prominent ancestors and those names are still used today as though they are still living.
— Tā Toby Curtis

This section relates to sites, areas and landscapes of significance to Te Arawa, located in and around our lakes. It also relates to those actions that enhance our collective identity as Te Arawa.

Cultural heritage encompasses both tangible (e.g. physical) and intangible values. It includes, but is not limited to, pa sites; marae; kainga (dwelling places); burial grounds; canoe landings; taonga (artefacts); as well as places where food, materials, and rongoā are gathered. It is unsurprising that many recorded archaeological sites are clustered around and near our lakes, rivers and wetlands. Some natural features around the lake shores, such as kohatu (rocks), repo (swamps), ana (caves), puna (springs), motu (islands) are also regarded as waahi tapu by hapū for many different reasons. Over centuries, and due to fluctuating lake levels, many of these sites are now above or below a lake surface or margin.

The protection of Te Arawa cultural heritage is closely aligned to our spiritual and cultural wellbeing. These areas tie us to our land and lakes, creating cultural reference points in, and around, our lakes. This relationship is an integral part of our collective identity as Te Arawa, and responsibilities as hungatiaki.

6.2 Other regulatory tools

Other regulatory tools that can impact landscapes include Crown pastoral lease discretionary consents, concessions for activities within public conservation areas and protective land covenants which we discuss below.

Crown pastoral lease discretionary consents

Both the Land Act 1948 and the Crown Pastoral Land Act 1998 play a significant role in the management of Crown pastoral lease land in the high country. This land is owned by the Crown, but is leased on a perpetually renewable basis.

Pastoral leases recognise the natural capital of the land, and in order to protect this, restrict the activities that can be undertaken by lessees. Lessees are granted a right to pasturage (pastoral farming) only. This includes the grazing of animals such as sheep and cattle on the land. Activities that disturb the soil; such as burning vegetation, increasing stock numbers (beyond the limit determined by the Land Settlement Board and specified in the lease)[25] or the cultivation of the soil, require permission from the Commissioner of Crown Lands under the Crown Pastoral Land Act. This process is referred to as the discretionary consent process.

In making a decision about whether to grant a discretionary consent, the Commissioner is required to take into account two potentially conflicting purposes:

1. The desirability of protecting the inherent values of the land concerned, in particular the inherent values of indigenous plants and animals, and natural ecosystems and landscapes
2. The desirability of making it easier to use the land concerned for farming purposes.[26]

Neither of these purposes contain clear outcomes or environmental bottom lines that must be given effect to. In addition, no hierarchy is included to guide the assessment of discretionary consents. On the face of the legislation alone, the two purposes are given equal weighting.

The legislation also requires that the Commissioner only 'take into account' rather than 'give effect to' the two purposes, enabling a broad degree of discretion. If the two considerations are in direct conflict, which is often the case, the Commissioner must exercise his or her judgment as to which takes priority, and how to mitigate any adverse effects. There is a legal requirement under section 18(1) for the Commissioner to 'consult' the Director General of Conservation, but there is no additional policy guidance as to how the two statutory purposes should be weighted and applied. This lack of legislative priority, combined with a lack of policy guidance, has resulted in the majority of consents applied for being granted, albeit with conditions attached.[27]

This has resulted in a system that has not delivered good ecological sustainability or natural capital outcomes.[28] For example, a study identified that from 2003 to 2018, 9,000ha of intensification was enabled on pastoral lease land in the Mackenzie Basin through the discretionary consent system. Much of this was on the ecologically significant and highly vulnerable Basin floor. This is a significant figure when compared to the 16,000ha that occurred on newly freeholded land. It is clear that decisions on discretionary consents have played a significant role in enabling the large amount of agricultural conversion that has occurred within the Basin and resultant negative impacts on its landscape values.[29]

Another failing of the discretionary consents process has been its narrow focus. The Commissioner has viewed applications for intensification activities in the Mackenzie Basin at a site or property-scale rather than a landscape-scale. Consents have been considered in isolation, and the statutory purposes balanced only within the area described in the application. This has resulted in the fragmentation of contiguous ecosystems and landscapes.[30]

The Crown Pastoral Land Act also provided a system for tenure review, whereby pastoral leasehold land was freeholded and divided between the Crown and the leaseholder. Much high country land has been freeholded under this system, with mixed results. In the Mackenzie Basin, it helped facilitate the establishment of the Ahuriri and Ruataniwha conservation parks. But most of the low latitude land on the Basin floor, which supports rare dryland ecosystems, is now in private hands.

In early 2019, the Government decided to end the tenure review process and the Crown Pastoral Land Reform Bill was introduced into Parliament in July 2020. As well as providing for the cessation of tenure review, the Bill sets out new outcomes to be achieved in the management of remaining pastoral leasehold land. These remove the balancing exercise under the current Act and make it clear that the inherent values across the Crown pastoral estate need to be maintained and enhanced while providing for pastoral farming. The outcomes also provide for the better recognition of Māori interests. If the Bill is enacted as drafted, this should help improve landscape outcomes from the discretionary consenting process.

Pivot irrigation in the Mackenzie Basin

Outcomes in the Crown Pastoral Land Reform Bill

4 Outcomes for decision-makers

(1) All persons performing or exercising functions, duties, or powers in relation to pastoral land under this Act or the Land Act 1948 must seek to achieve the following:

(a) maintaining or enhancing inherent values across the Crown pastoral estate for present and future generations, while providing for ongoing pastoral farming of pastoral land; and

(b) supporting the Crown in its relationships with Māori under te Tiriti o Waitangi; and

(c) enabling the Crown to get a fair return on its ownership interest in pastoral land.

(2) Subsection (1)(c) applies only to functions, duties, or powers that relate to rents, easements, or recreation permits in respect of pastoral land.

5 Māori interests

In achieving the purpose of this Act, the Crown—

(a) must recognise and provide for the relationship of Māori and their culture and traditions with their ancestral lands, water, mahinga kai, wāhi tapu, and other taonga in any case where—

(i) consent for a discretionary pastoral activity is sought; or

(ii) a recreation permit over pastoral land is sought; or

(iii) a stock limitation exemption or a variation or revocation of a stock limitation exemption is sought; or

(iv) a protective mechanism is being considered; and

(b) must consult Māori in accordance with this Act (*see* sections 22B (monitoring framework), 22D (strategic intentions document), and 100O (standards and directives)).

Concessions

Most commercial activities within conservation areas, such as accommodation facilities, guiding, commercial recreation and transport services, require a concession. Concessions are granted under section 17Q of the Conservation Act which states that "the Minister may grant a concession in the form of a lease, licence, permit, or easement in respect of any activity." The consideration of concession applications focuses on an assessment of effects within the framework of statutory management plans, similar to the approach taken to resource consents under the RMA. Applications must include a description of the potential effects of the proposed activity and any actions proposed to avoid, remedy or mitigate adverse effects. Applications can be declined if they do not comply with or are "obviously inconsistent with" a relevant conservation management strategy or conservation management plan which are prepared under the Conservation Act.[31]

Although the 'first in first served' model often applies to the allocation of concessions, the Act includes provision for different approaches to be applied. Under section 17ZG the Minister may tender the right to make applications, invite applications and carry out other actions to encourage specific applications. This section also enables the Minister to attach an obligation to the concession requiring the holder to undertake management of a conservation area or to reach a separate agreement with concession holders to do so. This enables positive conservation outcomes to be achieved through the grant of concessions and also for funds to be raised through a tendering process. Tendering processes, which are also referred to as reverse auctions, have been used successfully to increase conservation outcomes in other jurisdictions such as with the Australian BushTender programme which involves land owners tendering to protect areas of native vegetation.[32]

For the year ended 30 June 2020, there were 1100 'recreation' concessions managed by DOC which broadly equates to the number of tourism concessions. DOC also manages a range of other concessions for activities such as grazing, telecommunications and filming. Overall, the government generated $21 million from all its conservation concessions, leases and licences during that year. This was a reduction from $27 million the previous year primarily due to the impacts of Covid-19. Prior to Covid-19 there was close to an 80 per cent increase in revenue over four years from this source, partly due to the increase in visitor numbers from 2015 to 2018.[33]

Section 4 of the Conservation Act states that "This Act shall so be interpreted and administered as to give effect to the principles of the Treaty of Waitangi." In 2018, the Supreme Court issued a decision on the grant of concessions in the context of the Treaty provision in *Ngāi Tai Ki Tāmaki v Minister of Conservation*.[34]

The case concerned the grant of concessions on Rangitoto and Motutapu islands to the Fullers Group and Motutapu Island Restoration Trust. These grants were challenged by the Ngāi Tai Ki Tāmaki Tribal Trust which argued that no concessions should be granted to other operators in order to preserve the Trust's opportunities to develop tourism services on the islands.

The Court noted that section 4 requires more than procedural steps and that "substantive outcomes for iwi may be necessary including, in some instances, requiring that concession applications by others be declined." In addition, the court notes that "enabling iwi or hapū to reconnect to their ancestral lands by taking up opportunities on the conservation estate (whether through concessions or otherwise) is one way that the Crown can give practical effect to Treaty principles."[35] DOC is still working through the implications of the decision and has initiated a partial review of the Conservation General Policy and the General Policy for National Parks so that they better reflect DOC's Treaty responsibilities. The review will also consider the Waitangi Tribunal's 2011 recommendations on the Wai 262 claim in respect of flora and fauna.[36]

A different approach to concessions is being applied within Te Urewera. Concessions have been renamed 'Friendship Agreements' to emphasise a focus on values and relationships and "the fact the wellbeing of Te Urewera as a living system is at the heart of these relationships."[37] They operate within the framework of the Te Kawa O Te Uruwera (Te Uruwera management plan) which "describes the interrelationship between people and Te Uruwera".[38] This is starkly different to the effects-based approach under the Conservation Act and could be drawn on in any future review of the legislation.

Compliance has also been an issue. DOC is currently monitoring just under a quarter of its long-term tourism concessions each year. A recent compliance check undertaken of operators at Punakaiki, South Westland, Wanaka, Wakatipu and Te Anau found that about a quarter of concession holders were either breaching the conditions of consent or operating illegally.[39]

In our case study on Tourism and Landscape Protection we investigated, in some depth, the interface between the tourism industry and the concession system. We concluded that:

> The *Ngāi Tai* Supreme Court decision has prompted a pause and time of reflection for DOC on how it should manage the concession system. The legal framework within which concession decisions are made is now somewhat outdated with its focus on avoiding negative effects rather than on generating positive outcomes. The impacts of Covid-19 on the tourism industry has created some breathing space. This is an opportune time to review the concessions system from first principles in order to design a system that better reflects Treaty obligations, generates a substantive income flow for DOC, provides better flexibility and business opportunity for the tourism industry, and generates positive outcomes for biodiversity and landscapes more generally.[40]

Protective covenants

Covenants can be used to protect areas of land in private ownership. They are legally binding agreements between two or more parties to protect the covenanted land according to terms in the covenant agreement (which will set out the values to be preserved and how to preserve them). In Aotearoa New Zealand, there are a number of different types of covenants that can be used, with the main types summarised in Figure 6.2. It is noted that while the Reserves Act 1977 and the Conservation Act have different statutory purposes, the similarity in how they are applied warrants them being grouped together for the purpose of this discussion.

Aoraki Mount Cook National Park

	Conservation Covenant	Queen Elizabeth the Second Open Space Covenant	Ngā Whenua a Rāhui Kawenata
Legislation	Reserves Act 1977 (s77) or Conservation Act 1987 (s27)	Queen Elizabeth the Second National Trust Act 1977 (QEII Trust)	Reserves Act 1977 (s77A)
Purpose	To preserve the natural environment, landscape, amenity, wildlife or historical value of covenanted land	To covenant open space on private land for cultural, archaeological, landscape and biodiversity purposes	The protection of historical, cultural, spiritual and ecological values on private Māori-owned land
Duration	Generally, covenants are granted in perpetuity (but under the Reserves Act they can also be for a specified period of time)	Most covenants are granted in perpetuity but some may be registered for a limited period of time	Granted in perpetuity with terms and conditions reviewable every 25 years
Administration	DOC, local authorities or another body approved by the Minister (currently this is only the Banks Peninsula Conservation Trust)	QEII Trust	The Ngā Whenua Rāhui team, DOC
Monitoring and enforcement	Monitored either by DOC or local authorities. Many councils have specific monitoring schemes, with various requirements and accordingly there is not a nationwide monitoring programme. Costs associated with covenanting private land also vary depending on the contribution made by the council (in some cases this is nothing). Under the Reserves Act, once the conservation covenant is in place the enforcement provisions in Part 5 of the Reserves Act apply to the covenanted area as if it were a reserve. This is why covenants under the Reserves Act are largely preferred to those created under the Conservation Act.	Conditions on covenanted land include things such as: maintaining fences, pest and weed control and vegetation clearance. Monitoring occurs bi-annually, but compliance action is rare, even for egregious non-compliance.[41] Instead the Trust prefers to work with the landowners in these situations to inform them and share information about how to remedy the situation. Funding for the management of covenanted land is generally a mix between private funding and central government contributions. However, it is mostly funded by central government (the Vote Conservation budget).	The Ngā Whenua Rāhui is the only model that generally applies to multiple landholdings with multiple owners (instead of individual privately owned properties). A key feature of the Ngā Whenua Rāhui model is the ability for the landowner to retain rangatiratanga (ownership and control) over their land.[42] The landowning entity is paid a one-off consideration for undertaking the protection in perpetuity. Ngā Whenua Rāhui is the only covenanting model where the landowners are provided with a payment other than to secure the physical protection (eg toward fencing costs). DOC contributes to the funding of on-going operational costs, pest control and monitoring. There is also central government funding (from the Vote Conservation budget).

Figure 6.2 *Main types of land covenants*

Covenants are a useful tool as they create a legally binding agreement between the landowner and the covenanting agency which runs with the land. Land must be managed to preserve the values set out in the covenant agreement. Covenants can provide a high degree of protection, and often concerns about the use of covenants for landscape protection do not stem from the mechanism itself, but rather its implementation. In order to effectively protect landscape values on private land, covenants need to include detailed conditions for management and have regular monitoring to ensure compliance with the conditions.

Another overarching concern with the use of covenants as a landscape protection tool is their site-specific nature. Covenants focus on the property-scale and as such may not be suitable for achieving broad landscape-scale sustainability. However, there are options to address this. An integrated management plan could be developed to identify priority areas for the use of covenants. This could be accompanied by a condition included in all covenants, requiring landowners to submit information into a common information portal, which could then be used to track the contribution of covenants to achieving broader landscape goals.

Another tool that could have applicability, but has yet to be widely deployed, is the declaration of 'protected private land' under section 76 of the Reserves Act which can apply to private or leasehold land. The effect of the declaration is that DOC rangers can assist the owner in managing the land for conservation purposes.

Reserves Act 1977, Section 76, Declaration of protected private land

(1) The *owner of any private land or the lessee of any Crown land* may at any time apply to the Minister for his or her land or any part thereof to be declared to be protected private land under and subject to the terms of any agreement entered into between the owner or lessee and the Minister.

(2) The Minister, if satisfied that the land possesses such qualities of *natural, scientific, scenic, historic, cultural, archaeological, geological, or other interest* that its protection is desirable in the public interest, or that rare species of indigenous flora or fauna are on the land, and the preservation of such flora and fauna is in the public interest, and that the land is sufficiently fenced or is otherwise protected from damage by stock, may, by notice in the *Gazette*, declare the land to be protected private land for nature, scenic, historic, or scientific purposes, having regard to the provisions of sections 18 to 21 relating to the classification of historic, scenic, nature, and scientific reserves, and may in like manner revoke any such declaration.

6.3 Treaty-based legal tools

In recent years, more than 70 Treaty settlement agreements have been reached and the role and rights of Māori in relation to the land and landscapes has invariably been a focus in many of these negotiations. The settlements have helped drive the development of co-governance arrangements and have resulted in unique bicultural based frameworks and solutions being crafted. In addition, DOC has entered into more than 40 conservation protocols, accords, relationship agreements and memoranda of understanding with tribal authorities that recognise the core role that Māori play as mana whenua and kaitiaki (guardians) over natural resources in this country.[43] The recognition of the relationships and values iwi and hapū have with their landscapes is also growing within wider society.

The Te Awa Tupua (Whanganui River Claims Settlement) Act 2017, for example, has given legal personality to the Whanganui River and recognises the river not only as a person but as an ancestor, providing for a more relational and reciprocal based approach to its management and protection. Similarly, the Treaty settlement reached between the Crown and the tribal federation of Tūhoe in relation to Te Urewera National Park, acknowledges the living and spiritual force of that area (te mana me te mauri).

Tūhoe are described by the Waitangi Tribunal as holding a "central position in the cultural landscape of Te Urewera".[44] Both the importance of the cultural landscape to Tūhoe, and the strength of the ancestral relationship that Tūhoe has with Te Urewera, are recognised in the Purpose of the Te Urewera Act 2014 which is:

> ...to establish and preserve in perpetuity a legal identity and protected status for Te Urewera for its intrinsic worth, its distinctive natural and cultural values, the integrity of those values, and for its national importance, and in particular to—
>
> (a) strengthen and maintain the connection between Tūhoe and Te Urewera; and
>
> (b) preserve as far as possible the natural features and beauty of Te Urewera, the integrity of its indigenous ecological systems and biodiversity, and its historical and cultural heritage; and
>
> (c) provide for Te Urewera as a place for public use and enjoyment, for recreation, learning, and spiritual reflection, and as an inspiration for all.[45]

The new arrangements for Te Urewera illustrate the potential to transition away from the Western notion that land is a resource, towards a model that recognises and gives effect to Māori ontologies and ways of knowing that conceive of whenua as a 'person' with rights and mana.[46] Under the legislation, Te Urewera is

provided with "all the rights, powers, duties, and liabilities of a legal person".[47] This explicitly recognises and provides for the relational and associative connections between landscape and people. An important aspect of this arrangement is that, "by humanising nature as a person, we are accepting it as a participant in our community of justice and recognising specific duties towards it".[48]

Many Treaty settlement acts including the Te Urewera Act, Te Awa Tupua (Whanganui River Claims Settlement) Act 2017 and the Ngā Mana Whenua o Tāmaki Makaurau Collective Redress Act 2014 (which applies to the Auckland volcanic cones in Auckland) have created co-governance arrangements between the respective Crown agency or council, and iwi or hapū. These arrangements recognise the Treaty principle of partnership and enable mana whenua to fulfil their obligation to care as kaitiaki for their ancestral taonga and landscapes.[49]

For example, the Te Urewera Act creates new governance and management arrangements for the Te Urewera Forest. The Te Urewera Board was established to speak as the forest's voice and to provide governance and management. The Board is responsible for producing the management plan and a statement of priorities in accordance with the objectives of the Act. The iwi of Te Urewera forest (Tūhoe) have more governance representation on the Board than DOC. This reflects the strength of the relationship and obligation that Tūhoe has "with and for Te Urewera".[50] The arrangement empowers the iwi and enables the exercise of rangatiratanga and kaitiakitanga in respect of the protection and management of Te Urewera.

Similarly, in Tāmaki Makaurau, 14 ancestral maunga have been recognised for their significance to 13 tribal entities as taonga tuku iho (treasures handed down from the ancestors). A co-governance arrangement between those entities and Auckland Council enables an integrated approach to the management of the maunga stretching across the region, and therefore also an integrated approach to recognising and providing for that significant cultural landscape (see spotlight).

Maungauika North Head

A spotlight on the management of the Tūpuna Maunga of Tāmaki Makaurau

The Tūpuna Maunga (ancestral mountains) of Tāmaki Makaurau are both physical and cultural icons of the largest city in Aotearoa. The geographic formation of the maunga has shaped Tāmaki Makaurau's (Auckland's) identity. Māori were the first to occupy this area many centuries ago, establishing numerous kāinga and cultivating large areas on and surrounding the maunga. Being strategically located, with unobstructed views across the moana and whenua, fortified pā were also developed on many of the maunga. With its elevated areas, fertile soils and strategic maritime location, Tāmaki Makaurau became a popular region attracting many tribes. It was a place of complex inter-tribal relationships, transit, trade and (at times) dispute.

Over time, as Pākehā and others came to settle in the region, and as the city grew and developed as an urban centre, those same features became an "unparalleled network of open spaces that all Aucklanders draw a sense of identity from".[51]

A common creation narrative tells how the Tūpuna Maunga were created by the god Mataaho (guardian of Earth's secrets) and Rūaumoko (god of earthquakes and volcanoes). The wrath of Mataaho flowed deep within the earth, shaping and modifying the distinctive volcanic landscape of Tāmaki. Tāne Mahuta (god of flora and fauna) created the rich ecological diversity above the earth's surface. The Tūpuna Maunga stand paramount, integral to the historical, spiritual, ancestral and cultural identity of the mana whenua of Tāmaki Makaurau.

Various iwi and hapū hold customary interests within the region, creating a complex multi-layered Māori cultural landscape. While different tribes know the volcanic craters that comprise the Tūpuna Maunga by various names, the maunga are sacred places; being the ancestral sites of battles, marriages, births, burials, settlements and agriculture, and more recently, recreation. Collectively, the maunga comprise a shared Māori cultural landscape, and have special significance to Māori in terms of the intergenerational knowledge and sense of identity associated with them. According to te ao Māori values and responsibilities, there is a duty to care, honour, respect, protect and actively manage these places for the future. This protection is even more important given the increasing pressures of population growth, urban development and intensification within the city.

The landscape of the Tūpuna Maunga has been co-governed under Treaty settlement legislation since late 2014 (Ngā Mana Whenua o Tāmaki Makaurau Collective Redress Act). Ownership of the Tūpuna Maunga (except Māngere Mountain) was transferred to the 13 tribes of Ngā Mana Whenua o Tāmaki Makaurau through a collective legal entity – the Tūpuna Taonga o Tāmaki Makaurau Trust. The Tūpuna Maunga are held by the Trust for the common benefit of the iwi and hapū of Tāmaki Makaurau and the other people of Auckland. Prior to the Treaty settlement, each Tūpuna Maunga was classified as a reserve under the Reserves Act and this protection was re-applied by the Collective Redress Act.

The Tūpuna Maunga Authority is the statutory authority established under the Act to co-govern the 14 Tūpuna Maunga. The Authority comprises equal representatives from the three rōpū of the iwi and hapū of Tāmaki Makaurau (Marutūāhu, Ngāti Whātua and Waiohua-Tāmaki) and Auckland Council, together with Crown (non-voting) representation. Auckland Council is responsible for day-to-day management under the direction of the Tūpuna Maunga Authority. This co-governance arrangement is complex, because it involves so many tribal entities and affects multiple stakeholders and various interests. It serves to demonstrate how multiple entities can share responsibilities and actively protect and manage a landscape together, in a collaborative and integrated way, even though they hold different values, histories and narratives.

Most of the Tūpuna Maunga had a pre-existing reserve management or conservation plan. Due to the varying approaches of the different legacy councils before amalgamation into the unitary Auckland Council in 2010, these plans vary in age and content. They were wholly replaced by a new integrated management plan, prepared by the Authority, which unified the management approach across the Maunga. This enables the holistic care and protection of the Tūpuna Maunga as a whole, rather than as individual components.

The integrated management plan provides direction for the restoration, protection and management of the Tūpuna Maunga. It set out the 'values' to provide the tika (correct) framework for their care and protection and the 'pathways' that elaborate and give tangible expression to the values. The preparation of the plan was a collaborative process including input from the Tūpuna Taonga Trust, mana whenua, Auckland Council, local boards, volunteer groups, various stakeholders and the broader community. The plan underwent wide public consultation prior to being adopted in 2016. There is a companion strategies document. Individual Tūpuna Maunga plans are also being prepared.

According to a 2016 report by the Independent Māori Statutory Board, the key challenge for the protection of the Tūpuna Maunga over past decades has been the general lack of care shown towards them and the separate management regimes adopted by the legacy councils. The report argues that "a dedicated and integrated approach to restoring these mighty taonga has been the singular focus of the Tūpuna Maunga Authority since establishment" and "the guidance of the Tūpuna Taonga Trust has been crucial".[52]

Since its establishment, the Tūpuna Maunga Authority has successfully implemented many initiatives to actively protect the Maunga and to inform manuhiri (visitors) of the cultural importance of these taonga. This includes removing motor vehicles from the tihi (summit) of all those maunga with road access, developing innovative track designs that protect and preserve the cultural, archaeological and geological fabric of the maunga including a world-class boardwalk on Maungawhau Mt Eden, and activating a massive native restoration programme across the entire network of the maunga. The Authority has also opened a visitor centre, Te Ipu Kōrero o Maungawhau, on Maungawhau Mt Eden and launched a website to tell the story of the maunga as connected landscapes of national importance. The network-wide approach to restoration and education was unable to be achieved under the previous administrative arrangements.

Key elements of this approach is the use of specific legislation that recognises and protects the Tūpuna Maunga as significant cultural landscapes and appoints a dedicated co-governance board with responsibility for the development and implementation of an integrated management plan. The Authority provides a committed vehicle from which further avenues for protection and management can be sought. The integrated management plan operates as a separate overarching document that focuses on the health and wellbeing of the maunga. It operates in a similar way to a master plan that guides and shapes planning, strategy and practice.

The 'values' and 'pathways' in the integrated management plan also provide mechanisms for leaseholders and stakeholders to support sustainable land management initiatives. And furthermore, the plan empowers and enables mana whenua to govern and manage significant landscapes through te ao Māori and their own tikanga and mātauranga (practices and knowledge). Ultimately, the Independent Authority protects and actively manages the cultural values associated with the Tūpuna Maunga.

The Tūpuna Maunga are now on a tentative list for UNESCO World Heritage status, which is a priority goal for the Authority. A major part of a World Heritage bid is demonstrating that the unique heritage values are being properly protected. This underscores all the mahi (work) undertaken on the maunga. Significant advancements have already been made towards restoring the maunga as wāhi tapu, and this mahi continues.

In the marine and coastal space, there are likely to be implications for landscape protection from settlements made with iwi and hapū through the Marine and Coastal Area (Takutai Moana) Act 2011.[53] As well as protecting the interests of all New Zealanders in the marine and coastal area, the purpose of the Act is to "recognise the mana tuku iho exercised in the marine and coastal area by iwi, hapū, and whānau as tangata whenua", to provide for the exercise of customary interests in the area and to acknowledge te Tiriti o Waitangi.[54]

Under the Act, iwi, hapū or whānau groups can gain recognition of two types of customary interests: a customary marine title or protected customary right. Customary marine title over an area provides a wide range of rights to the title holding group as follows:

- A 'RMA permission right' which means the group can give or decline permission for activities that need resource consents or permits in the area
- A 'Conservation permission right' which means the group can give or decline permission for certain conservation activities in the area including establishing a marine reserve or granting concessions
- A right to be notified and consulted about applications for marine mammal watching permits in the area
- A right to be consulted about changes to the NZCPS
- A right to protect wāhi tapu and wāhi tapu areas
- Ownership of minerals other than petroleum, gold, silver and uranium which are found in the area
- Interim ownership of taonga tūturu found in the area
- Ability to prepare a planning document which sets out the group's objectives and policies for the management of resources in the area.[55]

At this stage, settlements are still being negotiated; very few are completed. However, potential redress outcomes for landscape protection could include protection of wāhi tapu (which might relate to landscapes) and protected customary rights, which should offer protection for those landscape features

that customary rights pertain to. Redress also offers iwi, hapū and whanau the ability to develop a marine management planning document that must be taken into account by local, regional and national government. These tribal planning documents are potential opportunities for marine and coastal landscape protection in particular. In addition, given the holistic and integrated nature of te ao Māori views and management approaches, there may be further opportunities to consider broader landscape protection values and mechanisms across land and seascapes through alignment or integration of these plans with other iwi planning documents (both coastal and other) that are lodged with central and regional authorities.

Ngāti Paoa waka

6.4 Non-regulatory approaches

There are a wide range of non-regulatory approaches applied to landscape management in Aotearoa New Zealand. Here we discuss the critical role of community trusts and collaborative initiatives.

Community trusts

Many localised activities contributing to landscape protection are facilitated by community trusts. In our landscape protection case studies we identified a number of trusts which were serving to bring people together, raise funds and initiate projects. They provide important support for positive land management and restoration work. This can very usefully supplement the regulatory framework that mainly focuses on stopping bad things happening. However, community trusts need to be adequately resourced to be effective.

For example, the Banks Peninsula Conservation Trust (BPCT) acts not only as a covenanting body under the Reserves Act, but also as an initiating and coordinating body for wider landscape-scale restoration initiatives. In doing so, the Trust has been able to leverage off the positive relationships it has developed with landowners on Banks Peninsula. The work of the BPCT has been supported and supplemented by two philanthropic trusts with significant capital behind them, the Rod Donald Banks Peninsula Trust which was set up by the former Banks Peninsula District Council and the Joseph Langer Trust which was established as a personal legacy.

Mackenzie Symposium

On Waiheke Island, the Hauraki Gulf Conservation Trust and the Waiheke Resources Trust have both worked to support environmental and restoration initiatives on the island. The Hauraki Gulf Conservation Trust was established by the Waikato Regional Council and it acts as umbrella funder for a range of community work. The Waiheke Resources Trust has been very active in wetland restoration and it acts as an umbrella funder for the Waiheke Collective which aims to bring together island people working on conservation initiatives so they can amplify the results of their efforts.

In the Mackenzie Basin, the Mackenzie Country Trust was formed in 2016 as a mechanism to implement the outcomes of a collaborative agreement which had been reached by the community over the future of the Basin. But unfortunately, in this case, the Trust was not adequately resourced for its work and it has therefore struggled to gain traction.

Collaborative initiatives

Our case study investigations highlighted the fact that much more can be achieved if people work together and combine their resources and effort. This was very evident on Banks Peninsula where statutory agencies (DOC, Christchurch City Council and Environment Canterbury) were not only working together, but working closely with rūnanga and local community trusts (described above) to achieve better biodiversity outcomes.

There are many other examples of successful collaborative initiatives in the case study areas. For example, the Te Manahuna Aoraki Project initiated by the NEXT Foundation is a large-scale predator control project in the upper Mackenzie Basin which has brought together DOC, rūnanga, runholders and a variety of funders to embark on the ambitious goal of controlling predators over 310,000ha of land.

Also in the Mackenzie Basin is the Mackenzie Basin Agency Alignment Programme. This is an alliance established in 2017 between the five agencies with statutory responsibilities within the Mackenzie Basin: DOC, Land Information New Zealand (LINZ), Environment Canterbury, Waitaki District Council and Mackenzie District Council. The alliance is designed to improve the way the agencies work together to achieve "enhanced environmental outcomes for the Mackenzie Basin."[56] It represents an acknowledgment by the agencies of past failings, and a fresh and welcome commitment to adopt a more joined-up approach to the Basin's management, in order to ensure more integrated and positive outcomes. It was the first initiative of its kind in Aotearoa New Zealand and has proved a successful approach which has helped to break down agency silos.

KEY MESSAGES

- There is currently no national direction under the RMA on landscape outside the coastal environment, meaning there is little consistency between approaches taken by councils to landscape protection.
- National direction on other matters impacts landscape outcomes, with some likely to be positive (eg the NPS on Freshwater Management) and others negative (eg the NESPF).
- RPSs could provide strong strategic direction for the identification and protection of landscape values of importance to each region, but this opportunity is often not utilised. RPSs should be used to map important landscapes within each region and set robust policy for their management.
- There is little clarity about the respective roles of regional councils, territorial authorities and tribal authorities in landscape protection.
- District plans are typically where rules that protect landscape values are found, as territorial authorities manage many of the land use activities which can degrade natural landscapes. But territorial authorities in areas with high natural landscape values are often the most poorly resourced.
- Tribal authorities have rights and interests in natural resource and landscape protection which span regional and district boundaries. They face similar resourcing and capacity issues, and even greater constraints in terms of influence on provisions, rules and decision making authorities.
- An NPS on Landscape could usefully set out the scale at which landscapes should be assessed, the assessment methodology to be used and the roles of the respective councils, iwi and hapū and their planning documents in providing protection. It could also establish environmental bottom lines for landscapes outside the coastal environment.
- Planning documents prepared by iwi and hapū can usefully provide councils with information on tribal issues, values and interests in relation to Māori cultural landscapes. However, they are not a substitute for council engagement with mana whenua.
- The legislative framework for the consideration of discretionary consents for activities on Crown pastoral leasehold land fails to give priority to the protection of landscape values. This has resulted in the degradation of landscapes in the high country. Proposed reforms should help address this issue.
- The concession system controls access to public conservation land and waters and operates within a now outdated legal framework. It is an opportune time to review the system.
- Treaty settlements are creating novel new models for managing our natural landscapes including creating legal personalities for natural systems and establishing co-governance bodies. These approaches could have wider utility within landscape protection in Aotearoa New Zealand.
- Marine and Coastal (Takutai Moana) settlements recognise and empower iwi, hapū and whanau with customary marine title and customary rights. Settlements may provide further opportunities for marine, coastal and potentially wider landscape protection.
- Protective covenants can be used to protect landscape values on private or leasehold land but need to be well designed and monitored to be effective. They focus on the property-scale and as such may be unsuitable for achieving broad landscape-scale sustainability.
- Community trusts can play an important role in facilitating community activities that contribute to landscape protection. However, they need to be adequately resourced.
- Much more can be achieved for landscape protection if agencies, iwi and hapū, and communities work together and collaboratively combine their resources and effort.

ENDNOTES

1 Resource Management Act 1991, s56
2 Peart, 2009, 216-217
3 *Environmental Defence Society Inc v Otago Regional Council* [2019] NZHC 2278 at [109]
4 Resource Management Act 1991, ss62(3), 67(3)(b) and 75(3)(b)
5 *Environmental Defence Society Inc v New Zealand King Salmon Company Ltd* [2014] NZSC 38 at [77]
6 *Environmental Defence Society Inc v New Zealand King Salmon Company Ltd* [2014] NZSC 38 at [24(b)] and [92]-[97]
7 *Environmental Defence Society Inc v New Zealand King Salmon Company Ltd* [2014] NZSC 38 at [132] and [153]
8 Department of Conservation, 2017, 21
9 Department of Conservation, 2017, 36
10 Department of Conservation, 2017, 36
11 Department of Conservation, 2017, 50
12 National Policy Statement for Freshwater Management 2020, Policy 3.2(1)
13 https://www.mfe.govt.nz/land/land-acts-and-regulations/national-environmental-standards-plantation-forestry/about-standards
14 Wright et al, 2019
15 Wright et al, 2019
16 Paul, 2015
17 *North Shore CC v Auckland Regional Council (PT A70/94)*
18 Peart and Reaburn, 2011, 30-31
19 Peart and Woodhouse, 2020b, 96-97
20 Resource Management Act 1991, ss63(1) and 72
21 Resource Management Act 1991, s31(1)(a)
22 Resource Management Act 1991, ss 61(2A)(a), 66(2A)(a) and 74(2A)
23 Saunders, 2017
24 Adapted from Saunders, 2017
25 For example the Simons Pass lease had a stocking rate of 3,500 sheep. A discretionary consent was granted by the Commissioner of Crown Lands in 2013 to increase this to 50,540 stock units (including up to 1,434 sheep and 7,581 dairy cows)
26 Crown Pastoral Land Act 1998, s18
27 Informant interview
28 Land Information New Zealand, 2019
29 Brower et al, 2018
30 *High Country Rosehip Orchards Ltd v Mackenzie District Council* [2011] NZEnvC 387, [419]
31 Conservation Act 1987, s17SB
32 Rolfe, Whitten and Windle, 2017
33 Department of Conservation, 2019, 77; Department of Conservation, 2020a, 153
34 *Ngāi Tai Ki Tāmak Tribal Trust* v *Minister of Conservation* [2018] NZSC 122
35 *Ngāi Tai Ki Tāmaki Tribal Trust* v *Minister of Conservation* [2018] NZSC 122 at [52]
36 Department of Conservation, 2020b
37 https://img.scoop.co.nz/media/pdfs/1705/Te_Kawa_o_Te_Urewera_QA_.pdf
38 https://www.ngaituhoe.iwi.nz/te-kawa-o-te-urewera
39 Department of Conservation, 2019, 4
40 Peart and Woodhouse, 2020, 30
41 Brown et al, 2015, 66-67
42 Snoyink, 2017
43 Lyver et al, 2019, 396
44 Waitangi Tribunal, 2009,18
45 Te Urewera Act 2014, s4
46 Joseph, 2018
47 Te Urewera Act 2014, s11(1)
48 Severinson, 2019, 163
49 Severinson 2019, 163
50 Greenaway, 2019
51 Tūpuna Maunga o Tāmaki Makaurau Integrated Management Plan 2016, 2
52 Independent Māori Statutory Board, 2016
53 Taylor, Te Whenua and Hatami, 2018, 17-21
54 Marine and Coastal Area (Takutai Moana) Act 2011, s4
55 Marine and Coastal Area (Takutai Moana) Act 2011, s62
56 https://mackenziebasin.govt.nz/about/

Summary of case studies

Akaroa Harbour

Five case studies were undertaken for our landscape protection project: Te Manahuna Mackenzie Basin, Te Pātaka o Rākaihautaū Banks Peninsula, Waitākere Ranges, Hauraki Gulf Islands and tourism and landscape protection. As indicated in Chapter 1, each case study report has been published in full and is available on the EDS website. Each case study area has its own particular context and story to tell. One of the key lessons from our project is that one size does not fit all. Each area has different historical trajectories, environmental pressures, and economic drivers for landscape change.

We briefly summarise the key characteristics of each case study in this section of the report. The case studies highlight a series of landscape management issues including weak policy and plans, misaligned government policies, a system which is slow to respond to pressures, a lack of check and balances, and a lack of resourcing for landscape management. They also highlight the positive benefits of partnership, collaboration and philanthropic endeavours. It is not possible to do justice to the depth of these case studies in a short summary. We therefore strongly recommend that readers access the full case study reports to obtain a more in-depth picture of the dynamics of landscape protection and change in each area.

We conclude this chapter by identifying the key lessons that can be drawn from the case studies collectively, and what insights they provide on elements which should be considered for inclusion in any future landscape management approaches applied in Aotearoa New Zealand.

7.1 Te Manahuna Mackenzie Basin

This case study tells a story of gradual and continual landscape and biodiversity degradation over the past 170 years, with recent acceleration due to the development of large-scale pivot irrigation. It highlights the fragmented management system which has struggled to adequately respond to landscape pressures.

Biophysical landscape

The Mackenzie Basin is an upper montane basin extending over some 269,000ha and ranging in elevation from around 800m to 375m. It is by far the largest and most complex intermontane basin in Aotearoa New Zealand and is one of only two basins that retain indigenous vegetation to any large extent, particularly on the valley floor.[1]

The landscape has been carved out by three periods of glacial activity. The broader glacial landscape includes rugged mountains to the west within the Southern Alps, extensive glacial moraine areas where rocks and gravel have been deposited by the massive sheets of ice, and large outwash areas where meltwater and rivers have carried and deposited fine sediment. There are three large natural lakes in the Basin (Tekapō, Pūkaki and Ōhau) and many intricate braided river systems.[2] The Mackenzie Basin is the only place in Aotearoa New Zealand where it is possible to see the entire intact glacial sequence from glaciers in the Southern Alps, through to moraines, outwash terraces and extensive plains.

The Mackenzie Basin supports a great variety of species, many which are rare, due to significant climatic and landform differences across its vast expanse which create numerous microclimes. The climate is extreme, featuring hot dry summers and cold, snowy winters. The outwash plains are among the most challenging and distinctive environments for plant growth in the country due to the very dry climate and shallow stony soils. They support a distinctive, endemic and often cryptic flora that is exceptionally drought tolerant.[3]

At least 81 plant species have been recorded on the Basin floor. It provides habitat for several locally endemic species and is a stronghold for many nationally threatened ones. The Basin also supports an impressive array of fauna, much of which has adapted to the harsh environment.[4]

Cultural landscape

Te Manahuna provided rich kai and other resources for Māori over the centuries which were mainly gathered through seasonal harvesting. A network of ancient trails (Ata Tawhito) connect over 160 known sites which were traditionally occupied for resource gathering purposes.[5]

Three papatipu rūnanga are mana whenua of Te Manahuna: Te Rūnanga o Arowhenua, Te Rūnanga o Waihao and Te Rūnanga o Moeraki. In 1848, the Crown purported to purchase a large part of the South Island from Ngāi Tahu under 'Kemp's Deed' including Te Manahuna. The boundaries of the purchase were not well defined and have always been contentious.[6] A settlement between Ngāi Tahu and the Crown to address Te Tiriti o Waitangi breaches was reached in 1998. It included statutory acknowledgements, dual place names and nohoanga (traditional resting places). These acknowledge the particular cultural associations that mana whenua has with areas within Te Manahuna including Aoraki Mount Cook, which is a sacred ancestral mountain central to mana whenua creation stories.[7] Te Manahuna, in its entirety, is a significant ancestral landscape to mana whenua.[8]

Early landscape change

Large scale fires sometime between 1150 and 1400 AD started a process of fundamental landscape change in the Mackenzie Basin.[9] Pastoral farming commenced during the 1850s. It was reliant on the native vegetation which was regularly burnt to clear woody vegetation and promote new growth that was more palatable for stock.[10] This changed the vegetation cover from scrub-tussock to open tussock grasslands and resulted in soil loss and decline in fertility.[11]

The adverse effects of pastoral farming were exacerbated by the introduction of rabbits to Aotearoa New Zealand during the 1860s. They rapidly bred and quickly reached plague proportions in the Basin, stripping the dry soils of pasture and uprooting tussocks. There has been an ongoing battle against rabbits ever since. The battle appeared won for a period when a rabbit haemorrhagic disease was illegally introduced into the country in 1997. This knocked down rabbits for a decade or more, but immunity has now developed in the rabbit population, which has meant a reversion back to various poisoning and shooting methods.

Topdressing and oversowing, which had become common by the 1970s, resulted in the replacement of native grass species with introduced grasses which flourished between the more robust short tussocks. By the 1970s, there was also an invasion of hawkweed (*hieracium*) which thrived in the depleted tussock areas and expanded over the bare soil.[12]

The loss of indigenous plant species in the Basin continues today under pastoral farming, despite a reduction in stocking levels when compared to historical levels. Grazing pressure has had the greatest impact on the dry outwash ecosystems, which are now extensively depleted compared to the moraine ecosystems, which have proved more robust and have more intact indigenous plant cover.

Hydroelectric power development, and its flow on effects, has had major impacts on the landscapes of the Mackenzie Basin of a magnitude second only to the changes wrought by pastoral farming. The Mackenzie Basin is the

Ferintosh Station, Mackenzie Basin

location of the country's largest network of hydroelectric power stations which were developed during the 1950s to 1980s. They involved raising the water levels of lakes Pūkaki and Tekapō and creating lakes Benmore and Ruataniwha. Extensive canals were also constructed over the landscape including the 27km long Tekapō-Pūkaki canal.[13]

These works resulted in the flooding of the village of Pūkaki, farm homesteads and much productive farmland. In addition, the canals divided some properties making stock movement difficult. The town of Twizel, now the largest settlement in the Basin, was originally built as a temporary works town, and was slated for removal once the hydro scheme was completed, until the community successfully fought to save it.

But more significantly, the hydro works resulted in the provision of water for irrigation within the Basin, thereby paving the path towards 'greening' the brown tussock landscape. In addition, beautification works associated with hydro development resulted in thousands of trees being planted, particularly along the raised lakesides. These have become major seed sources for the wilding pines which have become a serious threat to landscape and biodiversity values in the Basin.

Tenure review has also been a precursor to much landscape change within the Mackenzie Basin. Much of the land within the Basin was originally Crown pastoral leasehold land. Early tenure review agreements employed an 'altitude' model, where the high-elevation land became the basis of conservation parks, and the lower land (which was already more highly modified and had more productive value) was freeholded. The practical effect of this was to remove the option of grazing high-altitude tussock grasslands. This resulted in more grazing on the Basin floor. One third of the Basin floor was freeholded under tenure review.

In early 2019, the government decided to end the tenure review process, although those runholders who had accepted a substantive proposal by that time will be allowed to complete the process. Those who have commenced the tenure review process, but have not yet accepted a substantive proposal, are being dealt with on a case-by-case basis until the legislation changes. Simons Pass Station, arguably one of the most controversial leases in the Mackenzie Basin, falls into this category. A substantive proposal was accepted by the lessee in late 2020.

Current and future landscape pressures

Irrigation and associated farming intensification are major threats to the Mackenzie Basin's landscape and biodiversity values. There are strong economic drivers for the move towards developing irrigated pastures as a small amount of irrigation has the potential to double the profitability of Mackenzie Basin pastoral sheep and beef farms. This is through enabling the growth of winter feed during the dry summer months. This can be used to carry animals through their first winter, after which they are large enough go to the works where they generate much better returns. In addition to enabling stock to be 'finished' on the land, irrigation enables more flexible farm management, and can take pressure off paddocks during dry periods. Irrigation is a pre-requisite for dairy farming in the Basin as the activity cannot be undertaken in a dry, drought-ridden environment without it.

A large amount of water, close to 173 million m^3 a year, was allocated for irrigation in the Basin as part of the water rights granted for the hydro works in 1969.[14] Various irrigation consents were granted during the 1970s to utilise some of this water. However, the greening of the Basin (which spans the Mackenzie and Waitaki Districts) quickly accelerated during the 2000s with the establishment of the Benmore Irrigation Scheme within Waitaki District. This provided water to support the use of large pivot irrigators on around 4000ha of land running along both sides of State Highway 8 between Ōmarama and Twizel which is the main route through the Basin from Queenstown. As well as increasing production on sheep and beef farms, the scheme enabled the introduction of large dairy farms into the Basin.

The greening of the Mackenzie District side of the Basin was facilitated by an agreement reached in 2004 between Meridian Energy (which operated the hydro works in the Basin at the time) and a group of farmers who formed the Mackenzie Irrigation Company. The agreement almost doubled the amount of irrigation water available within the Basin by agreeing to make available a further 150 million m^3 a year.[15] This is the water being utilised by the Simons Pass Station dairy conversion proposal which at one stage was slated to run up to 15,000 cows.

Irrigation negatively affects a range of landscape values in the Mackenzie Basin. The greening caused by the application of water to the land detracts from the large, open, brown landscape and its natural values. The large centre-pivot infrastructure also intrudes into the largely natural (and unbuilt) landscape. In addition, irrigation fundamentally changes the flora and associated fauna of the area, eliminating indigenous species which have evolved to survive in a dry desert environment.

A second major threat to the landscapes of the Mackenzie Basin is wilding pines. This has resulted from a misplaced enthusiasm for planting trees in the Basin since the early 20th Century. The biggest plantings were associated with the hydro scheme, with 2.4 million trees being planted between 1958 and 1983. Around 20 per cent of the trees were conifers and they included species now known to create a wilding pine risk.[16] Conifers also feature in the Lake Tekapō

Regional Park and the Mackenzie District Council has seven commercial forestry plantations within the Basin which contain conifers.

By 2015, wilding pines had established on over 129,000ha of land comprising some 20 per cent of the main Basin area.[17] Two of the major seed sources for wildlings, at Pūkaki Downs and Mount Cook Station, have been accepted into the emissions trading scheme, meaning that removing the trees potentially creates an obligation to either replant or purchase carbon credits.

The Mackenzie Wilding Conifer Strategy, which sets out a strategic plan to address the wilding issue in the Basin, was completed in 2016. It identified five main seed sources which were the core of the problem and estimated that an additional $28 million was needed over a 15-year period to remove all the wilding spread mapped in 2016. The Strategy noted that front-loading the investment will save considerable cost.[18] Despite this, only $3.5 million additional funding was made available during the 2016/17 and 2017/18 years.[19] A further $2 million has recently been made available for wildling control in Canterbury (including at Ōhau and Tekapō) as part of the government's Covid-19 economic regeneration plan[20] but this is nowhere near the magnitude required to get on top of the problem. Currently, wilding conifers are probably the biggest risk to the landscape and biodiversity values of the Mackenzie Basin.

Before the arrival of Covid-19 into the country, and the closure of the borders, tourism was a substantial industry within the Mackenzie Basin generating over $140 million annually.[21] This was supported by the award of International Dark Sky Reserve Status to the area in 2013 and the opening of the Alps 2 Ocean cycle trail. The numbers of tourists started to overwhelm the available infrastructure, with some tourist hotspots such as the Church of the Good Shepherd in Tekapō, being overrun with vehicles and people. Many pastoral farms were involved in some kind of tourism activity including providing accommodation, horse trekking, helicopter flights, four-wheel drive tours and hunting. Proceeds from such tourism can help support sustainable land management.

Climate change will undoubtedly affect the Mackenzie Basin with consequential implications for land use. The climate of the Basin will likely become warmer, wetter and more variable. There is likely to be more rain overall (except in autumn), but this will be coupled with more dry days, more hot days, heavier rainfall, fewer cold nights and increased likelihood of water stress. Pasture production is likely to increase overall, mostly in winter and spring, due to warmer temperatures and a longer growing season. However, there will likely be a decrease in growth during summer due to hotter, drier conditions. Greater variability in weather is also likely to increase farming risks and more irrigation may be sought in response. The potential range of wilding pines is likely to significantly increase under a warming climate, leading to further invasion of tussock grasslands.[22]

Current management responses

The Mackenzie Basin has been identified as an ONL in the Canterbury RPS 2013, although its boundaries have not yet been mapped in that document. However they were mapped in the 2010 Canterbury Regional Landscape Study Review, which informed the RPS. The Review identified the entire Basin, including its valley floors and surrounding ridgelines, as an ONL. Despite this, no regional plan addresses landscape and biodiversity protection, with these matters being left to district councils.

In Canterbury, there are separate regional plans addressing water allocation and water quality. Allocation is dealt with in the Waitaki Catchment Water Allocation Regional Plan which has allocated an annual 275 million m^3 of water for agricultural and horticultural activities within the Mackenzie Basin. This has enabled the granting of numerous water rights for irrigation despite the cumulative landscape and biodiversity impacts that have and will result.

The Land and Water Regional Plan contains the nutrient management provisions for the region, and acts to control the intensification of farming by placing limits on the use of fertiliser, in particular nitrogen. It was initially very lenient when it came to nutrient loss in the Mackenzie Basin, potentially permitting significant intensification. However, since Plan Change 5 became operative in February 2019, the revised nutrient provisions effectively prevent further intensification of farming activities there.

The Mackenzie District Plan became operative in 2004 and PC13 was notified in 2007. PC13 identified the entire Mackenzie Basin within the Mackenzie District (aside from the township areas) as an ONL and sought to put in place stronger controls over rural-residential subdivision and housing. Prior to PC13, there were no controls over pastoral intensification and irrigation which were permitted activities under the Plan's provisions. The Plan Change was heard by Commissioners who rejected the ONL classification for the entire Basin area and recommended less stringent controls. These recommendations were adopted by the Council and this spurred the beginning of a series Environment Court and High Court appeals which lasted almost a decade. The resulting appeals turned PC13 into a much wider document, with a focus not only on built development and subdivision, but also on pastoral intensification.

The Plan Change was effectively written by the Environment Court, not the Council, and the rules only became operative in 2015, eight years after

first notification. During this considerable delay, there was a gold rush of development, with landowners intensifying their properties under the old, permissive district plan rules. Following PC13, pastoral intensification, irrigation or agricultural conversion can no longer occur as a permitted activity in the Mackenzie District. These activities are now largely discretionary. As most of the Basin has been identified as an ONL, and the effects of pastoral intensification and agricultural conversion have been identified as causing adverse effects on the landscape's values, it is highly unlikely that a discretionary consent for intensive farming would be granted. There is now robust protection for landscape values in this part of the Basin.

Wildling pines, Mackenzie Basin

It is a very different story for the Waitaki District portion of the Basin. There is 38,500ha of the Mackenzie Basin ONL (as identified in the Canterbury RPS) that is not identified in the Waitaki District Plan. This misalignment, that predominately excludes the gentle terrain of the Basin, has serious implications for how the land is managed and protected. In the Waitaki District Plan, designation as an ONL creates a much stronger rule framework than the rural scenic zone which currently applies to the Basin floor.

The District Plan requires that the "overall landscape qualities of the Rural Scenic Zone are retained" but also that "ongoing landuse change may occur" and irrigation "could well expand further in the Basin".[23] This clearly illustrates that further intensification is provided for in the Waitaki-portion of the Mackenzie Basin. Significantly, in the rural scenic zone, farming activities including the use of irrigation is permitted. This deficiency has enabled significant land use change. In 2014, the Waitaki District Council resolved to update its District Plan, but progress has been slow and a proposed Plan has yet to be publicly notified six years later.

The other legal regime that applies to the Mackenzie Basin landscapes is the Crown Pastoral Land Act. Pastoral leaseholders require discretionary consent from the Commissioner of Crown Lands before intensifying farming activities on their properties. As described in Chapter 6, although there is provision in the Act to protect 'inherent values' of the land, the consenting process has primarily been used to authorize a range of farming activities including indigenous vegetation clearance and irrigation. This is despite DOC frequently providing advice that inherent values would be compromised.

The development at Simons Pass and Simons Hill stations serves to illustrate how the regulatory system broke down under pressure. Roughly 80 consents were required to turn the property into a large-scale dairy enterprise for between 5,000 and 15,000 cows. They were applied for piecemeal, over a period of 13 years, with easements and discretionary consents granted by the Commissioner of Crown Lands, water permits granted by Environment Canterbury and land use consents and certificates of compliance granted by Mackenzie District Council. During this time there was little communication across agencies on the consents they were processing for the development and, as a result, no decision-maker was aware of the full picture.

There has also been very poor compliance monitoring and enforcement of the rules that do apply within the Basin. For example, Waitaki District Council only has one compliance officer for all matters administered by the Council over its entire district (of which the Mackenzie Basin is only a small part). Until recently, LINZ had largely failed to monitor pastoral lease activities in order to check compliance with consent conditions or the requirements of the Act more generally.

A range of other measures have been applied in the Mackenzie Basin with various levels of success. Conservation covenants have been used to protect ecologically significant areas on private property. These appear to have been effective when the conditions are well constructed and monitoring is undertaken to ensure they are complied with.

A collaborative process between 26 interested parties, established as a result of a Mackenzie Country Symposium convened by EDS in Twizel in November 2010, successfully led to an agreement in 2013 called the 'Mackenzie Agreement'. This set out a common vision for the future of the Basin and resulted in the establishment of the Mackenzie Country Trust in February 2016 as an implementation mechanism. However the Trust was not adequately funded and, in the interim, Environment Canterbury issued irrigation consents well in excess of the amount provided for in the Agreement.

The Agreement supported the concept of establishing a Mackenzie Basin Drylands Area. This concept has recently gathered the support of Government with the former Minister of Conservation announcing the legal protection of 11,800ha of new conservation land on the floor of the Mackenzie Basin, in September 2020, as part of the Tu Te Rakiwhanoa Drylands initiative. Previous conservation land in the Basin has been acquired through the tenure review process and also through purchase by the Nature Heritage Fund. DOC has been working with mana whenua to co-design a broader collaborative effort aimed at protecting larger areas of the Basin's landscape including those under private ownership.[24]

Another collaborative initiative in the Mackenzie Basin is the Te Manahuna Aoraki Project which was launched in November 2018 and which covers 310,000ha in the upper Basin. It is focused on predator control, with an emphasis on research and experimentation. The project was initiated by the NEXT Foundation which formed a partnership with DOC and the three rūnanga which have mana whenua status over the area. The project has also brought in the affected runholders who have agreed to provide access to their land for pest control purposes. The project has been well received by runholders and has helped to strengthen positive relationships between the various parties involved.

As also described in Chapter 6, the five agencies with statutory responsibilities within the Mackenzie Basin – DOC, LINZ, Environment Canterbury, Waitaki District Council and Mackenzie District Council, established the Mackenzie Basin Agency Alignment Programme in 2017. There are now regular meetings between staff from the various agencies. The agencies have developed a shared set of digital maps, are working together on predator control and are ensuring that there are joint hearings where appropriate. In addition a dedicated website has been established.

Lessons learned to inform future management responses

- There can be very strong economic drivers behind land use change and it is important to understand what these are in order to design effective policies and measures to counter their negative impacts. Voluntary measures or highly discretionary consenting systems will not generally work in the face of strong economic imperatives.
- Human-induced landscape change which is occurring today, can stem from earlier actions, and these need to be recognised in order to fairly identify who the contributors to the current problems are.
- Cultural landscapes are of great significance to mana whenua and they need to be partners in developing measures to protect them.
- Policy developed through a narrow lens can often result in unintended consequences. It is important to achieve policy alignment across central, regional and local government and tribal authorities if key objectives are to be met.
- Small, poorly resourced councils should not be left to protect important landscapes of national significance on their own. Central government needs to provide stronger policy and support. Adequate resourcing and funding need to be budgeted for and provided to councils and tribal authorities responsible for protecting important landscapes of national significance on behalf of the nation.
- There need to be robust checks and balances over council and government performance, including that of entities such as the Commissioner of Crown Lands.
- Any new system needs to be agile and able to quickly respond to developing events. The lag in achieving landscape protection in the Mackenzie Basin has meant that much landscape degradation occurred while policy was being developed and contested in the courts.
- Collaborative processes can bring various parties together to work towards a common goal. However, there need to be adequate mechanisms and resourcing to implement any agreements reached.
- Alignment between statutory agencies is important for achieving positive outcomes for important landscapes and this can be achieved through mechanisms such as regular meetings, co-location of staff and the use of common information sources.
- Similar alignment is also necessary between statutory agencies and respective mana whenua groups in order to achieve positive outcomes.

7.2 Te Pātaka o Rākaihautū Banks Peninsula

This is a story about early and devastating landscape change, with loss of virtually all the forest cover from Banks Peninsula, followed by a long process of gradually healing the land. Steep topography and lack of water has protected the Peninsula from many farming intensification pressures. However plantation forestry is a likely future threat to the area's outstanding landscape values.

Biophysical landscape

Banks Peninsula, the largest true Peninsula in Aotearoa New Zealand, is located on the east coast of the South Island within the Canterbury region. The Peninsula is around 50km long and 30km wide, covering an area of approximately 1,120km$^{2.}$ It is an iconic landscape featuring two ancient volcanic cones and associated lava flows. For most of the past 20 million years, Banks Peninsula was an island. It was only connected to the mainland around 20,000 years ago, when the gravels brought down by the eroding Southern Alps built up the Canterbury Plains, and eventually reached the old volcanoes. The Peninsula has a rugged topography with more than 100 short, and often very steep, catchments.[25]

Because of its long history as an island and steep, rugged topography creating numerous microclimates (with some areas being virtually frost free), Banks Peninsula supports unique indigenous biodiversity. There are high levels of endemism and a high proportion of species are at their distributional limits.[26]

Prior to human arrival, the Peninsula was almost entirely covered by forest. In the lowland areas the forest was dominated by tōtara, matai and kahikatea. On the higher peaks there were thin-barked tōtara and native cedars. A small area of beech forest established on the south-east corner of the Peninsula. Nīkau palms grew in warm coastal valleys and snow tussock on the bare rocky high ground.[27] The forest environments supported a vast array of fauna. In particular, Banks Peninsula is notable for its endemic invertebrates.[28]

It is estimated that there were around 100 species of bird on the Peninsula at the time of first human arrival.[29] These included numerous forest and seabirds. Petrels created nesting burrows all around the Peninsula from the mountain summits right down to the sea edge. Nesting penguins would have been present in their tens of thousands including the endemic little blue white flippered penguin which was thought to have reached around 20,000 pairs.[30] There would have been fur seals aplenty around the Peninsula's shores as well as the New Zealand sea lion. The sea around the Peninsula was also the stronghold for the little endemic Hector's dolphin.

Cultural landscape

Banks Peninsula landscapes embed a rich and diverse human history. Māori tradition records three waves of historic settlement of the area starting with the Waitaha people, led by Rākaihaitū, who arrived from the north on the *Uruao* canoe. There was a second and longer wave of settlement by the Ngāti Māmoe people, descendants of Hotu Māmoe who moved south from the Poverty Bay area. The most recent wave of settlement, from around 1700, was by the Ngāi Tahu people who gradually moved south from the east coast of the North Island.[31]

Settlements were established in most bays, from which the tribe harvested the Peninsula's rich forest, marine and freshwater resources. Kūmara cultivation was also undertaken, facilitated by the Peninsula's rich volcanic soils and relatively mild climate. Settled areas and resource harvesting sites were connected by a network of trails, tracks and waka passages.[32]

Akaroa harbour is of considerable historical importance. It was the location of the first contact between Europeans and Ngāi Tahu residents of the Peninsula in 1815 or 1816. It was also the location of an incident at Takapūneke which was a precursor to Britain deciding to assert sovereignty over Aotearoa. This involved a British sea captain, John Stewart, becoming complicit in a raid by Te Rauparaha and his warriors on the Ngāi Tahu settlement. In 1830, Stewart brought the warriors to Akaroa and then hid them in the hull of his ship while he lured the Ngāi Tahu chief Te Maiharanui, his family and other tribal members aboard the boat where they were slaughtered.[33]

From around 1835, whaling ships from America, France, England and Australia started frequenting the Peninsula and five shore whaling stations were established. Ngāi Tahu provided water, firewood, pork and potatoes to the whalers and in return received blankets, biscuits, firearms and alcohol. Many European whalers married Ngāi Tahu women and lived amongst the tribe.[34]

Akaroa is also the site of the only French settlement in Aotearoa New Zealand, which had its inception in the purported purchase of the entire Peninsula from its Māori owners by a French whaling ship commander, Jean Langlois. The French government supported a company subsequently formed to establish the settlement, but by the time the prospective settlers arrived at Akaroa from France, Britain had already declared sovereignty over the country. The settlers decided to stay and imparted a distinctive French architectural style on parts of Akaroa.[35]

Akaroa was also the site of the signing of several Deeds that purported to purchase much of the South Island from Ngāi Tahu including the Banks Peninsula.[36] In 1998, the Ngāi Tahu Claims Settlement Act was passed and a formal apology was given by the Crown at Ōnuku. The Act vested various land in Ngāi Tahu including land on the Banks Peninsula (Okains Bay, Te Waiomākua, Ōnawe Pā and Wairewa).

Banks Peninsula contains a rich tapestry of Māori history and stories, which can be found in place names, oral history and a vast array of archaeological remains in settlement and pā sites around the Peninsula. Four Papatipu Rūnanga have marae on the Peninsula: Te Hapū o Ngāti Wheke (Rāpaki), Te Rūnanga o Koukourārata, Ōnuku Rūnanga and Wairewa Rūnanga.

Early landscape change

800 years of Māori occupation had significant environmental impacts, although they pale in comparison with the subsequent European environmental onslaught described below. In the first phase of human habitation, around a third of the forest was cleared and a range of bird species (mainly those that were large and edible) became extinct.[37]

As whaling dwindled on the Peninsula during the mid-nineteenth Century, attention turned to the area's rich timber stands. The Peninsula was by far the largest forested area close to Christchurch, so it soon became the source of firewood and timber for the new settlement. Twenty-six sawmills were established before 1880 and they were a feature of most bays. As well as logging, fires also devastated much forest. In 1869, it was estimated that fires had swept through 12,000ha of bush during the previous 10 years. By the 1870s, the timber industry was on the decline.[38]

Many mill workers had made enough money to set themselves up on farms. The new farmers set about burning the cut-over areas and remaining bush and planting seed. One European grass variety did particularly well on the rich volcanic soils of the Peninsula and out-competed other grasses. It became known as 'Akaroa cocksfoot'. Rather than growing the grass for stock feed, farmers started to treat the grass as a crop itself, harvesting the seeds and selling them to the numerous farms being developed in the North Island as well as overseas.[39]

Cocksfoot seed production proved very lucrative. The booming industry meant that a living could be made from a combination of dairying and cocksfooting on lots as small as 12-16ha. This prompted further subdivision and close settlement of the rural area. In order to harvest the seed, stock had to be excluded from the paddocks, so farmers favoured the more rugged areas of their farms for its production which they cleared of native bush. This led to further deforestation, particularly on the north side of the Peninsula and around Little River, where there were warmer and more sheltered conditions favouring grass growth.[40]

By 1906, the remaining forested area of the Peninsula was estimated at around 320ha, or just 0.7 per cent of the area remaining in 1850[41] and perhaps 0.3 per cent of the forest existing prior to human settlement. The almost complete loss of forest was devastating for the indigenous species that lived on the Peninsula. It has been estimated that 20 plant species became extinct with several more species reduced to a few individuals or small vulnerable populations.[42] Many bird species also succumbed to the loss of forest habitat as well as the introduction of predators such as rats, cats, stoats and ferrets.

The cocksfoot industry provided fodder that supported a surge in dairying. This was the main industry on the Peninsula from around 1910 up until the Great Depression in 1930. Where the land was closely grazed by livestock, weeds were kept under control and the native bush was not able to regenerate. But where grazing was less intense, large areas of gorse and regenerating bush rapidly took over from the grassland and dominated the landscape.[43] The cocksfoot industry ceased in the 1960s, and by the 1980s there was only one dairy factory left on the Peninsula. Government farm subsidies during the 1970s encouraged farmers to bring marginal land back into development, resulting in much clearance of regenerating native bush and increased stocking rates.[44]

Despite the forest devastation, many scenic reserves were preserved by early conservation action and DOC now manages some 65 scenic, historic, recreation and other reserves on the Peninsula. Most of these are located along the spine of the Peninsula. They are small, isolated and fragmented and mostly on the upper shady slopes which were the areas least attractive for farming.[45]

Healing the land

The trend of land clearance and increased stocking during the 1970s was reversed during the economic reforms of the 1980s which, combined with a succession of drought years, resulted in considerable restructuring of farm systems. The cost of maintaining these new marginal pastures proved uneconomic, land was left to revert to scrub, and stocking rates were reduced. Many farms struggled for economic survival and farming families looked to diversify their income in order to survive. Overall, there was a 25 per cent reduction in livestock farmed on Banks Peninsula between 1980 and 1990. The coverage of indigenous forest increased rapidly reaching 2,600ha in 2002.[46] Nature was making a come-back.

The economic pressures of the 1980s were felt most markedly by the more economically marginal farms, those in the steeper, colder and more exposed parts of the Peninsula such as on the south-east coast. There, farmers were desperate to find new income sources so they could stay on the land. The farmers eventually developed a 4-day walking track which started in Akaroa and did a loop through their properties as well as public land. Accommodation was provided for the walkers in renovated farm buildings. The 'Banks Peninsula Track' opened in 1989 and was the first privately-owned track in Aotearoa New Zealand. It is still going 30 years later.[47]

As well as providing much needed additional income, the track started to change farmers' perceptions of how their land should be managed. It demonstrated, in a very tangible way, that there was money in conservation. People wanted to see nature on the track so that was an incentive for the farmers to look after it.[48] All of the landowners participating in the track are now involved in on-farm conservation initiatives.

Not long before the Track was established, botanist Hugh Wilson moved into the area to manage a rundown property that had been purchased by the Maurice White Forest Trust for the purpose of native forest regeneration and named the Hinewai Reserve. Hugh believed that, left to its own devices, nature would reassert itself. He saw gorse as assisting native regeneration by supressing the cover of grass and providing a protective canopy for native seedlings. He removed the goats and possums from the property and, to the horror of his farming neighbours who had spent their lives fighting gorse, left the gorse to run rampant.[49] In 1991, the reserve was significantly expanded through the purchase by the Nature Heritage Fund and Maurice White Native Forest Trust of a neighbouring farm. It now covers most of three catchments. Hinewai is the largest private reserve in the country covering over 1,250ha of land.[50] It has become an inspiring demonstration of what can be achieved on the land through letting it heal itself with minimal intervention.

The establishment of the BPCT in 2001 provided a real boost to farmer efforts to restore the land. The Trust was established by the former Banks Peninsula District Council in the wake of conflict over provisions in the proposed Banks Peninsula District Plan which were notified in 1997. In the face of a strong backlash from the rural community, the Council largely backed down on the statutory controls, instead establishing BPCT to support landowners in adopting voluntary measures. In 2003 the Trust was granted covenanting powers under the Reserves Act 1977.[51] It is still the only non-governmental organisation in the country to have such powers.

The BPCT covenanting programme has gone from strength to strength and often works in partnership with the QEII Trust. As at September 2020, the BPCT had entered into 76 covenants protecting 1,502ha of land.[52] Many of the covenants are small and dispersed, evidenced by the total area protected being similar to that within the privately-owned Hinewai Reserve. But covenanting can be a first positive step taken by a landowner that leads them down a path of engaging in conservation initiatives.

Although covenanting has been the focus of the BPCT for many years, the Trust has more recently played an important facilitation role in a range of landscape-scale conservation initiatives on the Peninsula. This has enabled it to leverage off the positive relationships it has built up with landowners. In 2017, BPCT launched a '2050 Ecological Vision' for Banks Peninsula which is "to create an environment in which the community value, protect, and care for the biodiversity, landscapes and special character of Banks Peninsula/Te Pātaka o Rākaihautū".[53]

Farmer interest in native regeneration and covenanting has been strongly supported by the changes in the economics of farming. The price of coarse wool has collapsed, now being as little as $1 kg,[54] whereas the price of red meat has increased. This has prompted farmers to shift from concentrating on wool production to prioritising sheep and beef meat production. There is now an increased focus on improving lambing percentages and growth rates through selective breeding and providing better nutrition for stock. In turn, this has resulted in lower stocking rates and the development of more nutritious pastures, fencing of land into smaller paddocks and a higher degree of animal management and husbandry. Farming is now generally focused on the more manageable and accessible areas of land, with less grazing pressure (and more indigenous bush reversion) in the steeper and more difficult areas.

Indigenous regeneration on Flea Bay Farm

The regeneration has been actively supported by Papatipu Rūnanga on Banks Peninsula who have been involved in numerous initiatives to restore Te Pātaka o Rākaihautū. As well as undertaking localised projects, rūnanga have been involved in many landscape-scale initiatives including the Wildside Project, the Pest Free Banks Peninsula Project, the recent Te Kākahu Kahukura project on the Southern Port Hills and the Whaka-Ora Heathy Harbour project focused on improving the health of Lyttelton Harbour Whakaraupō. Rūnanga have also been very active in seeking to restore marine biodiversity around Te Pātaka o Rākaihautū with the establishment of five mataitai and one taiapūre.

Restoration at scale

During the 2000s, restoration on the Banks Peninsula moved more strongly from individual property efforts towards landscape-scale initiatives where landowners, not-for-profit entities and government agencies pool their efforts. This has enabled results to be scaled up significantly.

An early example of this is the Wildside project which sought to support and scale up conservation work being undertaken by farmers on the south-east coastal flanks of the Peninsula. It was here that the best examples of remaining native biodiversity were found.[55] In addition, little blue white flippered penguins (endemic to Banks Peninsula) and yellow-eyed penguins frequented the area, and there were numerous seabirds nesting on the coastal cliffs. Several landowners in the area were already involved in protecting the penguins and seabirds that frequented their shores.

The Wildside project brought together landowners, councils, the rūnanga and science interests to combine their efforts in order to improve predator control and support native forest regeneration. The project has gone from strength to strength and now protects over 2,500ha of continuous regenerating forest with mammalian predators are being supressed over 7,000ha of land.[56] A Wildside committee oversees activities, which brings together local landowners, rūnanga representatives, people from the councils and DOC and other scientific specialists. It is chaired by a local landowner. The Wildside work has been supported by the Josef Langer Charitable Trust which has enabled a wide range of activities to be funded.[57]

The work of the Rod Donald Banks Peninsula Trust has also increased appreciation of the natural values of the Peninsula. The Trust was established in

2010, in the wake of the disestablishment of the Banks Peninsula District Council. The Council placed $3.5 million from the sale of endowment properties into the Trust where it could be used to create an enduring legacy for the Peninsula. The Trust works on a range of projects to foster public access, support biodiversity restoration and improve knowledge of the Banks Peninsula area. It has mapped all of the Peninsula's walks, regardless of land tenure, and has put this information together into a comprehensive directory. Its flagship project is Te Ara Pātaka – the Summit Walkway which is a 2.5-day tramping track linking the Lyttelton and Akaroa craters.[58] It has also led work investigating carbon offsetting and exploring how it can be used to support biodiversity regeneration and has supported the work of other organisations including funding the BPCT in its early stages.

Banks Peninsula has a wide variety of pest animals including possums, rats, mustelids, feral deer, goats, pigs, rabbits, hares, feral cats and hedgehogs. Canada geese are another emerging pest. In October 2018, BPCT brokered the signing of a memorandum of understanding between 14 agencies and organisations on the Peninsula who formed the 'Pest Free Banks Peninsula/Te Pātaka o Rākaihautū and Port Hills Partnership'. The goal of the Partnership is for the Peninsula to be "effectively free of pest animals" meaning they have been eradicated or are kept at such low levels that they do not threaten biodiversity values.[59]

In August 2020, the former Minister of Conservation announced that $5.11 million would be allocated to the Pest Free Banks Peninsula project. This will fund large-scale pest control over around 28,500ha on the expanded Wildside area as well as on the spit and dune systems at Kaitorete. The funding, which is sourced from the Government's Covid-19 'Jobs for Nature' package, is expected to create around 15 full-time jobs.[60]

In addition, in 2019, Environment Canterbury introduced a new Pest Free Banks Peninsula targeted rate which is providing a long-term sustainable funding source for pest control on the Peninsula. It is being combined with revenue from general rates to provide $600,000 a year for pest control activities in the area.[61]

There is also a current proposal to establish a 'geopark' on the Peninsula. This is based on the idea that the underlying geology of the Peninsula is the base layer on which all other significant features and elements sit. The geopark would inform locals and visitors about the region's geology, landscape, flora, fauna, archaeology, histories and communities. This would include the establishment of 'geopoints' at specific sites, 'geosites' which are collections of geographically related sites, and 'geotrails' which are pathways connecting the geopoints and geosites by walking, kayaking, boating or cycling. Interpretative panels would provide information about the locality, describing the physical location and what had occurred at the site.[62] This focus on geology could serve to compliment the other initiatives which are more targeted at the biodiversity that naturally drapes the geodiversity – both of which contribute to the natural landscape attributes.

A key theme that strongly emerged from these initiatives was the power of collaboration. There were many instances of organisations working together on Banks Peninsula and collectively achieving much more as a result.

Current and future landscape pressures

Farming systems on Banks Peninsula are closely linked to the natural attributes of the land, soil and climate. The volcanic soils on the upper slopes have high natural fertility as they are derived from basaltic volcanoes which are rich in phosphate. The lower levels of the Peninsula (mainly below 150m) are dominated by loess soils which have less natural fertility and are highly erodible.[63]

Unlike the Canterbury Plains, which sit over extensive gravel aquifers, it is not possible to irrigate farmland on the Peninsula as there are no large and reliable sources from which water can be drawn. The other main restriction on farming on the Peninsula is topography. Much of the land is very steep and is not readily accessible by tractor or other large farm equipment. This means that it cannot be cultivated and used for intensive pasture production or to grow crops.

All these factors have resulted in low intensity farming systems which are largely managed within natural constraints. There is little likelihood of a shift to intensive farming models or dairying in the future. Although farming clearly has some landscape impacts, farming systems are generally of low intensity with few inputs and are largely sympathetic to the landscape and natural values of the area.

There are currently only small pockets of exotic forestry on Banks Peninsula. The larger plantations are predominantly in the Lyttelton harbour area, where the trees are closer to the port, thereby reducing cartage costs. Many of these were established during the 1990s, when there was significant expansion in the area planted throughout Aotearoa New Zealand, as a result of the doubling of log prices overseas.[64] Pine and other exotic trees grow well on the same marginal land that is currently regenerating into native bush. There is also the possibility that exotic plantations could replace land currently in pasture if the economic incentives are strong enough, and this is already starting to occur.

There are currently three main pieces of government policy which impact the incentives for forestry on Banks Peninsula, the NESPF, the emissions trading scheme and the One Billion Trees Programme.

Subject to a few weak conditions relating to wilding control and setbacks (for shading, but not fire resilience), the planting of exotic pines and other commercial species is now provided for over most of the Banks Peninsula as of right under the NESPF, irrespective of landscape impacts. This is due to the severely restricted

mapping of ONLs in the district plan which are limited to only prominent ridgelines. The potential impact of this on the Peninsula's iconic landscape, in combination with the financial incentives discussed below, is frankly alarming and needs to be addressed through a review of the NESPF. This has been promised by the Government.

The economic future of farming on the Peninsula is uncertain and landowners are looking to derive some income off their marginal land. This has resulted in many turning to carbon credits under the emissions trading scheme. While we heard that many would like to do this through native forest regeneration, the impediments within the emissions trading scheme may see more people turning to exotic plantation forestry as a source of carbon credits. These impediments include the benefits of indigenous forest not being fully valued and the onerous methods required to prove that an area of regenerating indigenous forest is eligible to receive credits and to demonstrate the rate of growth. As the price of carbon credits rises, these impediments will likely provide a strong economic incentive for land use on the Peninsula to move from farming to exotic forestry, particularly for marginal land.

The One Billion Trees fund was launched in November 2018 to increase tree planting across Aotearoa New Zealand. It aims to double the current planting rate to achieve one billion trees planted by 2028. It was agreed by Cabinet that two-thirds of the new planting budget should be native trees. To date, this target has been largely met nationally but not in Banks Peninsula where two thirds of the money provided in direct grants has supported the planting of exotic trees.

Residential development on the Peninsula has been constrained by topography, and demand has been reduced by the Global Financial Crisis in 2007-08 and the Christchurch earthquake sequence in 2010-11. The Environment Court has been reluctant to approve residential development that significantly degrades landscape values. This means that urban development is not currently the major threat to the Peninsula's landscapes. However, there has been a steady stream of farmers carving off small lifestyle blocks and covenanting the balance lot.

Both Lyttelton and Akaroa have managed to retain much of their historic fabric. However, sadly, many heritage buildings were lost in Lyttelton as a result of the Christchurch earthquakes. There has been a long history of community activism

Regenerating indigenous forest at Hinewai Reserve

to protect the historic heritage of Akaroa, including the early French influence on the settlement of the town, particularly by the Akaroa Civic Trust.

After the Banks Peninsula District Council merged with Christchurch City Council in 2006, there was concern that the focus on protecting Akaroa's heritage might be diluted. We were told that, since amalgamation, less attention has been paid to heritage issues. For example, standard solutions for street furniture and traffic management, that had been applied in other parts of Christchurch, were being proposed for Akaroa. On the positive side, a new Akaroa Design Panel is being established to provide timely input into development proposals.

Tourism has become increasingly important to the economy of Banks Peninsula, particularly in Akaroa and surrounding rural communities. Many farming properties are involved in tourism initiatives. Tourism has been generally welcomed in the Akaroa area as a valuable addition to the rural economy. But over the last 10 years, the advent of cruise ships and freedom campers has threatened to overwhelm the small town and its struggling water and wastewater infrastructure. During the 2018-19 cruise season, there were 93 ship visits to Akaroa Harbour by 21 vessels carrying a total of 145,914 passengers and 64,564 crew. This impacted the sensitive environment of Akaroa Harbour where the use of thrusters for manoeuvring the ships resulted in plumes of sediment being driven up from the seabed. There has also been concern about the impacts of the vessels on wildlife including the Hector's dolphin and multiple threatened seabirds. [65]

Due to Akaroa's small resident population of less than 800, during a cruise ship visit day, there could be multiple tourists in town for every one resident. Statistics New Zealand has reported that, during the 2017/18 season, there were 210,478 cruise passengers and crew visiting Akaroa. They spent a total of close to $35 million while onshore (which may have been in Akaroa or any other destination that they visit during their stopover such as Christchurch) which is an average of $270 per passenger. This provides considerable additional income to the region. However the deluge of visitors, many who bus out of the town to visit Christchurch for the day, have undoubtedly impacted residents and the town's infrastructure. Guidance issued by Environment Canterbury in June 2020[66] will likely exclude large ships from the harbour in the future should cruise ships be permitted back into Aotearoa New Zealand. In addition, Lyttelton harbour's new $56 million cruise berth will be able to accommodate the larger vessels.[67]

Much like other places in Aotearoa New Zealand, tourism in Akaroa and the broader Banks Peninsula has been largely unmanaged. The Covid-19 pandemic, and diversion of future large cruise ships away from Akaroa, provides an opportunity for pause and rethink about the tourism model which might best fit the Banks Peninsula landscapes and communities. There appears to be considerable opportunity to strengthen the Peninsula's image as a sustainable tourism hub, where its remarkable geology, biota, landscapes, seascapes and cultural heritage is matched by community efforts to heal the land and farm sustainably.

Marine farming remains a small but viable industry on Banks Peninsula, with a focus on greenshell mussels, king salmon and pāua.[68] Interest in new species such as seaweed, including kelp, has the potential to put demand on limited space in the harbours of the Peninsula, and increase the potential for negative impacts on landscape and seascape values, and natural character.

Current management response

Over the last three decades, there have been a number of studies focused on the identification of significant landscapes on Banks Peninsula which have generated very different conclusions. In 1993, the entire Banks Peninsula was identified as a regionally outstanding landscape in a study carried out by Boffa Miskell Limited and Lucas Associates. However, this was neither formally adopted by Environment Canterbury nor incorporated into the Canterbury RPS.

As described in Chapter 5, the Banks Peninsula District Plan was notified in early 1997 and identified around half (some 50,000 of approximately 96,000ha) of the rural land in Banks Peninsula as either landscape protection areas (with high or moderate sensitivity) or coastal protection areas. These areas were subject to stricter controls on buildings and production forestry. No landscape assessment was carried out to directly inform the plan. Instead, the findings of a visual assessment undertaken in 1991, after some reanalysis, were used as a basis for identifying important landscapes on the district plan maps.

The proposed plan provisions for coastal and landscape protection areas were heavily criticised, particularly by the farming community mobilised under the umbrella of the North Canterbury Branch of Federated Farmers. Federated Farmers threatened High Court action if the Council continued processing these sections of the proposed plan. In response, the Council decided not to hear submissions on the rural chapter of the plan, but instead to establish a Rural Task Force to investigate ways of resolving the conflict. The Task Force recommended adopting interim provisions for landscape protection while a further and comprehensive landscape study was undertaken.

In response to this advice, Banks Peninsula District Council notified Variation 2 in 2002, which significantly reduced the areas of protection.[69] Under Variation 2, only 30,000ha of landscape and coastal protection areas were mapped. This resulted in a number of appeals; however, these were put on hold pending the finalisation of the landscape study.[70]

The 2007 Boffa Miskell landscape study sought to maximise the objective information which could be incorporated into layers, before assessing values, and integrating the layers into the various landscape protection categories.[71] It identified only some 23,000ha as protected landscape and coastal areas which were limited mostly to prominent upper ridgelines and the coastal edge.[72] This mapping was subsequently endorsed by the Environment Court (see spotlight in Chapter 5).[73]

Although comprehensive, the Boffa Miskell study made a fundamental error. In mapping the various landscape values, it failed to adopt a landscape-scale approach and instead focused on individual landscape units. In adopting this methodology, the study failed to recognise the significance of the larger landscape as an integral entity. The study mapped parts of a landscape but did not fully integrate them or recognise the landscape as a whole.

A 2010 review of the Canterbury Regional Landscape Study confirmed that Banks Peninsula and the Port Hills were an ONL. However, the earlier Banks Peninsula District Plan provisions were largely carried over into the Christchurch District Plan which was developed under fast track procedures. Most of the Peninsula is now identified, not as an ONL, but as a rural amenity landscape.

The earlier Banks Peninsula District Plan did not address cultural landscapes. However, it did provide for the protection of sites and features of significance to mana whenua through the identification of 'silent file' areas. These provisions were also largely brought through into the proposed Christchurch District Plan. However, Ngāi Tahu made a submission seeking a significantly more comprehensive approach to the identification and protection of sites of cultural significance. As described in the spotlight in Chapter 4, a range of cultural layers have now been incorporated into the District Plan.

The process for identifying significant ecological sites on Banks Peninsula has been particularly contentious. It started in 2008, as the result of an Environment Court consent order in 2007, which required the Christchurch City Council to "identify areas of significant indigenous vegetation and significant habitat of indigenous fauna" in the Banks Peninsula District Plan. In response, the Council established the Banks Peninsula Significance Ecological Steering Group to work alongside the Council planning team and landowners to identify sites. As part of the process, Council staff started talking to farmers on the Peninsula to obtain their approval to undertake ecological studies on their land. Many landowners agreed, with the expectation that they would work constructively with Council to identify the best ways to manage areas found to be of ecological importance.

However, as a result of the fast-tracking legislation for the preparation of the new Christchurch District Plan (described above), this process was abandoned and the council listed 58 significant ecological sites on the Banks Peninsula in the notified version of the Plan.[74] In many cases, these sites had not been ground-truthed and were based solely on an earlier ecological survey. The Plan provisions were challenged in front of the independent hearings panel by many of the landowners who had not been consulted on them. It was undisputed by all parties that there had been a failure by the Council to properly engage with landowners.[75]

Some of the most contentious issues were the requirements to take areas out of farm production and pay for the costs to fence them. Because many of the farms on the Peninsula are quite small, with some verging on being uneconomic in size, removing significant areas of land from production as significant ecological sites can impact the viability of the farm.

As a result of this contention, the District Plan now includes Schedules A and B. Schedule A includes significant ecological sites on public land or where, following collaboration, private landowners had agreed to their inclusion. Schedule B is included 'for information only' and identifies sites where the council intends to continue discussions with landowners about the ecological values that are present.[76] Rules only apply to the management of sites listed in Schedule A. The Council is required to initiate a plan change within 6 years (by 2022) to update Schedule A.[77] The Plan also includes an Appendix of areas outside Schedule A sites that potentially contain significant vegetation and habitat.[78] These are controlled through the use of vegetation clearance rules.

As part of the plan change to update Schedule A, it is proposed that the Appendix be removed. This is concerning, as it is important that the Plan retains rules for the protection of potentially significant vegetation and habitat, even if a full assessment of significance has not been undertaken. This also recognises that biodiversity is not static, and that some areas that have not been listed in Schedule A may be significant in the future, and should be managed as such.

In an effort to strengthen non-regulatory support to landowners, in 2017, Christchurch City Council set up a new biodiversity fund. The funding is only available for significant ecological sites with a management plan. It covers activities such as fencing, planting and pest control. The Council will fund up to $40,000 per property over a one-year period from a total contestable fund of $200,000 per annum. The fund only covers 50 per cent of costs but landowner contributions for the other half can be in kind.[79] This is a positive move to further support farmers to protect special areas.

Neither the Canterbury RPS nor the Canterbury Regional Coastal Environment Plan (notified in 1995 and operative November 2005) give effect to the 2010 NZCPS. A review of the Coastal Plan is now underway, with notification of a new Plan expected in 2023/24, but this is some 28 years after notification of the operative plan.

Lessons learned to inform future management responses

- It is important to identify the economic and social drivers of land use change in order to manage adverse effects on landscape values.
- Where economic drivers favour restoration, government policy should support this.
- Where economic drivers favour intensification of farming, such as on Kaitorete Spit, then clear and strong regulatory rules are required to manage its landscape and biodiversity effects.
- Central government policy can have unintended consequences at the regional and local levels, particularly it fails to take into account the local context.
- Mana whenua can play a strong kaitiaki role in spearheading restoration efforts and should be supported in doing this.
- Tourism can support restoration efforts through landowners generating value from nature. However, unmanaged tourism can swamp communities and infrastructure and result in environmental degradation.
- A 'one size fits all' approach by councils can serve to homogenise local natural and cultural landscapes so that uniqueness and diversity is lost. Use of localised input, such as through design panels, can help configure projects to be sensitive to the local context.
- Cultural landscape layers in planning documents, accompanied by protective rules, can serve to identify and protect values of significance to mana whenua within the landscape.
- Collaboration can effectively serve to jointly harness community resources to achieve common goals.
- Not-for-profit organisations and philanthropists can play a critical role in funding, supporting and brokering collaborative efforts

Akaroa waterfront

7.3 Hauraki Gulf islands

This case study focuses on three islands in the Hauraki Gulf with permanent settlement, Aotea Great Barrier Island, Waiheke Island and Rākino Island although the Gulf in its entirety could also be considered a (larger) landscape entity. These islands each have a unique character and settlement pattern and face different, but in some cases overlapping, challenges. Regional legislation, in the form of the Hauraki Gulf Marine Park Act 2000 (HGMPA), applies to their management alongside the national legislative framework. The case study examines how effective this piece of tailored legislation has been in protecting the landscapes of these islands.

Biophysical landscape

Aotea Great Barrier Island, located around 90km from Auckland, is the largest island in the Hauraki Gulf with a land area of 28,500ha. Its landscapes are predominately volcanic which give rise to a spectacularly rugged landscape and a large variety of coastlines. The west coast is covered in steep forest-covered ranges that lead down to sheltered bays and islands that are drenched in pōhutakawa. This is in contrast to the windswept east coast that has long white sand beaches, extensive dune systems, tidal creeks and wetlands. The Whangapoua estuary, located on the north-east of the island, is one of the least modified wetlands in the country and is valuable habitat for coastal and wetland birds.[80] The Kaitoke Swamp is the largest wetland in Auckland and supports orchids, mātātā North Island fernbird, pūweto spotless crake and matuku Australasian bittern.[81]

The island is one of the richest areas in Auckland for plant biodiversity with at least 75 species of regionally and nationally threatened vascular plants found there, including the endemic prostrate kānuka (*Kunzea ericoides* var. *micro ora*) and Great Barrier tree daisy (*Olearia allomii*).[82] The island is a sanctuary for threatened fauna including thirteen species of lizard, tāiko black petrel, tītī Cook's petrel, stronghold populations of North Island kākā and mohopereru banded rail, and the only island population of pepeketua Hochstetter's frog.[83] Possums, mustelids (stoats, ferrets and weasels) and Norway rats never reached the island. Just over half the island is public conservation land managed by DOC.

Waiheke is the second largest island in the Hauraki Gulf (at around 9,324ha) but is still only a third of the size of Aotea Great Barrier Island. It is located just 22 kilometres from Auckland. This island is rich in natural features and includes over 40km of white sandy beaches and large tracts of indigenous vegetation interspersed with areas used for vineyards, olive groves and farming. On the eastern end of the island, the landscape values associated with the sequence of rolling to steep hill country, rock coastline and large tracts of native forest intermixed with open pasture and vineyards have been identified as an ONL at a national scale. The western end of the island is heavily settled with a series of five villages. In stark contrast to Aotea Great Barrier Island, most of the island is privately owned.

It is likely that Waiheke was originally covered in coastal broadleaf podocarp forest, with dense kauri groves on the upper valley slopes and ridges. There were extensive wetland areas featuring raupo and flax. The western end of the island appears to have lacked the dense forest of the east, and this may be due to fire caused by the Rangitoto eruption followed by Māori clearance for gardening, and possibly accidental fires.[84] Little ash from the Rangitoto eruption reached the island itself and the clay soils are of low to medium fertility. Possums have never reached Waiheke Island. The island supports a number of seabirds including the 'at risk' kororā little blue penguin and ōi grey-faced petrel, and the very rare parekareka spotted shag.[85]

Rākino Island (Motuhurakina) is one of the smaller inhabited islands in the Hauraki Gulf. The island is approximately 146 ha in size and is some 22km from Auckland. Rākino is characterised by its small size and undulating topography with high ridgelines, sloping steep sea cliffs on the northern end and a number of sheltered bays on the western and southern sides of the island. It is an area with unique coastal character and high amenity value.

The landscape of Rākino has several notable characteristics: it has minimal tree cover and is largely covered in self-maintaining kikuyu grass. As well as reducing biodiversity, the grass creates considerable fire risk on the island, as the green top-growth of the plant covers an understorey of dead leaves which can easily ignite and facilitate the fast spread of the flames. There are pockets of coastal forest and regenerating bush as well as clusters of buildings along several of the ridges. There are also fringes of ancient pōhutukawa around the coast. The marine area surrounding the island is degraded and there are no marine reserves in the vicinity. Almost all of the island is privately owned.

Cultural landscape

To Māori the Gulf is known as Tīkapa Moana or Te Moananui ā Toi. Multiple iwi and hapū have interests in the Gulf, many of which are overlapping. "This is a product of a turbulent history and long competition for this most sought after place, and it's many resources, and the relationships between iwi that have resulted."[86] Different iwi or hapū refer to different features within the Gulf that have greater significance to them. Therefore, it is likely that there are multiple cultural landscapes with varying scales within the Gulf, which creates protection and management challenges, but also opportunities.

Aotea Great Barrier Island is the ancestral home of the Ngāti Rehua hapū of Ngātiwai. Ngāti Maru, Ngāti Tamaterā and Ngāti Whanaunga also have historic and cultural associations with the island.[87] The island was heavily settled over the centuries and the earthwork fortifications of numerous pā, terraced agricultural and settlement sites, and storage pits and middens are still visible in the landscape.

All of Aotea is sacred land to Māori. At its centre is Maunga Hirakimatā (Mount Hobson); a mountain of great significance in oral history and tradition due to it heralding "the foreboding omen depicting the death of Rehua" who is a Ngāti Rehua founding ancestor. The name Hirakimatā translates to "lightening striking the cliff".[88]

Most of the Ngāti Rehua landholding on the island was lost through a series of pre- and post-Treaty land transactions which reduced their land to just five per cent of the island. There are two marae on Aotea – Motairehe Marae and Kawa Marae. Ngāti Rehua have yet to settle their Treaty grievances. After lengthy negotiations, a Deed of Settlement was signed with the Crown, but has not yet been ratified.

Many different tribal groups have lived on Waiheke Island over the centuries. They include Arawa peoples who were displaced by Tainui peoples (Marutuahu, Ngāti Paoa and Ngāti Maru) around 1700. Patukirikiri, Ngāi Tai and Te Urikaraka were earlier arrivals who also have mana whenua status over the island.[89] Indicative of the density of Māori settlement on the island, there are some 600 archaeological sites recorded, which makes it "one of the most archaeologically rich areas in the country". These sites are more densely clustered on the western end of the island, and 45 of them are pā sites.[90] There was extensive Māori settlement in Matiatia and Church Bay and gardens at Blackpool. Ōwhiti Bay, on the north-east of the island is recognised as a particularly early settlement site.

During the nineteenth century, Māori on Waiheke Island provisioned European settlers after the establishment of Auckland in 1840. Regenerating mānuka and kānuka was cut for firewood, flocks of sheep established and flat areas on western Waiheke extensively cultivated. The eastern end of the island became a popular stopping off place for ships to take on water, supplies and kauri spars and early European boat builders established there to be near the rich kauri resource.[91]

From 1838 onwards, extensive areas of the island were sold to Europeans, who valued the island's proximity to the Waitematā harbour, and saw the potential for boat building and farming. By the late 1860s only one substantial block of Māori-owned land remained – Te Huruhi, on the western end of the island. This land was subsequently partitioned and by 1914 it had all been sold apart from Whetumatarau Point.

It was not until later that some land was returned to Māori. In 1989, the 830ha Waiheke Station near Onetangi was returned to Ngāti Paoa, after a hearing of the tribe's claim by the Waitangi Tribunal. In 1976, land was made available for the establishment of the Piritahi Marae on the shores of Te Huruhi Bay, close to the site of an earlier Ngāti Paoa village. Its first building was opened in 1982 and it remains the only marae on the island.[92]

Rākino is part of the chain of islands in the Hauraki Gulf collectively known as Nga Poitu o te Kupenga a Taramainuku (the floats of the net of Taramainuku), with Taramainuku being a descendant of Toi. Both Ngāi Tai and Ngāti Paoa have mana whenua status over the island. The island was ostensibly purchased from Te Irirangi of Ngāi Tai, Ngatai of Ngāti Paoa and others in January 1840 by Scottish trader and boat-builder Thomas Maxwell. During the mid-1860s, the island was used to intern Māori during the Aotearoa New Zealand Wars.[93]

Early landscape change

Aotea Great Barrier Island has supported a number of industries based predominately on the exploitation of the island's rich natural resources. They include kauri logging, copper and gold mining, kauri gum digging, whaling, fishing and farming. These were often short-lived boom and bust industries that collapsed following the resource being fully exploited.

Aotearoa New Zealand's first mine was established on Aotea Great Barrier Island in 1842 after copper was discovered near Miners Head. The area was extensively mined until the company collapsed in 1867. In 1892, silver and gold were discovered at Ōkupu. The minerals were mined until 1920 when the mine was abandoned. In 1983, the Mining Act was introduced banning all mining on the island.[94]

The kauri forests on the island were logged with increasing intensity between the early 1880s and 1940s, by which time most stands had been removed. Seven kauri dams were built on the Kaiarāra stream to drive the enormous logs downstream. Possibly the largest mill in the Southern Hemisphere was located in Whangaparapara Harbour in 1909 but closed just five years later in 1914.[95]

Following kauri logging, pioneering farmers burnt off the remaining bush and converted much of the land to exotic pasture. Extensive farming continued until the 1970s when the removal of fertiliser subsidies made farming the land economically marginal. High land value and transport costs provide an ongoing barrier to primary production, and today only around three operational dairy and beef farms remain on the island.[96]

Despite widespread historical deforestation, most of the island is now covered in regenerating indigenous forest. While much of this is due to natural regeneration following the collapse of the farming industry, it is also partly a result of the

150,000 kauri trees that were planted by the New Zealand Forestry Service during the 1970s and 80s.[97] A study conducted in 2010 indicated that kānuka and mānuka forest make up 54 per cent of the land cover and more mature indigenous broadleaf forest (including tōtara, rimu and kahikatea) another 35 per cent.[98]

Like the other islands in the Hauraki Gulf, Waiheke Island also has a history of extractive industries beginning with the logging of kauri trees for spars. As the availability of trees suitable for spars reduced, kauri was instead used to provide building timber for Auckland. By the 1880s, most of Waiheke Island's forests had been cut down and replaced with pasture.

In addition, the landscape of Waiheke Island has been shaped by the removal of shingle and sand for the use in concrete construction (of buildings, roads and bridges) in Auckland. The process was eventually stopped in the 1920s in order to protect the recreational values of the coastline. Despite this, coastal erosion from the removal of shingle is still visible at Ōwhanake and Hooks Bays today.[99]

Early European settlement on the island was predominately farming based, with a few landowners owning large areas of land. Pastoral farming began with cattle grazing on bush and fern land. By the 1880s the main pastoral properties on the island had emerged; and between 1900 and 1920 the process of converting forest into pasture for stock was completed.[100] Commercial vineyards were first established on the island in 1978 and have proliferated since.[101]

Waiheke Island has a history of subdivision and development dating back to the early 20th century. In 1915, large sections of land between Onetangi and Putiki were purchased by kauri gum merchants who then subdivided the land into smaller sections which formed the suburb of Ostend. This was quickly followed by those in Surfdale and Palm Beach. At this time, Waiheke had no form of local government and the subdivision was largely uncontrolled.[102]

In the 1990s, a new wave of subdivision occurred. Church Bay, a farm owned by the Johnstone family, was subdivided into lifestyle blocks ranging in size from 5 to 14ha. The subdivision was conditional on the provision of public walkways and the replanting of a minimum area with native species. This set a precedent for coastal subdivisions of this kind in Aotearoa New Zealand, and in 1995 the land between Matiatia and Owhanake Bay was subdivided with similar conditions in place.[103]

Overall, two distinctly different landscapes have been created at either end of Waiheke Island. The western end contains most of the island's residents and is characterised by a series of five villages: Oneroa, Blackpool, Surfdale, Onetangi and Palm Beach. Interspersed between these are areas of rural character and amenity forming greenbelts. As a result of conservation-lot subdivision, there are also large areas of regenerating bush that have been established to mitigate the adverse effects of built development. In contrast, eastern Waiheke is characterised as a rural working landscape of large scale agricultural and horticultural activities. It has a rural character and low residential population.

Much like Waiheke, Rākino Island also has a history of subdivision dating back to the 1960s. As already indicated, in 1840 the island was purchased from Ngāi Tai and Ngāti Paoa by a Scottish trader and boat builder, Thomas Maxwell. The island was then on-sold a number of times.

In 1963, the island was acquired by the United Peoples Organisation by which stage virtually all of the indigenous vegetation had been removed and the island was farmed. Maxwell Rickard had a philanthropic vision to create a retreat from the rush of modern life, and had planned to develop homes for unmarried mothers, orphans and the elderly.[104] But in 1965 he ran out of money and, after a failed attempt to sell the land to Auckland Council, he subdivided the island into 25 blocks 4 ha in size, as well as 125 smaller sections. This created two residential settlements, one overlooking Orange Bay and the other overlooking Māori Garden Bay. However, even with the money raised from subdividing the island, the company was unable to come up with the funding required for roads, and went into liquidation.

In 1969, Rākino Island was surrendered to the receivers and two-thirds of the island was brought by North Shore Ferries, whose owner intended to build a resort on the island. After financial difficulties, the island was subdivided further. These sections were then made available for purchase by the public, which opened Rākino Island up to the development of housing areas.[105] As Rākino Island did not come under the auspices of any council at the time, there were no controls over the location or design of sites and buildings. There are four intensively subdivided settlements which have been poorly designed – with houses largely clustered in groups along the ridgelines of the island and with many sections bearing little resemblance to the actual contours of the land. The rest of the island was divided into 4ha blocks, with many of these consisting of long slivers of land that run from the ridgetop along the centre of the island down to the sea.

Historically, Rākino Island suffered from Norway rat populations at pest proportions. The pests not only posed a nuisance to residents and bach owners (with reports of rats chewing through the walls of houses to reach food that was inside), but they also preyed on the eggs and young chicks of seabirds, like the little blue penguin. It is likely that the Norway rat was also responsible for the eradication of the grey-faced petrel from the island. Given the short swimming distance to the Noises Islands, those pest-free islands were constantly being re-invaded by rats from Rākino.[106]

In the early 2000s, a joint community eradication initiative between the Rākino Ratepayers Association and Auckland Council was established. This was eventually successful in 2002 and today the island is mammalian pest free and the birdlife is naturally returning.

Current and future landscape pressures

Unlike some of the other areas explored in the EDS Protected Landscapes project, the current landscape issues on Aotea Great Barrier Island largely stem, not from development pressures, but from historical degradation and ongoing environmental damage from introduced pests. The island is remote and rural, and contains just a small number of scattered settlements, largely isolated from each another. There is no central water or electricity supply and the island runs off solar renewable and petrol or diesel generated power. There are also no banks, supermarkets or public transport facilities.

During the 1980s, the island had one of Aotearoa New Zealand's fastest growing populations,[107] but today it is in decline. The 2018 census counted 936 permanent residents which is a slight decrease from five years earlier. Great Barrier is the only local board area in the Auckland region that has not seen any population growth over the past five years.

Houses on Aotea Great Barrier Island have historically been small and of simple construction, with many being self-built. A majority are now owned by those who do not live permanently on the island. In 2018, there were 1,125 private dwellings of which just over half (594 or 53 per cent) were unoccupied on census night.[108] Overall, the issues with housing on the island are not so much to do with the effects of subdivision and development, but rather housing affordability. Second home owners are driving up the cost of land on the island, meaning that local residents are being priced out of the entry-housing market.

Tourism has been a mainstay of the Aotea Great Barrier Island economy for some years with the small-scale businesses subject to seasonal peaks and troughs.[109] Despite an earlier decline, prior to the Covid-19 pandemic, tourism on the island was increasing. It comprised roughly equal numbers of domestic and international tourists. In January 2017 alone it was reported that there were 31,800 visitor arrivals (out of an annual total of 179,600), an increase of 57 per cent from the previous year.[110] However, these figures may overstate actual numbers due to the methodology used.

Conservation land, with its exceptional landscapes and natural heritage, forms a key part of the attraction for tourists coming to the island. A small number of concessions have been granted by DOC for guided walking and tramping. However, the potential of publicly-owned conservation areas to support the tourism industry on the island appears to be underutilised. We were told that concessions were mainly held by people who were based off-island and that local operators could not afford the price of the concessions.

The birdlife and vegetation of the island have been able to flourish due to the absence of many of the introduced pests that plague mainland Aotearoa New Zealand. The island has remained free of possums, mustelids (stoats, ferret and weasels) and Norway rats. In addition, there have been extensive programmes to eradicate wallabies, feral goats and fallow deer. However, there are still a number of problematic pest species on the island. Ship and kiore rats are present in abundance and, in conjunction with feral cats, pose a significant risk to the birdlife of the island. The impacts of these pests is one of the biggest environmental challenges facing the island.

By far the biggest pressure that Waiheke Island is facing is how to manage significant population growth within a fragile island environment. Waiheke is the most densely settled of all the Hauraki Gulf islands, with a resident population of over 9,350.[111] More than a quarter (28 per cent) of Waiheke residents who are employed, commute by ferry to Auckland for work.[112] The introduction of fast ferries in 1987 greatly reduced the commuting time from Auckland (ferries now take 35 minutes and leave every half hour from Downtown Auckland) and contributed to a 25 per cent population increase in the period between 1986 and 1991.[113] This increasing trend continued, and between 1996 and 2017 the island experienced population growth of a further 47 per cent. Given the island's accessibility to Auckland, this growth is expected to continue.

There were 5,862 private dwellings on Waiheke in 2018 of which 2,157 (37 per cent) were unoccupied and therefore potentially holiday homes or rentals. This was an increase of 303 over the five-year period since 2013. The pace of development appears to be increasing, with 115 new dwellings consented in 2018, and a steady increase in consents since 2011.[114]

Historically, Waiheke Island was seen as a vibrant, sustainable and politicised community but it is now being perceived much more as 'Auckland's playground'. As such, the island faces development pressures similar to those seen on the mainland. In particular, many small baches are being replaced with larger houses, thereby increasing the intensity of development in urban areas. There has also been pressure to remove the rural urban boundary which constrains urban development to the western end of the island and retains the rural character of the east.

On the eastern side of the island there has been development on some of the large lots – requiring extensive earthworks and excavation. The scale of these properties, even with mitigation planting, is arguably not appropriate and not

in keeping with the character of the island. On the western side of the island, development of small lots is having cumulative environmental effects.

Development on Waiheke Island is currently constrained by wastewater disposal. Other than a small plant servicing the commercial area of Oneroa and Matiatia wharf, the island does not have a reticulated wastewater system, and landowners are expected to manage wastewater on their own property through the installation of septic tanks. Much of Waiheke is not well suited to the use of septic tank technology due to the sleep slopes and slow percolation rates in the heavy clay soil. There has been pollution of waterways and beaches, particularly at Little Oneroa.

Waiheke Island has many narrow, steep catchments which historically drained into small wetland systems at the bottom of the valleys and near the coastal edge. Urban development on the island has resulted in the ongoing loss of these wetlands. Freshwater wetlands have now been reduced to only 1.3 per cent of Waiheke Island's land area. Many of those remaining are important regionally and nationally because they have intact native vegetation sequences. These sometimes extend from coastal ridges all the way to the sea including coastal forest, freshwater wetlands, saltmarshes and mangroves. Most of these sequences have been lost on the mainland.[115] Although many smaller wetlands are being lost on the island, restoration work is being undertaken on the larger wetland systems.

Ridgetop development, Waiheke Island

Climate change is a significant issue for Waiheke Island because many of the island's coastal areas have been developed on the relatively flat coastal shelf, which creates risk of impacts from sea level rise as well as storm surge inundation. For example, in 2017, after a heavy downpour there were massive slips particularly around areas which had urban development.[116] Many of the roads are built on compacted mud rock and so are particularly prone to slippage.

Tourism is a major industry on the island. Prior to Covid-19, Waiheke's popularity as a tourist destination was rapidly increasing. In the 2016/17 year the island received around 1.3 million visitors, increasing 18 per cent from the previous year.[117] Up to 40,000 visitors were attracted to the island each day during the summer peak,[118] meaning that for every resident on the island there were four visitors. Many of these were day-trippers (66 per cent).[119] Cruise ship passengers were also visiting the island in large groups.

The impacts of the burgeoning tourism industry prompted considerable community concern. Community surveys identified the top concerns of Waiheke residents as being the increasing and poorly managed tourism and a lack of infrastructure to handle it; and the lack of managed environmental protections to protect the 'special character' and natural beauty of the island from increases in both tourism and permanent populations. Specific concerns included that the roads were dangerous, there was a lack of parking availability, the transport system was not adequate to deal with the level of growth in permanent residents and visitors, and the ferries were overcrowded.[120]

The residential population on Rākino Island has always been small. It fluctuates at around 16 to 20 permanent residents, most of retirement age. The population can swell in the summer months to around 100-200 people. As access can be difficult, the island is not under the same pressure as nearby Waiheke Island. There are approximately 120 dwellings on the island with the large bulk of the houses being holiday homes. Most of the land on the island is owned by the holders of the 25 4ha blocks. There are only small pockets of public land.

It is very expensive to get building materials out to the island and also to build a permitted dwelling, as a building inspector needs to regularly visit the island during the build at the cost of the property owner. Collectively this means that there are still many unpermitted sheds, caravans and other structures that are used as dwellings.

There are limited places that visitors can go to on the island. There are two small conservation reserves and access to the rest of the island's coastline is largely precluded as a result of the pattern of subdivision which resulted in there being one main road along the ridgeline with properties that run all the way down to the coast. There is also a general lack of council-owned facilities on the island.

Illegal dumping of inorganic waste has been a significant issue on Rākino Island. In the year 2015-16 alone there were 12 abandoned vehicles removed from the island.[121] There has been a practice of bringing old unwarranted and unregistered cars to the island and parking them on the public roadside leading to the wharf (for use when non-resident homeowners arrive on the island) creating an eye-sore on the landscape. In addition, the increase in road traffic threatens pedestrians due to the narrow unpaved roads and lack of footpaths on the island.

Current management responses

An extra layer of protection over the Hauraki Gulf islands is potentially provided by the HGMPA. This has the overall purpose of better integrating the management of the natural, historic and physical resources of the Hauraki Gulf, its islands and catchments and establishing common management objectives; establishing the Hauraki Gulf Marine Park and Hauraki Gulf Forum; and recognising the deeply rooted relationships that exist between tangata whenua and the Gulf.[122] The purpose of the Hauraki Gulf Marine Park includes "to recognise and protect in perpetuity the international and national significance of the land and the natural and historic resources within the Park."[123]

Currently the Park consists primarily of publicly-owned land, although there is provision within the Act to include other areas. Local authorities can add their reserves to the Marine Park and private landowners with protected areas (eg land subject to a covenant) can also include their land. These areas remain under the ownership and control of the original landowner but are managed to give effect to the purposes of the Act.[124] To date, reserves have been contributed by the Waitākere City Council, Royal Forest and Bird Protection Society and Sir Rob Fenwick– the only private landowner who contributed over 300ha of covenanted land at Te Matuku Bay, Waiheke Island in 2002.[125]

Section 7 of the HGMPA provides for, as matters of national importance, the recognition of the interrelationship between the Hauraki Gulf, its islands and catchments, and the ability of that interrelationship to sustain the life-supporting capacity of the environment of the Hauraki Gulf. The life-supporting capacity of the Gulf includes the capacity to provide for the historic, traditional, cultural, and spiritual relationship of tangata whenua with the Gulf and its islands; the social, economic, recreational and cultural wellbeing of people; the use of the resources of the Gulf for economic activities and recreation and to maintain the soil, air, water and ecosystems of the Gulf. The management objectives to provide this recognition are set out in section 8. No prioritisation between the different objectives is provided.

Given the national importance of the Hauraki Gulf Marine Park, these management objectives are treated as being of an equivalent status as a NPS under the RMA.

The provisions of the HGMPA must be given effect to in RPSs and regional and district plans. In addition, consent authorities must have regard to the sections when considering an application for resource consent.

However, the balancing of interests inherent within the Act, has ultimately led to it having little weight in judicial decisions. This means that, although in law the HGMPA provisions should have considerable weight within RMA decision-making, their broad wording has meant that this is not the case in practice. It is a salutary lesson on the importance of using clear and directive language if protection of important values is sought to be achieved.

The HGMPA also established the Hauraki Gulf Forum, a new entity to oversee and integrate the management of the Hauraki Gulf and its catchments. The members of the Forum consist of representatives of local and central government as well as tangata whenua of the Hauraki Gulf and its islands. The Forum was not intended to duplicate the responsibilities of these agencies, but was instead tasked with encouraging them to exercise their responsibilities in a manner that was consistent with the management objectives contained in the Act. In addition, the Forum prepares a report every three years on the state of the Hauraki Gulf. Through these reports, the Forum highlights issues and promotes and advocates for the integrated and sustainable management of the Gulf. The Forum also instigated the Sea Change Tai Timu Tai Pari project which developed a marine spatial plan for the Hauraki Gulf.

While Waiheke, Aotea Great Barrier and Rākino islands come within the jurisdiction of Auckland Council, they are currently subject to an unusual planning regime. Unlike the other Auckland Council district plans that were incorporated into the Auckland Unitary Plan in 2016, the Hauraki Gulf Islands District Plan remains separate. However, the Hauraki Gulf islands are subject to the RPS, regional coastal plan and regional plan provisions in the Auckland Unitary Plan.

In the Auckland Unitary Plan, Aotea Great Barrier Island is largely classified as an ONL. Waiheke and Rākino islands both have ONL designations but these are not nearly as extensive. In particular, the protection on Waiheke Island is largely restricted to the less developed eastern end and very little of Rākino Island is designated as an ONL.

Aotea Great Barrier Island is assigned its own strategic management area, in recognition of the island's individual character and issues. The District Plan indicates that decision-makers are to provide for the economic, social and cultural wellbeing of the community while ensuring the protection of the historic heritage, natural character of the landscape, the natural features, ecosystems and visual amenity of the island. To achieve this, natural character and features are to be protected and the relationship between Ngāti Rehua and their ancestral lands recognised and provided for.

The District Plan seeks to provide a framework that enables the community of Aotea Great Barrier to undertake activities that will assist in maintaining an economy while also recognising that the island's key asset is its natural environment. Within existing settlement areas, residential and commercial activities are to be provided for; however, development outside of these areas is restricted to protect the natural character of the landscape. All development is to be of an appropriate scale, form and location.

There are eight settlement areas on the island: Tryphena, Claris, Ōkiwi, Ōkupu, Whangaparapara, Medlands, Āwana and Port Fitzroy. These contain a mix of residential and low-scale commercial development. However, due to the sheer scale of the landscape behind the settlements, the natural character of the island remains the dominant and defining feature. Some settlement areas are recognised as having capacity to expand (such as within Tryphena, Ōkupu, Ōkiwi and Claris) whereas others with more sensitive surrounding environments are not. The District Plan appears to have operated reasonably well in managing the small amount of development on the island. However, there are currently no design guidelines in the District Plan and these would be helpful in the context of a trend towards bigger and more prominent houses being built. In addition, more flexibility in building standards could be provided to enable local innovations in design and use of local materials.

DOC currently manages 56 per cent of the land on Aotea Great Barrier Island, comprising some 16,000ha. Despite the large proportion of the island being managed for conservation purposes there are no marine reserves along its coast. In July 2014, the Aotea Conservation Park was approved. This consolidated and reclassified 12,109ha of stewardship land (around 43 per cent of island) as conservation land. The designation sought to provide an additional layer of protection to the area and provide for a single management and marketing entity.[126] It was hoped that the conservation park status could help encourage more visitors to the island to support the tourism industry. An advisory committee for the Park has been established comprising ten representatives from mana whenua and the local community.

DOC's total annual funding for biodiversity management on Aotea Great Barrier Island is currently around $180,000. In the context of the area of land to be managed, this averages out at around $11 per hectare. This limited budget is due to DOC's national prioritisation system in which Aotea Great Barrier Island does not rank highly. This severely hampers DOC's ability to effectively manage the island's public conservation land.

DOC's conservation activities on the main island are targeted primarily towards the pāteke brown teal and tāiko black petrel. Since 2002, DOC has undertaken control of feral cats on Ōkiwi Station primarily to protect pāteke but also the banded rail and nihi taniwha chevron skink. Despite these efforts, the number of pāteke appear to be decreasing.[127] The Department also undertakes feral cat trapping on Maunga Hirakimatā to protect the tāiko which only breed on Aotea Great Barrier Island and Te Hauturu-o-Toi Little Barrier Island. They are ranked as nationally vulnerable and are the seabird species at greatest risk from commercial activity in the country. Despite DOC's efforts, there is still evidence of cat predation on tāiko.[128]

There is no other predator control currently undertaken by DOC in the Conservation Park. As rats have not been identified as the main threat to the two target species (which are more threatened by feral cats) DOC has not funded any rat control on Aotea Great Barrier Island, even though rats are a serious threat to indigenous birds. DOC did recently eradicate pests from Rakitū Island and the Motu Kaikoura Trust, which manages publicly-owned Kaikoura Island, has reduced rat numbers there to very low levels.

The monitoring of conservation outcomes on the island is also weak. Unlike other private conservation groups on the island that are undertaking extensive monitoring (for example, Windy Hill Sanctuary) this is not happening on public conservation land. DOC's monitoring is undertaken by a centralised monitoring group that is responsible for monitoring biodiversity at a national level. This means there is a disconnect between those working on the ground and those holding the information about biodiversity states and trends.

Whangaparapara, Aotea Great Barrier Island

There are many voluntary initiatives on the island with two notable conservation efforts being at the Windy Hill sanctuary and on the Kotuku Peninsula. At Windy Hill, pest management is undertaken over a total of 770ha extending over 15 properties and engaging 55 landowners.[129] The Kotuku Peninsula contains a 240ha predator-fenced sanctuary which consists of scenic reserve land managed by DOC, an Auckland Council Regional Park and land owned by three private entities.

The Great Barrier Island Environmental Trust, which was formed in 2002, has also been supporting the eradication of rats and feral cats from the island, which would allow for the re-introduction of previously resident native species including the North Island kōkako.

The Great Barrier Island Visitor Strategy seeks to guide the development of tourism on the island and constitutes a commitment by the community to enhance the visitor experience (and therefore increase visitor spend and yield). The Strategy recognises the key challenge to a successful tourism industry on the island as being to gain economic value while conserving the island's key asset – its natural environment. The Strategy was instigated by the local board and was designed to give the residents some control over what was happening on the island.

Much like the planning framework for Aotea Great Barrier Island, the Gulf Island District Plan addresses issues specific to Waiheke Island under the Waiheke Island strategic management area. These issues differ slightly from those on Aotea Great Barrier Island and include those relating to assisting economic growth and local employment, providing a range of living and working environments on the island, maintaining the characteristics of both eastern Waiheke (large scale rural activity) and western Waiheke ('village' style development) and ensuring that there is sufficient land available for residential use without adversely impacting the rural economy or the environment. One of the drivers of the provisions in the District Plan was the desire to encourage restoration of natural vegetation and incentivise people by providing bonus lots on subdivision.

The District Plan enables, for most land units, visitor accommodation for up to 10 people as a permitted activity. This is a potential loophole in the plan, as through enabling another self-contained unit to be built on a property, the intensification of the built environment can occur. It can also be a precursor to subdivision of land, where an application is made to divide the property so that the main house and visitor accommodation are on separate lots. Once that is approved, then new visitor accommodation can be built on each smaller lot as of right. This can result in a cascade of development intensification, particularly in the landscape amenity zones where there are large properties which can be subdivided down to 3.5ha lots as a restricted discretionary activity. At some stage, a tipping point can be reached where the loss of the landscape values is so great that they do not warrant protection and intensive urban development can follow.

During interviews, we received much feedback about the implementation of the District Plan on the island. Interviewees were concerned about the incremental degradation of natural values including visual impacts, loss of wetlands and sedimentation entering the marine area. Although there are ridgeline protections and coastal setbacks provided for in the District Plan, we were told that they were infringed quite often without sufficient mitigation. There was also concern about the types of development that could be undertaken in urban areas, and that rather than being sensitive to the fragile island environment in which they were located, they were more attuned to mainland Auckland. Quite simply, the District Plan was not resulting in the outcomes that had been expected.

The Plan was seen as too discretionary and there was a need to provide more clarity about what types of activities might or might not be permitted. Most applications are now being non-notified so the community no longer has a say in how the District Plan provisions are being applied. People only become aware that an application had been made after construction has started, and at that stage it is too late to influence the outcome. An analysis of data obtained from Auckland Council confirmed that out of 46 subdivision consent applications received in the past three years (from April 2017) all have been non-notified.

On the positive side, the replanting conditions attached to lifestyle block development has resulted in regeneration of native vegetation. However, although the earlier rural-residential subdivision on the western headland area of the island provided for public access, this is not being provided in the newer developments.

There is a wealth of community initiatives on Waiheke Island. 'Essentially Waiheke' is a strategic document setting out a community approved framework for Waiheke's development. It was refreshed in 2016 and outlines some recommended strategic outcomes to address key community concerns. These include monitoring the island's capacity and managing growth in a manner that retains the island's rural and village nature; addressing housing supply and affordability and developing Waiheke as a sustainable tourism destination.

This last outcome arises as a result of specific concerns about the state of Waiheke's environment and the pressures that population growth and tourism are having on water quality, biodiversity, wastewater, litter and pollution. Project Forever Waiheke was established in 2017 to identify the impacts of tourism on the island (both positive and negative) and to feed into the local board's development of a sustainable tourism strategy. The initiative has since developed a strategic framework for sustainable community and tourism development.

In Chapter 6, we described the work of the Hauraki Gulf Conservation Trust. The Waiheke Collective was formed in September 2017. It arose from a community meeting to discuss the Predator Free Waiheke initiative and the potential application to Predator Free 2050 for funds. Although people working in conservation on the island had diverse views, they agreed that they could and should work together for the benefit of the island, thus amplifying their efforts. This resulted in the establishment of the Collective, with an initial focus on raising the funds for the Predator Free Waiheke Initiative, which was ultimately successful. More recently it established a marine sub-group which is focused on the Waiheke Marine Project. As part of this project, DOC and mana whenua co-facilitated a three-day Future Search event on behalf of the Collective in November 2020, designed to bring together approximately 70 people "to dialogue, reach common ground and agree action to protect and regenerate the marine environment of Waiheke Island."[130]

In September 2018, Predator Free Waiheke was awarded $2.6 million from Predator Free New Zealand. This led to the establishment of a separate trust, Te Korowai o Waiheke, to run the project. Auckland Council provided further funding of $2.85 million and Foundation North $875,000 giving the project a total of $5.9 million in cash. There are also considerable volunteer contributions including landowners and Auckland Council undertaking pest control on their properties at their own cost. The first stage of the Te Korowai o Waiheke project was launched in September 2018 and it seeks to eradicate all mustelids (stoats, weasels and ferrets) across the whole island using a mixture of traps and enclosed bait stations. In addition, initial work has been undertaken on identifying community attitudes to, and actions on, rat eradication. A pilot rat eradication programme is also being developed.

Under the 1996 Hauraki Gulf Islands District Plan, Rākino Island was not subject to its own land unit and was instead largely covered by the previous Land Unit 20 (landscape protection) and Land Unit 11 (traditional residential). Traditional residential sites were consolidated along the ridgelines and development was subject only to the general rules in the Plan. Landscape protection sites were predominately 4ha lots consisting of unproductive pasture. On these sites, it was a controlled activity to erect, alter or add to any building(s). While ridgeline controls now exist for significant ridgelines in the Auckland district, none are mapped for Rākino Island. As a result of community concerns about unconstrained development, expressed during the 2006 District Plan review, there are now tighter planning controls in the operative plan and it is a restricted discretionary activity to construct or alter a new dwelling.

Unlike Aotea Great Barrier and Waiheke islands, Rākino is not subject to its own strategic management area. It instead comes under the 'Other Islands' strategic management area, which covers the other islands used for conservation and/or recreational activities, or those that have been developed and are generally in private ownership (such as Rākino Island).

The objective for these areas is to provide for recreation, conservation and other activities, including agriculture and horticulture, while ensuring the protection of the historic heritage, natural character of the landscape, ecosystems, visual amenity and the natural features of the other islands.[131] This is to be achieved by protecting the natural landscape and features of the islands, and providing for the use and development of land for public open space, conservation and recreation. Activities such as agriculture and horticulture are also to be provided for where they are of an appropriate nature, scale, form and location.

Rākino Island also has a specific island-based land unit (Rural 3: Rākino amenity) which provides a planning framework to manage activities on the island, while ensuring the landscape character and natural features are protected. This is in recognition that the coastal environment of Rākino Island is particularly sensitive to the impacts of development. Most of the island is covered by this land unit, with some areas of Conservation and Island Residential 1 (traditional residential) around existing development.

On a simplified analysis, taking into account the minimum lot sizes for both Island Residential 1 and Rural 3 Zones, there is the capacity for 117 further subdivisions on the island. If all these sites were developed, it would roughly double the number of houses. While there are a number of development constraints, that arise both from the district plan overlays (such as building within areas of outstanding natural character or landscape or soil warning areas) and also the complications and associated costs of building on a remote island, this level of development would cause a significant change to the character of Rākino. As such, it is important that design guidelines be implemented to ensure any future development is sympathetic to the landscape.

There has been considerable regeneration of indigenous forest on the island through replanting efforts. There is a local native nursery which helps to avoid the biosecurity risks of bringing seedlings onto the island. However much remains to be done. There are extensive grassed areas which could be replanted with a resultant reduction in fire risk, increase in habitat for birds and other indigenous species, and enhancement of the island's landscape and natural values.

Lessons learnt to inform future management responses

- Imprecise language, and the balancing of interests, in local legislation such as the HGMPA, can lead to it having little weight in protecting environmental or cultural interests in judicial decisions.
- Design guidelines in the District Plan can be helpful to reduce landscape impacts when there is a trend towards bigger and more prominent houses being built.
- More flexibility in building standards, within landscape protection rules, could provide for local innovations in design and use of local materials.
- Tourism needs careful management to ensure that the local environment, population and fragile infrastructure does not become overwhelmed nor natural or cultural landscapes adversely affected.
- Where tourism is an important component of local economies, the provision of a local quota for concessions could be considered.
- The management of public conservation land can have significant impacts on landscape values and associated communities, and where DOC lacks the funds to properly manage the land, other options could be explored.
- Co-governance approaches to the management of public conservation land could enable a bicultural management approach to be developed and the cultural heritage of the area to be celebrated.
- Clear and strong district plan provisions are needed to control the cumulative effects of incremental subdivision and development in places where there is strong development pressure. This includes reducing council officer discretion and notifying applications so the community can have input.
- In island environments, more integrated planning across the land-sea interface is needed to ensure that the impacts of land development and infrastructure (such as kerbing, channelling and piping stormwater) don't adversely impact on the marine area. This could be achieved through the use of a precinct overlay which includes both land and the sea.
- Island environments are also excellent places to experiment with new styles and forms of building that use local materials and draw inspiration from the local landscape.
- Engaged communities supported by independent not-for-profit entities can make considerable progress on restoration and pest eradication initiatives.
- Restoration efforts on land need to be extended into the adjacent marine area for a whole of landscape (and seascape) protection approach.

7.4 Waitākere Ranges

This case study examines the effectiveness of the Waitākere Ranges Heritage Area Act 2008 (WRHAA) which was enacted largely in response to concerns that the planning regime was insufficient to protect the landscape characteristics of the area. The WRHAA applies to regional parkland and adjacent private land in the Waitākere Ranges. It created the Waitākere Ranges Heritage Area, which consists of 27,720ha of land spanning the eastern foothills, the ranges and the coastal settlements on the west coast.

Biophysical landscape

The Waitākere Ranges sit on an uplifted dissected plateau comprised of resistant volcanic material creating a landscape of gorges, bluffs, waterfalls, ridges, slopes and gullies. There are a series of large catchments terminating in the dunes and exposed coastal beaches to the west and a less rugged landscape along the Manukau Harbour.[132]

The western coastal landscapes include patterns of windswept beaches, dunes, streams, wetlands, ridges, headlands, cliffs, islands and extensive coastal forests, all of which contribute to the wildness of the west coast. This contrasts with the northern Manukau Harbour's sense of sheltered openness with expansive tidal flats, cliffs and headlands, broad estuaries, beaches, streams, wetlands and coastal forest.[133] Rolling pastoral foothills frame the bush-covered Ranges from the east, providing a buffer between the Ranges proper and the sprawl of metropolitan Auckland.

These landscapes support a wide range of flora and fauna. The principal characteristic of the Ranges proper is its bush-clad nature – typically dominated by kauri interspersed with pockets of rimu and northern rātā. In summer, the coastlines drip with the red and yellow colours of kōwhai and pōhutakawa flowers. The Ranges comprises one of the largest blocks of contiguous indigenous

vegetation in Auckland, with approximately 21,200ha of native bush. The diverse habitat of the Ranges attracts a variety of both indigenous and exotic birdlife.

Sixty per cent of the Waitākere Ranges Heritage Area comprises the Waitākere Ranges Regional Park which was founded on reserves set aside from the late 1890s. It incorporates the Auckland Centennial Memorial Park, Aotearoa New Zealand's first regional park which was established in 1941. Over time, the initial 7,852ha that was purchased has been supplemented with gifts from private landowners and the transfer of some Crown land to more than double the Park's original size.[134]

Cultural landscape

The earliest human occupation of the Waitākere Ranges dates back around 800 years, with oral traditions referring to the earliest occupiers as Tino o Maruiwi, descendants of Chief Maruiwi. When the people of Toi came to occupy the Waitematā area in the 13th century, the Tino o Maruiwi peoples were pushed into the Ranges. They were joined by a number of Toi refugees (now referring to themselves as Ngaoho) and these peoples integrated to become the Kawerau tribe. The tribe primarily settled in the Te Henga Bethells Beach area and the Waitākere River Valley – a strategic location midway between the Manukau and Kaipara Harbours with abundant food resources.[135]

There are many Te Kawerau ā Maki tohu or place-names and landmarks that extend over all parts of the land and surrounding seas. "The many peaks extending down the Waitākere Ranges from Muriwai to the Manukau Harbour entrance became known as 'Nga Rau Pou ā Maki', or 'the many posts of Maki'."[136] "The traditional name for the wider west Auckland area is Hikurangi, while the name for the huge forest that once covered the area is Te Wao nui ā Tiriwa – the great forest of Tiriwa."[137]

Today, both Te Kawerau ā Maki and Ngāti Whatua are recognised as mana whenua of the Waitākere Ranges Heritage Area, and this is expressly recognised in the WRHAA. Ngāti Whātua is referred to as residing in and using the resources within the Waitākere Ranges Heritage area for about 400 years.[138]

In the early-mid 1850s, the Crown, under pressure from the early European settlers, began to buy up land in the Waitākere Ranges. By December 1856, the Crown had purchased most of the western Ranges, leaving Te Kawerau ā Maki with only two reserves: Waitākerei (the name used for the area around the mouth of the Waitākere River – 1180ha) and Piha (725ha). The Crown then began dividing the land up into civil parishes and districts for sale to settlers.[139]

In 2014, Te Kawerau ā Maki finalised its Treaty settlement with the Crown,[140] which has since been legislated in the Te Kawerau ā Maki Claims Settlement Act 2015.[141] The cultural redress includes statutory acknowledgements of Te Kawerau ā Maki's cultural, historical and traditional association with 11 west Auckland areas.[142] This is reflected in the WRHAA which provides the opportunity for Te Kawerau ā Maki and Ngāti Whatua to form a 'deed of acknowledgement' with the Crown or Council to acknowledge their historical, traditional, cultural, or spiritual relationship with that land.[143] While such deeds do not have any legal weight, they identify opportunities for contribution by tangata whenua to the management of the land concerned.[144] To date, no deed of acknowledgement has been entered into.

Early landscape change

Between 1840 and 1925, kauri was logged extensively in the area. Driving dams were used to flush the logs down to the lower holding dams. Logs left standing were removed by later harvests or cleared by fire in attempts to create pasture for farming. Despite the Waitākere Ranges now being covered in indigenous vegetation, only a small proportion of this has not been modified by felling or burning.

The forestry industry influenced the pattern of European settlement in the Waitākere Ranges which took the form of several small townships, with the largest early communities being located near to the timber mills, on the west coast and at Henderson and Titirangi. Starting during the 1930s, large blocks of land were purchased by farmer settlers who felled the trees and cleared the land in order to establish farming operations.

Land around Titirangi was subdivided during the mid to late 1800s and much of the land was cultivated. Most of the farms were purchased by the Auckland City Council in the early 1890s to provide a water supply for the city and this allowed the bush to start regenerating. Land around Piha started to be subdivided after the 1930s when road access was created.

Current and future landscape pressures

Kauri dieback is now one of the biggest threats to the Ranges. Despite the historical onslaught, the Waitākere Ranges still contain approximately 2,571ha of dense podocarp forest, as well as many small stands, and extensive areas containing individual trees including some of the largest kauri in the Auckland region.[145] In 2006, kauri trees in the regional park were first identified as being affected by a soil and water borne pathogen which causes kauri dieback.

The recorded distribution of kauri dieback relative to the kauri area within the regional park has more than doubled from 7.9 per cent infected and a further 2.7 per cent possibly infected in 2011 to 18.95 per cent infected and a further 4.65 per cent possibly infected in 2016.[146] As at 2017, of the distinct areas of kauri forest within the Park over five hectares in size, 58.3 per cent were exhibiting

symptoms of dieback.[147] Kauri dieback disease is spread almost exclusively by soil disturbance and people are one of the biggest causes of this in the Ranges.

With no known cure for kauri dieback, the loss of the kauri forest ecosystem is the biggest threat presently facing the Heritage Area. As an 'ecosystem engineer', the loss of kauri would have a cascade effect on the plants and animals that depend upon the trees, including orchids and ferns that only grow under kauri, and the nationally vulnerable long-tailed bat whose preferred roosts are mature kauri trees.[148] In addition, the Waitākere Ranges is heavily infested with pest plants and nuisance weeds which are a major threat to the terrestrial and aquatic ecosystems of the Heritage Area.[149]

The population of the Waitākere Ranges has grown steadily over the years. In 2005, approximately 21,000 people lived in the Waitākere Ranges area (outside the Regional Park). That number had doubled by 2017 with the population reaching 51,650. By 2051 this is expected to reach 56,000.[150] Although this growth rate is substantially less than that of the rest of Auckland, an increase in population necessitates additional housing and infrastructure, both of which can significantly impact the landscapes of the Heritage Area.

In addition to population increase, the Waitākere Ranges is experiencing more tourism from a growing Auckland metropolitan area and (prior to Covid-19) from overseas. The Ranges are a major visitor attraction within the Auckland region, with the Regional Park containing over 140 tracks covering approximately 250km.[151] Visitor numbers doubled between 2000 and 2010 and this trend is likely to continue due to the close proximity of the Auckland metropolitan area to the Ranges.[152] Parts of the regional park are under pressure from increasing recreational activity, including commercial activities related to tourism, film production and sporting events.

A corollary to population increases within the Heritage Area is an increase in subdivision and development to accommodate those additional people, and to upgrade the living standards of those already in situ. Since 2000, the number of subdivision, land use and building consents sought within the Heritage Area has been declining, with a significant drop off following the enactment of the WRHAA.[153] These statistics suggest that the WRHAA has had some impact in slowing subdivision and development pressure in the Heritage Area since its inception. The actual extent of that effect will be difficult to ascertain given external influences, including most significantly, the 2007-08 Global Financial Crisis which stymied development and growth throughout the country.

Pleasingly, landscape assessments of the Heritage Area indicate that, overall, there have been only minor or very minor negative changes to the landscape character and heritage features of the Heritage Area as a result of subdivision and development during the first decade of the WRHAA.[154]

Current management responses

Te Kawerau ā Maki iwi have played a paramount role in protecting the Waitākere Ranges bushland from kauri dieback. In 2017, the iwi declared a rāhui (customary prohibition) over the Waitākere Ranges to protect the bush from kauri dieback. The purpose of the rahui is to quarantine the forest and enable the environment to recuperate and regenerate without the presence and impacts of humans. For Te Kawerau ā Maki, who are the mana whenua of Waitākere, the death of the forest is an existential threat.[155]

The rāhui covers the entirety of the Heritage Area, extending further than the track closures originally put in place by Auckland Council. Since it was declared, central government and Auckland Council have subsequently enforced the rāhui with official track closures and the use of a Controlled Area Notice. Auckland Council now work alongside Te Kawerau ā Maki when determining whether upgraded tracks have reached the required standard and can be reopened. Tracks cannot be reopened without the agreement of iwi. This illustrates the strength of adopting a bicultural approach to management of landscape threats.

Almost the entire Ranges is subject to the ONL overlay in the Auckland Unitary Plan, with the exception of pockets of coastal settlement and some of the more developed foothills. The WRHAA provides a further protective layer of regulation. It has the status of a NPS under the RMA and must therefore be given effect to in the Auckland Unitary Plan. In addition, consent authorities must have regard to it when considering an application for resource consent

Piha

The WRHAA had its genesis in the 1970s following a proposal to develop a rubbish tip at Te Henga. The proposal was met with significant pushback from the community, with the Waitākere Ranges Protection Society leading the charge. Although the proposal was eventually dropped, it highlighted the need to increase the environmental protection in the relevant planning instruments.

In 1998, a pro-development political grouping called Go Waitākere, campaigned on the basis of restricting the 'extreme' development controls in the district scheme. Following success at the election, Go Waitākere sought to weaken the district plan. It declared that there was no basis for the restrictive controls on 7,000ha of private land in the foothills and coastal villages and sought the creation of 12,000 new titles and dwellings.[156] Following the election of Go Waitākere, subdivision consents were granted with unprecedented intensity.

In 2003, Waitākere City Council and Auckland Regional Council set up a working group to discuss the concerns of the local community and others in regard to subdivision pressures in the Waitākere Ranges. There was concern that the administration of district plan rules had not been rigorous enough or consistent, and that applications for non-complying consent were often granted leading to negative cumulative effects. Many people considered that the Ranges needed 'something extra' to protect them and in November 2004 the process of drafting a Bill began. Once presented to Parliament, the Bill took a further two years to be passed into law.

The purpose of the WRHAA is twofold: to recognise the national, regional and local significance of the Heritage Area and to promote the protection and enhancement of its heritage features for present and future generations. It lists the heritage features that contribute to the significance of the Heritage Area including features relating to terrestrial and aquatic ecosystems of prominent indigenous character and the different classes of natural landforms and landscapes, as well the importance of community wellbeing, culture and recreation. It then goes on to provide a comprehensive list of objectives for establishing and maintaining the Heritage Area and protecting, restoring and enhancing the listed heritage features.

Councils must give effect to the purpose and objectives of the WRHAA when preparing or reviewing a RPS, regional plan or district plan that affects the area and must have particular regard to them when considering an application for a resource consent for a discretionary or non-complying activity. They also become matters over which the Council has reserved its control when considering controlled or restricted discretionary activities. Private plan changes can be rejected if they are not consistent with the purpose and objectives of the WRHAA.

The provisions of the WRHAA were incorporated into the Waitākere Section of the Auckland District Plan. These provisions were largely carried over into the Auckland Unitary Plan although they were simplified and references to the local area plans were removed. The Rural Urban Boundary largely aligns with the edge of the Heritage Area with the exception of land zoned residential large lot in Titirangi and Laingholm.

The WRHAA requires the Council to produce a five-yearly report monitoring the state of the environment of the Heritage Area, as well as reporting on progress made towards achieving the objectives of the WRHAA and the funding impact arising from activities to be undertaken specifically to give effect to the Act. To date there have been two reports: the first in 2013 and the second in 2018. They were informed by landscape assessments which both concluded there had been only minor or very minor negative changes to landscape values[157] indicating that the WRHAA was largely achieving its aims.

The WRHAA also provided for the preparation of local area plans which were designed to recognise the different character of various areas within the Heritage Area. Five local plans have been prepared. While the plans are helpful guides for the aspirations of defined areas, they have no legal effect under the RMA or Local Government Act 2002 (under which council budgets are prepared) and have not been referred to in the Auckland Unitary Plan.

Overall, it is clear that the WRHAA has been influential in protecting the landscapes of the Waitākere Ranges. The trend leading up to the passing of the WRHAA was that subdivision and development was rampant, often badly designed and unsympathetic to the landscape. This pattern of development is

Te Henga Bethells *(Craig Potton)*

no longer occurring, although the extent to which this can be attributed to the existence of the legislation cannot be determined.

The WRHAA has had direct impact on private plan changes which can be rejected if they are not consistent with the purpose and objectives of the Act. This same level of regulation does not however apply to resource consent applications which are only required to have 'particular regard to' the purpose and objectives of the WRHAA. The Act does leave open the possibility that consents resulting in minor landscape change, which could cumulatively become significant, might be granted.

Another strength is the requirement that policy and plans give effect to the purpose and objectives of the WRHAA. This has resulted in robust policy and plan provisions specifically addressing heritage features and their protection. This has been the main success of the Act to date. In the future it will continue to play an important role when vetting plan changes and reviews, in assessments of resource consent applications, and in broader local government planning contexts. It also has the potential to be significant in terms of mana whenua contributing to the management of land within the Heritage Area via deeds of acknowledgement.

Compared to the HGMPA, the WRHAA is much more specific and directive – and this has also likely contributed to its success. It contains a comprehensive and clear list of heritage features to be protected, setting the scene for decision-making. One downside of this, however, is that all features are given equal weighting which can cause tension when deciding between competing objectives. This is particularly so when it comes to consideration of community and economic wellbeing versus environmental values. Given the importance of the natural environment in the Waitākere Ranges these features should be given priority status in decision-making.

Whilst the WRHAA has been successful in preventing urban development and the threat of creeping subdivision, this report has highlighted that other pressures have not been dealt with so well. In particular, the Act appears to have failed at promoting positive changes and restoring the landscape. The ecology of the Waitākere Ranges is in a perilous state, particularly as a result of kauri dieback disease. The prevalence of weeds and pests in the area is still largely uncontrolled, notwithstanding objectives in the WRHAA that have resulted in stronger pest management controls.

The failure to promote restoration of the Waitākere Ranges may also be, in part, due to the failure of local area plans. These plans were intended to localise the WRHAA's objectives and recognise the different characteristics of each area, as well as the opportunities and constraints in each community. They could have been a driver to bring together the community initiatives already in place and fill the gaps where needed. However, the Waitākere Ranges Local Board lacked funding to properly achieve this.

Te Kawerau ā Maki and Ngāti Whātua have identified similar issues, expressing concern about the need to improve management of kauri dieback disease, the spread of pest plants and animals, the ecological quality of lakes and the growth of tourism and recreation in sensitive areas. They have also called for the allocation of more funding so that it is proportionate to the national significance of the area. To address their concerns they have recommended that: a co-governance and co-management steering group be established for the heritage area; a strategic plan be developed for the area to integrate and co-ordinate activities; baseline information gaps be identified and the measurement and monitoring processes be re-designed to align with both western science and tikanga Māori; and progress be made to complete deeds of acknowledgement with Te Kawerau ā Maki and Ngāti Whātua as enabled by the WRHAA.[158]

Lessons learned to inform future management responses

- Māori approaches to resource management, including the use of rāhui, can be effective in addressing landscape threats.
- Bicultural approaches which combine traditional management approaches with legal mechanisms can be particularly effective.
- Local legislation can be effective in protecting landscape values from development pressures so long as the objectives are clear and explicit and the Act is well implemented.
- Such local legislation can be tailored to recognise the specific landscape values of the area and to address the specific pressures on them. It needs to directly relate to other legislation such as the RMA.
- Local area plans could become strategic documents to guide restoration of local landscapes, harnessing existing and future community efforts in a collaborative manner. However, to be effective, they need dedicated funding for implementation.
- Any future reforms, such as to the RMA, need to provide for ongoing linkages with local protective legislation.
- Strong advocates for landscape protection are important.

7.5 Tourism and landscape protection

This case study was different to the other four, in that it didn't focus on a particular place, but on a particular economic sector – tourism. The tourism industry is underpinned by Aotearoa New Zealand's natural landscapes but has largely failed to help protect them. It is an industry that (prior to Covid-19) outgrew the social and physical infrastructure that supported it. With the current downturn in tourism due to border closures, there is an opportunity to develop new tourism models that are more sympathetic to the landscapes and natural environments that tourists come to see. Now is the time to reflect on domestic tourism potential, virtual tourism and increased understanding and promotion of our cultural landscapes which interconnect with our natural environments.

Tourism industry

Aotearoa New Zealand tourism has its roots in the traditional Māori practice of manaakitanga or hosting guests which was extended to manuhiri (visitors) including early European explorers.[159] Well-to-do British, Australian and American travellers started visiting Aotearoa New Zealand in ocean-going steamships during the 1870s. This prompted the government to become involved in the tourism industry with the Department of Tourist and Health Resorts established in 1901.[160]

It took many decades for the international tourism industry to develop due to Aotearoa New Zealand's long distance from major markets, the high cost of travel, and a lack of interest and investment by the private sector. Government bought out many failing companies and so became a major tourism operator in its own right. There was a turnaround in the industry during the 1960s supported by the advent of passenger jets. Wide-bodied jets brought in greater numbers of visitors during the 1980s. As the industry became more profitable, government exited from commercial involvement in the sector.[161]

The Tourist Department was disbanded in 1990, leaving in place a smaller policy unit and establishing Tourism New Zealand to market Aotearoa New Zealand overseas. The highly successful '100% Pure New Zealand' brand was developed during the early 2000s. On its establishment in 1987, DOC was given the task of managing visitor activities on public conservation land and waters, along with a wide range of other regulatory responsibilities under the Conservation Act.

Until the impact of Covid-19, tourism was Aotearoa New Zealand's biggest export earner, contributing 20.4 per cent of the country's foreign exchange earnings. International tourists spent $17.2 billion and domestic tourists $23.7 billion. There were over 3.9 million annual visitor arrivals into the country with the greatest number being from Australia (40%), followed by China (10%) and the USA (9%).[162]

The tourist industry consists of a large number of small businesses, as well as some medium-sized and a few large entities. Prior to Covid-19, over 229,000 people were directly employed in the tourism industry, comprising 8.4 per cent of national employment. Another 164,000 were indirectly employed meaning that more than 1 of 7 jobs in the country have been supported by tourism.[163] The past two decades has seen a significant increase in Māori investment in tourism, particularly following Treaty settlements, with possibly around 1,200 Māori tourism enterprises involved.[164]

Institutional arrangements

Institutions that support tourism include, at a national level, the Tourism Policy Unit in the Ministry of Business, Innovation and Employment, Tourism New Zealand, New Zealand Māori Tourism, DOC and Tourism Industry Aotearoa. Regional tourism organisations operate variously at regional and local levels around the country.

At the central government level, the Tourism Policy Unit provides policy advice and commissions around $4.8 million of tourism-related data each year.[165] There are several notable gaps in the information gathered, particularly concerning domestic tourists and the value of landscape to tourism. Tourism New Zealand is a Crown Entity established under the New Zealand Tourism Board Act 1991 with the sole purpose of marketing Aotearoa New Zealand overseas, which it has done very successfully. New Zealand Māori Tourism supports the large network of Māori tourism businesses. DOC is heavily involved in the tourism industry with public conservation land and waters providing a major contribution to the tourism offering in Aotearoa New Zealand. Despite all these government agencies involved in tourism, as visitor numbers burgeoned in recent years, there was a noticeable lack of proactive planning on how to manage such growth at a central government level.

Tourism Industry Aotearoa is an industry body representing around 1,650 businesses and organisations in the tourism sector, which comprise some 85 per cent of total industry turnover.[166] It is a membership-funded body (unlike primary sector industry bodies which are funded through statutory levies), and this has severely limited its budget. Its small team focuses on advocacy, communication and strategy.

There are 31 regional tourism organisations located around the country of greatly varying structures and sizes. They have mainly focused on destination marketing rather than management.[167] There is a general lack of economies of scale and connection between the work of these organisations and other council functions such as infrastructure provision and environmental protection.

The fragmented nature of the tourism industry and its institutional arrangements has been a barrier to involvement in broader landscape issues.

Impacts of tourism

Tourism can have impacts on local communities, infrastructure provision and the environment more generally. Tourism impacts small communities by overwhelming them with people, overloading local infrastructure, increasing property prices out of the reach of locals and reducing the availability of housing through conversion of homes to AirBnBs and the like. On the other hand, tourism can benefit such communities through providing much needed jobs and income.

A stronger focus on 'place', and the aspirations of the community at a local level when planning for tourism, could help ensure that landscape values of importance to local communities are better protected. However, this may require linking such tourism plans to statutory processes, such as plans under the RMA, to give them 'teeth'.

Prior to Covid-19, the country's rapidly increasing visitor numbers had outstripped available infrastructure in many areas. As well as detracting from the visitor experience, a lack of adequate tourism infrastructure can result in environmental degradation through problems such as overflowing sewerage systems and waste left along tracks and in informal campsites.

There have been various government funds established over the years designed to provide assistance to councils in funding infrastructure improvements. However, such initiatives appear ad hoc and reactive, seeking to address serious problems after they have developed. More permanent funding arrangements, accompanied by better planning, might help to avert such crises before they occur.

Tourism can have significant impacts on the environment due to the large numbers of people travelling, engaging in outdoor activities, and being accommodated and provisioned. A degraded environment can also significantly impact the tourism industry although there is little quantification of this. The downturn in tourism due to the Covid-19 pandemic should provide some short to medium term relief for the environment. It also provides the opportunity to rebuild a more resilient and environmentally aware tourism sector which is able to grow in the future without causing the damaging environmental impacts recently experienced.

Opportunities to create a closer link between tourism and landscape protection

There are a number of opportunities to create positive synergies between tourism and landscape protection. These include undertaking destination management, reconfiguring concessions on the conservation estate, utilising tourism levies and charges and moving towards 'impact' and 'slow' tourism models.

Destination management has traditionally been understood as the coordinated management of the visitor experience in a specific place including marketing the area, managing its values and attractions (including people), and providing suitable visitor access and infrastructure. The aim is to ensure a quality visitor experience and therefore likely repeat visitation.[168] The concept is rapidly evolving and broadening out from purely considering the tourist experience to include the local community and environment. It can include aims such as ensuring tourism is community led, enriches the environment and is financially sustainable. It can also link to broader regional funding initiatives.

Historically, there has been a strong emphasis in Aotearoa New Zealand on marketing to bring tourists into the country rather than on managing the places that tourists visit. Management of destinations at a regional and local level, if it happens at all, is particularly fragmented with the role undertaken by a variety of entities including territorial authorities, regional tourism organisations, economic development agencies and macro-regional marketing alliances.

Destination management planning provides an excellent opportunity to identify landscape values of importance to mana whenua, visitors and local communities.

Tourists at Church of Good Shepherd, Tekapō

Landscape protection measures could be incorporated into statutory planning by linking them with policies and plans under the RMA and conservation management strategies and plans under the Conservation Act.

As discussed in Chapter 6, most commercial activities within conservation land require a concession granted under the Conservation Act. A recent review of Tourism New Zealand, along with interviewees from the tourism industry, were critical of the current concessions system. Some suggested DOC should be more entrepreneurial and facilitate more tourism activity if it is well managed and can contribute significant funds to conservation activities. Interviewees also noted that the concessionaire relationship with DOC was largely transactional and based on financial matters rather on exploring opportunities for mutual conservation and business gain. Mechanisms for managing the allocation of limited concession opportunities were also seen as fraught.

These comments from the industry reflect the inherent tension between the aspirations of the tourism industry and DOC's statutory role, which is to manage conservation land in the broader public interest. The challenge is to design a system that both supports tourism, which is an important industry for local communities and the nation, and generates positive outcomes for biodiversity and landscapes with the broader conservation management system.

As also discussed in Chapter 6, the recent *Ngāi Tai Ki Tāmaki Tribal Trust* v *Minister of Conservation* Supreme Court decision[169] is likely to support greater involvement by iwi and hapū in tourism activities on conservation land and water. However, it has created some uncertainty for the broader tourism industry with expiring concessions currently being rolled over for short terms and reviews of some park plans being put on hold.

A range of tourism charges have been developed over recent years, including the accommodation provider targeted rate in Auckland and the International Visitor Conservation and Tourism Levy which came into force in July 2019. The latter was particularly promising as it was to generate considerable funds (estimated to be $450 million over 5 years) for both conservation as well as tourism provision.[170] It therefore provided a very tangible mechanism for tourism to support biodiversity and landscape protection. But with the current closure of the borders, and likely paucity of international visitors over the next few years, the funds raised through this mechanism are likely to be negligible over the short to medium term.

'Impact' tourism, is tourism which provides both economic and environmental benefits to the locality within which it takes place. It is a way of reframing the industry in a way that moves away from just making money towards making a broader contribution to Aotearoa New Zealand based on the principle of aroha or caring.

Many of the country's tourism operators are already shifting towards 'impact tourism' and are seeking to directly contribute to biodiversity protection through funding or involving tourists directly in actions such as replanting and predator control. This approach also creates opportunities for Māori through potentially connecting investors with hapū and whanau interested in tourism that is positive for the people and the environment. The challenge is how to scale this activity up to a landscape level, and in the post-Covid-19 context, this will initially need to be on the back of domestic tourism.

Experiences such as cycle trails and wine trails help to slow down tourist travel and they facilitate engagement with mana whenua, the local community and businesses such as on-farm accommodation and local cafes. A focus on promoting 'slow' tourism, including developing experiences where tourists have a deeper engagement with Aotearoa New Zealand's landscapes, Māori cultural associations with them and local communities which derive a livelihood from them, could help draw a closer linkage between tourism and landscape protection.

Lessons learned to inform future management responses

- There is considerable opportunity for tourism to be a positive contributor to conservation and landscape protection. The industry is already moving in this direction but would benefit from improved institutions, data and support to help achieve it.
- The institutions supporting tourism are currently weak and fragmented and need to be rationalised and strengthened.
- A new tourism model should emphasise value over volume, support 'slow' tourism and create stronger links with the landscape, nature, Māori culture and local communities in a meaningful way.
- Destination management planning is a tool that could be used to identify opportunities to develop more positive connections between tourism, mana whenua, communities and their landscapes. They could usefully link with statutory plans such as those prepared under the RMA and the Conservation Act and regional funding initiatives.
- The relationship between the tourism industry and public conservation land and water, could be reorientated towards a deeper 'friendship' with mutual obligations as well as benefits.

ENDNOTES

1 Head, 2016, 13
2 Relph, 2010
3 Walker, 2016, 9
4 Head, 2016, 14
5 Waaka-Home, 2016, 7-9
6 Waaka-Home, 2016, 12-13
7 Waaka-Home, 2016, 16-17
8 Te Rūnanga o Arowhenua et al, 2008, executive summary
9 McGlone, 2001, 11
10 Connor and Vucetich, 1964, 325; O'Connor, 1982, 100
11 O'Connor, 1982, 101; Treskonova, 1991, 65
12 Duncan, Webster and Jensen, 2001, 35-36
13 O'Connor, 1976, 73; Sheridan, 1995, 23
14 Brown and Harris, 2005, 1
15 McKnight, 2013
16 Young, 2016, 8
17 Young, 2016, 9
18 Young, 2019, 11-12
19 Environment Canterbury, undated
20 https://www.beehive.govt.nz/release/biosecurity-and-conservation-jobs-redeployed-workers
21 Gawith, 2019
22 Rutledge et al, 2017
23 Waitaki District Plan 2004, Objective 16.8.2 and 16.8.5 Explanation and Reasons
24 https://www.doc.govt.nz/our-work/tu-te-rakiwhanoa-drylands/
25 Weaver et al, 1985
26 Appelton, 2015, para 7.9
27 Wilson, 2013, 35-38
28 Wilson, 2013, 71
29 Wilson, 2013, 86
30 Banks Peninsula Conservation Trust, 2019
31 Christchurch District Plan, 2015, Te Pātaka o Rākaihautū/Banks Peninsula Landscape Study – Ngāi Tahu Values Addendum; Olgivie, 2010, 12
32 Christchurch District Plan, 2015, Te Pātaka o Rākaihautū/Banks Peninsula Landscape Study – Ngāi Tahu Values Addendum; Olgivie, 2010, 12
33 Ngāi Tūāhuriri Rūnanga et al, 2013, 40; Christchurch City Council, 2012, 23
34 Ogilvie, 2010, 5
35 Olgivie, 2010, 23-24
36 https://ngaitahu.iwi.nz/our_stories/banks-peninsula-deeds-of-purchase/
37 Wilson, 2013, 40
38 Wood and Pawson, 2008, 453-454
39 Wood and Pawson, 2008, 453-454
40 Boffa Miskell, 2007, 32
41 Wood and Pawson, 2008, 461; Olgivie, 2010, 7
42 Wilson, 2013, 41
43 Boffa Miskell, 2007, 32
44 Boffa Miskell, 2007, 33
45 Department of Conservation, 2016, 108
46 Boffa Miskell, 2007, 33
47 Harvey, 2020
48 https://www.avenues.net.nz/all-stories/2019/7/26/saving-banks-peninsula-penguins
49 Wilson, 2002, 8
50 Wilson, undated
51 Peart, 2004, 77
52 Banks Peninsula Conservation Trust, 2018, 26
53 Banks Peninsula Conservation Trust, 2018, 4
54 Burry, 2020
55 Burleigh, 2019
56 Banks Peninsula Conservation Trust, 2017, 15-20
57 Burleigh, 2019
58 Rod Donald Banks Peninsula Trust, 2020
59 Banks Peninsula Conservation Trust, 2019, 24
60 Minister of Conservation, 2020
61 Banks Peninsula Water Zone Committee, 2019
62 Te Pātaka o Rākaihautū/Banks Peninsula Geopark Trust, 2020
63 McMurtrie et al, 2017
64 Rhodes and Novis, 2002
65 Johnston, 2019, 45
66 Environment Canterbury, 2020
67 Hayward, 2019
68 https://www.mbie.govt.nz/immigration-and-tourism/tourism/tourism-funding/international-visitor-conservation-and-tourism-levy/how-the-ivl-works/
69 Christchurch City Council, 2015, 4
70 *Briggs v Christchurch City Council* C045/08 at [12]
71 *Briggs v Christchurch City Council* C045/08 at [130]
72 *Briggs v Christchurch City Council* C045/08 at [22] and Annexure 1:E
73 *Briggs v Christchurch City Council* C045/08
74 Hogan, 2015, [9.1]
75 Independent Hearings Panel, 2016, [25]
76 Independent Hearings Panel, 2016, [146] – [148]; Christchurch District Plan, Policy 9.1.2.2.1(a)
77 Christchurch District Plan, Policy 9.1.2.2.5
78 Christchurch District Plan, Appendix 9.1.6.6
79 https://ccc.govt.nz/culture-and-community/community-funding/christchurch-biodiversity-fund/
80 Department of Conservation, undated
81 Department of Conservation, 2014
82 Department of Conservation, 2014, 73
83 Department of Conservation, 2014, 72
84 Monin, 2012, 10-11

85 Waiheke Local Board, 2018

86 Sea Change Tai Timu Tai Pari, 2017, 26

87 Department of Conservation, 2013, 12

88 Department of Conservation, undated(a); Deed of Settlement between Ngāti Rehua - Ngātiwai ki Aotea and the Crown, 2016

89 Monin, 2012, 17

90 Monin, 2012, 7

91 Monin, 2017, 22-24; Peart, 2016, 129 and 132

92 http://www.piritahimarae.net.nz/about-us

93 Coster and Spring-Rice, 1984, 10

94 Great Barrier Island Environmental Trust, 2010

95 Great Barrier Island Environmental Trust, 2010

96 Strateg.Ease and Capital Strategy Limited, 2006, 12

97 Department of Conservation, 2004

98 Great Barrier Island Environmental Trust, 2010, 6-2

99 Hauraki Gulf Islands District Plan, Annexure 1a, 5

100 Hauraki Gulf Islands District Plan, Annexure 1a

101 Baragwanath, undated

102 Peart, 2016, 182

103 https://discover.stqry.com/v/7-farming-and-fishing-history-of-waiheke-island/s/be803b29-84f2-4093-8b4c-7e1d3258dc14

104 Peart, 2016, 177

105 https://www.rra.nz/Rākino/; Peart, 2016, 177

106 Peart, 2016, 178; Auckland Regional Council, 2002

107 Great Barrier Island Environmental Trust, 2010, Chapter 5

108 Auckland Tourism, Events and Economic Development, 2020a

109 Strateg.Ease and Capital Strategy Limited, 2006

110 New Zealand Tourism Research Institute, 2018a, 5; New Zealand Tourism Research Institute, 2018b

111 Allpress and Tuatagaloa, 2018, 1

112 Butler and Allpress, 2019, 3

113 Logie, 2016, 219

114 Auckland Tourism, Events and Economic Development, 2020a

115 Bennett et al, 2012, 3-4

116 Owen, 2017

117 Turbott, 2019, 17

118 Auckland Tourism, Events and Economic Development, 2020b

119 Waiheke Local Board, 2020, 8

120 Project Forever Waiheke, 2019, 5

121 Waiheke Local Board, 2016

122 Hauraki Gulf Marine Park Act 2000, s3

123 Hauraki Gulf Marine Park Act 2000, s32

124 Hauraki Gulf Forum, 2009

125 Peart, 2016, 325; Hauraki Gulf Forum, 2009

126 Westbrooke, 2013

127 Aotea Conservation Park Advisory Committee, 2020

128 Bell et al, 2018, 7

129 Gilbert, 2019

130 https://www.waihekemarineproject.org/future-search-hui

131 Hauraki Gulf Islands District Plan, 3.4.3

132 Auckland Regional Council, 2010, 376

133 Melean Absolum Limited, 2017, 8

134 Auckland Regional Council, 2010, 375; Waitākere Ranges Protection Society, 1978, 5

135 Taylor, undated, 8; Murdoch, undated, 31

136 Auckland Council, 2018a, 9

137 Auckland Council, 2018a, 9

138 Auckland Council, 2018a, 10

139 Harvey and Harvey, 2006, 214

140 https://www.govt.nz/treaty-settlement-documents/te-kawerau-a-maki/

141 Ngāti Whatua have an agreement in principle to settle historical claims with the Crown but, as yet, have not signed a Treaty agreement with the Crown.

142 Te Kawerau a Maki Claims Settlement Act 2015, Schedule 1

143 Waitākere Ranges Heritage Area Act 2008, s29

144 Waitākere Ranges Heritage Area Act 2008, ss29 and 30

145 Auckland Council, 2017, 13; Auckland Regional Council, 2010, 378

146 Auckland Council, 2017, 4

147 Auckland Council, 2017, 4

148 Auckland Council, 2018a, 13 and 61; Kauri Dieback Governance Group, 2019, 11

149 Waitākere Ranges Local Board, 2015, 11

150 Auckland Council, 2018b, 56; Auckland Council, 2018a, 148

151 Auckland Regional Council, 2010, 376

152 Auckland Regional Council, 2010, 377

153 Auckland Council, 2018a, 137

154 Melean Absolum Limited, 2017, 31; Melean Absolum Limited, 2013, 29

155 http://waitakererahui.org.nz/the-rahui

156 Waitakere Ranges Protection Society, forthcoming, 64

157 Melean Absolum Limited, 2013; Melean Absolum Limited, 2017, 31

158 Auckland Council, 2018c, 4

159 Parliamentary Commissioner for the Environment, 2019, 28

160 McClure, 2004, 8-11; Parliamentary Commissioner for the Environment, 2019, 32

161 McClure, 2010

162 Kantar, 2019; https://www.tourismnewzealand.com/about/about-the-tourism-industry/

163 https://www.tourismnewzealand.com/about/about-the-tourism-industry/

164 The actual number of Māori tourism operators can only be very broadly estimated due to a paucity of relevant data

165 New Zealand Productivity Commission, 2019, 268

166 Ministry of Business, Innovation and Employment, 2019, 14

167 Pearce, 2015, 7

168 Department of Conservation, 2011a

169 *Ngāi Tai Ki Tāmaki Tribal Trust* v *Minister of Conservation* [2018] NZSC 122

170 https://www.mbie.govt.nz/immigration-and-tourism/tourism/tourism-funding/international-visitor-conservation-and-tourism-levy/how-the-ivl-works/

International protected landscape models

Schorfheide-Chorin Biosphere Reserve, Germany

In earlier chapters of this report, we considered the concept of landscape and landscape protection regimes within Aotearoa New Zealand. This chapter takes a broader purview and investigates some of the more notable landscape protection models that have been employed in other countries including the United Kingdom, Australia, Canada, the United States of America and several European Union countries.

The models discussed in this chapter operate across a range of landscapes, from large-scale river basins and coastal regions, to areas that contain extensive forest habitat, unique geological features and mountain ranges. Substantial settlements and cities, extractive industries and agricultural landscapes frequently operate within their boundaries. All of the frameworks described represent attempts to balance a complex array of competing interests and land uses. They seek to provide robust protection of important wildlife, habitat and resources while addressing the frequently conflicting economic, social, cultural and environmental pressures at play.

The models were selected to illustrate the range of regulatory and non-regulatory approaches taken to landscape protection internationally. Some models, such as biosphere reserves, have inherent flexibility to enable deployment across a range of landscapes and contexts, while others have been crafted to address very specific local circumstances and pressures. The objective of this review is not to identify the 'perfect model', but to provide an understanding of some of the key elements that have been employed elsewhere, their associated advantages (or disadvantages) and lessons that can be drawn from them of relevance to Aotearoa New Zealand's unique context.

8.1 Bicultural and co-governance models

In recognition of our country's rich Māori heritage and founding document Te Tiriti o Waitangi, we have started this international review with a consideration of how the status, rights, role and needs of indigenous or first peoples have been addressed in other jurisdictions. Around the world, indigenous peoples are asserting their rights to manage their own land. At the same time, there is also increasing awareness on the part of state actors that traditional landscape models, particularly in the conservation context (such as national parks) have often operated as a mechanism for dispossession of these indigenous rights. Historical land alienation has both fragmented the land and severed cultural linkages to it.

The drive for greater innovation, openness and inclusivity in relation to indigenous peoples is motivated not only by the obvious equity issues at play, but also out of practical concern to better protect the environment. For example, one recent study conducted in 2019, found that indigenous-managed lands in Brazil, Australia and Canada have *higher* biodiversity levels than other protected areas in those countries.[1] Studies like this demonstrate the environmental benefits of working with indigenous peoples. Indigenous peoples' demands for greater acknowledgement of their special rights and push to obtain a greater say in the management of the environment, is often in response to specific threats (such as mining or logging) or out of concern that more robust and comprehensive management and improved standards are required. Understandably, given the interconnection between indigenous peoples and the natural environment, many of the most innovative and biculturally unique landscape protection models that exist have evolved in the context of *protected areas*.

Although our project is more broadly focused on *landscape* protection over a range of land tenures, an investigation into protected area regimes enables a detailed examination of the cultural complexities at play and the governance arrangements possible. We have examined five specific frameworks in more detail, which represent some of the more innovative initiatives seen in Australia and Canada (see Figure 8.1).

First Nations Inukshuk, Canada

Model	Key features of governance arrangements
Gwaii Haanas Protected Area (Canada)	Shared co-governance (50/50) Recognition and management under state and Haida law Management board has planning and decision-making powers Express acknowledgement of dispute over land title
Thaidene Nëné (Canada)	Shared co-governance Managed under federal, territorial and Denesoline law Co-governed land tenure
Wehexlaxodiale (Canada)	Indigenous governance Tlicho government has the power to enact laws over their land and resources and develop land use plans (including exclusion areas). Formal Crown/State recognition as a protected area
Uluru-Kata Tjuta National Park (Australia)	Crown governance with statutory oversight Statutory board with aboriginal majority prepares management plans Ministerial oversight of management Aboriginal land title restored with lease back to Crown
Nitmiluk National Park (Australia)	Shared co-governance with statutory oversight Statutory board with Jawoyn nation majority The board has broad planning, enforcement and decision-making powers and full ownership of all commercial tourism Aboriginal land title restored with lease back to Crown

Figure 8.1 *Summary of governance arrangements for bicultural and co-governance models*

In Canada, a raft of exciting developments and initiatives are being progressed. An important first step taken by the Canadian government has been to intentionally move away from what has been called the 'fortress'[2] model of conservation protection (which excludes indigenous populations and traditional land uses and aims to preserve pristine wilderness areas) to prioritise, recognise, support and maintain traditional linkages of indigenous peoples to their lands and resources. This paradigm shift has been driven, in part, by the 2010 Aichi targets set under the United Nations Convention on Biological Diversity, of which Aotearoa New Zealand is also a signatory.

The United Nations Convention on Biological Diversity

Aichi Target 11: To protect at least 17 per cent of terrestrial and inland water areas and 10 per cent of coastal and marine areas by 2020. Target 11 requires that that these areas be 'equitably managed' with the full participation of indigenous communities and recommends the inclusion of indigenous conserved areas in national conservation frameworks and legislation.

Aichi Target 18: That by 2020, the traditional knowledge, innovations and practices of indigenous and local communities relevant for the conservation and sustainable use of biodiversity, and their customary use of biological resources are respected and integrated nationally with the full and effective participation of indigenous and local communities.

Ōrakei Marae

In Canada, indigenous peoples' land often overlaps with existing protected areas and contains significant remaining intact ecosystems. Greater collaboration with indigenous peoples was necessary to comply with the Aichi targets and to be able to reach the protected area goals set. In 2016, federal, provincial, territorial and local governments in Canada came together in partnership with First Nations to launch an initiative known as 'Pathway to Canada Target 1'. This established the goal of protecting 17 per cent of land and 10 per cent of the marine and coastal area by 2020. A National Advisory Panel and Indigenous Circle of Experts was established to provide advice and make recommendations.

A spotlight on the Indigenous Circle of Experts

The Indigenous Circle of Experts is made up of representatives from indigenous nations, provincial governments and government agencies. The body has 20 members, with indigenous persons filling the roles of chair and co-chair. The vision is for Canada's protected areas system to be managed in partnership with indigenous governments, in line with the United Nations Declaration on the Rights of Indigenous Peoples.

First Nations totem poles, Vancouver

Canada's Indigenous Protected and Conserved Areas

Like Aotearoa New Zealand, a large number of individually negotiated settlements, agreements and co-governance arrangements are in place in Canada. Areas where indigenous peoples have asserted a leading role in decision-making *and* where there is a long term commitment to environmental and/or conservation can now be officially recognised as 'Indigenous Protected and Conserved Areas'.[3] The Canadian government has made a commitment to explore this concept on an ongoing basis, to clarify its contribution to national biodiversity and conservation priorities, and to work collaboratively with First Nations.[4] The objective is twofold: to ensure that reconciliation and settlements with indigenous groups are proactively pursued and to lay down the foundations for a coordinated, consistent and integrated pan-Canadian approach.

> "IPCAs [Indigenous Protected and Conserved Areas] are lands where indigenous governments have a primary role in protecting and conserving ecosystems through indigenous laws, governance and knowledge systems. Culture and language are at the heart and soul of an IPCA."[5]
>
> 'Indigenous self-government' is the "formal structure through which Indigenous communities may control the administration of their people, land, resources and related programs and policies, through agreements with federal and provincial governments."[6]

Indigenous governments are responsible for setting the conservation standards in these areas. Every arrangement is unique as it reflects the needs, priorities and concerns of each community. Part of the role of the Indigenous Circle of Experts is to support such initiatives and provide advice and recommendations regarding their development.

In 2018, the Indigenous Circle of Experts reviewed the range of governance options that exist in Canada and identified the most effective arrangements. We examine three of these: the shared governance models of the Gwaii Haanas in British Columbia and Thaidene Nëné in the Northwest Territories, and a wholly indigenous based framework – Wehexlaxodiale.

The Gwaii Haanas National Park Reserve is considered one of the most effectively co-managed parks in the country and has been held up by numerous commentators as a best practice example of participatory governance.[7] The scale of this framework is significant. The Gwaii Haanas Agreement brought together eighteen pre-existing conservation areas and heritage sites under co-management, including a National Park Reserve and National Marine Conservation Area. Collectively the sites cover more than 500,000ha.

The Agreement is focused on genuine power sharing. It recognises both Canadian and Haida designations and interests, and the sites are actively managed by the Haida Nation under Haida Stewardship Law and recognised under British Columbia's Park Act.[8] The Agreement sets out the parties' mutual commitment to the conservation and protection of Gwaii Haanas, and the goal is to protect native species and habitats while meeting the Haida Nation's food, health needs and wellbeing. The Archipelago Management Board is comprised of three representatives of the Haida Nation and three from the Canadian Government. It has full management, planning and decision-making powers.[9] Decisions are made on a consensus basis with provision for disputes to be referred jointly to the Canadian Government and Council of the Haida Nation for resolution. The jointly developed plan relies on both science information and traditional ecological knowledge, with management being based upon *yahguudang* – a concept reflecting 'respect for all living things'.

One area of ongoing dispute is in relation to land title and sovereignty. Although the land remains legally vested with the Crown, the Agreement expressly acknowledges the continued existence of Haida title and importantly – the ongoing disagreement over sovereignty. Express recognition of this dispute appears to have been key in enabling progress to be made, despite conflict on the issue.

A second Canadian example of a shared co-governance arrangement is at Thaidene Nëné,[10] which covers an area of over 2.6 million ha and incorporates a 1.4 million ha National Park Reserve. The governance arrangements here have contextual similarities to Aotearoa New Zealand. The regime at Thaidene Nëné is underpinned by 'Treaty 8' signed in 1899. Treaty 8 was one of the last treaties signed in Canada. It set aside reserves and land allocations for First Nations families, provided for payments in return for the surrender of land, and provided for the sharing of land, resources and associated benefits. It also guaranteed First Nations the right to pursue traditional practices, such as hunting, trapping and fishing, throughout the area. Oral promises were also made to take care of the poor, provide medical care and not interfere with their way of life.

> "The protection and stewardship of Thaidene Nëné is the sacred responsibility of the Łutsël K'é Dene First Nation, as passed down to us through the generations from our Elders. Achieving the protection of Thaidene Nëné for the Łutsël K'é Denesǫłine is a decades-long dream, and is a critical step towards ensuring our way of life can be maintained and shared with all Canadians."[11] Chief Darryl Marlowe, Łutsël K'é Dene First Nation.

The negotiations were lengthy and it took more than 50 years to get agreement. Three factors complicated the process; there were competing First Nations

claims, industry groups opposed protections that prevented extractive resource use and development,[12] and initial proposals for the creation of the national park reserve would have imposed strict limitations on traditional activities in contravention of Treaty 8.[13] The ultimate success demonstrates what can be achieved even where significant complexities like these exist.

The final Agreement provides for joint governance between Parks Canada, the Government of the Northwest Territories and Lutsel K'e Dene First Nation. Indigenous knowledge forms the foundation for protecting the ecosystem. The Agreement allows traditional practices to continue, and indigenous peoples to remain in residence in the reserve, and makes provision for the economic development of local indigenous communities. The latter is supported through a trust fund that enables investment in staff and management training, tourism activities, research partnerships and youth activities. These investments have generated significant economic, environmental, social and cultural returns.[14] A range of ancillary agreements were signed with other First Nations to secure their rights to continue to hunt, trap and fish in the park area.[15]

A final Canadian example is Wehexlaxodiale. This is one of the few examples of a completely indigenous led and managed conservation management framework which has been formally recognised by the Canadian Government. Signed in 2005, the Tłı̨chǫ Agreement gives the Tłı̨chǫ government the power to enact laws in relation to the use, management, administration and protection of their lands and the resources they contain. With the agreement of the Tłı̨chǫ government, Wehexlaxodiale has been included in the country's Conservation Areas Reporting and Tracking System as a protected area.

Australia's Indigenous Protected Areas regime

Turning to the Australian context, the 'Indigenous Protected Areas' regime is interesting. These are areas of aboriginal-held land, that are voluntarily declared protected areas, and are managed with government assistance. This framework has enabled western science and traditional indigenous ecological knowledge to operate side by side. The Australian Government now formally recognises these areas as part of the National Reserve System of Protected Areas. The federal government has established budgets for the programme and extends considerable technical and managerial support to participants.

The regime has proven highly successful. The demand from indigenous landowners to develop and declare protection over their land exceeded all government expectations and has driven significant growth in Australia's National Reserve System.[16] Some 48 million ha have been designated under the regime which now constitute more than 35 per cent of Australia's reserve system.

> "The Australian experience with Indigenous Protected Areas demonstrates that there are no inherent barriers to indigenous governed and managed protected areas being recognised as part of national protected area systems. Assessments of governance and management processes and outcomes from Indigenous Protected Areas, while generally lacking depth, consistently report empowerment of the Aboriginal owners, increased economic participation and opportunity, and improvements in local capacity for biodiversity conservation".[17]

Australian co-governance arrangements[18]

Uluru-Kata Tjuta National Park is one of the best known co-governance frameworks in place in Australia. Although the framework has been described as providing a strong joint management arrangement, aspects of it remain controversial. Title over the National Park was granted to its aboriginal owners by the Crown on condition that it be leased back to the government for national park purposes for a term of 99 years. While governance is vested in a statutory board with an Aboriginal landowner majority, governmental oversight has been retained. The management board is responsible for preparing a management plan and can make management decisions that are consistent with this. The plan itself must be developed in conjunction with the Director of National Parks and is subject to Ministerial oversight.

A research paper that canvassed Māori views of the co-management models in Australia found that Māori were highly critical of the extent to which the state retained significant and final decision-making authority at Parks like Uluru (and also Kakadu National Park). They also questioned the degree to which traditional ecological knowledge had been employed in park management at a practical and conceptual level.[19] Rather, in the Australian context, Māori pinpointed the Nitmiluk National Park model with the Jawoyn Nation as the most progressive and innovative approach.

The distinctions between Nitmiluk National Park and Uluru are subtle but important. Nitmiluk is jointly managed by the Jawoyn Nation, the Northern Territory government and the Northern Parks and Wildlife Commission. Like Uluru, governance is vested in a statutory board with an Aboriginal majority. The Board prepares management plans and makes decisions consistent with those plans. It therefore has much broader decision-making power over the management plan, as well as over the protection and enforcement of Aboriginal use rights.

While decisions outside the scope of these matters are subject to direction by the Chief Minister, the wider scope of powers gives the Board an ability to act largely independently of government. In addition, the Jawoyn Nation has full ownership of all commercial tourism operations in the park and runs a number of joint ventures.

These generate substantial economic benefits to the Aboriginal owners and have resulted in a much higher number of aboriginal people working both in the park and in management positions, in contrast to what has been possible at Uluru.

Key lessons

Importance of acknowledging historical injustice and grievances

In order to make progress, it is important to formally recognise and if possible address, historical grievances and conflicts (such as land confiscation, exclusion from governance, prohibitions on traditional practices or degradation of local resources). Even where debate is ongoing, acknowledgment of contested matters helps achieve clarity and mutual understanding. It is seldom possible to resolve all issues at the outset – the arrangements represent a starting rather than end point. Express recognition of a disagreement as to sovereignty in the Gwaii Haanas Agreement helped the parties move past a key issue of contention to progress matters where there *was* significant alignment of purpose. This is important for moving forward from historical grievances and into a more positive, collaborative and transformative bicultural space.

The need for highly contextualised solutions

To be successful, any framework must address and reflect very specific local cultures, histories and relationships, and navigate complex political and cultural contexts. For example, the Agreement for Thaidene Nëné had to address a raft of competing claims and incorporate the needs of many parties.

The need for true power sharing relationships

The most successful and enduring frameworks empowered and supported indigenous communities and provided for genuine partnerships and power sharing. Key features included:

- Equal or majority representation of indigenous actors on governance boards and/or recognition and support for indigenous-led governance arrangements and models.
- Management bodies that have regulatory and decision-making powers, rather than simply advisory ones. The extent to which indigenous peoples have decision-making power is typically reflective of both the degree of trust that exists (ie the strength of the Crown-First Nation relationship) and the capacity of indigenous actors (ie how effectively resourced and supported indigenous communities are to take on this role).
- Utilisation of consensus based decision making and provision of clear pathways for dispute resolution where agreement cannot be reached.
- Recognition of multiple (indigenous and state) legal frameworks, systems and institutions, often demonstrating a high degree of legal pluralism.

A bicultural approach to landscape protection

The inclusion and empowerment of indigenous peoples within landscape protection frameworks fundamentally alters the nature of those regimes: the purposes and approach, the institutions, the mechanisms in place and the narrative surrounding them. Such frameworks protect not just the landscape but the cultural connections to those landscapes – the traditional ways of life, institutions and world views held in relation to them. They recognise and enable indigenous peoples access and user rights to their lands and resources. They employ traditional holistic (including ecological) knowledge alongside western scientific frameworks. They also utilise traditional management practices and concepts to inform planning.

Rethinking the past – developing ways forward

A bicultural approach to landscape protection requires a reconsideration of many traditional western notions about nature and our relationships with it, and the adoption of a less formulaic approach. It requires a willingness to be more inclusive in governance arrangements. It also requires recognition that other ways of being, organising and viewing the world exist, and have legitimacy and standing.

> The experience in both Canada and Australia has shown that being more open, innovative and inclusive not only enables more land and biodiversity to be protected but also facilitates better monitoring and management. Indigenous communities have a long history and association with their lands and resources, and their traditions and practices are based on that connection. In the context of historical land fragmentation, the acknowledgment and protection of traditional cultural linkages facilitates greater connectivity across large landscapes.

Capacity building, equity and redress

To be effective, co-governance and bicultural models for landscape protection rely on strong indigenous leadership and capacity. It is important that institutions are appropriately supported and resourced. A priority identified in all the frameworks discussed, is to provide for the economic development (which cannot be isolated from wider development aspects) of indigenous groups, particularly where (usually through historical land alienation) significant opportunities had previously been removed.

Shifting from reactive to proactive engagement

Perhaps the biggest lesson that can be learnt from the Australian and Canadian experiences is the value of putting structures and mechanisms in place that enable a more proactive rather than passive or reactive role to be taken by government. Australia's Indigenous Protected Areas framework has driven significant growth in the national reserve system and provided a direct mechanism for the independent growth and inclusion of indigenous governed and protected landscapes. In Canada, the creation of the Indigenous Circle of Experts has helped bridge the gap between the Crown and First Nations and played an invaluable role in laying the groundwork for a much more proactive engagement with indigenous communities.

8.2 Biosphere reserves

Biosphere reserves are large-scale areas designated under the UNESCO Man and the Biosphere Programme and are designed to protect natural *and* cultural landscapes. The model allows for a gradation of human interactions – including landscapes on a continuum – from those that are wholly natural to those shaped by human activities. Both modified and natural landscapes are treated as valuable, and this additionally enables the long-standing character of an area and its land-use practices (such as agriculture) to be enhanced, protected and retained. The purpose of the designation is to achieve a balanced relationship between humankind and the biosphere.

> "Biosphere reserves are designed to deal with one of the most important questions the world faces today: How can we reconcile the conservation of biodiversity, the quest for economic and social development and the maintenance of associated cultural values?"[20]

To be designated as a biosphere reserve an area must have a mosaic of ecosystems representative of the region; it must be significant for conservation; it must provide opportunities to explore approaches to sustainable development on a regional scale; and it must be of an appropriate size to fulfil the relevant functions through appropriate zonation (core, buffer and transition zones).[21] Biosphere reserves must also fulfil three core functions: (1) conservation of genetic resources, species, ecosystems, habitats and landscapes; (2) fostering sustainable economic and human development; and (3) a logistical function to support research, monitoring and education. All of these are equally important – conservation is not given priority over sustainable development, research or education.

Participation in the biosphere reserves programme is voluntary and the land remains within the control and management of its owners. Potential biosphere reserves are nominated by the relevant country and, if the designation is successful, the landscape becomes part of the world network governed by the Statutory Framework for Biosphere Reserves. Standards exist and are assessed through periodic reviews, but the United Nations has no powers of enforcement. Implementation of the model is also flexible. Specific legislation can be enacted and it can be included within a country's existing protected areas networks, or a biosphere reserve can even exist as 'soft law' only.

A defining feature of the biosphere reserve model, and one of the keys to its success, is the use of zones. There is a *core zone* of undisturbed, natural habitat that is strictly regulated to protect its biodiversity, surrounded by a *buffer zone* where low impact uses (such as farming, agriculture and forestry) can occur provided they are consistent with the overall functions of the biosphere reserve. Finally, a *transition zone* exists that can be used for a variety of purposes; including human settlements where the local communities work together to sustainably develop the area's resources.

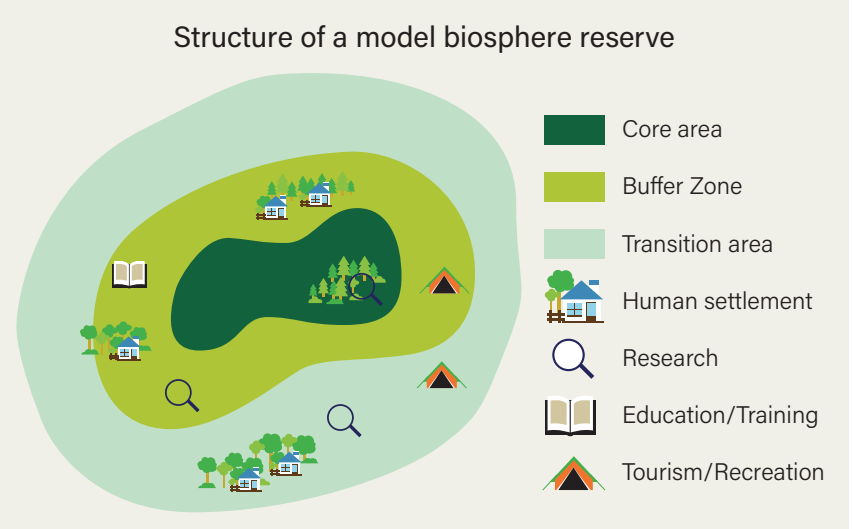


Figure 8.2 *Structure of a model biosphere reserve*[22]

The zones can be organised in a large number of ways, allowing a diverse range of flexible configurations. Multiple core sites may exist and need not be connected, and the relative size of the core, buffer and transition zones varies dramatically in practice. Ownership arrangements are also variable. Generally core areas are

publicly owned and protect pre-existing parks and reserves, while buffer and transition zones tend to be largely privately owned.

We have looked at four specific biosphere reserves, one each from Canada, Germany, Australia and the United Kingdom. The Canadian example, the Niagara Escarpment Biosphere Reserve, includes a vast core of almost 200,000ha of wilderness area, surrounded by a buffer zone (66,000ha) and smaller transition zone (14,000ha). In contrast, Germany's Schorfheide-Chorin Biosphere Reserve is situated in a primarily agricultural setting and is designed to foster new sustainable farming initiatives and environmentally friendly tourism opportunities. The bulk of the zonation is designated as transitional (100,000ha) and with much smaller buffer (24,426ha) and core (2,901ha) zones.

Two of the four biosphere reserves operate in a more urban context. Urban biosphere reserves are a more recent development. The fact that the model still operates effectively in such contexts demonstrates the utility and flexibility of the regime and its ability to provide protection for high value biodiversity areas even where significant development pressures exist. The Brighton and Lewes Downs Biosphere Reserve in England aims to protect a range of historical, cultural, heritage and archeological sites in the area, some dating back to the Neolithic period. The Mornington Peninsula and Western Port Biosphere Reserve near Melbourne has over 670,000 people living within the reserve. The biosphere reserve status provides additional protection for important sites that exist in close proximity to urban areas and helps to raise awareness of those high value sites. Urban Biosphere Reserves have smaller core zones and more substantial buffer and transition zones.

Key lessons

Biosphere reserves promote an integrated approach, where the development needs of local communities are a central component alongside the conservation of biodiversity. Designation of a biosphere reserve can assist in creating a regional identity and lead to increased opportunities for tourism and/or additional funding from government and non-governmental sources. Their effectiveness varies and is dependent on the model being applied in the right context and having broad community support. Reviews of the biosphere reserve network take place every ten years. Recent reviews have seen a range of Australian-based reserves focused around national parks removed from the framework. Conversely, reserves operating in more diverse, multiple land use contexts appear to be expanding and thriving.[23] This indicates that the model may be most suited to landscapes where competing land use needs coexist in proximity to each other – rather than a more conservation focused setting. It is in the management of this interface, where biosphere reserves are proving to be most appropriate and effective.

In Aotearoa New Zealand, biosphere reserves have been discussed in relation to a number of areas, from the Mackenzie Country and Marlborough Sounds to Taranaki and Waiheke Island. However, the country has yet to establish a biosphere reserve.

Schorfheide-Chorin biosphere reserve, Germany

8.3 IUCN Category V protected areas

Another type of protected area, that allows for significant human land and resource use, is Category V (5) IUCN Protected Areas. The objective in these areas is to safeguard landscapes of high ecological, biological, cultural or social character. The focus is on maintaining the relationship between people and nature, recognising that many 'lived in', working landscapes are important for nature conservation, particularly those having valuable habitats and rare species whose continued existence may depend on the survival of traditional forms of land use.[24] Category V landscapes can contain human settlements and typically provide for activities such as forestry and agriculture.

The United Kingdom has developed a range of models that fall within this category, including national parks, Areas of Outstanding Natural Beauty (AONB) and Heritage Coasts. Because the governance and implementation of each of those models is different, we have reflected on the differences between them, and the impact this has had on the robustness of the protection in place. These frameworks were recently subject to an independent review by the UK Government. 'The Glover Review' was published in 2019 and its findings and recommendations have been incorporated into our analysis.

United Kingdom national parks

The United Kingdom's national parks regime was established following a nationwide survey to identify particularly beautiful and relatively wild country. A broad values-based approach to landscape protection was taken employing a conception of 'natural beauty' that encompassed an area's historical character or heritage. This has enabled protection to be extended to include human-made features such as stone walls, barns and houses, as well as natural features. It also means that traditional farming and forestry practices seen as contributing to the identity of an area, are also protected and preserved. The result is that many national parks contain human settlements, much land is under private ownership, and farms, villages and towns are protected alongside the landscape and wildlife.

National parks are managed by independent statutory-based National Park Authorities under the Environment Act 1995. National Park Authorities perform all the functions of a local authority for the area. They are required to develop a Management Plan (setting out their objectives and policies), a Local Plan (like a district plan), and a Local Development Framework. They actively consider all planning applications and have the power to prosecute breaches. National parks are fully funded by central government.

What makes the model particularly interesting is the way in which the framework manages the competing pressures and uses. National parks have two core purposes; the conservation and protection of the natural beauty and heritage of the area, and promotion of recreational enjoyment and use by the public, as well as a *secondary* duty to foster economic and social wellbeing. A clear hierarchy therefore exists. Where additional conflicts exist, these are dealt with according to the Sandford Principle.

The Sandford Principle

If a conflict exists between conservation and recreation within a national park, the national park authority is required to attach greater weight to the purpose of conserving and enhancing the natural beauty, wildlife and cultural heritage of the area.[25]

Major developments (eg wind farms and large housing developments) are not prohibited, but are only permitted in national parks in exceptional circumstances. Their approval is governed by the Silkin Test.

The Silkin Test

Major developments in national parks:

- must be in the national interest;
- there must be no practicable alternative; and
- they must be built in a way that minimises any detrimental effects on the environment.

Areas of Outstanding Natural Beauty[26] and Heritage Coasts

AONBs are also selected for their beauty and for the outstanding quality of their landscape. Although they are generally smaller and relatively less wild than national parks, they are far more numerous. Governance of each AONB is in the form a non-statutory Joint Advisory Committee, made up of relevant stakeholders and local authority members. The Committee is responsible for developing a management plan and setting policies for the AONB. However, they are only advisory bodies. Planning and development control remains with local authorities and is implemented through local planning provisions. AONB management plans cannot override local plans and local authorities are only required to 'have regard to' the purposes of the AONB. AONBs also receive much less central government funding.

Heritage Coast designations are even more nebulous. Heritage Coasts are areas of coastline recognised for their natural beauty, wildlife and heritage. The objective of the designation is to maintain the character of undeveloped coast, protect its distinctive landscapes and improve public access and enjoyment. A Heritage Coast designation is non-statutory. Protection is provided for through agreements between local authorities, landowners and Natural England (the government's advisor for the natural environment in England) and is far less binding and robust.

Contrasted governance, protection and performance

When the Glover Landscapes Review considered the protection and effectiveness of these designations in 2018, the importance of empowering governance bodies within the planning system, and giving management plans statutory recognition became clear. The Review concluded that the national parks model provided far more robust protection than either AONB or Heritage Coast designations. This was attributed not only to their increased funding allocation, but also to their statutory basis, and the fact that national park plans could not be overridden by local plans and authorities.

Previous studies reinforce these findings. An earlier study of Heritage Coast areas found robust protection of inshore coastal environments was only achieved where additional marine designations, such as marine reserves, had been established.[27] In contrast, a 2014 study found that the quality of environmental protection afforded by a national parks designation was such, that it was serving to stimulate local economic activity and positively affect business performance in those areas, creating a number of additional synergies.[28]

Exmoor National Park, United Kingdom

Despite the historical success of the national parks model, the Glover Review also found that the pressures on parks had been steadily increasing. Movement towards more intensive farming practices, increasing urbanisation and the climate change crisis were all putting additional stress on the system. The Review noted that, despite the success seen in national parks, habitat decline and species loss continued nationally. On these bases, the Review recommended the national park framework should be both *strengthened and extended*.

The Glover Review recommended:

- The establishment of a centralised Natural Landscapes Service to help coordinate conservation boards; provide additional support, expertise and services; improve reporting and oversight (to achieve more consistency in approach); set more ambitious targets; and give them a bigger voice.
- A review of existing AONBs and redesignation of some as national parks. Remaining AONB governance bodies should be given a statutory basis and increased status in the planning system, stronger purpose provisions and more secure funding. Their name should be changed to 'national landscapes' to better communicate their value and importance.
- Additional areas be identified and added to the network to improve connectivity and linkages between sites in order to assist habitat and species recovery.
- A new, less complex financial model was needed.
- Greater diversity was needed on management bodies to increase representation.

Key lessons

The landscape protection mechanisms in place in the United Kingdom underscore the importance of:

- Empowering governance bodies within the planning system and ensuring that their plans and policies have statutory status over other plans and instruments.
- Providing strong and clear purpose provisions to guide decision-making and planning and, in particular, indicating the *relative status* of stated purposes. The Sandford Principle and Silkin Test are good examples of the sorts of mechanisms that can be utilised to provide additional security and clarity regarding the relationship between purposes.

- Language and terminology used in relation to protected landscapes signalling the status of a framework and fostering broader support. Names are influential. They symbolise and convey meaning. They tell a story and build identity, and consequently community.
- Providing for greater connectivity between protected areas and managing them as a network rather than as isolated islands.
- Ensuring protective frameworks are sufficiently resourced with secure and reliable funding mechanisms to enable long term and large-scale planning and projects to succeed.
- Ensuring membership of governance bodies refelcts the diverse communities they serve, in order to deliver inclusive and fair outcomes.

8.4 Urban areas and landscape protection

Cities are our biggest carbon emitters, and with more than half the people on the planet living in urban areas (and with this figure set to increase to two-thirds by 2050),[29] there is increasing interest internationally in green infrastructure and greener cities, as well as rewilding projects to increase the biodiversity and habitat in those settings. Studies show that even urban wastelands can make a valuable contribution to urban biodiversity, harbouring more species than other urban greenspaces.[30]

In an increasingly modified world, distinctions between 'wild' and 'not wild' are losing relevance. This is leading to a more nuanced understanding of, and provision for, biodiversity and landscape protection. Our remaining green and

Yorkshire Dales National Park, United Kingdom

semi-wild spaces are of increasing importance and need to be viewed as part of a wilderness continuum. 'Relative wilderness mapping' is becoming an important planning tool, especially when considering landscape protection in and around urban settings.[31]

Many of the world's most vulnerable areas in need of protection, now contain significant human settlements, and are frequently under private ownership. It is in these highly managed and mixed urban and rural contexts that interest in concepts such as urban national parks has sprung up. This is a relatively new concept and, as a model, it remains rather poorly articulated. The specific mechanisms by which such parks might be best realised, and their point of difference from other initiatives (such as biophilic cities or green infrastructure planning), still lacks clarity.

Urban national parks attempt to facilitate the rebuilding of connection between communities, biodiversity and wildlife conservation. It is often noted that people who engage with nature will be more likely to support interventions to protect it[32] and studies show that people and populations that have a positive relationship with wildlife and biodiversity also have higher wellbeing,[33] and develop healthier immune systems.[34]

> Our urban environments "could be wilder, full of nature, more beautiful and much more accessible" and "our cities could begin to surround themselves with woodlands and wetlands, alongside well-designed new development".[35]

Practical examples of this concept include the London National Park City and Sweden's Royal National City Park in Stockholm. In London, the National City Park's Strategy has set a raft of new targets, including the phasing out of fossil-fueled vehicles and a zero emissions target for road transport by 2050. Boroughs will also be required to achieve net gains in biodiversity, with the goal of increasing London's green cover by more than 50 per cent by 2050. Some commentators have been critical of the project, describing it as more of a rebranding exercise than anything new or novel.[36] However, what the initiative *is* attempting to do – is to change the narrative around cities and to articulate a different way of viewing sites, parks and wildlife that live within them. It also constitutes a concerted attempt to drive a 'rewilding', not just of existing green space, but of the urban environment.

Established in 1995, Stockholm's Royal National City Park, provides greater insights into what can be achieved in this setting, and the results are inspiring. Stockholm is the largest city in Sweden and the region is the largest metropolitan region in northern Europe, being both a national centre for research and development and the main driver of the national economy. The 2,700ha park abuts and is connected to the city and is viewed as providing 'ecosystem services' to the area; cleaning the air, dampening noise and supporting microclimate stability, recreation and recuperation from stress. It has become Sweden's most visited greenspace and has enabled greater protection of green and relatively wild areas in close proximity to a major urban centre. It is now possible to spot rare birds, butterflies and insects right in the city. The model is also held up for the high degree of community support and involvement it has achieved, and for the strong connection and working relationships it has created between public authorities, research partners and environmental NGOs.[37]

Green belts or green wedges?

Green wedges are increasingly preferred over green belts in urban contexts, because rather than circling and so encasing a city like a green belt does, green wedges provide a series of green corridors that extend outwards. From above, the green wedge model looks more like a spiral. The wedge shape is particularly useful, as it can more easily follow water courses and allows greater connectivity. These connected green spaces can then be used as the basis for developing trail networks. They also help ensure that all urban areas are in close proximity to significant green space. Conversely, the areas between the wedges can be utilised as transportation corridors and be extended to enable the city to grow and expand.

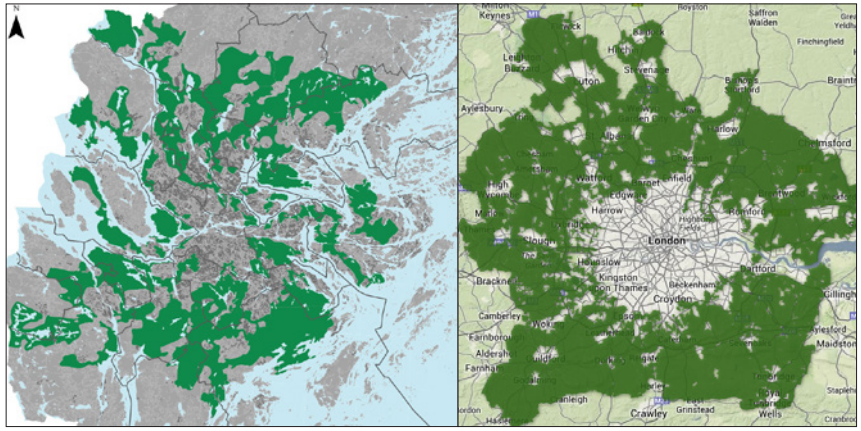


Figure 8.3 *Stockholm's Green Wedge compared with London's Green Belt*[38]

While green belts have been criticised for creating disparities through gentrification in areas closer to green space, and of overly restricting development and creating congestion, they remain an effective tool for preventing urban sprawl and can work well where a compact city approach is preferred. Green belts are an especially effective means of enabling towns and small village communities to preserve their character – so that they do not continually expand and merge into one another.

Supporting mechanisms

A range of support mechanisms can be employed in conjunction with more comprehensive landscape protection models and frameworks. The most common example is land trusts and conservancies. These have proven invaluable, especially in contexts where significant areas remain in private ownership. They are a common protection mechanism employed around the world. Land trusts may purchase land directly or fund other groups to acquire land, work to persuade conservation minded individuals to acquire land, and facilitate negotiations with other non-profit organisations or public agencies to acquire land. Conservation easements may also be employed to permanently restrict development and ensure the protection of the land's conservation values. Many land trusts also provide environmental education programmes and participate in local planning initiatives. The work and progress made by the UK's National Trust and the Land Trust Alliance in the United States is impressive. The National Trust is now Europe's largest conservation charity, and one of the biggest landowners in the United Kingdom, providing effective protection for extensive areas of countryside and coastline. Similarly, in the United States, Land Trust Alliance accredited trusts now steward over 22 million ha of land.[39]

Rewildling at Gelderse Poort, The Netherlands

The non-statutory Regionaal Landschaps model has been applied in Europe. Regionaal Landschaps are partnerships between stakeholders (farmers and conservationists) and regional authorities. They seek to promote regional character, nature recreation, education and management of landscapes. This is through acquiring, restoring and managing land and heritage buildings while also encouraging community consultation and cooperation on private land (such as working bees). Regionaal Landschap organisations provide private landowners with a number of tools to manage their land, ranging from advice to practical assistance. In some countries, they receive government funding, but most are resourced through members and charitable donations. Some have large membership bases (over 500,000) which provide increased funding capacity and can generate significant political lobbying power. While such partnerships can make a valuable contribution to landscape protection, it is important that such mechanisms are paired with a strong regulatory framework, in order to effectively protect a landscape from development.

The provision of subsidies and incentives to farmers through agricultural policy and land management schemes also impacts landscapes. Such schemes typically pay landowners to maintain their land in good condition or create environmental benefits (eg the EU's Common Agricultural Policy). While these regimes incentivise protection, they have been criticised for relying on voluntary efforts, and they often result in the easiest and least costly (rather than the most effective) responses to problems being adopted. Other commentators have raised concerns that subsidy schemes can sometimes operate to perpetuate unsustainable practices that would not be viable without additional financial inputs.

8.5 Bespoke landscape protection models

The practical reality is that there is frequently no 'one size fits all' model that can be easily applied to an area. Just as no two natural landscapes will ever be identical, the pressures and the environmental, social, economic and cultural context in any area will always be unique. Sometimes, a more bespoke (custom-made) approach will be preferable. We have investigated five different models of landscape protection which have been developed and applied to specific areas (see Figure 8.4). While none are perfect, all have been widely identified by commentators as successful in achieving what they set out to do – although those objectives vary widely.

All of the areas are important landscapes which include significant habitat, wildlife and natural resources. All additionally encompass a mixture of natural and human settlement areas where a range of development, agricultural, resource extraction, tourism, social and economic pressures exist.

Model	Landscape	Pressures	Approach
Oak Ridges Moraine (190,000ha)	Geological landform (ridgeline) containing the headwaters of 65 river systems and significant forest.	The area is highly flood prone. Five million people live nearby and 90 per cent of the land is under private ownership.	Overlay plan
California Coastal Commission & Conservancy (607,000ha)	1770km of coast encompassing significant land and marine areas. A high biodiversity area encompassing important sites for migratory species.	The area is at high risk from sea level rise and coastal erosion. There are significant populations so strong urban development and beach access pressures. There are also controversial offshore drilling operations.	Overlay regulatory authority + land trust/ conservancy
Adirondack Park (2.4 million ha)	Adirondack mountains and associated drainage basins and forests.	Half of the land is under private ownership and the area attracts over 12 million visitors a year. There are subdivision, forestry and mining pressures.	Overlay plan + regulatory authority
Tennessee Valley Authority (777,000ha)	Large river valley, with vast network of lakes and rivers that sits across seven states.	The valley is flood prone and in an historically impoverished region. Past oil and forestry extraction has left the environment degraded.	Overlay plan + regulatory authority (quango)
Hudson River Valley Greenway (1.2 million ha)	A natural watershed and national heritage area. The Hudson river flows through 13 counties.	A highly populated area, with significant agricultural areas and development pressures.	Co-operative management + land trust/ conservancy

Figure 8.4 *Summary of key features of bespoke landscape protection models*

Oak Ridges Moraine, in Ontario Canada, covers some 190,000 ha. It contains the headwaters of 65 river systems and its nutrient-rich soils support extensive old growth forests (approximately one third of the land is upland forest) and tall grass prairies, and provides important habitat to many at risk species. The pressures in the area are many. Not only is the Moraine located on the outskirts of a major city (Toronto), around 250,000 people reside within its boundaries, and *ninety per cent* of the land is under private ownership. The area's gravel deposits are also a significant source of materials for the aggregate industry. Protection is in the form of a Conservation Plan promulgated by the state government under the Oak Ridges Moraine Conservation Act 2001. Much like the biosphere reserve model, zoning provides for a graduation of areas with natural/core, linkage, countryside and settlement designations. These designations and their statutory basis have been highly effective at reducing urban sprawl and providing robust protection to environmentally sensitive areas.

> The vision is for "a continuous band of green rolling hills that provides form and structure to south-central Ontario, while protecting the ecological and hydrological features and functions that support the health and well-being of the region's residents and ecosystems."[40] *Oaks Ridge Moraine Conservation Plan*

This protection has been achieved even in the absence of a special purpose regulatory agency. The main challenge under this framework has been ensuring consistent implementation of the Conservation Plan across local municipality land use plans, and coordination of the Conservation Plan with the three provincial plans that exist in adjacent areas – with which there is a degree of overlap. In recent years, the government has aligned the review of the four provincial plans to address this problem and has worked to the increase consistency of terminology and definitions used across the various planning documents.[41]

A non-regulatory agency, the Oak Ridges Moraine Foundation, was established in 2002 with an initial grant of $US15 million from the provincial government to support protection of the Moraine. However, without formal statutory recognition and secured funding, the Foundation ran into financial difficulty and has now ceased its land acquisition and project grants programme. This has significantly reduced the amount of trail development and conservation work possible and underpins the fragility of non-statutory bodies that are not themselves sufficiently supported and resourced.[42]

Oak Ridges Moraine can be contrasted with the arrangements that are in place in California to assist with coastal management, where there is a strong oversight agency, but no purpose-built overlay plan. The California Coastal Commission is an independent, statutory based, quasi-judicial state agency responsible for ensuring compliance with the California Coastal Act 1976. Local authorities remain the primary planning bodies. However, they must prepare a Local Coastal Programme (including a land use plan) and the Commission reviews and approves the plans. The Commission is also responsible for issuing coastal area permits and has appellate authority over development approved by local government. It also directly regulates development from the high tide line out to the three nautical mile boundary.

This regime, and the legislation that underpins it, was forced through by a public initiative, similar in nature to a referendum. This was in response to controversial coastal developments including offshore oil drilling, nuclear power plants and proposed desalination plants, as well as concern to protect public access to beaches threatened by the emergence of large gated communities along the coast. This means that, from the outset, the framework had a strong social mandate behind it. Half of the Commission's membership is comprised of locally elected officials with the other half appointed from the public at large.

San Diego coast, managed by the California Coastal Commission

The Commission is supported in its work by the California Coastal Conservancy, a statutorily based state agency that purchases environmentally valuable coastal property and runs a number of coastal related projects. The Conservancy's purpose is to protect and restore the coast, increase public access and improve resilience to climate change. It is supported through state bonds and appropriations. Its 2019/20 budget was almost $US90 million.[43] The Conservancy also receives funds through innovative schemes such as the sale of specialty California state 'whale tail' car license plates.

The pressures in this area are significant. Recent research indicates that 25 per cent of conservation land along the coast and 60 per cent of the area's beaches and estuarine marshes, could be lost to climate change. Erosion is an ongoing problem. California also has the highest biodiversity in the continental United States. The Californian coast provides important nesting and pupping habitat, resting areas for migratory species, and feeding grounds for critical populations found nowhere else in the world. Much of the coastal area is highly urbanized and development-related pressures are also high. There is an urgent need for more contiguous blocks of land to be acquired in order to address many of these issues. However, provision of compulsory powers of acquisition remains too controversial.

Both the Commission and Conservancy have, in the past, been relatively well resourced and supported and this has enabled them to be responsive to the pressures that exist. In 2019, the Commission introduced an environmental justice policy to allow explicit consideration of the equitable distribution of environmental benefits and impacts caused by the loss of beach access in low-income and minority communities. The Commission has also officially recognised the role of human activity in climate change and adopted policy directions that require planning and permitting decisions to be driven by science.[44] In 2013, the Conservancy launched a climate-ready grant programme to assist smaller towns and cities in being more proactive in this arena. Since its inception, the grant rounds have been hugely over-subscribed.

Reductions in federal allocations and funding for climate change planning and environmental protection have made implementation more difficult in recent years. There have also been moves to override state decision making on offshore oil and gas drilling plans and permits. This demonstrates that even the most robust frameworks will always be, to some extent, vulnerable to changes at the broader political level. [45]

A contrasting landscape protection model is that of Adirondack Park[46] in New York State. This is interesting for a number of reasons. First, it combines features of the first two models discussed, with a strong statutory based governmental regulatory agency (the Adirondack Park Agency) that has robust planning powers. The Agency is responsible for developing plans for both public and private lands within the park. Second, it is a landscape protection model that operates on significant scale. The jurisdiction of the Adirondack Park Agency, which roughly corresponds to the Adirondack mountains, spans an area of some 2.4 million ha, making it the largest protected public lands in the contiguous United States. It contains five major drainage basins and thousands of lakes and rivers. Its forests and wetlands filter and provide clean water for communities as far away as New York City. These areas are home to a host of species including river otter, beaver, black bear, bob cat, moose and bald eagle. It also represents one of the oldest state reserves, being established in 1885, so has a long and proven track record.

Historically, strong governmental involvement served to ensure robust environmental protection and successfully prevented urban sprawl. However, with over 130,000 permanent residents, some 100 towns and villages and more than 12 million visitors a year, the development pressures are significant and increasing. From its inception, the Park was controversial. With over half the land under private ownership, the restrictions in place have effectively prevented many from developing their land, and this is increasingly articulated as a 'taking of private property'. These conflicts have had direct implications for the governance of the Park since appointments to the Agency are by the Governor with approval from the senate. The current Governor campaigned on an economy-centred platform and an impasse over new appointments now exists between the Governor and Senate. This has attracted criticism that appointments are now being made on the basis of politics rather than on the candidates' skills and expertise in the role.[47] Adirondack Park highlights how adversarial and polarized issues can become, when there is insufficient public participation, and the importance of maintaining social license. [48]

A further model is one that most New Zealanders will be familiar with – where the regulatory authority is a joint public-private government-owned corporation, or quango. The Tennessee Valley Authority was established as part of Roosevelt's New Deal legislation. By the 1930s, over exploitation in the valley had left large areas of it derelict, and the local population ranked as one of the poorest in the United States.[49] The framework put in place represents an attempt to address the multiple problems that existed in the area by taking a large-scale river basin approach to managing what is also an extensive and highly flood prone area. The Authority manages approximately 777,000ha which is one of the wettest and most biologically diverse areas in North America. It also encompasses 163 counties over seven states. The Tennessee Valley Authority Act grants the Board broad multi-resource conservation and regional planning powers, with the power to build its own projects. Environmental stewardship, economic development and affordable energy are the three core focuses of the Authority. This has meant that,

from the outset, it was recognised that the economic and energy related goals of the region were inherently dependent on a healthy resource base and required sustainable management to achieve.

The high level of federal funding and support (the Authority is run by a Board of Directors nominated by the President and confirmed by the Senate) underpinning this framework enabled management and projects of scale to be undertaken. Early projects transformed the river system. A series of lakes held back by 54 dams (nine major) was created. These have provided greater protection from flooding, made the rivers more navigable and are used to produce hydroelectricity. All river control, including dams and reservoirs, are owned, controlled and operated by the Authority, which supplies electricity to local power companies that serve over 10 million people. The environmental impact of these projects would have been significant at the time. Conversely, today these lakes provide habitat for a range of species (particularly fish and shellfish) and 82 sites (comprising 4,330ha) are classified as 'Natural Areas' in recognition of their important habitat, ecology and wildlife. Integrated development of the watershed's resources, combined with the Authority's institutional capacity, have been credited with transforming the Tennessee valley from one of the poorest regions in the United States into a region with a strong, diversified economy and healthy environmental base.[50]

Federal appropriations ended in 1998 and since then the Authority has been required to self-fund essential stewardship activities such as dam safety and maintenance, navigation, flood control and natural resource programmes. Following several major incidents the quality of the Authority's environmental management, and in particular its coal-based plants, has been highly criticised. This model highlights the importance of independent oversight and checks on compliance, particularly in the context of private, more market-based models.[51] To its credit, the Authority has acknowledged issues exist and it is now seeking to retire a number of its coal plants in favour of a focus on renewables.[52] Ironically, it is now the federal government that is actively opposing this closure. As in Adirondack, this demonstrates how vulnerable many of these models are to high level political and policy based changes.[53]

The final bespoke landscape protection model examined is one that is entirely cooperative and voluntary based. It serves to highlight what can be achieved outside of more formal management frameworks. The Hudson River Valley Greenway is one of the most ambitious river-based greenway efforts in the United States. Its objective is to preserve the rich scenic, natural, historic, cultural and recreational resources of the Hudson River Valley. The valley is a natural watershed with abundant fish species, bounded by spectacular mountains, and it has also been designated a national heritage area. The Greenway was founded out of concern that inconsistent land use laws adopted by the 13 counties (and over 250 communities) bordering the Hudson River were failing to protect the unique landscape.

The Hudson River Valley Greenway Act 1991 creates a process for voluntary regional cooperation and establishes two supporting agencies. The Greenway Communities Council prepares non-binding regional plans and promotes these to local government through a Greenway Compact programme. Counties and local authorities sign up to the regime and enter into a compact to implement consistent and collaborative planning and zoning. The Council also provides expert and technical advice, assists communities in the planning process, and has capacity to provide funding for local planning and projects that implement the plan. Complementary to this, the Hudson River Greenway Heritage Conservancy operates as a land trust and purchasing agency. Its aim is to preserve the "natural scenic beauty, heritage, natural resource values or conditions enhancing regional qualities of the Hudson river valley" and establish the framework for a greenway trail connecting cities and countryside within a green network.[54] The Conservancy produces a management plan for the Greenway and is charged with promoting the Valley as a tourism destination. The New York State's Environmental Protection Fund allocates approximately US$500,000 annually to the Conservancy towards a grant programme that funds trail design, planning and construction; rehabilitation and improvement projects; and education work.

While its voluntary nature is an obvious limitation, it also reflects the practical realities: the Hudson river flows through diverse communities, each with their own budgetary constraints and specific development and urbanisation pressures. The level of cooperation and capacity varies significantly and by necessity areas are at different stages in their adoption and implementation of the concept. The framework has been able to provide a path forward, where wide scale agreement from the outset was not possible, preserving local decision-making authority while expanding the opportunities for regional cooperation. It has also provided a vision and regional identity that encourages community participation. It has been remarkably successful at fostering partnerships to enable the acquisition of key land parcels.

Originally highly controversial within the business community (concerned at the impacts of possible development restrictions), opposition reduced as the Greenway developed into a marketable brand, and its tourism and economic value became self-evident. Today, 84 per cent of municipalities have designated themselves as 'Greenway Communities' and the Greenway provides recreational facilities and opportunities to hundreds of thousands of people who hike its many trails each year.[55]

Key lessons

Bespoke governance

The examples discussed above, all reflect landscape protection models operating at considerable scale. In particular, they highlight that:

- State government controlled regulatory authorities can be highly effective at implementing policy and planning at scale, but they can also be vulnerable to political changes that can impact funding and resourcing, as well as governance and direction.
- The quango model (Tennessee Valley Authority), where the regulator is a government owned corporation, can also produce impressive economic, as well as conservation outcomes, by providing the institutional capacity required to undertake projects at significant scale. However, in order to deliver robust environmental management, such entities must have clear statutory purposes, secure funding streams and sufficient checks and balances in place to ensure compliance and accountability.
- Where greater power is given to local authorities (Oak Ridges Moraine and Hudson Valley Greenway), capacity is often problematic, and while there may be enhanced local participation in planning, implementation and standards will be typically be more variable across areas, particularly where there is no overlay plan in place.

Whatever the governance arrangements are, it is important that they are crafted for the specific context in which they will operate. Where there is significant public support, an overlay regulatory authority that provides for public participation will usually be the best mechanism of delivering on the goals set (eg California Coastal Commission). However, more voluntary based frameworks (eg Hudson Valley Greenway), that allow individual local authorities to selectively opt in (or out), can be useful where wider agreement would not otherwise be possible. This approach provides a useful stepping-stone type of solution towards a more durable and effective model.

Supporting agencies

Most of the models discussed had an associated supporting agency that does not regulate but whose purpose is to purchase land, provide grants, run or fund projects, and undertake community and educational initiatives. The contribution of such agencies is unquestionable, but their long-term success and viability is variable. The most successful (eg the California Coastal Conservancy and Hudson Greenway Heritage Conservancy) are statutory based, with clear purposes and funding provision. Conversely, the decline of the Oak Ridges Moraine Foundation demonstrates how fragile such arrangements can be without these institutional support mechanisms.

Fostering regional identity

The examples canvassed also highlight the important role landscape protection frameworks can play in crafting a sense of community and regional identity. It is difficult to protect what people and communities do not value. The most successful models provide a vision, they make people aware of the natural heritage around them, and forge connections between landscapes and communities. They reflect local values and landscapes and foster a sense of regional pride. Where these synergies exist, they in turn also serve to enhance the resilience and longevity of the protections they put in place. Bespoke landscape protection models have the advantage of being developed to respond to the specific context, values and pressures at play.

A sense of identity, and increased community support and involvement, can be actively fostered in a number of ways including: broad and open public participation mechanisms within regulatory frameworks; provision of funding for community-led projects and educational initiatives; and purpose provisions that aim to support and improve public access to, and recreational use of natural areas. Many of the specific projects associated with the models discussed involve funding trail development and enhanced access to wilderness camping sites, beaches, lakes and rivers. These projects not only have tangible economic (tourism) benefits, but also highlight the value of these resources and landscapes to the local community.

Lastly, it is also important to remember that not all groups have equal access to green and recreational spaces, beaches and wilderness sites. The California Coastal Commission's adoption of an 'environmental justice policy' is an important initiative to highlight. It is an example of how express provision can be made to ensure such issues are considered and incorporated into the planning and policy making process.

8.6 Lessons learned from international models

All of the international models examined in this chapter are nested within a specific, often highly unique context. The aim was not to search for complete answers that could be simply transplanted to Aotearoa New Zealand, but rather to highlight patterns and trends, and discuss novel approaches and the potential benefits (or failings) they demonstrate. We have identified some of the lessons that might be learned and ideas that may be used to stimulate new conversations and thinking. Our investigation of overseas models has highlighted a number of

elements that have proven themselves to be particularly important for an effective landscape protection system.

Conflicting interests and pressures must be directly addressed

Landscapes serve many needs and mean many different things to different people and communities. They provide clean drinking water, food, energy and resources; a space for recreation; and a place for neighbourhoods, communities and wildlife to flourish. They are homelands and precious taonga to indigenous peoples; they provide us with a sense of identity, connection and heritage; and they support vital environmental services important to the health of our planet. They support not only the essential habitat necessary to protect biodiversity, they also connect to and support the social, cultural and economic bases of our communities.

Each of the models discussed in this chapter represents an attempt to respond to the range of conflicting interests and concerns, that invariably exist within any significant landscape, in order for protection of the values of those places to be achieved. They span the detailed agreements and co-governance arrangements with indigenous groups, the development of bespoke landscape protection models and the adoption of an inherently flexible protection models, such as biosphere reserves.

It is not always easy to balance the many competing needs and values that exist in these areas. As many of the models discussed in this chapter highlight, inherent tensions as well as synergies exist, and conservation and environmental protection often comes into conflict with personal property rights, as well as economic interests. Debates over land use can become highly polarised, particularly where issues have never been squarely addressed and frameworks are not flexible or sufficiently responsive. Whether a strong regulatory approach or a more cooperative and voluntary based regime is appropriate will depend on the specific pressures and context that exists in any area.

The need (and utility) of providing express recognition of the conflicting interests at play, and complex histories and relationships that exist, is especially underpinned in the discussion of biocultural models and indigenous peoples. This highlights the typically deeper equity issues at play in areas where significant and ongoing debate and conflict exists. That indigenous co-governance models are able to be developed in these contexts, demonstrates what can be achieved if clear lines of communication are kept open, and competing interests and claims squarely addressed. These models highlight the importance of having hard conversations, and being open to rethinking how things have been done in the past, to create new and unique responses to contemporary issues.

Buffer zones (such as those seen in the biosphere models) and transitional areas (as evidenced in Oak Ridges Moraine) have been useful mechanisms for allowing flexibility. Voluntary based opt in schemes (such as the Hudson River Valley Greenway and Land Trusts examples), especially in relation to private land-owners, have also shown some success. Providing for true power sharing relationships with indigenous groups and the establishment of specialist advisory groups, such as the Indigenous Circle of Experts in Canada, has also proved invaluable in ensuring indigenous voices are heard and that frameworks are inclusive.

Frameworks that are open and inclusive, and where the interests and perspectives at play have been clearly articulated and considered, are most likely to deliver the type of broad community acceptance that is necessary to ensure they are enduring.

An effective regulatory framework

What the models discussed also highlight is the value of having a robust regulatory framework at the heart of any environmental protection designation. The United Kingdom's national parks have a strong regulatory basis, with dedicated national park authorities and central government funding. In the UK context, they have been generally effective in protecting important landscapes from the negative impacts of development in comparison with AONBs. This is due to the weaker regulatory framework for AONBs, lack of central government funding and local authorities still making planning decisions.

Whatever the regime, the purposes of the regulatory framework need to be clearly articulated. Where purposes may be in conflict, guidance needs to be given as to the appropriate balance sought. The UK's Sandford test is an example of one such mechanism. Successful co-management regimes between government and indigenous actors also owe much of that success to an agreement on purposes and statutory direction. Without this clarity, implementation becomes difficult and impacts on outcomes.

Similarly, without powers to enforce protective provisions, inappropriate development is often enabled. Therefore, designation alone or an overlay policy without 'teeth', is unlikely to work. It is also key, that where multiple layers of planning exist, the relationship between plans is clear and there is an alignment in definitions and terminology.

Responsive governance

To be effective, the governance regime must also be fit for purpose. Frameworks imposed from 'outside or above' tend to generate adversarial and highly litigious environments that can undermine implementation. The correct approach is often difficult to strike and all models have their advantages and disadvantages.

For example, frameworks with a high degree of state involvement often enjoy increased capacity and resourcing, and high level government involvement helps ensure decisions are informed by the broader national interests that exist. However, without sufficient mechanisms for local input, they more easily fall out of touch with local concerns. They are also vulnerable to changing political support and direction, which can undermine large scale or long-term planning.

Conversely, devolved governance arrangements will be more responsive to the local context and community, but without sufficient policy and oversight mechanisms, there is an increased risk of local interest group capture and that inappropriate development (against the public and environmental good) will occur. Whatever the governance regime, its ability to achieve the correct balance is often facilitated by the detail – the clarity and robustness of the policy, processes, checks and oversight mechanisms that accompany it.

There is no question that the most successful governance bodies are those that have the capacity, community support and resourcing to adopt a proactive, problem solving approach. This support is what enabled the California Coastal Commission, in the face of rising conflict and opposition to projects and initiatives, to take measures to *increase* their community and stakeholder engagement in advance of projects. The Commission was able to respond, when public access to beaches became an issue in poorer neighbourhoods, by introducing a clear environmental justice policy to ensure that equitable distribution of environmental benefits and impacts of developments was being expressly considered. In contrast, agencies that lack sufficient funding, support, resources and capacity will struggle simply to keep a regulatory regime on track and more innovative responses to problems will usually not emerge.

Resourcing and funding

Not only is it important that governance agencies have adequate support, funding and institutional capacity to achieve the objectives of any framework – that funding must also be structured and guaranteed, to ensure that it is sufficiently reliable to support long term projects and planning initiatives. This is especially

Dorset AONB, England

important for the implementation of initiatives that are significant in size or scale. Environmental protection is often expensive, and in dealing with issues such as climate change adaptation, significant governmental support and financial contributions is frequently required.

The quality of management is one of the first aspects of a framework to fail without adequate funding. This is seen through a number of the case studies, including the Tennessee River Valley Authority, the Oak Ridges Moraine Foundation (which impacted on the amount of trail development and conservation work possible) and the AONB model in the UK. Conversely, well-resourced and reliable funding has been a cornerstone of the success across a number of the frameworks considered including the California Coastal Conservancy, Niagara Escarpment and Thaidene Nëné.

The importance of supporting agencies and mechanisms

Many of the most successful overseas models placed a particular focus on public education and involvement in landscape protection, providing opportunities for community action and engagement and cultivating a sense of local identity in the process. Non regulatory support agencies (including land trusts and conservancies) can often be useful to deploy in this role and several of the bespoke landscape protection models discussed operate in conjunction with such a body. However, to be effective in this role, they themselves must be properly supported and resourced.

Non-regulatory approaches such as public education, outreach and support programmes; providing technical assistance; and initiatives that respond to existing needs, can work well in fostering the protection of landscapes on private land. In providing these services, and fulfilling landowner needs, they are often able to form more robust partnerships. Their strength is that they play to what people, communities, organisations, business, industry and governmental agencies have in common rather than what separates them. Simply by focusing on commonalities, rather than differences, significant progress can often be achieved.

In all the overseas models that were investigated, getting community and especially landowner buy-in to the management approach, was a crucially important success factor.

Coordination and integration of landscape protection

No model for landscape protection operates in isolation. For example, in Ontario, large-scale watershed planning is undertaken and this requires the coordination of the Oak Ridges Moraine, Niagara Escarpment and Greenbelt plans. It is often by virtue of broader cooperation and connections that more coordinated large-scale landscape protection is achieved. More detailed purpose-built plans enable directed and contexualised responses, within a broader framework, while also providing the scaled planning, management and decision-making coordination necessary.

It is crucially important that landscape protection models are able to work in conjunction with other associated (and sometimes overlapping) planning and protection frameworks, and that where they sit within that broader scheme is clear. Supporting national policy direction will also be key. Unfortunately, it was not possible to delve into the intricacies of such planning relationships in detail in our review, but this is an aspect that must always be borne in mind when discussing any model or approach.

Relevance to Aotearoa New Zealand

At present there are no biosphere reserves in Aotearoa New Zealand, although the possibility for establishing these have been discussed in several contexts including the Mackenzie Basin, Marlborough Sounds and Waiheke Island. Aotearoa New Zealand also has relatively few bespoke landscape protection models. Examples that do exist include the Waitākere Ranges Heritage Area and Hauraki Gulf Marine Park, both which were investigated in the case studies. Much more work and investigation into the value and opportunities provided by such frameworks needs to be undertaken in order to take a more proactive (rather than reactive) approach to landscape protection.

In contrast, Aotearoa New Zealand does have a large number of co-management arrangements and several co-governance arrangements in place with iwi, albeit with limited scope and limited devolution of authority and decision-making powers. Many are the result of Treaty Settlement legislation. A range of new, exciting and highly innovative responses to landscape protection have been crafted in this context, including provision for the Whanganui River and Te Urewera to obtain legal personality. These initiatives have been held up internationally, by both indigenous peoples and environmental groups, as constituting significant progress towards increased and more holistic based environmental protection and as examples of much more authentic bicultural landscape protection models.

The progress being made in Canada underlines the value and importance of increased engagement with First Nations, not only for expanding the amount and quality of the landscapes under protection, but for resolving historical grievances and addressing some of the equity issues that arise from them. The establishment of the Indigenous Circle of Experts in Canada has been central to assist change in this area.

In the next chapter we draw on these lessons and those identified in earlier chapters, to identify a new approach to landscape management in Aotearoa New Zealand.

Reflooding drained wetlands in Cairngorms National Park, Scotland

ENDNOTES

1 Schuster, 2019, 1-6

2 The term 'fortress conservation' was coined by Brockington. See Brockington, 2002

3 This is the case whether the protection is explicitly provided for or simply exists as an understood goal. See Tran, Ban and Bhattacharyya, 2020

4 Canadian Federal, Provincial and Territorial Department for Parks, Protected Areas, Conservation, Wildlife and Biodiversity, 2018

5 Indigenous Circle of Experts, 2018, 5

6 https://www.thecanadianencyclopedia.ca/en/article/aboriginal-self-government

7 See case studies of Thomlinson and Crouch, 2012 and Gavin et al, 2018

8 Park Act, RSBC 1996, Clause 344

9 Gwaii Haanas Agreement 1993, 3

10 http://www.landoftheancestors.ca/

11 Nature United, 2019

12 The Park size was reduced by over 7,000km^2 to address industry concerns. For discussion of the negotiations and issues raised, see Thomson, 2019

13 Indigenous Circle of Experts, 2018, 84

14 Thomson, 2019

15 Blake, 2019

16 Taylor, 2015

17 Lyver et al, 2014, 102

18 See Lyver et al, 2014 and Ross et al, 2009

19 Lyver et al, 2014, 97

20 United Nations Educational, Scientific and Cultural Organisation, 1996, 3

21 Statutory Framework of the World Network of Biosphere Reserves 1996, Art 4

22 United Nations Educational, Scientific and Cultural Organisation, 2018

23 Bridgewater, 2019

24 https://www.iucn.org/theme/protected-areas/about/protected-areas-categories/category-v-protected-landscapeseascape

25 United Kingdom Environment Act 1995, s62(2)

26 They are designated under the Wildlife and Countryside Act 1981

27 Land Use Consultants, 2006

28 Silcock, Kieboom and Rayment, 2014

29 Xuan, 2020, 430

30 See Corlett, 2016, Jorgensen and Keenan, 2012 and Bonthoux et al, 2014

31 Müllera et al, 2018

32 Townshend et al, 2017, 206

33 See Mayer and Frantz, 2004 and Adjei and Agyei, 2015

34 See Li et al, 2008 and Rook G, 2013

35 Glover, 2019, 124

36 Townsend et al, 2017

37 https://wwf.panda.org/?204459/Stockholm-National-City-Park

38 Orveland, 2019

39 https://www.landtrustalliance.org/2018-annual-report

40 https://www.ontario.ca/page/oak-ridges-moraine-conservation-plan-2017

41 Watt, 2016

42 https://www.barrietoday.com/local-news/oak-ridges-moraine-greenbelt-foundations-join-forces-in-historic-move-2125168

43 Coastal Conservancy, 2019

44 https://www.coastal.ca.gov/climate/slrguidance.html

45 These changes have had a significant impact since 10 per cent of the Coastal Commission's budget is through coastal zone management grants to the state. Numerous new commentators have discussed these impacts, see for example Chase, 2017 and Phillips and Xia, 2019

46 https://adkreviewboard.com/mission/

47 Leon, 2019

48 For fuller discussion of the conflicts over the operation of the Park see Sandrow et al, 2018 and O'Donnell, 2015

49 Steiner, Young and Zube, 2014, 78

50 Miller and Reidinger, 1997, 7

51 See Environmental Integrity Project, 2009, 15-21

52 Allen at al, 2017

53 https://www.washingtonpost.com/national/health-science/tva-defies-trump-votes-to-shut-down-two-aging-coal-fired-power-plants/2019/02/14/6cc0c9f8-3077-11e9-86ab-5d02109aeb01_story.html

54 Schuyler, 2018, 168

55 https://hudsongreenway.ny.gov/news/hudson-river-valley-greenway-awards-232775-grants

A new landscape protection model

Mackenzie Basin

9.1 Pulling the threads together

Our review of landscape protection in Aotearoa New Zealand has highlighted the range of challenges that affect our landscapes. And although we have a range of methods in our toolbox to manage these challenges, they are generally underutilised. Here, we draw together the threads from the previous chapters to posit what a new landscape protection model in Aotearoa New Zealand might look like.

We began in Chapter 2 with an exploration of what landscape is in Aotearoa New Zealand. We concluded that a new system needs to be broad enough to include a range of world views including those stemming from te ao Māori and New Zealand European cultures, and it needs to more deeply recognise the deeply intertwined relationships between people and landscapes.

In Chapter 3 we reviewed some of the pressures on our landscapes. We concluded that pressures have increased markedly over the past two decades and that many of our landscapes are still in decline. The pressures are becoming more multifaceted, and they play out in different ways in different places. This means that any new landscape protection system needs to be robust in order to manage strong and increasing pressures. It needs to address a wide range of threats and be flexible enough to address the unique contexts of particular localities, potentially with bespoke solutions.

In Chapter 4 we reviewed the case law on landscape under the RMA. We charted the evolution of the concept of landscape, which is recognised as being much more than aesthetics, and includes biophysical, sensory/perceptual and associative values. Case law has also recognised that all our landscapes have been modified to some extent, so that in order for a landscape to be 'natural' and worthy of protection, it does not need to be pristine. The Courts have increasingly recognised and protected Māori cultural landscapes but have been reluctant to protect heritage landscapes. Neither cultural nor heritage landscapes are explicitly referenced in the RMA nor granted adequate protection.

Coastal landscapes have greater protection than those elsewhere in the country, solely because there is stronger national direction, the mandatory NZCPS that applies to the coast. Policy within the NZCPS is strong and directive, requiring avoidance of adverse effects where high values are at stake. This indicates the potential usefulness of strong protective national direction on landscape management. We concluded that, reflecting on case law, planning documents need to have clear and directive provisions in them if landscape values are to be guarded.

In Chapter 5 we charted the evolution of the landscape profession and approaches to landscape assessment. Various methodologies have been developed over the years, in a rich evolution of practice. There has been a more recent convergence of approach, with the NZILA developing new Landscape Assessment Guidelines with input from the Māori landscape architect collective Te Tau-a-Nuku. These show promise in developing a bicultural approach that encompasses both te ao Māori and non-Māori worldviews which could help inform legislative and policy development in this area.

In Chapter 6 we reviewed current landscape protection tools. We noted that the lack of RMA national direction on landscapes outside the coastal environment has resulted in little consistency between councils in their approach to managing landscape values. We highlighted the often unintended consequences of other national policy that can result in either positive or negative landscape outcomes. At a council level, we observed that it is unclear what the respective roles of regional councils, territorial authorities, iwi and hapū are in landscape protection under the RMA, but note that the heavy lifting is often left to small impecunious district councils and under-resourced tribal authorities.

In general, we lack clear landscape policy under the RMA. We suggested, in the context of the current legislative regime, that a NPS on Landscape could usefully set out at what scale landscape should be assessed, the assessment methodology to be used and the roles of the respective councils and their planning documents in providing protection. It could also establish environmental bottom lines for landscapes outside the coastal environment.

We also highlighted the potential role of regional policy statements in providing stronger direction for the identification and protection of important landscape values in each region. Sub-chapters within regional policy statements could then detail management considerations for specific landscapes. Within island landscapes, in particular, it is important that integrated planning spans the land and sea and embraces *sea*scapes as well as *land*scapes.

We discussed the role of iwi planning documents and the strong contribution they can make to the identification and protection of Māori cultural landscapes, particularly if followed by co-designed RMA policy and plan provisions. Ngā Tūtohu Whenua (cultural landscapes),[1] cultural layers, precincts and the like can be used to recognise mana whenua values.

Outside of the RMA, we discussed other statutory management frameworks that impact our landscapes. In the high country this includes the discretionary consenting regime for Crown pastoral leasehold land. The legislative framework for such consenting is particularly weak but there are promising proposals to address this through amendments to the Crown Pastoral Land Act.

We also looked at the concession system, which brokers commercial access to public conservation land, and noted that the legislative framework is outdated and in need of review in order to ensure positive conservation outcomes.

We highlighted the exciting innovations that have been achieved through Treaty settlements with novel approaches to managing natural landscapes and features including creating legal personalities for them and establishing co-governance bodies to oversee their management. We noted the potential for utilising these approaches more widely in a new landscape guardianship system for Aotearoa New Zealand.

Finally, in Chapter 6, we profiled the importance of brokers such as well-resourced community trusts. We noted the critical role of collaboration between central government agencies, local government, mana whenua, philanthropists, landowners and the community more generally in achieving positive landscape outcomes, particularly when undertaking restoration initiatives.

Chapter 7 brings together the rich lessons from the five in-depth case studies undertaken for this project. Foremost, the case studies highlight the difference between localities, where different pressures played out in different contexts on different landscapes. They strongly impressed upon us that a 'one size fits all' approach is unlikely to be successful and consideration should always be given to bespoke approaches. However, there were some key themes that ran across them all.

The case studies showed that clear and directive regulation is required when there are strong economic drivers for land use change which degrades cultural and landscape values, whether it be urban intensification and sprawl, agricultural

Kenepuru Sound

intensification or forestry. Voluntary measures or highly discretionary consenting systems will simply not work on their own in this situation.

However, where the economic drivers favour restoration, such as potentially on parts of the Banks Peninsula, supportive measures and incentives can be more effective than regulation.

Local legislation can be very effective in creating strong policies to protect a specific landscape, but the values to be protected need to be clearly set out in the legislation, and they need to be prioritised where values conflict. Such legislation also needs to have close statutory linkages with the RMA and other relevant national legislation to ensure effective implementation. We found a clear difference between the local legislation applying to the Waitākere Ranges on one hand, and the Hauraki Gulf on the other, with the former being much more effective for the reasons we have outlined.

Tourism can be a positive or negative force for landscape protection. We encountered communities that had been overwhelmed by the sheer number of tourists, with inadequate infrastructure in place, and too little of the economic benefits flowing to the local community and people managing the land. But we also encountered tourism operators who were proactively engaged in predator control and restoration work and landowners who were generating income from on-site tourism enterprises supported by on-farm nature restoration work. It was clear that careful planning and management of the industry is required. Destination management has the potential to help encourage 'slow' tourism and deeper engagement with the country's landscapes and would likely be more effective if linked to regulatory plans under the RMA and Conservation Act.

The management of public conservation land can have a significant impact on landscape values and local communities, particularly when it is the dominant land tenure in the area. When this is the case, alternative management approaches could be explored, including co-management arrangements with mana whenua, particularly when DOC lacks the capacity to care for the land properly.

Policy developed through a narrow lens can often have negative unintended consequences on landscapes. In particular, we found this in the way regional water policy affected the Mackenzie Basin and forestry policy is affecting Banks Peninsula. Alignment between government agencies and their policies is critical if positive outcomes are to be achieved. In many areas we found multiple policies being deployed, all driving in different directions.

Any system needs to be agile and able to respond quickly to emerging pressures. Otherwise a 'gold rush' situation can occur, when much landscape degradation occurs before rules can be put in place to stop it. This is what happened in the Mackenzie Basin with PC13. The system also needs to be robust, with strong checks and balances over council and government agency performance embedded within the system, so that remedial action is taken in a timely fashion when agencies fail to perform their statutory duties. In addition, effective monitoring, compliance and enforcement of the rules is critical.

From the case studies we learnt about the power of collaboration in bringing together multiple parties to achieve common ends. Community social entrepreneurs, such as community leaders and charitable trusts, often play a critical role in brokering such collaborative efforts. We also learnt about the important role of strong community advocates for landscape protection.

It is clear that there needs to be adequate resourcing for the landscape management system if it is to work effectively. Leaving management of the country's most iconic landscapes to small impecunious councils is a recipe for disaster. There should be central government funding for the management of landscapes of national significance. In addition, positive community initiatives need to be supported and resourced. This is now happening more through the likes of Predator Free 2050 and Jobs for Nature which is a positive development.

In Chapter 8 we reviewed a wide range of international models for landscape protection. We drew a number of lessons from these that might help craft an effective landscape guardianship model for Aotearoa New Zealand. Many of these lessons reflect what we also learnt from our five case studies.

The bicultural models we examined in Australia and Canada were instructive in highlighting the need for highly contextualised solutions and true power sharing relationships coupled with resourcing and capacity building. This has similarities

Banks Peninsula Track

with the approach taken in recent Treaty settlements in Aotearoa. Indeed, the international survey reinforced for us that the Te Urewera Forest and Whanganui River post-settlement approaches here are world-leading in their innovation.

It was clear from the international models, which traversed different countries with greater population and development pressures than here, that there are conflicting needs, values and threats in any landscape. These need to be recognised and directly acknowledged even if they cannot be resolved. This necessitates a management framework that is open and inclusive and provides for a range of approaches and tools.

There is no question that the most successful governance bodies are those that have the capacity, community support and resourcing to adopt a proactive and adaptive approach. In contrast, agencies that that lack these will struggle to even keep a regulatory regime on track, and more innovative responses to problems will usually not emerge. Resourcing must be sufficiently reliable to support long term planning and projects.

It was clear from the international examples (reinforcing our conclusions from the local case studies) that there needs to be a robust regulatory framework at the heart of any landscape protection system. This was demonstrated by the comparison between the United Kingdom national parks which have generally been effective and AONBs which have not. The national parks have a strong regulatory basis, with dedicated national park authorities and central government funding whereas there is a weaker regulatory framework for AONBs, less funding and local authorities still making planning decisions.

Whatever the regime, the purposes of the regulatory framework need to be clearly articulated and, where purposes conflict, it needs to be clear which takes priority. Similarly, without powers to enforce protective provisions, landscapes are often compromised and inappropriate development is often enabled. Therefore, designation alone or an overlay policy without 'teeth' is unlikely to work.

The regulatory regime needs to be supported by initiatives to involve landowners, mana whenua and the public in landscape management and restoration, providing opportunities for community action and engagement and cultivating a sense of local identity. Land trusts and conservancies have usefully played this role in a number of the international models, as similar organisations have in our Aotearoa New Zealand case studies. It is notable that, in all the successful international models we reviewed, getting indigenous, community and especially landowner buy-in to the management approach was crucially important.

While our context is of course different, there have been instructive lessons from our international comparative analyses.

9.2 Is the currently regulatory framework adequate?

When considering a new landscape protection model for Aotearoa New Zealand it is pertinent to query whether the current regulatory framework is up to the task. Our case studies highlighted that, where there are intense development pressures, the current system is not working well. We saw this most vividly demonstrated in the Mackenzie Basin, where the introduction of irrigation and intensification of agriculture, has significantly degraded landscape and biodiversity values despite a regulatory system intended to protect it. Demonstrably, the regulatory system failed when put under pressure.

One of the key problems has been weak regional and district plans. For example, until recently, agricultural intensification was a permitted activity on most of the Mackenzie Basin floor. And there is often too much discretion left to council officers, resulting in death by a thousand cuts as we saw on Waiheke Island. There has also been inconsistency between the approach taken to landscape in different areas with, for example, 97 per cent of the rural land in the Queenstown Lakes District being identified as an ONL and only 24 per cent on Banks Peninsula.

More recent Māori cultural planning provisions, such as those included in the Auckland Unitary Plan for Te Akitai Waiohua, show promising potential. However, this level of protection of cultural landscapes should be the norm rather than the exception.

The current system also lacks effective oversight when agencies fail to adequately fulfil their statutory functions. This means there is no certainty that national direction, or the provisions of the RMA itself, will be properly applied by councils. For example, our Banks Peninsula case study highlighted Environment Canterbury's failure to give effect to the NZCPS, which came into force a decade ago (although a review is finally underway). The Waitaki District Plan is still failing to give effect to the Canterbury RPS and section 6(b) of the RMA in terms of the protection of landscapes in the Mackenzie Basin.

In addition, there is poor integration between landscape protection under the RMA and other management and policy regimes. In the Mackenzie Basin we saw discretionary consents granted by the Commissioner of Crown Lands under the Crown Pastoral Land Act authorising major agricultural intensification. In the Banks Peninsula case study we saw the likely impact of the emissions trading scheme and the One Billion Trees programme in incentivising exotic plantation forestry in preference to indigenous regeneration with its multiple benefits.

Finally, rules can be effective at stopping things happening but cannot generally make things happen, and many landscapes need active management if their values are to be protected and enhanced.

We think Aotearoa New Zealand's landscapes deserve better.

9.3 Key elements of a new landscape protection model

Our investigation of overseas protected landscape models, and insights from our five case studies, indicate that there are four key elements of a national landscape protection system that need to be considered.

The first element is the key role of mana whenua in landscape management and restoration. The case studies demonstrated the strong leadership role that mana whenua already play in achieving improved landscape outcomes in many places. Any new landscape model needs to give effect to Treaty principles and enable the relationship of mana whenua with their ancestral lands to be better recognised and strengthened on a landscape scale. This can be achieved through moving towards a bicultural partnership model for landscape management.

The second element is the role of land managed for conservation purposes such as public conservation land managed by DOC and regional park land managed by regional councils. In some landscapes, public conservation land comprises a large part of the landscape (such as on Aotea Great Barrier Island), whereas in other landscapes (such as Rakino Island) there is very little. Where such conservation land is lacking, strategic land acquisition by government and others can help to establish a strong conservation core within the landscape. This has happened with the Tū Te Takiwhanoa Drylands within the Mackenzie Basin, where DOC recently acquired an additional 11,800ha of land to provide a stronger core of protected land on the Basin floor.[2]

Māori and privately-owned land managed for conservation purposes can also make a significant contribution to landscape restoration and enhancement. We saw this with the work of Koukourārata Rūnanga in replanting thousands of indigenous plants on tribal land around local streams. Privately-owned Hinewai Reserve changed the thinking of farmers and others about land management on the Banks Peninsula. Similarly, Windy Hill Sanctuary showed what could be achieved on Aotea Great Barrier Island using local initiative. Covenanting can also played a strong role in protecting private land for conservation purposes, particularly if the tool is used strategically to create corridors and refuges for indigenous species.

Hinewai Reserve, Banks Peninsula

Such public and privately-owned conservation land can provide a core area for pest control and regeneration work within the landscape. It can provide a sanctuary for plants and animals which do not thrive in a working landscape. We saw the importance of this in the Mackenzie Basin, where endemic plant species are still being lost, even under low intensity grazing regimes. When privately owned, such land can also generate an income, through mechanisms such as eco-tourism, honey production and carbon credits.

The third element is effective protective regulation for significant working (usually agricultural) landscapes to ensure that incompatible activities, intensification and harmful land use changes do not occur. Our case studies and international review highlighted the need for strong regulation to head off incompatible activities when there are strong economic drivers for change. We saw this particularly on Waiheke Island and the Waitākere Ranges with creeping urban development and in the Mackenzie Basin with agricultural intensification. A similar situation is emerging on the Banks Peninsula with forestry, which is why it is particularly concerning that regulation of forestry has recently been considerably weakened on the Peninsula.

Exotic forestry in Peraki Bay, Banks Peninsula

The fourth element of a landscape protection system is creating an effective partnership with landowners and others to support and encourage sustainable land use and management. This can be fostered through the adoption of measures such as education, research and provision of funding and expertise. Community trusts and other not-for-profit entities can play a significant role in supporting landowner efforts. They can help broker agreements, coordinate effort and bring people together. We saw this with the BPCT, Waiheke Collective and Mackenzie Country Trust amongst many others. Such entities can also help pursue research into more sustainable land management models for the area, innovative pest control methods, use of economic incentives such as the emissions trading scheme and the like. They can help to educate landowners and the public about the values in the landscape and create new opportunities to engage with it, as shown by the work of the Rod Donald Banks Peninsula Trust and Te Pātaka o Rākaihautū Banks Peninsula Geopark Trust.

In the Te Manahuna Mackenzie Basin case study we floated the idea of creating a new legislative tool in the form a Heritage Landscape Order or similar in order to provide an extra layer of regulatory protection for important landscapes. We discuss this proposal in more detail in section 9.5. This would provide a longer term solution to the challenge of providing effective and long-lasting landscape protection.

Since developing those recommendations, the report of the RMR Panel, has been released. This contains proposals for a significant strengthening of the current resource management system which provide a more immediate option for strengthening the regulatory framework around landscape management. We explore this option in more detail in section 9.4.

9.4 Prospective RMA reforms

The Government's proposed reform of the resource management system provides a significant opportunity to improve the way we currently provide for landscape protection. The RMR Panel, which reported to the Minister for the Environment in June 2020, has recommended fundamental changes to the current system. This includes replacing the RMA with new legislation and promulgating a new Act to provide for strategic spatial planning at a regional level (see spotlight). The RMR Panel's recommendations are providing the basis for the Government's reform programme which is being rolled out during 2021 and beyond.

We analyse the key recommendations in the context of landscape protection below and identify how they could serve to improve landscape management and how they might be strengthened to this end.

A spotlight on the Resource Management Review Panel's key recommendations

New legislation

- Create a new Strategic Planning Act and require the preparation of regional spatial strategies encompassing both land and the coastal marine area. These strategies are to align functions across other statutes, including the new Natural and Built Environments Act, the Local Government Act, the Land Transport Management Act 2003 and the Climate Change Response Act 2002.
- Repeal the RMA and replace it with a new Natural and Built Environments Act with a revised purpose and principles which signal a shift from managing environmental effects to achieving positive outcomes.
- Enact a dedicated Managed Retreat and Climate Change Adaptation Act, which would provide for managed retreat and for the establishment of a climate change adaptation fund.

Māori matters

- Require decision-makers to give effect to the principles of Te Tiriti o Waitangi.
- Incorporate the overarching concept of Te Mana o te Taiao in the purpose statement of the new Natural and Built Environments Act.
- Establish a National Māori Advisory Board to monitor the performance of central and local government in giving effect to Te Tiriti and provide for an integrated partnership process between mana whenua and councils.

Policy and planning

- Require national direction to be made on a range of core matters, and combine such direction into a coherent suite of instruments that clearly resolve conflicts and relationships between them.
- Require the establishment of environmental bottom lines and targets.
- Reformulate existing RMA plans into combined unitary plans for each region, reducing the 100 or so current plans to just 14.
- Reform the planning process, including the establishment of joint planning committees comprising regional council, territorial authority and mana whenua representatives.
- Require an audit of plans by the Ministry for the Environment before they are notified.

Resource consenting

- Alter how the notification framework operates, including removing the "no more than minor" threshold for notification of consents.
- Remove non-complying activity status.
- Provide an alternative dispute resolution pathway for minor matters.
- Strengthen the overall role of the Environment Court.
- Provide more flexibility to review existing resource consents to create a more responsive system.

Oversight, monitoring, compliance and enforcement

- Establish a nationally coordinated environmental monitoring system led by the Ministry for the Environment.
- Expand the role of the Parliamentary Commissioner for the Environment to provide a stronger auditing and oversight role of the resource management system.
- Establish regional hubs for compliance, monitoring and enforcement.
- Strengthen offences and penalties for non-compliance.

Other matters

- Strengthen the framework for water conservation orders.
- Provide for greater use of economic instruments to drive behaviour change.

Role of regional spatial strategies in landscape protection

The RMR Panel proposed a new Strategic Planning Act "as the key mechanism for improving strategic integration across the resource management system". It is to be designed to help improve strategic integration at a regional level across multiple statutes, functions, outcomes and agencies.[3]

The Strategic Planning Act would require spatial strategies to be developed for each region, and they would encompass land, freshwater and the coastal marine area. This means that they would encompass seascapes as well as landscapes. They are to provide a long term view, setting a strategic direction for at least the next 30 years and cover a wide range of matters including:[4]

> long-term objectives to improve the quality of the natural and built environments, provide sufficient development capacity, promote Māori interests and values, promote the sustainable use of rural land, protect historic heritage, address natural hazards and climate change.

The regional spatial strategies are to be developed jointly by central government, local government and mana whenua through consensus decision-making. This, importantly, enables mana whenua to participate in spatial planning as partners and "to better reflect Te Tiriti partnerships and incorporate mātauranga Māori knowledge".[5] It should enable Māori values associated with specific landscapes and Māori cultural landscapes to be more fully reflected in the planning documents.

The RMR Panel proposes that regional spatial strategies would include a range of environmental matters (alongside identifying the location of future development capacity and infrastructure) such as "regionally significant ecological areas, landscapes and recreational space that should be protected or enhanced", "areas of historic heritage values and areas of significance to mana whenua that should be protected and enhanced", "areas where significant change in land use is required to reduce impacts of land use and development in lakes, rivers, wetlands and the marine environment", "areas for enhancement and restoration, such as wetlands

Mussel farm near Coromandel

and green corridors" and "areas that may be affected by climate change or other natural hazards, and measures that might be necessary to address such issues".[6]

Therefore, as proposed by the RMR Panel, such regional spatial strategies would have the ability to address a range of environmental matters at the regional level including landscape change, protection and restoration. They would be useful mechanisms through which important landscapes could be identified, protected and enhanced in collaboration with Māori and communities at a regional level.

The RMR Panel recommended that central government should have the ability to develop a 'national priorities statement' under the Strategic Planning Act which would "signal its intention to address specific nationally significant issues through regional processes." This would be used to set out "particular nationally significant issues central government wishes to resolve at a regional level" amongst other things. Therefore, a national priorities statement could be a mechanism through which central government sets out its long-term national priorities for landscape protection.[7]

Regional spatial strategies are to be accompanied by 'implementation agreements' which will include more detailed planning for "certain infrastructure or environmental remediation projects" and apportion funding responsibility between central and local government. These could then be linked to the budgeting process for each government body, thereby helping to ensure that funding is made available to implement the regional spatial strategies. They could include funding provision for the protection and restoration of important landscapes.

The RMR Panel's recommendations also address the potential linkage between regional spatial strategies and other resource management legislation through which the provisions of the spatial strategies would primarily be implemented. Our landscape case studies indicated the importance of making a close link between policy and the regulatory tools needed for effective implementation. The Panel proposed that regional spatial strategies should be "consistent with" the purposes of the new Natural and Built Environments Act (NBEA) which will replace the RMA, Local Government Act and Land Transport Management Act. It also proposed that they be consistent with national direction including NPSs, NESs, the national adaption plan and government policy statements on land transport and housing and urban development. This highlights the importance of having a strong NPS on landscape in place to guide the development of regional spatial strategies.

In turn, plans developed under the NBEA, the Local Government Act and the Land Transport Management Act would need to be consistent with the regional spatial strategy.[8] This should help ensure that alignment runs both ways. Such plans include long-term plans and annual plans which incorporate local government budgets, and this could help to ensure that local government expenditure is aligned with the regional spatial strategy's provisions on landscape protection and restoration.

The scope of the RMR Panel's review was limited to a primary focus on the RMA and its interface with the Local Government Act, Land Transport Management Act and Climate Change Response Act rather than the resource management system as a whole. The Panel's recommendations therefore relate only to these pieces of legislation. Our landscape case studies indicate that the scope needs to be broader in order to more fully integrate across statutory frameworks impacting on landscape matters.

Rangihoua Pā, Bay of Islands

The Strategic Planning Act should also apply to the Crown Pastoral Land Act, which applies to the management of Crown pastoral leasehold land and the Conservation Act which applies to the management of public conservation land amongst other things.

Overall, the proposed Strategic Planning Act could make a significant improvement to the way important landscapes are managed in Aotearoa New Zealand. However, in order to achieve this, it needs to be drafted to specifically require the identification of significant landscapes at a regional level, and provision for their protection and restoration. The national priorities statement needs to address landscape matters. The new Act should also provide for implementation agreements to address funding for the management and restoration of landscapes. And its scope should be expanded to include the Crown Pastoral Land Act and Conservation Act in the first instance.

Role of Natural and Built Environments Act in landscape protection

The bulk of the RMR Panel's recommendations focus on replacing the RMA with new legislation provisionally titled the 'Natural and Built Environments Act'. One of the driving forces behind the perceived need to replace the RMA was its failure to protect the natural environment alongside inadequate recognition of urban development matters. Identified issues with managing the natural environment included lack of national direction, inadequate resources at central and local government levels to implement the legislation, the application of an overall broad judgement approach which weakened environmental limits, and a legislative purpose which failed to address environmental restoration.[9] The Panel's report also notes the failure under the RMA to enable mana whenua to engage meaningfully in the resource management system.[10] All these issues have affected landscape protection.

The proposed NBEA seeks to create greater alignment between te ao Māori and non-Māori world-views and approaches to resource management through incorporating the concept of Te Mana o te Taiao in the purpose statement of the new Act. It has a much stronger emphasis on achieving positive outcomes (rather than primarily mitigating adverse effects). This is through the establishment of a series of environmental limits, outcomes and targets. As the RMR Panel explains in its report, relying on limits alone risks creating a 'race to the bottom', whereas setting outcomes and targets shifts the focus to continuous environmental improvement:[11]

> ... where a healthy and flourishing environment is sought, rather than one that can merely endure human modification. Outcomes are intended to be high-level enduring goals reflecting a desired future state. Targets are time-bound steps for improving the environment and moving towards achieving outcomes.

The RMR Panel's proposed drafting of a new Part 2 identifies a series of outcomes to be provided for the natural environment, built environment, tikanga Māori, rural, historic heritage and natural hazards and climate change (see below). Unlike the RMA, which created a hierarchy between section 6 and section 7 matters, there is no priority given to this list of outcomes. Any conflicts are to be reconciled through national direction, or failing that, through the provisions of the combined plans.

The outcomes include specific reference to the "protection and enhancement" of "outstanding natural features and outstanding natural landscapes" thereby largely carrying over the provisions of section 6(b) of the RMA but with some important changes. That section did not refer to enhancement. It also contained a reference to protection "from inappropriate subdivision, use and development" and this qualifier has been removed in the revised proposed drafting. There is a direct reference to "cultural landscapes", and the protection and restoration of the relationship of iwi, hapū and whanau with them, in proposed section 7(j). This should raise the prominence of cultural landscapes within the resource management system. The revised drafting also includes new references to the enhancement and restoration of ecosystems and restoration of viable populations of indigenous species. These are all positive changes.

The refence to protection of "historic heritage" has been brought over from the RMA, with the same definition of the term, but with a new qualifier of "significant" replacing the earlier qualifier of "from inappropriate subdivision, use and development". This seems reasonable. However, no direct reference to heritage landscapes is proposed, which may perpetrate the lack of recognition of such landscapes in places such as Ihumātao as profiled in section 4.3 above. Consideration should be given to including heritage landscapes in this section either through direct incorporation into section 7(p) or through amending the definition of 'historic heritage' to explicitly include heritage landscapes.

The RMR Panel proposals also remove any reference to 'amenity values' which was included in section 7(c) of the RMA as the "maintenance and enhancement of amenity values". Instead, there is a broader reference to "enhancement of features and characteristics that contribute to the quality of the natural environment" in the new proposed Part 2. Amenity values are an important component of landscape in Aotearoa New Zealand. As noted in Section 4.4 above, amenity landscapes are significant landscapes, but those that are not quite natural enough to qualify as an ONL under section 6(b). They often form a buffer between ONLs and landscapes with no particular values meriting management or urban areas. We consider that reference to amenity landscapes, and the need to carefully manage them, should be included in the new proposed Part 2. A NPS on landscape (or its equivalent) could then follow to flesh out the policy parameters and obligations.

Section 7 Outcomes (as proposed by the RMR Panel with our proposed changes in bold)

To assist in achieving the purpose of this Act, those exercising functions and powers under it must provide for the following outcomes:

Natural environment

(a) enhancement of features and characteristics that contribute to the quality **and amenity** of the natural environment;

(b) protection and enhancement of:

  (i) nationally or regionally significant features of the natural character of the coastal environment (including the coastal marine area), wetlands, lakes, rivers and their margins:

  (ii) outstanding natural features and outstanding natural landscapes:

  (iii) areas of significant indigenous vegetation and significant habitats of indigenous fauna:

(c) enhancement and restoration of ecosystems to a healthy functioning state;

(d) maintenance of indigenous biological diversity and restoration of viable populations of indigenous species;

(e) maintenance and enhancement of public access to and along the coastal marine area, wetlands, lakes, rivers and their margins;

Built environment

(f) enhancement of features and characteristics that contribute to the quality of the built environment;

(g) sustainable use and development of the natural and built environment in urban areas including the capacity to respond to growth and change;

(h) availability of development capacity for housing and business purposes to meet expected demand;

(i) strategic integration of infrastructure with land use;

Tikanga Māori

(j) protection and restoration of the relationship of iwi, hapū and whanau and their tīkanga and traditions with their ancestral lands, cultural landscapes, water and sites;

(k) protection of wāhi tapu and protection and restoration of other taonga;

(l) recognition of protected customary rights;

Rural

(m) sustainable use and development of the natural and built environment in rural areas;

(n) protection of highly productive soils;

(o) capacity to accommodate land use change in response to social, economic and environmental conditions;

Historic heritage

(p) protection of significant historic heritage and **heritage landscapes**;

Natural hazards and climate change

(q) reduction of risks from natural hazards;

(r) improved resilience to the effects of climate change including through adaptation;

(s) reduction of greenhouse gas emissions;

(t) promotion of activities that mitigate emissions or sequestrate carbon; and

(u) increased use of renewable energy.

One of the recognised weaknesses in the implementation of the RMA has been the lack of national direction and it is notable that there still is no NPS on landscape matters outside the coastal environment close to 30 years after the RMA came into force. The RMR Panel proposes to significantly strengthen the role of national direction through making it mandatory for a range of matters. It would retain the mandatory NZCPS. It would also include a mandatory NPS on "how the principles of Te Tiriti will be given effect through functions exercised under this Act".[12] This links to the revised Treaty clause proposed in a new section 6 which refers to "giving effect" to the principles of Te Tiriti o Waitangi[13] as opposed to merely taking them into account as provided for under the current section 8 of the RMA. This should significantly strengthen the Treaty compliance of management agencies under the new legislation.

NPSs will also be required to set targets (either binding or non-binding) "in order to achieve continuous improvement in the outcomes specified in section 7" which includes the protection and enhancement of ONLs and ONFs as described above. They will also need to state objectives and policies to achieve the targets. This means that national direction on landscape will be mandatory which is a positive change. This provides the opportunity to establish a robust national framework for the management of landscapes at the regional and/or local levels.

The RMR Panel's recommendations also propose to considerably strengthen resource management policies and plans. Over 100 RMA plans nationwide will be consolidated into just 14 combined plans which will include all the provisions of RPSs, regional plans and district plans within a region. This should enable better integration across the region and between the various statutory provisions in the different documents. It should, for example, help avoid the disjunct between regional water planning and district landscape planning that we found in the Mackenzie Basin. The mapping of important landscapes should be a mandatory component of combined plans.

The combined plans will be developed by a joint committee of delegates from the regional council and constituent territorial authorities within the region, a representative from DOC and representatives from mana whenua. With mana whenua having a seat at the table, rather than being a consultee or submitter, the new combined plans should better reflect mana whenua aspirations for landscape management. As well as this more strategic role for Māori in planning, the RMR Panel's recommendations include provision for an integrated partnership process between mana whenua and local authorities to agree on a wide range of matters including: processes to provide for aspects of iwi management plans to be taken into account in regional spatial planning and combined planning processes; opportunities to implement power sharing mechanisms; and funding for mana whenua participation in resource management processes.[14] This should enable bespoke arrangements to be developed for the active leadership and engagement of mana whenua in the identification and management of important landscapes.

Under the RMR Panel recommendations, the proposed combined plan will be audited by an expert reviewer appointed by the Minister for the Environment before it is notified. The audit will address alignment with national direction and the regional spatial strategy. This should help avoid the situation, such as we found in Waitaki District, where the district plan did not give effect to the RPS or section 6(b) of the RMA itself.

Submissions on the combined plans will be heard by an independent hearing panel appointed by the Principal Environment Court Judge to avoid any perceptions that the appointment process may be politicised.[15] All these new mechanisms should help ensure that future plans are much more robust, clearer and more directive, and meet the requirements of the new Act including those relating to landscape protection and enhancement. They should also help with resourcing to support higher quality planning, with the combined resources of multiple agencies contributing to the development of the combined plan. Overall, this should help resolve the problem of weak plans and small impecunious councils being left to manage nationally important landscapes on their own.

The RMR Panel also addressed compliance, monitoring and enforcement, an area we found to be particularly weak, especially in areas such as the Mackenzie Basin which is remote from the head offices of management agencies. The Panel recommends the regionalisation of these functions into stand-alone organisations or hubs that would be better resourced and located at arms-length from political interference. They also propose a range of other legislative amendments to strengthen this function. These are all positive changes which should help address lack of compliance with the rules.

Landscapes of national importance

Overall the RMR Panel's recommendations, if well implemented, should help to strengthen landscape management in Aotearoa New Zealand. They help address the consistent identification of important landscapes, the role of mana whenua in their identification and management, strategic integration across different management regimes affecting them, linkages with clear and directive regulation to protect important values, placing a focus on restoration and enhancement and improving the linkage with funding mechanisms to support sustainable land management and restoration work.

However, to further raise the profile of important landscapes within Aotearoa New Zealand we also suggest that regional spatial strategies be required to identify

a category of landscapes which we provisionally call 'landscapes of national importance' but leave open the option of using an appropriate Māori name. On this point, we note the proposal of Ngāi Tahu to include 'Ngā Tūtohu Whenua' in the Christchurch District Plan to denote landscapes of particular cultural importance.

Landscapes of national importance would be entire landscapes areas that include significant natural, Māori, cultural and/or heritage landscape values. They would include, but be broader than, ONLs which would continue to be identified and protected under the NBEA. They could include places like the Mackenzie Basin, Banks Peninsula, Wakatipu Basin, Waitākere Ranges, Marlborough Sounds and Aotea Great Barrier Island. A Landscape NPS could provide specific policy for these nationally important landscapes to prioritise the protection and restoration of their values. They could also receive priority for existing government initiatives and funding. This could include conservation land acquisition by the Natural Heritage Fund, support for covenanting by the QEII Trust and others, money from the International Visitor Conservation and Tourism Levy when international tourism restarts, and government science and innovation funding for predator control and research into sustainable land management. It could also include funding to support collaborative restoration initiatives and a focus for philanthropic investors.

The identification as a landscape of national importance could also be used to support the development of a collective brand for the area. This brand could assist with marketing strategic tourism opportunities and could also create a quality mark for products produced in the area. Similar to how tourists are attracted to the 'Great Walks' on DOC conservation land, they could also be attracted to the network of landscapes of national importance around Aotearoa New Zealand, where they could be assured of having high quality experiences of the country's outstanding natural and cultural landscapes.

Overall, approaching the resource management reforms through a landscape lens, could help ensure that the new system provides for stronger protection of Aotearoa New Zealand's special places for the benefit of current and future generations.

9.5 Heritage Landscape Order

As earlier suggested, the development of 'Heritage Landscape' Orders is a potential mechanism which could provide greater statutory protection over landscapes of national importance. We leave open the question as to what the most appropriate name for such Orders would be, but have initially proposed 'Heritage Landscape' as a term that encapsulates the concept of caring for and passing down through the generations Aotearoa New Zealand's important natural and cultural landscapes.

A Heritage Landscape Order could be confirmed by the making of an Order in Council under the RMA, a process described further below. However, we do not rule out other alternatives to provide longer-term regulatory protection for important natural and cultural landscape values. There is also the option of enacting bespoke legislation (such as the WRHAA or HGMPA), generic legislation (such as a Heritage Landscape Act) or coming up with a hybrid process to achieve the outcomes sought.

Designation as a 'Heritage Landscape' under the RMA or other legislation (such as the Conservation Act) through a Heritage Landscape Order could be used to create a stronger protective purpose for high value landscapes. Heritage Landscapes could cover both public and private land which would be managed in a manner that was sympathetic to the landscape, cultural and ecological values present. In developing our thinking, we have taken inspiration from Water Conservation Orders, which have provided water bodies with a very high level of legal protection under the RMA. A similar approach could be taken to Heritage Landscapes.

Drawing on the process provided for the establishment of Water Conservation Orders, and considering how a new approach could fit into the existing resource management framework, we propose that Heritage Landscapes could be implemented in the present legislative regime as follows:

- The RMA would set out a broad purpose of what Heritage Landscapes should achieve, and the national values to be protected (you could think of these as designation criteria).
- There would be provision for any party to apply to have a landscape designated as a Heritage Landscape. In order to be considered, the landscape would need to meet the national criteria specified in the RMA. An applicant would also need to be able to demonstrate local community and iwi support.
- Applications could be made to a suitable person/entity, who would then either reject the application (on the basis that it does not meet the purpose specified in the Act) or accept it in a pro forma way. Such an entity could be a specifically established co-governance Heritage Landscape Board, which could have a broader role of oversight over the management of important natural and cultural landscapes. It could also be given the ability to support the development of worthy applications through a Heritage Landscape Fund. Alternatively a suitable Minister of the Crown could perform this role.
- If the application is accepted, a specialist tribunal would be established to publicly notify the application, hear submissions and either provide a draft Heritage Landscape Order or recommend the application be declined.

- The Minister would then accept or reject the recommendations. If rejected, the recommendations would be referred back to the Tribunal for reconsideration. The Minister would then make a final determination on the Heritage Landscape proposal.
- This determination could then be challenged through a de novo appeal in the Environment Court by any party which made submissions, thereby providing independent oversight of the process.

It is envisaged that the cost structure for applying for a Heritage Landscape Order would be similar to that for Water Conservation Orders, namely that the applicant (whether that is an individual, group or governmental body) would need to pay a prescribed fee with the application. Funding could be made available from central government for the development of applications in order to encourage use of the tool. All further costs, for special tribunals and the promulgation of the Order, would be borne by central government.

This process would provide a bottom-up approach to protection. It would enable iwi, hapū, NGOs and local communities as well as agencies such as DOC, to initiate the process (particularly if government funding was available to support the development of applications). The provision for independent review by the Environment Court is important so that those with properties that may be affected by the Heritage Landscape have the opportunity to challenge the proposal if they wish to do so.

If the application is granted, a new Heritage Landscape Order would be created for the specific landscape, for example a Heritage Landscape (Te Manahuna Mackenzie Basin) Order. This Order would include the landscape, cultural and ecological values to be protected in that specific landscape, and activities that must be restricted or prohibited to protect these values. Resource consents that are granted, once the Order is in place, could not be contrary to these restrictions or prohibitions; and regional and district planning documents could not be inconsistent with the Order.

The provisions of the Order could also provide for the establishment of a co-governance steering group, comprising representatives from mana whenua and the relevant statutory agencies (such as the regional council, territorial authorities, DOC and LINZ), to provide oversight of management of the area and to ensure that aligned and joined-up management approaches are applied to the landscape. Support could also be provided for community trusts and the like to facilitate restoration work. Funding would need to be made available to cover the operational costs of these groups.

There is provision in the RMA to revoke or amend a Water Conservation Order. This means that it is not necessarily permanent. However, unless the amendment will have no more than minor effects, or is agreed between the original applicant and regional council; the application to amend or revoke goes through the same process as designation. To date, this has not happened. As such, Water Conservation Orders have offered more permanent protection than planning documents. They also provide an apolitical approach – anybody can apply to designate an area and anybody can apply to revoke an order, and in both circumstances the application must be decided on its merits.

In addition to its protective purpose, a Heritage Landscape designation could be used to support the development of a collective brand for the area. This brand could assist with marketing strategic tourism opportunities and could also create a quality mark for products produced in the area. Similar to how tourists are attracted to the 'Great Walks' on DOC conservation land, they could also be attracted to the network of 'Heritage Landscapes' around New Zealand, where they could be assured of having high quality experiences of New Zealand's outstanding natural and cultural landscapes.

West coast of the Waitākere Ranges

ENDNOTES

1 Pauling, 2016, [4.1]-[4.6]
2 https://www.doc.govt.nz/our-work/tu-te-rakiwhanoa-drylands/
3 Resource Management Review Panel, 2020, 129
4 Resource Management Review Panel, 2020, 142
5 Resource Management Review Panel, 2020, 146-147
6 Resource Management Review Panel, 2020, 142-143
7 Resource Management Review Panel, 2020, 138
8 Resource Management Review Panel, 2020, 139-140
9 Resource Management Review Panel, 2020, 52-53
10 Resource Management Review Panel, 2020, 88
11 Resource Management Review Panel, 2020, 66
12 Resource Management Review Panel, 2020, 486
13 Resource Management Review Panel, 2020, 483
14 Resource Management Review Panel, 2020, 106
15 Resource Management Review Panel, 2020, Chapter 8

Conclusions and recommendations

Lake Wanaka

This landscape project has involved an in-depth review of the case law and landscape assessment practices. It has taken us overseas to see how other countries approach similar challenges. It has brought us around the country to investigate how landscapes are being managed on the ground in various regions and rohe. We have been extremely impressed by the level of innovation and dedication of many land managers, mana whenua and communities in looking after their special places. But we found increasing anthropogenic pressures being placed on them. We documented a landscape management system often not up to the task.

The current resource management reform process provides an excellent opportunity to remedy this situation in the shorter term. However we also recommend that a more enduring approach be developed in the longer-term. We can learn from the mistakes of the past to develop a landscape management system fit for the future.

Drawing on our international and national case studies and examples, we identified four aspects critical for a new landscape protection model. The first element was the key role of mana whenua in landscape management and restoration and the need to give effect to Treaty principles. This includes enabling the relationship of mana whenua with their ancestral lands to be better recognised and strengthened on a landscape scale. The second element was the important role of land managed for conservation purposes such as public conservation land and sympathetically managed privately-owned land. The third element was effective protective regulation for significant working (usually agricultural) landscapes to ensure that incompatible activities, intensification or land use changes do not occur. The fourth element was creating an effective partnership with landowners and others to support and encourage sustainable land use and management.

We have considered a range of new approaches to landscape management throughout this report and set out below a summary of the key recommendations.

10.1 Short-medium term recommendations

Proposed Strategic Planning Act

1. Require regional spatial strategies to address landscape matters, including identifying 'landscapes of national importance' and providing for their protection and restoration.
2. As proposed by the RMR Panel, require regional spatial strategies to encompass land, freshwater and the coastal marine area to enable landscapes and seascapes to be managed in an integrated manner.
3. Ensure the national priorities statement addresses landscape protection including criteria for the identification and protection of 'landscapes of national significance'.
4. As proposed, require mana whenua to be included as equal decision-making partners with government entities in the development of regional spatial strategies to enable Māori rights, interests and values associated with specific landscapes, and Māori cultural landscapes, to be fully reflected in the strategies.
5. Ensure implementation agreements include funding provision to support mana whenua, land managers and community organisations to protect and restore important landscapes particularly within landscapes of national significance.
6. Provide a strong statutory connection between regional spatial strategies and plans developed under the RMA (and its successor) to ensure important landscapes are mapped and protected through robust policies and rules, and plans developed under the Local Government Act to ensure budgetary provision for landscape management and restoration.
7. Apply the Strategic Planning Act to the Crown Pastoral Land Act and Conservation Act to better align these regimes across landscapes.

Proposed Natural and Built Environments Act

8. Retain the proposed reference to the protection and enhancement of outstanding natural features and outstanding natural landscapes.
9. Implement the proposed requirement for mandatory national direction on section 7 outcomes including those relating to landscape protection and enhancement.
10. Prioritise the development of a NPS on Landscape to establish environmental outcomes, targets and bottom lines for the country's landscapes including outstanding and amenity landscapes. The NPS should also set out at what scale landscape should be assessed, the assessment methodology to be used, and the roles of mana whenua and respective councils and their planning documents in providing protection.
11. Include reference to heritage landscapes and amenity in the new legislation.
12. Implement the proposal to develop combined plans, including the inclusion of mana whenua representatives on the planning committee, an independent audit before notification and independent scrutiny of submissions.
13. Require combined plans to map important landscapes within each region.

14. Utilise RPSs to provide strong direction for the protection of important landscapes in each region and use RPS sub-chapters to detail considerations for specific landscapes.
15. Provide greater recognition for iwi planning documents in the management of Māori cultural landscapes and support for the co-design of policy and planning provisions to protect their values.
16. Implement the proposal to establish regional hubs for compliance, monitoring and enforcement and other proposals to strengthen this function.

Other matters

17. Review the NESPF to ensure that it is not permitting plantation forestry in areas of outstanding or high landscape values such as the Banks Peninsula and Te Manahuna Mackenzie Basin.
18. Review the emissions trading scheme to ensure that it incentivises indigenous vegetation regeneration as opposed to further expansion of exotic plantation forestry.
19. Establish a dedicated biodiversity enhancement fund to support indigenous forest restoration.
20. Provide support to strengthen iwi, hapū and community collaborative initiatives focused on landscape restoration.
21. Proceed with the proposed reforms set out in the Crown Pastoral Land Reform Bill.
22. Undertake a fundamental review of the concessions system under the Conservation Act.

10.2 Longer-term recommendation

23. Building on the concept of Heritage Landscapes develop a new tool for landscape protection, in partnership with Māori and through wide consultation, to provide enduring protection for Aotearoa New Zealand's important natural and cultural landscapes.

Tahunanui Beach, Nelson

REFERENCES

Adjei P and F Agyei, 2015, 'Biodiversity, environmental health and human wellbeing: Analysis of linkages and pathways', *Environmental Development and Sustainability*, 17(5), 1085-1102

Allen M, T Wilbanks, B Preston, S Kao and J Bradbury, 2017, *Assessing the costs and benefits of resilience investments: Tennessee Valley Authority case study*, report for the US Department of Energy, Oak Ridge National Laboratory, Tennessee

Allpress J and P Tuatagaloa, 2018, *Waiheke community survey: Results from a 2018 survey of Waiheke residents*, Technical Report 2018/014, Auckland Council, Auckland

Aotea Conservation Park Advisory Committee, 2020, *Minutes of meeting*, 6 March, Department of Conservation, Auckland

Appleton C, 2015, *Statement of evidence of Clive Appleton on behalf of Christchurch City Council*, In the matter of the proposed Christchurch Replacement Plan: Ecology

Auckland Council, 2017, *Kauri dieback report 2017: An investigation into the distribution of kauri dieback, and implications for its future management, within the Waitākere Ranges Regional Park* (version 2), Auckland Council, Auckland

Auckland Council, 2018a, *State of the Waitākere Ranges Heritage Area 2018*, Auckland Council, Auckland

Auckland Council, 2018b, *Auckland plan 2050*, Auckland Council, Auckland

Auckland Council, 2018c, *State of the Waitākere Ranges Heritage Area 2018: Summary report*, Auckland Council, Auckland

Auckland Regional Council, 2002, 'Rakino rats on the way out', press release, 3 December

Auckland Regional Council, 2010, *Regional parks management plan: Volume 1: Management policies*, Auckland Council, Auckland

Auckland Tourism, Events and Economic Development, 2020a, *Local Board economic overview*, ATEED, Auckland

Auckland Tourism, Events and Economic Development, 2020b, *Waiheke Local Board economic overview*, ATEED, Auckland

Banks Peninsula Conservation Trust, 2017, *Annual report 2017*, Banks Peninsula Conservation Trust, Christchurch

Banks Peninsula Conservation Trust, 2018, *Annual report 2018*, Banks Peninsula Conservation Trust, Christchurch

Banks Peninsula Conservation Trust, 2019, *Annual report 2019*, Banks Peninsula Conservation Trust, Christchurch

Banks Peninsula Water Zone Management Committee, 2019, *Agenda 18 June*, Christchurch City Council, Christchurch

Baragwanath L, undated, *The Waiheke project: Overview of tourism, wine and development in Waiheke Island*, University of Auckland, Auckland

Bell E A, D Burgin, J Sim, K Dunleavy, A Fleishman and R P Scofield, 2018, 'Population trends, breeding distribution and habitat use of black petrels (*Procellaria parkinsoni*) – 2016/2017 operational report', *New Zealand Aquatic Environment and Biodiversity Report No. 198*

Bennett M, C Webb and D Dromgoole, 2012, *Priorities for the restoration of significant wetlands within the Waiheke Local Board*, Auckland Council, Auckland

Blake E, 2019, 'Thaidene Nëné National Park Reserve Becomes N.W.T.'s Newest Protected Area', *CBC News,* 20 August, at https://www.cbc.ca/news/canada/north/thaidene-nene-finalized-1.5253685

Board of Inquiry into the Proposed New Zealand Coastal Policy Statement, 2009, *Board of Inquiry report on the New Zealand Coastal Policy Statement*, Department of Conservation, Wellington

Boffa Miskell, 2007, *Banks Peninsula landscape study*, Boffa Miskell, Christchurch

Boffa Miskell, 2010, *Northland regional landscape review: Review of the landscape assessment methodologies used by district councils for identifying Northland's outstanding natural features and landscapes*, Northland Regional Council, Whangarei

Boffa Miskell and Lucas Associates, 1993, *Canterbury Regional landscape study*, Canterbury Regional Council, Christchurch

Bonthoux S, M Brun, F Di Pietro, S Greulich and S Bouché-Pillon, 2014, 'How can wastelands promote biodiversity in cities? A review', *Landscape and Urban Planning,* 132, 79–88

Borthwick J E, 2016, 'OUTstanding landscapes: Is the RMA protecting New Zealand's heritage?', paper presented to the Resource Management Law Association conference, September

Bridgewater P, 2019, 'The four stages of biosphere reserves in Australia', in M G Reed and M F Price (eds), *UNESCO biosphere reserves: Supporting biocultural diversity, sustainability and society,* Routledge, London

Brockington D, 2002, *Fortress conservation: The preservation of the Mkomazi Game Reserve, Tanzania*, James Currey, Oxford

Brower A, R Sprague, M Vernotte and H McNair, 2018, 'Agricultural intensification, ownership, and landscape change in the Mackenzie Basin', *Journal of New Zealand Grasslands*, 80, 47-54

Brown S, undated, *The assessment and management of amenity values*, New Zealand Association for Impact Assessment, at https://www.nzaia.org.nz/stephen-brown.html

Brown S, 2011, *Coromandel Peninsula landscape assessment*, Brown NZ Limited, Auckland

Brown S, 2019, 'A question of identity', paper presented to the Environmental Defence Society Landscape Conference, August 2019, Auckland

Brown A and S Harris, 2005, *Environmental, economic and social impacts of irrigation in the Mackenzie Basin*, Ian Brown Consulting Limited, Dunedin

Brown M, T Stephens, R Peart and B Fedder, 2015, *Vanishing nature: Facing New Zealand's biodiversity crisis*, Environmental Defence Society, Auckland

Burleigh R, 2019, 'Conservation journey of Robin Burleigh: The man who coined the term "Wildside"', Joseph Langer Charitable Trust, at https://www.jlct.org.nz/robin-burleigh-wildsie-and-community-conservation/

Burry M, 2020, 'Rock bottom crossbred wool prices pose dilemma for farmers', *Radio New Zealand*, 1 July

Butler R and J Allpress, 2019, *An evaluation of the Waiheke governance pilot: Progress after the first 18 months*, Technical Report 2019/020, Auckland Council, Auckland

Canadian Federal, Provincial and Territorial Department for Parks, Protected Areas, Conservation, Wildlife and Biodiversity, 2018, *One with nature: A renewed approach to land and freshwater conservation in Canada,* Canadian Government, Ottawa

Chase A, 2017, 'Trump Budget Slashes Coastal and Ocean Funding', Natural Resources Defense Council Blog, 24 May, at https://www.nrdc.org/experts/alison-chase/trump-budget-slashes-coastal-and-ocean-funding

Christchurch City Council, 2015, *Stage 3, Section 32, Chapter 9, Natural and cultural heritage, Appendix 3.4, Banks Peninsula Landscape Study 2007*, Christchurch City Council, Christchurch

Christchurch District Council, 2012, *Takapūneke conservation report*, Christchurch City Council, Christchurch

Coastal Conservancy, 2019, *Coastal Conservancy annual financial report*, Coastal Conservancy, Oakland

Connor H E and C G Vucetich, 1964, 'Tussock grassland communities in the Mackenzie Country, South Canterbury, New Zealand', *New Zealand Journal of Botany*, 2(4), 325-351

Corlett T, 2016. 'The role of rewilding in landscape design for conservation', *Current Landscape Ecology Reports*, 1(3), 127–133

Coster J and W Spring-Rice, 1984, *History, archaeology and site management on Motutapu and Rangitoto*, Department of Lands and Survey, Auckland

Department of Conservation, 2004, 'Great Barrier Island historic heritage', factsheet, at https://www.doc.govt.nz/globalassets/documents/conservation/historic/by-region/auckland/great-barrier/gbi-historic-factsheet.pdf

Department of Conservation, 2011a, *Destination management framework – a new approach to managing destinations*, DOC, Wellington

Department of Conservation, 2011b, *Natural character and the NZCPS 2010, national workshop – summary of discussion and outcomes*, Department of Conservation, Wellington

Department of Conservation, 2013, *Aotea Conservation Park proposal: Public discussion paper and invitation to comment*, Department of Conservation, Wellington

Department of Conservation, 2014, *Conservation management strategy Auckland 2014-2024, Volume 1*, Department of Conservation, Auckland

Department of Conservation, 2016, *Conservation management strategy: Canterbury (Waitaha) 2016, Volume 1*, Department of Conservation, Christchurch

Department of Conservation, 2017, *Review of the effect of the NZCPS 2010 on RMA decision-making: Overview and key findings*, Department of Conservation, Wellington

Department of Conservation, 2019, *Annual report for year ended 30 June 2019*, Department of Conservation, Wellington

Department of Conservation, 2020a, *Annual report for year ended 30 June 2020*, Department of Conservation, Wellington

Department of Conservation, 2020b, *Giving better effect to the principles of the Treaty of Waitangi: Partial reviews of the Conservation General Policy and the General Policy for National Parks: A background paper about the process and main issues*, Department of Conservation, Wellington

Department of Conservation, undated, 'Great Barrier Island/Aotea', pamphlet, at https://www.doc.govt.nz/documents/parks-and-recreation/places-to-visit/auckland/great-barrier-brochure.pdf

Department of Conservation, undated (a), 'Aotea Conservation Park Proposal, Factsheet retrieved from: https://www.doc.govt.nz/globalassets/documents/getting-involved/consultations/current-consultations/auckland/aoteaconservationpark/aotea-conservation-park-factsheet.pdf

Department of Conservation, undated (b), 'NZCPS 2010 Guidance note, Policy 15: Natural features and natural landscapes'

Duncan R P, R J Webster and C A Jensen, 2001, 'Declining plant species richness in the tussock grasslands of Canterbury and Otago, South Island, New Zealand', *New Zealand Journal of Ecology*, 25(2), 35-47

Environment Canterbury, undated, *National wilding conifer control programme key messages*

Environment Canterbury, 2020, *Cruise ships in Akaroa: Guidance issued on permitted activity and sea-bed disturbance*, Environment Canterbury, Christchurch

Environmental Integrity Project, 2009, *Outside the law: Restoring accountability to the Tennessee Valley Authority*, Environmental Integrity Project, Washington

Fairclough G, I Sarlöv Herlin and C Swanwick (eds), 2018, *Routledge handbook of landscape character assessment: Current approaches to characterisation and assessment*, Routledge, London

Fairweather J, B Newton, S R Swaffield and D G Simmons, 2001, *Visitor's and local's experiences of Westland, New Zealand*, Tourism Recreation Research and Education Centre, Report No. 23, Lincoln University

Fairweather J R and Swaffield S R, 1999, *Public perceptions of natural and modified landscapes of the Coromandel Peninsula, New Zealand*, Agribusiness and Economics Research Unit, Research Report No. 241, Lincoln University

Fairweather J R and Swaffield S R, 2004, *Pubic perceptions of outstanding natural landscapes in the Auckland Region*, Agribusiness and Economics Research Unit, Research Report No. 273, Lincoln University

Fairweather J R, Swaffield S R and D G Simmons, 1998, *Understanding visitors' experiences in Kaikoura using photographs of landscapes & Q Sort*, Tourism Research and Education Centre, Report No. 5, Lincoln University

Fairweather J R, Swaffield S R and D G Simmons, 2000, *Understanding visitors' and locals' experiences of Rotorua using photographs of landscapes & Q Sort*, Tourism Recreation Research and Education Centre, Report No. 13, Lincoln University

Falconer G, 2015, *Living in paradox: A history of urban design across kainga, towns and cities in New Zealand*, Blue Acres Press, Auckland

Gavin M, J McCarter, F Berkes, A Te Pareake Mead, E Sterling, R Tang and N Turner, 2018, 'Effective biodiversity conservation requires dynamic, pluralistic, partnership-based approaches', *Sustainability* 10, 1846

Gawith D, 2019, *Private land economics – agriculture, tourism and t*rees, unpublished paper prepared for the Environmental Defence Society, Auckland

Gilbert J, 2019, *Windy Hill Rosalie Bay Catchment Trust twenty years of ecological restoration: A review*, Windy Hill Sanctuary, Great Barrier Island

Glover J, 2019, *Landscapes Review* (Final Report), at https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/833726/landscapes-review-final-report.pdf

Great Barrier Island Environmental Trust, 2010, *Great Barrier Island (Aotea) state of environment report 2010*, Great Barrier Island Environmental Trust, Ōkiwi

Greenaway A, 2019, *Friendship for (and with) Te Urewera: Insights from interviews in 2019*, Manaaki Whenua – Landcare Research, Auckland

Grieder T and L Garkovich, 1994, 'Landscapes: The social construction of nature and the environment', *Rural Sociology*, 59(1), 1-24

Harvey W, 2020, 'Trailblazing Banks Track: New Zealand's first private hike marks 30 years', *Stuff*, 1 January

Harvey B and T Harvey (ed), 2006, *Waitākere Ranges: Nature, history, culture*, Waitākere Ranges Protection Society, Auckland

Hauraki Gulf Forum, 2009, 'First private land added to Hauraki Gulf Park', *Scoop*, 9 May

Hayward M, 2019, 'Cruise ships returning to Lyttelton Harbour in 2020', *Stuff,* 2 May

Head N, 2016, *Statement of evidence of Nicholas John Head on behalf of the Director-General of Conservation,* Environment Court, 9 September

Hogan D, 2015, *Statement of evidence of Deborah Jane Hogan on behalf of Christchurch City Council*, In the matter of the proposed Christchurch Replacement Plan: Planning – Indigenous Biodiversity and Ecosystems

Hoskins R, K Wixon, R Rauna, R Te Nana, J Scott, C Wilson, S Roberts, L Henry and W Knox, 2008, *Te aranga – Māori cultural landscape strategy* (second edition), at https://tearanga.maori.nz/cms/resources/TeArangaStrategy28Apr08_lr.pdf

Independent Hearings Panel, 2016, *Christchurch Replacement District Plan Decision 50: Natural and cultural heritage, sub-chapter 9.1: Indigenous biodiversity and ecosystems*, Christchurch City Council, Christchurch

Independent Māori Statutory Board, 2016, *The Māori report for Tāmaki Makaurau 2016*, Independent Māori Statutory Board, Auckland

Indigenous Circle of Experts, 2018, *We rise together: Achieving pathway to Canada Target 1 through the creation of indigenous and conserved areas in the spirit and practice of reconciliation,* Northwest Territories, Canada

Johnston O, 2019, *Akaroa Harbour cruise ship visits: Preliminary ecological risk assessment*, report prepared for Environment Canterbury, Cawthron Institute, Nelson

Jorgensen A and R Keenan, 2012, *Urban wildscapes*, Routlege, London

Joseph R, 2018, 'The Treaty, tikanga Māori, ecosystem-based management, the RMA and power sharing for environmental prosperity', in G Severinson and R Peart, *Reform of the resource management system: The next generation, working paper 3,* Environmental Defence Society, Auckland

Kantar, 2019, 'Mood of the nation: New Zealanders' perceptions of international visitors', Kantar, Auckland

Kauri Dieback Governance Group, 2019, *National (kauri dieback) pest management plan proposal to meet requirements of section 61 of the Biosecurity Act*, Ministry for Primary Industries, Wellington

Kenderdine S, 2005, 'Heritage landscapes: Developing legislative frameworks which allow for protection and change', paper presented at the New Zealand Institute of Landscape Architects conference, 28-30 April, Dunedin

Knight C, 2018, *Beyond Manapouri 50 years of environmental politics in New Zealand*, Canterbury University Press, Christchurch

Land Information New Zealand, 2019, *Crown pastoral land regulatory system: Regulatory system assessment*, Land Information New Zealand, Wellington

Land Use Consultants, 2006, *Review and evaluation of heritage coasts in England: Report to the Countryside Agency*

Lau N and A Wong, 2016, Topic 081f Rezoning and precincts (geographical areas), Joint evidence report on submissions by Nicholas Lau and David Alan Wong on behalf of Auckland Council Puhinui Precinct, 28 January

Leon S, 2019, 'Depleted Adirondack Park board leaves questions about management', *Adirondack Explorer,* 5 November available at https://www.adirondackexplorer.org/stories/depleted-adirondack-park-board

Li Q, K Morimoto, M Kobayashi, H Inagaki, M Katsumata, Y Hirata, K Hirata, H Suzuki, Y Li, Y Wakayama, T Kawada, B Park, T Ohira, N Matsui, T Kagawa, Y Miyazaki and A Krensky, 2008, 'Visiting a forest, but not a city, increases human natural killer activity and expression of anti-cancer proteins', *International Journal of Immunopathology and Pharmacology*, January, 117-127

Logie M J, 2016, 'Waiheke Island: An island paradise facing an uncertain future', *New Zealand Geographe*r, 72(3), 219-225

Lyver P, J Davies and R Allen, 2014, 'Settling Indigenous Claims to Protected Areas: Weighing Māori Aspirations Against Australian Experiences' *Conservation and Society* 12(1): 89-106

Lyver P, J Ruru, N Scott, J Tylianakis, J Arnold, S Malinen, C Bataille, M Herse, C Jones, A Gormley, D Peltzer, Y Taura, P Timoti, C Stone, M Wilcox and Henrik Moller, 2019, 'Building biocultural approaches into Aotearoa – New Zealand's conservation future', *Journal of the Royal Society of New Zealand*, 49(3), 394-411

Marsden M, 2003, *The woven universe: Selected writings of Rev. Māori Marsden*, edited by Te Ahukaramu Charles Royal, Estate of Rev. Māori Marsden, Otaki

Mayer S and C Frantz, 2004, 'The connectedness to nature scale: A measure of individuals' feeling in community with nature', *Journal of Environmental Psychology*, 24(4), 503–515

McClure M, 2004, *The wonder country: Making New Zealand tourism*, Auckland University Press, Auckland

McClure M, 2010, 'Tourist industry', Te Ara – the Encyclopedia of New Zealand at http://www.TeAra.govt.nz/en/tourist-industry

McGlone M S, 2001, 'The origin of the indigenous grasslands of southeastern South Island in relation to pre-human woody ecosystems', *New Zealand Journal of Ecology*, 25(1), 1-15

McKnight T, 2013, *Planning for intensive dairying in the Mackenzie Basin: An inquiry into social and collaborative planning*, Masters of Planning thesis, University of Otago

McLachlan L, 2019, 'Mana whenua reach decision on Ihumātao land', *RNZ*, 18 September

McMurtrie S, T Adamson, H Goslin and P Hopwood, 2017, 'Erosion control treatment trials on loess soils', paper presented to Water New Zealand's Stormwater Conference, 3-5 May, Auckland

Mead H M (ed), 1984, *Te Māori – Māori art From New Zealand collections*, Heinemann, Auckland

Melean Absolum Limited, 2013, *Waitākere Ranges Heritage Area 2013 monitoring report: Landscape*, Unpublished report prepared for Auckland Council, Melean Absolum Limited, Auckland

Melean Absolum Limited, 2017, *Waitākere Ranges Heritage Area 2018 monitoring report: Landscape*, unpublished report prepared for Auckland Council (draft), Melean Absolum Limited, Auckland

Menzies D, 2017, 'Assessing the Māori cultural landscape', paper prepared by Te Tau a Nuku for the New Zealand Institute of Landscape Architects Māori cultural landscape meeting, 10 November

Menzies D and Ruru J, 2011, "Chapter 2 Indigenous Peoples' Right to Landscape in Aotearoa New Zealand" in *The Right to Landscape: Contesting Landscape and Human Rights*. Routledge, London.

Miller B and R Reidinger, 1997, *Comprehensive river basin development: The Tennessee Valley Authority,* World Bank Technical Paper No. 416

Ministry for the Environment and Statistics New Zealand, 2019, *Environment Aotearoa 2019 summary report*, Ministry for the Environment and Statistics New Zealand, Wellington

Minister of Conservation, 2020, 'Major funding boost for Predator Free Banks Peninsula', Press release, 9 August

Ministry of Business, Innovation and Employment, 2019, *Optimising Tourism New Zealand's future role and contribution to New Zealand: The Tourism New Zealand report*, Ministry of Business, innovation and Environment, Wellington

Moewaka Barnes H, 2019, 'Healthy land, water and people: Indigeneity, relationships and planetary health', paper presented to International Union for Health Promotion and Evaluation Conference, April, Rotorua

Moewaka Barnes H and T McCreanor, 2019, 'Colonisation, hauora and whenua in Aotearoa', *Journal of the Royal Society of New Zealand*, 49(sup1), 19-33

Monin P, 2012, *Matiatia: Gateway to Waiheke*, Bridget Williams Books, Wellington

Müllera A, P Bøcher, C Fischer and J Svenning, 2018, ''Wild' in the city context: Do relative wild areas offer opportunities for urban biodiversity?', *Landscape and Urban Planning,* 170, 256-265

Murdoch G, 2015, *Māori cultural landscapes*, background report for Auckland Council, Auckland Council, Auckland

Murdoch G, undated, *Bethells/Te Henga/Waitākere Valley background report: Part 2 human heritage*, at: http://www.piha.co.nz/wpcontent/uploads/2013/06/backgroundrpt-part2-humanheritage.pdf

Nature United, 2019, *Nature United celebrates the signing of the Thaidene Nëné establishment agreements and commits to supporting indigenous co-governance,* at https://www.natureunited.ca/explore/newsroom/thaidene-nene-establishment-agreements/

New Zealand Productivity Commission, 2019, *Local government funding and financing*, New Zealand Productivity Commission, Wellington

New Zealand Tourism Research Institute, 2018a, *Visitor strategy: Aotea/Great Barrier Island 2018 -2023*, report prepared for Great Barrier Local Board, Auckland University of Technology, Auckland

New Zealand Tourism Research Institute, 2018b, *Aotea/Great Barrier Island visitor strategy: Visitor and community survey report (high season)*, Auckland University of Technology, Auckland

Ngāi Tūāhuriri Rūnanga, Te Hapū o Ngāti Wheke (Rāpaki), Te Rūnanga o Koukourārata, Ōnuku Rūnanga, Wairewa Rūnanga and Te Taumutu Rūnanga, 2013, *Mahaanui iwi management plan*, at https://mahaanuikurataiao.co.nz/wp-content/uploads/2019/08/Full-Plan.pdf

New Zealand Institute of Landscape Architects, 2010a, *The Aotearoa-New Zealand landscape charter*, New Zealand Institute of Landscape Architects, Wellington

New Zealand Institute of Landscape Architects, 2010b, *Best Practice Note – Landscape Assessment and Sustainable Management 10.1*, New Zealand Institute of Landscape Architects, Wellington

Nightingale T and P Dingwall, 2003, *Our picturesque heritage: 100 years of scenery preservation in New Zealand*, Department of Conservation, Wellington

Nolan D and Somerville R, 2003, *The Resource Management Amendment Act (No 2)*, New Zealand Law Society, Wellington

O'Connor K F, 1976, *An introduction to the Waitaki*, New Zealand Man and the Biosphere Report No. 1, Tussock Grasslands and Mountain Lands Institute, Lincoln

O'Connor K F, 1982, 'The implications of past exploitation and current developments to the conservation of South Island tussock grasslands', *New Zealand Journal of Ecology*, 5, 97-107

O'Donnell J, 2015, *Collaboration and conflict in the Adirondack Park: An analysis of conservation discourses over time,* Master of Science in Natural Resources thesis, University of Vermont

Ogilvie G, 2010, *Banks Peninsula: Cradle of Canterbury* (revised edition), Phillips & King Publishers, Christchurch

Orveland F, 2019, *The green wedges of Stockholm – past present and future, at* https://cobhamgreenbelt.org.uk/

Owen C, 2017, 'Flash floods, landslides and power cuts strike Waiheke Island, causing widespread damage', *Stuff*, 8 March

Parliamentary Commissioner for the Environment, 2001, *Managing change in paradise: Sustainable development in peri-urban areas*, Parliamentary Commissioner for the Environment, Wellington

Paul T, 2015, *Guidelines for the use of the decision support system: Calculating wildling spread risk from new plantings*, No. S0019, Scion, Wellington

Pauling C, 2016, Statement of supplementary evidence of Craig Aaron Pauling on behalf of Christchurch City Council, In the matter of the proposed Christchurch Replacement Plan Chapter 9.5: cultural values – Ngāi Tahu values and the natural environment

Pearce D G, 2015, 'Destination management in New Zealand: Structures and functions', *Journal of Destination Marketing & Management*, 4, 1-12

Peart R, 2004, *A place to stand: The protection of New Zealand's natural and cultural landscapes*, Environmental Defence Society, Auckland

Peart R, 2009, *Castles in the sand, what's happening to the New Zealand coast?*, Environmental Defence Society, Auckland

Peart R, 2016, *The story of the Hauraki Gulf*, David Bateman Limited, Auckland

Peart R and P Reaburn, 2011, *Strengthening second generation regional policy statements*, Environmental Defence Society, Auckland

Peart R and C Woodhouse, 2020, *Tourism and landscape protection*, Environmental Defence Society, Auckland

Peart R and Woodhouse C, 2020b, '*Te Manahuna-Mackenzie Basin and landscape protection*', Environmental Defence Society, Auckland

Pere L and A Barnes, 2009, 'New Learnings from old understandings: Conducting qualitative research with Māori', *Qualitative Social Work*, 8(4), 449-467

Phillips J, 2007, 'Perceptions of the landscape', *Te Ara – the Encyclopedia of New Zealand*, at http://www.TeAra.govt.nz/en/perceptions-of-the-landscape/print

Phillips A and R Xia, 2019, 'Trump might limit states' say in offshore drilling plan. Here's how', *Los Angeles Times,* 21 March

Pollock K, 2014, 'Landscape architecture', Te Ara - the Encyclopedia of New Zealand, at http://www.TeAra.govt.nz/mi/landscape-architecture/print

Project Forever Waiheke, 2019, *Waiheke Island sustainable community tourism strategy 2019-2024*, report developed for the Waiheke Island Local Board and the Waiheke Community, Project Forever Waiheke, Waiheke Island

Rankin J, 2019, 'Second stage of New Zealand's biggest wind farm to go ahead', *Stuff*, 12 November

Read M, 2005, *The 'construction' of landscape: A case study of the Otago Peninsula, Aotearoa/New Zealand*, Doctoral thesis, Lincoln University

Read M, 2012, *Landscape boundaries: An examination of the influence of Environment Court decisions on the practice of landscape assessment in Aotearoa/ New Zealand*, Masters of Resource and Environmental Planning thesis, Massey University

Reid J, K Taylor-Moore and G Varona, 2014, 'Towards a social-structural model for understanding current disparities in Māori health and well-being', *Journal of Loss and Trauma,* 19(6), 514-536

Relph D, 2010, *The Mackenzie Country: A fine plain behind the Snowy Range*, David Ling Publishing, Auckland

Renata A, G Lawson, D Cushing and D Menzies, 2015, 'Indigenous cultural landscapes: Equitably defining the "authenticity" of the intangible', in S Lira, R Amoeda and C Pinheiro (eds), *Sharing cultures 2015 – Proceedings of the 4th International Conference on Intangible Heritage,* Green Lines Institute for Sustainable Development, Barcelos, 365-371

Resource Management Review Panel, 2020, *New directions for resource management in New Zealand*, Ministry for the Environment, Wellington

Reynolds F, 2016, *Fight for beauty: Our path to a better future*, Oneworld Publications, London

Rhodes D and J Novis, 2002, *Impact of incentives on the development of plantation forest resources in New Zealand*, Food and Agriculture Organisation, Rome, at http://www.fao.org/3/ae535e/ae535e09.htm

Rod Donald Banks Peninsula Trust, 2020, *Striding forward/hikoi whakamua strategic plan 2020-2030*, Rod Donald Banks Peninsula Trust, Christchurch

Rolfe J, S Whitten and J Windle, 2017, 'The Australian experience in using tenders for conservation', *Land Use Policy*, 63, 611-620

Rook G, 2013, 'Regulation of the immune system by biodiversity from the natural environment: An ecosystem service essential to health', *Proceedings of the National Academy of Sciences of the United States of America*, 110(46), 18360–18367

Ross H, C Grant, C Robinson, A Izurieta, D Smyth and P Rist, 2009, 'Comanagement and Indigenous Protected Areas in Australia: achievements and ways forward', *Australasian Journal of Environmental Management,* 16(4), 242–252

Rutledge D T, W T Baisden, N Cradock-Henry, E Keller, N Mason, B Mullen, J Mc C Overton, A Sood, S Stuart, A Tait, T Timar, V Vetrova, C Zammit and A Dunningham, 2017, *Upper Waitaki catchment synthesis report (RA2) uplands case study: Climate change impacts and implications (CCII) for New Zealand to 2100*, MBIE Contract C01X1225

Salmond A, 2005, 'Ancestors in the land: New Zealand's heritage landscapes', paper presented at the New Zealand Institute of Landscape Architects conference, 28-30 April, Dunedin

Sandrow C, A Feldpausch-Parker, E Vidon and I Parker, 2018, 'Anything but a walk in the park: Framing analysis of the Adirondack State Park land classification conflict', *Frontiers in Communication,* 3(42), 1-10

Saunders W, 2017, *Setting the scene: The role of iwi management plans in natural hazard management*, GNS Science, Lower Hutt

Schuster R, R Germain, J Bennett, N Reo and P Arcese, 2019, 'Vertebrate biodiversity on indigenous-managed lands in Australia, Brazil, and Canada equals that in protected areas', *Environmental Science & Policy*, *101*, 1-6

Schuyler D, 2018, *Embattled river: The Hudson and modern American environmentalism*, Cornell University Press, London

Sea Change Tai Timu Tai Pari, 2017, *Hauraki Gulf marine spatial plan*, Environment Waikato, Hamilton

Severinsen G, 2019, *Reform of the resource management system – a model for the future: Synthesis repor*t, Environmental Defence Society, Auckland

Sheridan M, 1995, *Dam dwellers: End of an era*, Sheridan Press, Twizel

Silcock P, E Kieboom and M Rayment, 2014, *The economic contribution of protected landscapes*, Final report for the Department for Environment Food and Rural Affairs, Cumulus Consultants, Worcestershire, England

Snoyink N, 2017, *Nga whenua rāhui: A way forward for New Zealand's cherished landscapes?*, unpublished paper

Steiner F, G Young and E Zube, 2014, 'Ecological planning: Retrospect and prospect', in F Ndubisi (ed) *The Ecological Design and Planning Reader*, Island Press, Washington DC

Stephenson J, H Bauchop and P Petchey, 2004, *Bannockburn Heritage landscape study*, Department of Conservation, Wellington

Strateg.Ease Limited and Capital Strategy Limited, 2006, *Economic influences on the future of Great Barrier Island: Report to City Planning*, Auckland City Council, Auckland

Strecker A, 2012, "Chapter 15 The Human Dimension to Landscape Protection in International Law". In *Cultural Heritage, Cultural Rights, Cultural Diversity*. Leiden, The Netherlands: Brill | Nijhoff. doi: https://doi.org/10.1163/9789004228382_016

Swaffield S, 1993, 'Naming the rose: Observations on "landscape" usage and professional identity', *Landscape Research* 18(2), 58-64

Swaffield S, 1999a, 'Landscape assessment in New Zealand: Background and current issues', *Landscape Review*, 5(1), 3-16

Swaffield S, 1999b, 'Landscape assessment: Means and ends', *Landscape Review*, 5(1), 1-2

Swaffield S, 1999c, 'A framework for landscape assessment', *Landscape Review*, 5(1), 45-51

Swaffield S and K O'Connor, 1986, 'Perceiving, conceiving, protecting and using New Zealand landscape systems', paper presented at the International Conference on Landscape of the Southern Hemisphere, 2-7 September, Adelaide

Taylor G, undated, *A proposal for a Waitākere Ranges Authority,* Auckland

Taylor L, Te Whenua T and Hatami B, 2018, '*How current legislative frameworks enable customary management and ecosystem-based management in Aotearoa New Zealand: the contemporary practice of rāhui*', Discussion paper

Taylor P, 2015, 'Valuing people in the landscape: Re-thinking conservation approaches', *USDA Forest Service Proceedings*, 187-194

Te Pātaka o Rākaihautū/Banks Peninsula Geopark Trust, 2020, *Te Pātaka o Rākaihautū/Banks Peninsula Geopark business plan*, Te Pātaka o Rākaihautū/Banks Peninsula Geopark Trust, Christchurch

Te Rūnanga o Arowhenua, Te Rūnanga o Waihao, Te Rūnanga o Ngā Tahu and Boffa Miskell Limited, 2008, *Cultural impact assessment of the proposed plan change 13 to the Mackenzie District Plan*, Boffa Miskell, Tauranga

Thomlinson E and G Crouch, 2012, 'Aboriginal peoples, Parks Canada, and protected spaces: A case study in co-management at Gwaii Haanas National Park Reserve', *Annals of Leisure Research* 15(1), 69-86

Thomson J, 2019, 'Thaidene Nëné heralds a new era of parks', *The Narwhal,* 12 October, at https://thenarwhal.ca/thaidene-nene-heralds-new-era-parks/

Townsend T, M Roe, C Davies and Q Qin, 2017, 'National park city: Salutogenic city?' *Sustainable Development and Planning* 10: 203 -211

Tran T, N Ban and J Bhattacharyya, 2020, 'A review of successes, challenges, and lessons from indigenous protected and conserved areas', *Biological Conservation,* 241, 108271

Treskonova M, 1991, 'Changes in the structure of tall tussock grasslands and infestation by species of *Hieracium* in the Mackenzie Country, New Zealand', *New Zealand Journal of Ecology*, 15(1), 65-78

Turbott C, 2019, *Tourism topic paper: Draft Waiheke area plan*, Auckland Council, Auckland

United Nations Educational, Scientific and Cultural Organisation, 1993, Tongariro National Park Advisory Body evaluation, at https://whc.unesco.org/en/list/421/documents/

United Nations Educational, Scientific and Cultural Organisation, 1996, *Biosphere reserves: The Seville Strategy and the Statutory Framework of the World Network*, UNESCO, Paris

United Nations Educational, Scientific and Cultural Organisation, 2018, *Man and the Biosphere Programme 2018 – 2019*, pamphlet

Waaka-Home M, 2016, *Statement of evidence on behalf of Te Rūnanga o Ngāi Tahu and Te Rūnanga o Arawhenua,* In the matter of appeals in relation to decisions on Plan Change 13 to the Mackenzie District Plan, Environment Court, 9 September

Waiheke Local Board, 2016, Attachment to meeting of 15 December 2016: 'Rakino Subsection – Hauraki Gulf Islands Waste Plan', at https://infocouncil.aucklandcouncil.govt.nz/Open/2016/12/WHK_20161215_AGN_7024_AT_files/WHK_20161215_AGN_7024_AT_Attachment_50634_2.PDF

Waiheke Local Board, 2018, *State of Auckland terrestrial report card*, at https://knowledgeauckland.org.nz/media/1262/soa-rpt-card-2018-terrestrial-waiheke.pdf

Waiheke Local Board, 2020, *Draft Waiheke area plan: Working together for a better future – a 30 year vision*, Auckland Council, Auckland

Waitākere Ranges Local Board, 2015, *Waitākere Ranges strategic weed management plan*, Auckland Council, Auckland

Waitākere Ranges Protection Society, 1978, *Wainamu – Te Henga: A study,* Waitākere Ranges Protection Society, Auckland

Waitākere Ranges Protection Society, forthcoming, *A place to stand,* Waitākere Ranges Protection Society, Auckland

Waitangi Tribunal, 2009, *WAI 894: Te Urewera pre-publication part 1*, Waitangi Tribunal, Wellington

Walker S, 2016, *Statement of evidence on behalf of the Mackenzie Guardians Incorporated,* In the matter of appeals in relation to decisions on Plan Change 13 to the Mackenzie District Plan, Environment Court, 9 September

W Watt K, 2016, *Successes and challenges of implementing the Oak Ridges Moraine conservation plan,* Master of Arts thesis, University of Toronto

Weaver S D, R J Sewell and C J Dorsey, 1985, *Extinct volcanoes: A guide to the geology of Banks Peninsula*, Geological Society of New Zealand, Guidebook No. 7

Westbrooke L, 2013, 'Conservation land for GBI D.o.C land, *Environmental News,* Issue 32

Williams M, undated, Landscape management in the new world order: Time to lift our game, New Zealand Association of Impact Assessment, at https://www.nzaia.org.nz/uploads/1/2/3/3/12339018/impact_connector_6_-_landscapes_-martin_williams.pdf

Wilson H, undated, *Hinewai information and map*, pamphlet, Maurice White Native Forest Trust, Akaroa

Wilson H, 2002, *Hinewai: The journal of a New Zealand naturalist*, Shoal Bay Press, Christchurch

Wilson H, 2013, *Natural history of Banks Peninsula,* revised edition, University of Canterbury Press, Christchurch

Wilson S, 2019, 'The tragedy of Ihumātao', New Zealand Herald, 24 July

Wood V and E Pawson, 2008, 'The Banks Peninsula forests and Akaroa cocksfoot, Explaining a New Zealand forest transition', *Environment and History*, 14(4), 449-468

Wright M, S Gepp and D Hall, 2019, *A review of the Resource Management (National Environmental Standards for Plantation Forestry) Regulations 2017*, Environmental Defence Society, Auckland

Xuan P, 2020, 'Models of green parks of several developed cities in the world: Visionary recommendations for Ho Chi Minh City', *Advances in Economics, Business and Management Research* 115: 430

Young R, 2016, *Mackenzie wilding conifer strategy 2016-2030*, Te Manahuna Consulting, Twizel

Young R, 2019, Submission on 'Enduring stewardship of Crown pastoral land', Te Manahuna Consulting, Twizel

Aotearoa New Zealand's distinctive natural and cultural landscapes are an integral part of our individual and collective wellbeing. But despite the importance of these landscapes, and this being recognised in law for close to 30 years, we are still seeing poor outcomes. This prompted the Environmental Defence Society to initiate a project to investigate why the landscape management system is failing and how it might be strengthened.

This Synthesis Report brings together the large body of work we have undertaken on landscape management. It includes the findings of four case studies focusing on landscape management in Te Manahuna Mackenzie Basin, Te Pātaka o Rākaihautū Banks Peninsula, the Waitākere Ranges and the Hauraki Gulf Islands. A fifth case study investigated potential linkages between tourism and landscape protection. The Report also includes an in-depth review of the case law and landscape assessment practices in Aotearoa New Zealand and an investigation into the application of a te ao Māori approach to landscape. In addition, we have included the findings of an in-depth international review of innovative landscape protection models.

We found that the pressures on our landscapes have increased markedly over the past two decades and many landscapes are still in decline. The pressures are becoming stronger, more multifaceted, and they play out in different ways in different places. We found that in the face of intense development pressures, the current system is not performing well.

We conclude the report with identifying how the current proposed reforms to the resource management system could be better configured to provide more robust landscape protection. We also propose a more enduring landscape protection model to better protect Aotearoa New Zealand's landscapes for our future generations.

ISBN 978-0-9951186-3-8

EDS
ENVIRONMENTAL DEFENCE SOCIETY